

Tightly Secure Inner-Product Functional Encryption Revisited: Compact, Lattice-based, and More

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Abstract. Currently, the only tightly secure inner-product functional encryption (IPFE) schemes in the multi-user and multi-challenge setting are the IPFE scheme due to Tomida (Asiacrypt 2019) and its derivatives. However, these tightly secure schemes have large ciphertext expansion and are all based on the matrix decisional Diffie-Hellman (DDH) assumption.

To improve the efficiency of tightly secure IPFE and enrich the diversity of its underlying assumptions, we construct a set of tightly secure IPFE schemes, with very compact ciphertexts and efficient algorithms, from the matrix DDH, the decisional composite residuosity (DCR), and the learning-with-errors (LWE) assumptions, respectively. In particular,

- our DDH-based scheme has about $5 \times \sim 100 \times$ *shorter* public/secret keys, $2 \times \sim 2.9 \times$ *shorter* ciphertexts and about $5 \times \sim 100 \times$ *faster* algorithms than all existing tightly secure IPFE schemes, at the price of 3~7 bits of decreased security level, resolving an open problem raised by Tomida (Asiacrypt 2019);
- our DCR-based scheme constitutes the *first* tightly secure IPFE scheme from the DCR assumption;
- our LWE-based scheme is the *first* almost tightly secure IPFE scheme from lattices, hence achieving post-quantum security.

We obtain our schemes by proposing a generic and flexible construction of IPFE with a core technical tool called *two-leveled* inner-product hash proof system (TL-IP-HPS). Specifically, our IPFE construction is in a compact design, and to achieve tight security, we propose an economic proof strategy to reuse the spaces in such compact IPFE. Interestingly, our new proof strategy also applies to the loosely secure IPFE schemes proposed by Agrawal, Libert and Stehlé (Crypto 2016), and yields a *tighter* bound on their security loss.

1 Introduction

Functional encryption (FE) [16, 55, 45] is an advanced cryptographic primitive that enables fine-grained access control on ciphertexts, rather than the “all-or-nothing” access pattern induced by the traditional public key encryption schemes. The access control capability of FE is characterized by a family of functions $\mathcal{F}$: the holder of (master) secret key sk can generate functional secret keys sk_f for functions $f \in \mathcal{F}$, and with sk_f , one can decrypt a ciphertext ct of message x to obtain the function value $f(x)$ and nothing else.

INNER-PRODUCT FUNCTIONAL ENCRYPTION. In this work, we focus on inner-product functional encryption (IPFE) [2, 8, 13, 61], which is an important type of FE that reveals only the inner product values of encrypted messages when decrypted using functional secret keys. Namely, the function is indexed by a vector $\mathbf{y} \in \mathbb{Z}^m$, and the message is a vector $\mathbf{x} \in \mathbb{Z}^m$ as well. Decryption of a ciphertext ct of $\mathbf{x}$ using the functional key $sk_{\mathbf{y}}$ associated with $\mathbf{y}$ yields the inner product $\langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z}$. In fact, inner product is an extremely powerful functionality that can be used for a variety of statistical computations like conjunctions, disjunctions and weighted sums, which make IPFE suitable for many practical applications such as biometric authentication, nearest-neighbour search and statistical analysis [2, 22, 43].

The standard security requirements for IPFE is indistinguishability against chosen-plaintext/ciphertext attacks (CPA/CCA security for short). CPA security stipulates the hardness of distinguishing a ciphertext of $\mathbf{x}_0$ from a ciphertext of $\mathbf{x}_1$, even if an adversary $\mathcal{A}$ can choose $\mathbf{x}_0, \mathbf{x}_1$ adaptively and have access to a functional key generation oracle: each time $\mathcal{A}$ submits a vector $\mathbf{y}$ and obtains the corresponding functional secret key $sk_{\mathbf{y}}$, as long as $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$ holds. CCA security is stronger and also allows $\mathcal{A}$ to have access to a decryption oracle.

It is Agrawal et al. [8] who proposed the first IPFE schemes achieving (adaptive) CPA security from the standard decisional Diffie-Hellman (DDH), decisional composite residuosity (DCR) or learning-with-errors (LWE) assumptions. Their schemes are pretty efficient and have very compact ciphertexts. Later, Zhang et al. [66] and Benhamouda et al. [13] constructed the first (adaptively) CCA-secure IPFE schemes based on the DDH and DCR assumptions, respectively. Meanwhile, IPFE schemes are extended with other attractive properties, like function-hiding IPFE [14, 22, 62, 43] and multi-input IPFE [5, 4, 20, 1].

TIGHT SECURITY. The security of IPFE (and also other primitives) is usually established by security reductions, which turn an efficient adversary $\mathcal{A}$ breaking the security of the IPFE scheme into an efficient algorithm $\mathcal{B}$ solving the underlying computationally hard problem, with approximately the same running time. The ratio of $\mathcal{A}$'s advantage $\epsilon_{\mathcal{A}}$ to $\mathcal{B}$'s advantage $\epsilon_{\mathcal{B}}$ is defined as the *security loss* factor $\ell := \epsilon_{\mathcal{A}}/\epsilon_{\mathcal{B}}$, which measures the quality of the security reduction.

If ℓ is a polynomial in the number of users n and/or ciphertexts Q in the cryptographic system, the security reduction is called a loose one. When instantiating the scheme in a theoretically sound manner, one has to compensate the security loss ℓ by increasing key lengths or group sizes, which will slow the algorithms of

IPFE and enlarge the sizes of keys and ciphertexts. Moreover, it may not be clear at the time of deployment how many users will be involved and how many ciphertexts will be generated in the lifetime of the system. If the estimation is too small, the provided security guarantee will not be backed by the security proof. Consequently, it is desirable to have *tight* or *almost tight* security where ℓ is only a small constant or a small polynomial in the security parameter λ , and in particular, in both cases, ℓ should be independent of the number of users n and ciphertexts Q . Due to its importance, tightly secure cryptographic schemes have become an active area in the last decade, ranging from basic primitives like public-key encryption and digital signature, to more advanced ones like identity-based encryption and key exchange [40, 18, 15, 27, 35, 28, 6, 29, 34, 44, 23, 36, 58, 57] (to name a few).

TIGHTLY SECURE IPFE. For IPFE, most of the existing schemes established their security only in the single-user and single-challenge setting. In the more realistic setting of multiple users and multiple challenge ciphertexts, their security will suffer from a large loss factor of $n \cdot Q$ due to a hybrid argument. As shown in [18, 15, 37, 61], n and Q can be as large as 2^{20} or even 2^{30} in realistic settings. For example, when using IPFE for DNA analysis, Q can be as large as 2^{27} (e.g., GenBank operated by the National Center for Biotechnology Information has more than 2^{27} DNA sequences [30]). Consequently, the security loss $n \cdot Q$ is large enough to make it worthwhile to consider the tight security of IPFE.

Up to now, the IPFE schemes in [61, 48, 49] are the only ones achieving tight CPA/CCA security in the multi-user and multi-challenge setting (mCPA/mCCA for short). These schemes are all based on the (matrix) DDH assumptions and stem from the tight mCPA secure IPFE scheme due to Tomida [61].

However, the efficiency of the existing tightly secure IPFE schemes is far from satisfactory, and tight security from other assumptions remains to be exploited.

- **Efficiency:** Among all the tightly secure IPFE schemes [61, 48, 49], the DDH-based Tomida scheme [61] is currently the most efficient one. However, its ciphertext consists of $3m$ group elements when encrypting a vector $\mathbf{x} \in \mathbb{Z}^m$ of length m , showing that the ciphertext is about $3\times$ the size of the input vector. In comparison, the loosely secure DDH-based scheme proposed by Agrawal, Libert and Stehlé (ALS) [8] has only $m + 2$ group elements in ciphertext. If we define *ciphertext expansion* as the ratio of the ciphertext length to the input vector length, then the Tomida scheme has a ciphertext expansion as large as 3, in sharp contrast to the ciphertext expansion of $1 + \frac{2}{m}$ in the ALS scheme. Moreover, all algorithms of the Tomida scheme are about $m\times$ slower than those of ALS. So the tightly secure Tomida scheme would be much less efficient than the loosely secure ALS scheme even after the compensation of security loss, and it was left as an open problem by Tomida [61] to construct more compact tightly secure IPFE schemes.
- **Diversity of security assumptions:** All existing tightly secure IPFE schemes [61, 48, 49] have only instantiations under the (matrix) DDH assumptions. It is desirable to extend the diversity of the computational assumptions that tightly secure IPFE schemes can be based on. Moreover, the

developing speed of post-quantum cryptography makes the diversity extension to lattice-based tightly secure IPFE especially welcomed.

These raise the following question:

Can we construct more compact and efficient IPFE schemes with tight security and enlarge the basket of assumptions with DCR and especially LWE?

1.1 Our Contributions

In this work, we revisit tightly secure IPFE in the multi-user and multi-challenge setting, and answer the above question affirmatively.

Table 1. Comparison of our (almost) tightly secure IPFE schemes with the most efficient tightly secure scheme Tomida [61] in the standard model prior to our work. Here m denotes the vector length. The columns $|\mathbf{mpk}|$, $|\mathbf{msk}|$, $|\mathbf{sk}_y|$ and $|\mathbf{ct}|$ count the size of master public key $\mathbf{mpk}$, master secret key $\mathbf{msk}$, functional key $\mathbf{sk}_y$ and ciphertext $\mathbf{ct}$, respectively, in terms of numbers of group elements, lattice vectors and matrices, where we do not count y in $\mathbf{sk}_y$ since it can be made public. The column **Ciphertext Expansion** shows the ratio of ciphertext length to input vector length, i.e., $\frac{|\mathbf{ct}|}{m}$, where l is an LWE parameter. The column **Security Loss** shows the security loss factor of the mCPA security reductions in the multi-challenge setting, where λ denotes the security parameter. The column **Assumption** shows the computational assumption on which the security is based. Jumping ahead, our schemes are parameterized by a positive integer $L \in \mathbb{N}$, and here we present our schemes under a general L as well as under two concrete settings $L = 5$ and $L = 100$. We approximate $\lceil \frac{m}{L} \rceil$ as $\frac{m}{L}$ for ease of exposition when counting the parameter sizes.

Tightly Secure IPFE Scheme	$ \mathbf{mpk} $	$ \mathbf{msk} $	$ \mathbf{sk}_y $	$ \mathbf{ct} $	Ciphertext Expansion	Security Loss	Assumption
Tomida [61]	$m^2 + 2$	$2m^2$	$2m$	$3m$	3	$O(1) = 3$	DDH
Ours $\begin{cases} \text{general } L \\ L = 5 \\ L = 100 \end{cases}$	$\frac{m^2}{L} + 2$	$\frac{2m^2}{L}$	$\frac{2m}{L}$	$m + \frac{2m}{L}$	$1 + \frac{2}{L}$	$O(1) = 3L$	DDH
	$\frac{m^2}{5} + 2$	$\frac{2m^2}{5}$	$\frac{2m}{5}$	$m + \frac{2m}{5}$	1.4	$O(1) = 15$	
	$\frac{m^2}{100} + 2$	$\frac{2m^2}{50}$	$\frac{m}{50}$	$m + \frac{m}{50}$	1.02	$O(1) = 300$	
Ours $\begin{cases} \text{general } L \\ L = 5 \\ L = 100 \end{cases}$	$\frac{m^2}{L} + 1$	$\frac{m^2}{L}$	$\frac{m}{L}$	$m + \frac{m}{L}$	$1 + \frac{1}{L}$	$O(1) = 3L$	DCR
	$\frac{m^2}{5} + 1$	$\frac{m^2}{5}$	$\frac{m}{5}$	$m + \frac{m}{5}$	1.2	$O(1) = 15$	
	$\frac{m^2}{100} + 1$	$\frac{m^2}{100}$	$\frac{m}{100}$	$m + \frac{m}{100}$	1.01	$O(1) = 300$	
Ours $\begin{cases} \text{general } L \\ L = 5 \\ L = 100 \end{cases}$	$\frac{m}{L} + 1$	$\frac{m}{L}$	$\frac{m}{L}$	$\frac{m}{L} + 1$	$1 + \frac{1}{L}$	$O(\lambda^2) = O(L\lambda^2)$	LWE
	$\frac{m}{5} + 1$	$\frac{m}{5}$	$\frac{m}{5}$	$\frac{m}{5} + 1$	$1 + \frac{1}{5}$	$O(\lambda^2) = O(5\lambda^2)$	
	$\frac{m}{100} + 1$	$\frac{m}{100}$	$\frac{m}{100}$	$\frac{m}{100} + 1$	$1 + \frac{1}{100}$	$O(\lambda^2) = O(100\lambda^2)$	

GENERIC APPROACH TO TIGHTLY SECURE IPFE WITH COMPACT DESIGN.

We propose a generic construction for tightly mCPA secure IPFE in the standard model. The main technical tool in our construction is a new variant of hash proof system [21] called *two-leveled* inner-product hash proof system (TL-IP-HPS), where the outer-level HPS can trigger an inner-level core hash function

information-theoretically during the security proofs, and this plays an important role for the tight security of the resulting IPFE. Our IPFE construction is in a compact design, and to achieve tight security, we propose an economic proof strategy to reuse the constrained spaces in such compact IPFE.

Moreover, our generic construction is very flexible and equipped with a parameter $L \in \mathbb{N}$ controlling the trade-off between the efficiency and loss factor of the resulting IPFE scheme. By configuring L , one can obtain tightly secure IPFE schemes with desired (constant) security loss factors. As L increases, the security loss ℓ grows with $\ell = 3L$ and this will decrease the security level of the resulting IPFE by at most $\log L$ bits. This is worthwhile, as it brings a big benefit that the resulting IPFE becomes roughly $L \times$ more efficient. Besides, as long as L is a small constant (e.g., $L = 5$ or $L = 100$), the security is still in the realm of tight security with a constant security loss $\ell = 3L$.

By instantiating our generic construction, we are able to obtain tightly secure IPFE schemes from (matrix) DDH or DCR, and almost tightly secure IPFE schemes from LWE. We refer to Table 1 for comparison of our (almost) tightly secure IPFE schemes with the Tomida scheme.

It is interesting to note that our construction is the *first* generic construction for IPFE that has instantiations under a variety of assumptions including DDH, DCR and LWE, no matter for tightly secure IPFE or loosely secure IPFE.

MORE EFFICIENT TIGHTLY SECURE IPFE FROM (MATRIX) DDH. Thanks to our compact design and novel proof strategy, we obtain tightly secure (matrix) DDH-based IPFE schemes with very compact ciphertexts and efficient algorithms, resolving the open problem raised by Tomida [61].

In particular, our tightly secure DDH-based IPFE scheme outperforms the Tomida scheme [61], which was serving for the most efficient tightly secure IPFE scheme prior to our work: our DDH-based scheme has about $5 \times \sim 100 \times$ *shorter* public/secret keys, $2 \times \sim 2.9 \times$ *shorter* ciphertexts and about $5 \times \sim 100 \times$ *faster* algorithms than the DDH-based Tomida scheme. The only price is the $5 \times \sim 100 \times$ larger (yet still constant) security loss, which will decrease the security level by at most $3 \sim 7$ bits. Nevertheless, even after the compensation of the extra constant security loss, all algorithms of our scheme are still about $5 \times \sim 100 \times$ *more efficient* than theirs. This suggests that our schemes significantly improve the efficiency of tightly secure IPFE.

In terms of ciphertext expansion, our tightly secure DDH-based scheme has very compact ciphertexts with ciphertext expansion only $1.02 \sim 1.4$, which approaches the ideal factor 1 and is much smaller than the ciphertext expansion 3 of the Tomida scheme [61]. In some situations, our scheme even has shorter ciphertexts than the ALS scheme after the compensation of security loss (see Appendix A for an example).

THE FIRST (ALMOST) TIGHTLY SECURE IPFE SCHEMES FROM DCR/LWE. Thanks to the generality of our construction, we are able to obtain tightly secure IPFE schemes under not only the (matrix) DDH assumption but also the DCR and LWE assumptions, by instantiating TL-IP-HPS accordingly.

In particular, our DCR-based scheme is the *first* tightly secure IPFE scheme from the DCR assumption, while our LWE-based scheme is the *first* almost tightly secure IPFE scheme from lattices, both of which enjoy very compact ciphertexts and efficient algorithms. Similar to our DDH-based scheme, our DCR-based and LWE-based schemes may even have shorter ciphertexts than the ALS schemes in certain scenarios (see Appendix A for more details).

Notably, to obtain our LWE-based IPFE scheme, we follow a recent paradigm [39] which generalizes HPS to probabilistic HPS to handle the noises inherent in LWE. In our case, we slightly adapt our generic approach by allowing the two levels of TL-IP-HPS to be *probabilistic* outer-level HPS and *randomized* inner-level core hash function, and give LWE-based instantiations of such TL-IP-HPS. The security loss factor is quadratic in the security parameter λ , which essentially stems from the loss factor of the random self-reducibility of LWE (i.e., the reduction from LWE to Multi-secret LWE) [9, 39], and thus our LWE-based IPFE scheme achieves almost tight security.

MORE APPLICATIONS. We note that our tightly mCPA secure IPFE schemes and our new proof strategy can find more applications.

Firstly, by applying existing transformations [48, 49, 61, 4, 46, 47], we can upgrade our tightly mCPA secure IPFE schemes to tightly secure IPFE schemes of other types, like mCCA secure IPFE, function-hiding IPFE and multi-input IPFE, and thus improve the efficiency of such tightly secure IPFE schemes.

For example, Liu et al. [48, 49] converted the tightly mCPA secure Tomida scheme [61] to tightly mCCA secure IPFE schemes by adding tightly secure non-interactive zero-knowledge (NIZK) proofs to reject ill-formed ciphertexts. Besides, Tomida [61] presented tightly function-hiding secure IPFE and tightly secure multi-input IPFE schemes by adapting the generic transformations in [4, 46, 47] to the tightly mCPA secure Tomida scheme.

Now by replacing the tightly mCPA secure Tomida scheme [61] with our more efficient and compact tightly secure IPFE schemes, we are able to get more efficient and compact schemes, including tightly mCCA secure IPFE schemes, tightly function-hiding secure IPFE schemes and tightly secure multi-input IPFE schemes, via basically the same (semi-)generic transformations in [48, 49, 61, 4, 46, 47]. In particular, the public/secret keys and ciphertexts of the resulting IPFE schemes will be much shorter than those of the existing tightly secure schemes, benefiting from the improved efficiency of our tightly mCPA secure IPFE schemes.

Secondly, our economic proof strategy of reusing the constrained spaces in a compact scheme can also be applied to the ALS scheme¹ [8] to improve its loose mCPA security loss factor from $O(Q)$ to $O(m)$, where Q denotes the number of ciphertexts and m is the length of message vectors. This broadens our understanding of the ALS scheme, and we are optimistic that our proof strategy might find more applications in building efficient and compact schemes and giving tighter analysis on their security loss factors.

¹ To be more precise, our proof strategy applies to the DDH-based and the DCR-based ALS schemes to improve their loss factors, but not to the LWE-based ALS scheme.

2 Technical Overview

In this section, we provide a technical overview of our new techniques in improving the efficiency and enriching the diversity of the underlying assumptions for tightly secure IPFE schemes. We firstly review the ALS scheme [8] and the tightly secure Tomida scheme [61] in Subsect. 2.1. Then in Subsect. 2.2, we explain our main technical tool, namely *two-leveled* inner-product hash proof system (TL-IP-HPS), and show how to instantiate it from the (matrix) DDH, DCR and LWE assumptions, respectively. Finally, in Subsect. 2.3, we give a high-level overview of our generic IPFE construction, and explain the main ideas in our novel economic proof strategy for achieving tight security. Besides, in Appendix B, we compare our techniques with existing IPFE frameworks from (variants of) HPS, and discuss simulation-based security of IPFE in the multi-challenge setting. For the ease of exposition, here we focus on the security in the single-user setting.

2.1 Recap: The ALS Scheme and The Tomida Scheme

Below we review the DDH-based ALS scheme and the DDH-based Tomida scheme, and explain the difficulties in constructing tightly secure IPFE schemes in a compact way and from assumptions other than (matrix) DDH. Let $\mathbb{G}$ be a cyclic group of prime order p . We use $\llbracket \mathbf{A} \rrbracket$ to denote implicit representation [26] of matrix $\mathbf{A}$ in $\mathbb{G}$ (see Appendix E.1 for more details).

THE ALS SCHEME [8]. Let $\mathbf{A} := \begin{pmatrix} 1 \\ a \end{pmatrix} \in \mathbb{Z}_p^{2 \times 1}$ with $a \leftarrow_s \mathbb{Z}_p$. The DDH-based ALS scheme for vectors of length m works as follows.

- $\text{mpk} := (\llbracket \mathbf{A} \rrbracket, \llbracket \mathbf{KA} \rrbracket)$ and $\text{msk} := \mathbf{K}$ with $\mathbf{K} \leftarrow_s \mathbb{Z}_p^{m \times 2}$.
- $sk_{\mathbf{y}} := (\mathbf{y}^\top \mathbf{K}, \mathbf{y})$, where $\mathbf{y} \in \mathbb{Z}^m$.
- $ct_{\mathbf{x}} := (\llbracket \mathbf{Aw} \rrbracket, \llbracket \mathbf{KA} \cdot w + \mathbf{x} \rrbracket)$ with $w \leftarrow_s \mathbb{Z}_p$, where $\mathbf{x} \in \mathbb{Z}^m$.
- Decryption of $ct_{\mathbf{x}}$ using $sk_{\mathbf{y}}$ solves the discrete logarithm of $\mathbf{y}^\top \cdot \llbracket \mathbf{KA} \cdot w + \mathbf{x} \rrbracket - \mathbf{y}^\top \mathbf{K} \cdot \llbracket \mathbf{Aw} \rrbracket = \llbracket \langle \mathbf{x}, \mathbf{y} \rangle \rrbracket$.

We briefly recall its single-challenge security proof. The adversary $\mathcal{A}$ can get mpk and a collection of functional secret keys $\{sk_{\mathbf{y}}\}_{\mathbf{y} \in \mathcal{Q}}$ with vectors $\mathbf{y}$ in certain set $\mathcal{Q}$, from which $\mathcal{A}$ learns the following information about $\text{msk} = \mathbf{K}$:

$$\text{mpk} : \llbracket \mathbf{K} \cdot \mathbf{A} \rrbracket, \quad \{sk_{\mathbf{y}} : \mathbf{y}^\top \cdot \mathbf{K}\}_{\mathbf{y} \in \mathcal{Q}}. \quad (1)$$

Moreover, $\mathcal{A}$ obtains a single challenge ciphertext $ct^* := (\llbracket \mathbf{Aw} \rrbracket, \llbracket \mathbf{KA} \cdot w + \mathbf{x}_\beta \rrbracket)$ that encrypts $\mathbf{x}_\beta \in \{\mathbf{x}_0, \mathbf{x}_1\}$. Its security proof goes with two main steps.

- **Computational Argument:** By the DDH assumption, ct^* can be changed to

$$ct^* := (\overset{\text{-----}}{\llbracket \mathbf{c} \rrbracket \leftarrow_s \mathbb{G}^2}, \llbracket \mathbf{K} \cdot \mathbf{c} + \mathbf{x}_\beta \rrbracket). \quad (2)$$

- **Statistical Argument:** Conditioned on (1), there remains some entropy left in $\text{msk} = \mathbf{K}$, which suffices to randomize the challenge ciphertext in (2).

More precisely, let $\tilde{\mathbf{x}} := \mathbf{x}_1 - \mathbf{x}_0 \in \mathbb{Z}^m$ and $\mathbf{A}^\perp := (-a, 1) \in \mathbb{Z}_p^{1 \times 2}$, we redefine $\text{msk} = \mathbf{K}$ as

$$\mathbf{K} := \tilde{\mathbf{K}} + \boxed{r} \cdot \tilde{\mathbf{x}} \cdot \mathbf{A}^\perp, \quad (3)$$

where $\tilde{\mathbf{K}} \leftarrow_s \mathbb{Z}_p^{m \times 2}$ and $\boxed{r} \leftarrow_s \mathbb{Z}_p$. Then by the facts that $\mathbf{A}^\perp \cdot \mathbf{A} = 0$ and $\{\mathbf{y}^\top \cdot \tilde{\mathbf{x}} = 0\}_{\mathbf{y} \in \mathcal{Q}}$ (since $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$ holds for all $\mathbf{y} \in \mathcal{Q}$ that $\mathcal{A}$ obtains functional secret keys), it is not hard to see that $\boxed{r} \in \mathbb{Z}_p$ is completely hidden in (1). As for the challenge ciphertext in (2), now we have

$$\llbracket \mathbf{K} \cdot \mathbf{c} + \mathbf{x}_\beta \rrbracket = \llbracket \tilde{\mathbf{K}} \cdot \mathbf{c} + \boxed{r} \cdot \tilde{\mathbf{x}} \cdot (\mathbf{A}^\perp \cdot \mathbf{c}) + \mathbf{x}_\beta \rrbracket.$$

Since $\llbracket \mathbf{c} \rrbracket \leftarrow_s \mathbb{G}^2$ is uniformly chosen, the scalar $\mathbf{A}^\perp \cdot \mathbf{c} \in \mathbb{Z}_p$ is non-zero with overwhelming probability. Consequently, the term $\boxed{r} \cdot \tilde{\mathbf{x}} \cdot (\mathbf{A}^\perp \cdot \mathbf{c})$ is uniformly distributed over the space spanned by $\tilde{\mathbf{x}}$ from $\mathcal{A}$'s point of view, and perfectly hides the information about β contained in $\mathbf{x}_\beta = \beta \cdot \tilde{\mathbf{x}} + \mathbf{x}_0$.

The above statistical argument works well in the single-challenge setting. However, in the setting of multi-challenge, there are multiple challenge ciphertexts and involve multiple $\tilde{\mathbf{x}} := \mathbf{x}_1 - \mathbf{x}_0 \in \mathbb{Z}^m$, and we cannot redefine $\text{msk} = \mathbf{K}$ suitable for all challenge ciphertexts simultaneously as in (3). Consequently, as noted by Tomida [61], this proof strategy needs a hybrid argument for each challenge and incurs a security loss of $O(Q)$, with Q the number of challenge ciphertexts.

THE TOMIDA SCHEME [61]. To achieve tight security, Tomida [61] enlarges the sizes of keys and ciphertexts. Below we recall the DDH-based Tomida scheme. Our description takes a different form, but is essentially the same as the original Tomida scheme.

- $\text{mpk} := (\llbracket \mathbf{A} \rrbracket, \llbracket \mathbf{K}_1 \mathbf{A} \rrbracket, \dots, \llbracket \mathbf{K}_m \mathbf{A} \rrbracket)$ and $\text{msk} := (\mathbf{K}_1, \dots, \mathbf{K}_m)$ with $\mathbf{K}_1, \dots, \mathbf{K}_m \leftarrow_s \mathbb{Z}_p^{m \times 2}$.
- $sk_{\mathbf{y}} := (\mathbf{y}^\top \mathbf{K}_1, \dots, \mathbf{y}^\top \mathbf{K}_m, \mathbf{y})$.
- $ct_{\mathbf{x}} := (\llbracket \mathbf{A} w_1 \rrbracket, \dots, \llbracket \mathbf{A} w_m \rrbracket, \llbracket \sum_{i=1}^m \mathbf{K}_i \mathbf{A} \cdot w_i + \mathbf{x} \rrbracket)$ with $w_1, \dots, w_m \leftarrow_s \mathbb{Z}_p$.
- Decryption solves the discrete logarithm of $\mathbf{y}^\top \cdot \llbracket \sum_{i=1}^m \mathbf{K}_i \mathbf{A} \cdot w_i + \mathbf{x} \rrbracket - \sum_{i=1}^m \mathbf{y}^\top \mathbf{K}_i \cdot \llbracket \mathbf{A} w_i \rrbracket = \llbracket \langle \mathbf{x}, \mathbf{y} \rangle \rrbracket$.

By taking advantage of the enlarged spaces, Tomida proposed a clever tight proof strategy that introduces entropy to multiple challenge ciphertexts adaptively and simultaneously during the security proof. However, the price for tight security is the sacrifice of efficiency — the Tomida scheme is about $m \times$ less efficient than the ALS scheme [8]. Moreover, the Tomida scheme is a direct construction from (matrix) DDH and its tight security proof relies on specific properties of (matrix) DDH.

It is unclear whether and how the techniques can be adapted under other assumptions. To solve this problem, we will provide a generic approach to tightly secure IPFE by proposing a new technical tool.

2.2 Technique 1: Two-Leveled Inner-Product Hash Proof System

NEW TECHNICAL TOOL: TL-IP-HPS. Originally, the DDH-based ALS scheme [8] is implicitly built on hash proof system (HPS) [21] while the Tomida scheme

[61] is a direct construction from (matrix) DDH. We abstract both the ALS and the Tomida schemes, and put forward a new technical tool called *Two-Leveled Inner-Product Hash Proof System* (TL-IP-HPS), which is made up of two levels.

- **Outer-level: Inner-Product HPS (IP-HPS).** The outer-level is a new variant of HPS called Inner-Product HPS, which enables computations of inner products of hash values with the help of extra functional keys.

More precisely, similar to traditional HPS [21], IP-HPS is associated with an NP-language $\mathcal{L} \subseteq \mathcal{X}$ that has a hard subset membership problem (SMP) $\{c \leftarrow_s \mathcal{L}\} \stackrel{e}{\approx} \{c \leftarrow_s \mathcal{X}\}$, a projection function α that maps a hashing key $sk \in \mathcal{SK}$ to a projection key $pk = \alpha(sk)$, and has two ways of computing hash values $\mathbf{A}_{sk}(c)$: a public evaluation $\text{Pub}(pk, c, w) = \mathbf{A}_{sk}(c)$ for instance $c \in \mathcal{L}$ with witness w , and a private evaluation $\text{Priv}(sk, c) = \mathbf{A}_{sk}(c)$ for $c \in \mathcal{X}$.

In addition, we introduce a functional key generation function $\mu(\cdot)$ to compute functional keys $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$ w.r.t. vectors $\mathbf{y}$, and a functional private evaluation algorithm $\text{FPriv}(fk_{\mathbf{y}}, c) = \langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$ to compute the inner product of $\mathbf{A}_{sk}(c)$ (which is required to be vector) and $\mathbf{y}$ using $fk_{\mathbf{y}}$.

- **Inner-level: Core Hash Function (co-Hash).** The inner-level is a keyed hash function $\Gamma_s(\cdot)$ called core hash function, which has input space $\mathcal{X}$ (the same as the outer-level IP-HPS) and is indexed by an inner-level hashing key $s \in \mathcal{SK}_{\text{co}}$. We stress that the inner-level hash values $\Gamma_s(c)$ may not be efficiently computable (cf. Remark 1 in Sect. 4 for more explanations).

The two levels of TL-IP-HPS interplay with each other through the following properties of *function smoothness* and *multi-key-extracting*. Roughly speaking, function smoothness allows the outer-level IP-HPS to trigger the inner-level co-Hash and transfer some entropy of $\mathbf{A}_{sk}(c)$ to $\Gamma_s(c)$ information-theoretically; multi-key-extracting makes the entropy in those $\Gamma_s(c)$ amplified in a computational indistinguishable way. These properties will play important roles in the tight security proof of our IPFE construction.

- **Functional Smoothness.** Note that the smoothness of traditional HPS [21] requires the hash value $\mathbf{A}_{sk}(c)$ of instance $c \leftarrow_s \mathcal{X}$ retain some entropy conditioned on $pk = \alpha(sk)$.

For TL-IP-HPS, we formalize a statistical property called functional smoothness. Roughly speaking, it requires that the (outer-level) IP-HPS hash value $\mathbf{A}_{sk}(c)$ of $c \leftarrow_s \mathcal{X}$ is able to *trigger* the (inner-level) core hash value $\Gamma_s(c)$ such that $\mathbf{A}_{sk}(c) \stackrel{s}{\approx} \mathbf{A}_{sk}(c) + \Gamma_s(c) \cdot \tilde{\mathbf{x}}$ for a vector $\tilde{\mathbf{x}}$, even in the presence of both $pk = \alpha(sk)$ and multiple functional keys $\{fk_{\mathbf{y}} = \mu(sk, \mathbf{y})\}_{\mathbf{y} \in \mathcal{Q}}$ with vectors $\mathbf{y}$ in certain set $\mathcal{Q}$. Here $\tilde{\mathbf{x}}$ is required to be a vector orthogonal to all those $\mathbf{y} \in \mathcal{Q}$, since in the presence of the functional keys $\{fk_{\mathbf{y}} = \mu(sk, \mathbf{y})\}_{\mathbf{y} \in \mathcal{Q}}$, the inner products $\{\langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle\}_{\mathbf{y} \in \mathcal{Q}}$ of $\mathbf{A}_{sk}(c)$ and $\mathbf{y} \in \mathcal{Q}$ are completely determined via $\text{FPriv}(fk_{\mathbf{y}}, c) = \langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$.

In fact, we consider *multiple* instances $\{c^{(j)} \leftarrow_s \mathcal{X}\}_j$ and require that their (outer-level) hash values $\{\mathbf{A}_{sk}(c^{(j)})\}_j$ can trigger (inner-level) hash values $\{\Gamma_s(c^{(j)})\}_j$ simultaneously, i.e., $\{\mathbf{A}_{sk}(c^{(j)})\}_j \stackrel{s}{\approx} \{\mathbf{A}_{sk}(c^{(j)}) + \Gamma_s(c^{(j)}) \cdot \tilde{\mathbf{x}}\}_j$.

Moreover, we allow the set $\mathcal{Q}$ for which functional keys are computed and the orthogonal vector $\tilde{\mathbf{x}}$ to be *adaptively* determined by an unbounded adversary. We refer to Def. 7 in Sect. 4 for the formal definition of this property.

- **Multi-Key-Extracting.** It is a computational property for co-Hash that stipulates the pseudorandomness of multiple core hash values $\{\Gamma_{s_i}(c_{i,j})\}_{i,j}$ for multiple instances $\{c_{i,j} \leftarrow \mathcal{X}\}_{i,j}$ under multiple keys $\{s_i \leftarrow \mathcal{SK}_{\text{co}}\}_i$. Similar properties were formalized in [38, 37] for other variants of HPS.

DDH-BASED INSTANTIATION TLHPS_{MDDH}. It is not hard to extract a (matrix) DDH-based TL-IP-HPS instantiation from the ALS and the Tomida schemes:

- The language is $\mathcal{L} = \{[\mathbf{c}] = [\mathbf{A}w] \mid w \in \mathbb{Z}_p\} \subseteq \mathcal{X} = \mathbb{G}^2$ with $\mathbf{A} = \begin{pmatrix} 1 \\ a \end{pmatrix} \in \mathbb{Z}_p^{2 \times 1}$.
- The (outer-level) IP-HPS has hashing key $sk = \mathbf{K} \in \mathbb{Z}_p^{m \times 2}$, projection key $pk = [\mathbf{K}\mathbf{A}]$ and hash values $\mathbf{A}_{sk}([\mathbf{c}]) = [\mathbf{K}\mathbf{c}]$: the public evaluation is $\text{Pub}(pk, [\mathbf{c}], w) = [\mathbf{K}\mathbf{A} \cdot w]$, the private evaluation is $\text{Priv}(sk, [\mathbf{c}]) = [\mathbf{K} \cdot \mathbf{c}]$.
In addition, the functional key generation function $\mu(\cdot)$ computes functional keys w.r.t. vectors $\mathbf{y}$ via $fk_{\mathbf{y}} = \mu(sk, \mathbf{y}) = \mathbf{y}^\top \mathbf{K}$, and the functional private evaluation algorithm FPriv is defined by $\text{FPriv}(fk_{\mathbf{y}}, [\mathbf{c}]) = [\mathbf{y}^\top \mathbf{K} \cdot \mathbf{c}]$ which equals the inner product $\langle \mathbf{A}_{sk}([\mathbf{c}]), \mathbf{y} \rangle = \langle [\mathbf{K}\mathbf{c}], \mathbf{y} \rangle$.
- The (inner-level) co-Hash has hashing key $s \in \mathbb{Z}_p$, and defines hash values $\Gamma_s([\mathbf{c}]) = [s \cdot \mathbf{A}^\perp \cdot \mathbf{c}]$, where $\mathbf{A}^\perp = (-a, 1) \in \mathbb{Z}_p^{1 \times 2}$.²

DCR-BASED INSTANTIATION TLHPS_{DCR}. Moreover, we also propose a DCR-based TL-IP-HPS instantiation by adapting the DCR-based ALS scheme [8]. More precisely, let $N = pq$ be a product of safe primes p, q where $p = 2p' + 1$ and $q = 2q' + 1$ for primes p', q' , and let $N' = p'q'$. Then the multiplicative group $\mathbb{Z}_{N^2}^*$ can be decomposed as an internal direct product $\mathbb{Z}_{N^2}^* = \mathbb{G}_N \cdot \mathbb{G}_{N'} \cdot \mathbb{G}_2 \cdot \{-1, 1\}$, where $\mathbb{G}_N$ (resp., $\mathbb{G}_{N'}$, $\mathbb{G}_2$) is a cyclic group of order N (resp., N' , 2). $\mathbb{G}_N$ has a generator $g_N := 1 + N$, while a generator of $\mathbb{G}_{N'}$ can be sampled via $g_{N'} := g^{2N} \bmod N^2$ with $g \leftarrow \mathbb{Z}_{N^2}^*$ (see Appendix F.1 for more details).

- The language is $\mathcal{L} = \mathbb{G}_{N'} = \{c = (g_{N'})^w \bmod N^2 \mid w \in \mathbb{Z}\} \subseteq \mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$.
- The (outer-level) IP-HPS has hashing key $sk = (k_1, \dots, k_m)^\top \in \mathbb{Z}^m$, projection key $pk = (g_{N'}^{k_1}, \dots, g_{N'}^{k_m}) \bmod N^2$ and hash values $\mathbf{A}_{sk}(c) = (c^{k_1}, \dots, c^{k_m}) \bmod N^2$: the public evaluation is $\text{Pub}(pk, c, w) = ((g_{N'}^{k_1})^w, \dots, (g_{N'}^{k_m})^w) \bmod N^2$ while the private evaluation is $\text{Priv}(sk, c) = (c^{k_1}, \dots, c^{k_m}) \bmod N^2$.
In addition, the functional key generation function $\mu(\cdot)$ computes functional keys w.r.t. vectors $\mathbf{y} = (y_1, \dots, y_m)^\top$ via $fk_{\mathbf{y}} = \mu(sk, \mathbf{y}) = \sum_{i=1}^m y_i \cdot k_i \in \mathbb{Z}$, and the functional private evaluation algorithm FPriv is defined by $\text{FPriv}(fk_{\mathbf{y}}, c) = c^{\sum_{i=1}^m y_i \cdot k_i} \bmod N^2$ which equals the inner product $\langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle = \prod_{i=1}^m (c^{k_i})^{y_i} \bmod N^2$.
- The (inner-level) co-Hash has hashing key $s \in \mathbb{Z}_{N'}^*$, and defines hash values $\Gamma_s(c) = c^{s \cdot N'} \bmod N^2$.

² Here the instantiation of $\Gamma_s(\cdot)$ is slightly different from but equivalent to that in the main body. The same applies to the following DCR- and LWE-based instantiations.

LWE-BASED INSTANTIATION TLHPS_{LWE}. Furthermore, we propose an LWE-based TL-IP-HPS instantiation. Note that due to the LWE noises, the private and public evaluations of all existing LWE-based HPS schemes [32, 42, 12, 65, 41] are close but not perfectly consistent. However, in many applications of HPS (e.g., in building public-key encryptions [21]), the security proofs need to switch between the two evaluation modes in an indistinguishable way, so the inconsistency of the two evaluations cannot guarantee their indistinguishability. To solve this problem, a recent work [39] extended HPS to *probabilistic* private and public evaluations which are required to be statistically indistinguishable. We propose an LWE-based TLHPS_{LWE} inheriting this probabilistic feature, by allowing Pub and Priv of the (outer-level) IP-HPS to be *probabilistic* algorithms and allowing the (inner-level) co-Hash $\Gamma_s(\cdot)$ to be a *randomized* function. More precisely, let $n, q, l \in \mathbb{N}$ be LWE parameters and $B, B', B'' \in \mathbb{N}$ be error bounds. Below we give a simplified description of our TLHPS_{LWE} for better readability.³

- The language is $\mathcal{L} = \{\mathbf{c} = \mathbf{A}\mathbf{w} + \mathbf{e} \mid \mathbf{w} \in \mathbb{Z}_q^n, \mathbf{e} \in [-B, B]^l\} \subseteq \mathcal{X} = \mathbb{Z}_q^l$ with $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{l \times n}$.
- The (outer-level) IP-HPS has hashing key $sk = \mathbf{K} \in \mathbb{Z}^{m \times l}$ (of bounded norm), projection key $pk = \mathbf{KA} \in \mathbb{Z}_q^{m \times n}$ and hash values $\Lambda_{sk}(\mathbf{c}) = \mathbf{Kc} \in \mathbb{Z}_q^m$: the *probabilistic* public evaluation is $\text{Pub}(pk, \mathbf{c}, \mathbf{w}) = \mathbf{KA} \cdot \mathbf{w} + \mathbf{e}' \in \mathbb{Z}_q^m$ with $\mathbf{e}' \leftarrow_s [-B', B']^m$, and the *probabilistic* private evaluation is $\text{Priv}(sk, \mathbf{c}) = \mathbf{K} \cdot \mathbf{c} + \mathbf{e}' \in \mathbb{Z}_q^m$ with $\mathbf{e}' \leftarrow_s [-B', B']^m$. Here $\mathbf{e}'$ is a bigger noise that can smudge the small difference (i.e., $\mathbf{Ke}$) between the two probabilistic evaluation modes, thus making them statistically indistinguishable.
- In addition, the functional key generation function $\mu(\cdot)$ computes functional keys w.r.t. vectors $\mathbf{y}$ via $fk_{\mathbf{y}} = \mu(sk, \mathbf{y}) = \mathbf{y}^\top \mathbf{K} \in \mathbb{Z}^{1 \times l}$, and the functional private evaluation algorithm FPriv is defined by $\text{FPriv}(fk_{\mathbf{y}}, \mathbf{c}) = \mathbf{y}^\top \mathbf{K} \cdot \mathbf{c}$ which equals the inner product $\langle \Lambda_{sk}(\mathbf{c}), \mathbf{y} \rangle = \langle \mathbf{Kc}, \mathbf{y} \rangle$.
- The (inner-level) co-Hash has hashing key $s = \mathbf{s} \in \mathbb{Z}^{l-n}$, and defines *randomized* core hash function $\Gamma_s(\mathbf{c}) = \mathbf{s}^\top \cdot \mathbf{A}^\perp \cdot \mathbf{c} + e'' \in \mathbb{Z}_q$, where $\mathbf{A}^\perp \in \mathbb{Z}_q^{(l-n) \times l}$ satisfies $\mathbf{A}^\perp \cdot \mathbf{A} = \mathbf{0}$ and $e'' \leftarrow_s [-B'', B'']$.

We stress that our LWE-based TLHPS_{LWE} is non-trivial, and the main technical challenge is to design a suitable (inner-level) co-Hash function:

- On one hand, the co-Hash should be “*weak*” enough, i.e., containing less entropy, so that it can be triggered by the outer-level IP-HPS. To this end, we sample $sk = \mathbf{K} \leftarrow_s \mathbb{Z}^{m \times l}$ according to discrete Gaussians, and give a careful analysis of the distribution of $sk = \mathbf{K}$ conditioned on $pk = \mathbf{KA}$ and multiple functional keys $\{fk_{\mathbf{y}} = \mathbf{y}^\top \mathbf{K}\}_{\mathbf{y} \in \mathcal{Q}}$. Then we rely on the conditional distribution of $sk = \mathbf{K}$ to derive the form of co-Hash that can be triggered by the outer-level IP-HPS hash values $\{\text{Priv}(sk, \mathbf{c}^{(j)}) = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)}\}_j$.

³ More precisely, the following instantiation that $\mathcal{X} = \mathbb{Z}_q^l$ and the dimension of $s = \mathbf{s} \in \mathbb{Z}^{l-n}$ are not accurate, and we refer to the main body for the accurate descriptions.

In particular, this part involves a fine-grained statistical analysis of discrete Gaussians, and serves for the proof of functional smoothness of our $\text{TLHPS}_{\text{LWE}}$ (i.e., Theorem 3).

- On the other hand, the co-Hash should be “*strong*” enough, i.e., containing sufficient entropy, so that it can be amplified to uniformly random elements by reducing to the underlying computational assumption, namely the LWE assumption in this case. To this end, we carefully choose the form of co-Hash which is closest to LWE, so that we can establish its multi-key-extracting property based on the LWE assumption. This motivates us to relax LWE-based co-Hash to be *randomized*, by adding a small noise e'' .

Moreover, to achieve *tight* multi-key-extracting, we rely on several (almost) tight reductions among LWE and variants of LWE [33, 10, 9, 31, 39], including tight reductions from LWE to multi-instance LWE with unbounded samples and to multi-instance LWE with noisy secrets. This part serves for the proof of tight multi-key-extracting of our $\text{TLHPS}_{\text{LWE}}$ (i.e., Theorem 4).

Overall, we prove the functional smoothness and the multi-key-extracting properties of the above three instantiations by statistical analysis and by the random self-reducibility of DDH, DCR and LWE⁴, respectively.

With the new technical tool of TL-IP-HPS, we are ready to present our generic construction of IPFE and prove its tight security based on the functional smoothness and the multi-key-extracting properties of TL-IP-HPS (as well as the hardness of SMP) in a clean and unified way.

2.3 Technique 2: Compact IPFE Design & Economic Proof Strategy

GENERIC IPFE CONSTRUCTION IN A COMPACT DESIGN. Our construction of IPFE is designed in a generic and flexible manner. Specifically, our IPFE does not only use a TL-IP-HPS scheme $\text{TLHPS} = (\mathcal{L}, \mathcal{X}, \alpha(\cdot), \mathbf{A}(\cdot), \text{Pub}, \text{Priv}, \mu(\cdot), \text{FPriv}, \Gamma(\cdot))$ as the core building block, but also is parameterized by a positive integer $L \in \mathbb{N}$, thus is denoted by $\text{IPFE} = \text{IPFE}[\text{TLHPS}, L]$. For vector length m , we define $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$.

- $\text{mpk} := (pk_1, \dots, pk_{\tilde{m}})$ and $\text{msk} := (sk_1, \dots, sk_{\tilde{m}})$, where $sk_i \leftarrow_s \mathcal{SK}$ and $pk_i := \alpha(sk_i)$ for $i \in [\tilde{m}]$.
- $sk_{\mathbf{y}} := (fk_{1,\mathbf{y}}, \dots, fk_{\tilde{m},\mathbf{y}}, \mathbf{y})$, where $fk_{i,\mathbf{y}} := \mu(sk_i, \mathbf{y})$ for $i \in [\tilde{m}]$.
- $ct_{\mathbf{x}} := (c_1, \dots, c_{\tilde{m}}, \mathbf{e})$, where $c_i \leftarrow_s \mathcal{L}$ with witness w_i for $i \in [\tilde{m}]$, and $\mathbf{e} := \sum_{i=1}^{\tilde{m}} \text{Pub}(pk_i, c_i, w_i) + \mathbf{x}$.
- Decryption computes $\langle \mathbf{e}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \text{FPriv}(fk_{i,\mathbf{y}}, c_i) = \langle \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i) + \mathbf{x}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \langle \mathbf{A}_{sk_i}(c_i), \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$.

⁴ As we mentioned in Subsect. 1.1, the random self-reducibility of LWE (i.e., the reduction from LWE to Multi-secret LWE) currently has a loss factor $O(\lambda^2)$ [9, 39], which is directly delivered to the security loss of our LWE-based instantiations, making our LWE-based IPFE scheme almost tightly secure.

In fact, the idea of constructing IPFE from TL-IP-HPS is very natural in itself: a hash value $\text{Pub}(pk, c, w) = \mathbf{A}_{sk}(c)$ can be used to mask the vector $\mathbf{x}$ during encryption; then during decryption, since TL-IP-HPS supports the computation of inner product $\langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$ of the hash value by using the functional private evaluation algorithm FPriv , the inner product $\langle \mathbf{x}, \mathbf{y} \rangle$ can be recovered.⁵ The main novelty of our construction is that, we choose to employ $\tilde{m} = \lceil \frac{m}{L} \rceil$ independent copies of TL-IP-HPS (instead of a single copy), and this is crucial for us to improve the efficiency of tightly secure IPFE, as we will see below.

From a high-level perspective, the DDH-based ALS scheme [8] can be interpreted as $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}, L = m]$ (so that $\tilde{m} = 1$) and the DDH-based Tomida scheme [61] can be interpreted as $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}, L = 1]$ (so that $\tilde{m} = m$), which are two extreme cases of the DDH-based instantiations of our generic construction. Nevertheless, we stress that our construction works for any $L \in \mathbb{N}$ and also enables instantiations from DCR by using the DCR-based $\text{TLHPS}_{\text{DCR}}$ and from LWE by using the LWE-based $\text{TLHPS}_{\text{LWE}}$.

Clearly, our IPFE construction gains efficiency with larger L . As L increases (say $L = 5$ or $L = 100$), $\tilde{m} = \lceil \frac{m}{L} \rceil$ is about $L \times$ smaller, and consequently, all algorithms of our IPFE are about $L \times$ *faster* and all parameters except ciphertext are about $L \times$ *shorter* than the Tomida scheme (which corresponds to $L = 1$). Moreover, the ciphertext of our IPFE has an expansion factor of about $1 + \frac{2}{L}$ (for the DDH-based instantiation) or $1 + \frac{1}{L}$ (for the DCR-based instantiation) or $1 + \frac{1}{L}$ (for the LWE-based instantiation), approaching the ideal factor 1. This suggests that our IPFE has very compact ciphertexts.

ECONOMIC PROOF STRATEGY FOR ACHIEVING TIGHT SECURITY. As for security proof, we can show that our IPFE construction has security loss $\ell = 3 \cdot L$, thus achieving tight security when L is a constant.⁶ In the security experiment, the adversary $\mathcal{A}$ gets mpk and a collection of functional keys $\{sk_{\mathbf{y}}\}_{\mathbf{y} \in \mathcal{Q}}$, from which $\mathcal{A}$ learns the following information about $\text{msk} = (sk_1, \dots, sk_{\tilde{m}})$:

$$pk_1 := \alpha(sk_1), \dots, pk_{\tilde{m}} := \alpha(sk_{\tilde{m}}), \quad \{fk_{1,\mathbf{y}} := \mu(sk_1, \mathbf{y}), \dots, fk_{\tilde{m},\mathbf{y}} := \mu(sk_{\tilde{m}}, \mathbf{y})\}_{\mathbf{y} \in \mathcal{Q}}. \quad (4)$$

$\mathcal{A}$ can also obtain multiple (say Q) challenge ciphertexts, where the j -th ($j \in [Q]$) challenge ciphertext is

$$ct^{*(j)} := (c_1^{(j)} \leftarrow_{\$} \mathcal{L}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_{\$} \mathcal{L}, \mathbf{e}^{(j)} := \left[\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) \right] + \mathbf{x}_{\beta}^{(j)}) \quad (5)$$

that encrypts $\mathbf{x}_{\beta}^{(j)} \in \{\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)}\}$. Inspired by [61], in order to hide the information about the challenge bit β in (5), it suffices to extract enough entropy from the term $\left[\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) \right]$ for all challenge ciphertexts. The Tomida scheme

⁵ Indeed, there were some frameworks for constructing IPFE from (variants of) HPS [66, 13, 17] in the literature. However, none of these frameworks achieve tight security in the multi-challenge setting or have instantiations from lattices. We refer to Appendix B for more explanations and a comparison of our techniques with theirs.

⁶ For our LWE-based instantiations, the security loss is $\ell = O(L \cdot \lambda^2)$ since the computational properties of $\text{TLHPS}_{\text{LWE}}$ have loss factor $O(\lambda^2)$, as noted in Footnote 4.

[61] accomplishes this by exploiting its large enough sizes of keys and ciphertexts. However, our construction is more compact and cannot make use of the same proof strategy, so we propose a novel *economic* proof strategy to reuse the constrained spaces in our IPFE to extract entropy gradually.

The main ideas in our economic proof strategy are as follows. Roughly speaking, all the Q pairs of $\{\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)}\}$ have difference vectors $\tilde{\mathbf{x}}^{(j)} := \mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)} \in \mathbb{Z}^m$ for $j \in [Q]$, which span a space of dimension at most m . Our strategy is to divide this space into L subspaces of dimension $\tilde{m} = \lceil \frac{m}{L} \rceil$ (these subspaces are adaptively determined by those $\tilde{\mathbf{x}}^{(j)}$). Next, we first make use of the $\tilde{m}$ copies of TL-IP-HPS in our IPFE to hide the challenge bit β for those challenge ciphertexts whose $\tilde{\mathbf{x}}^{(j)}$ are in the *first* subspace. Then we reuse the $\tilde{m}$ copies of TL-IP-HPS to hide the challenge bit β for those challenge ciphertexts with $\tilde{\mathbf{x}}^{(j)}$ in the *second* subspace, and so on. After L hybrids, the challenge bit β in all Q challenge ciphertexts are perfectly hidden. This yields a security loss of $O(L)$.

More precisely, our proof strategy goes with L hybrids, where each hybrid contains four main steps. We first define some useful notations according to [61].

- **Notations and Observations:** For each $j \in [Q]$, let $\tilde{\mathbf{x}}^{(j)} := \mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)} \in \mathbb{Z}^m$, and let V_j be a space spanned by $\{\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)}\}$. Then a random element in V_j suffices to hide $\beta \cdot \tilde{\mathbf{x}}^{(j)}$ contained in $\mathbf{x}_\beta^{(j)} = \beta \cdot \tilde{\mathbf{x}}^{(j)} + \mathbf{x}_0^{(j)}$ and hence the information about β in the j -th challenge ciphertext (5).

Denote ϕ_j as the dimension of V_j . It is clear to see that $\phi_j \leq m$ holds. For all $i \in [m]$, we also define $\psi_i \in [Q]$ as the first query index τ such that $\phi_\tau = i$. It is not hard to confirm that $\{\tilde{\mathbf{x}}^{(\psi_i)}\}_{i \in [\phi_j]}$ forms a basis of V_j .

In the first hybrid, we extract entropy from $\boxed{\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})}$ as follows.

- **Language Switching via SMP:** By the hardness of (multi-fold) SMP,⁷ for all $j \in [Q]$, the instances in the j -th ciphertext $ct^{*(j)}$ can be changed to

$$\boxed{c_1^{(j)} \leftarrow \mathcal{X}, \dots, c_{\min\{\phi_j, \tilde{m}\}}^{(j)} \leftarrow \mathcal{X}}, c_{\min\{\phi_j, \tilde{m}\}+1}^{(j)} \leftarrow \mathcal{L}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}.$$

Namely, the first $\min\{\phi_j, \tilde{m}\}$ instances $c_1^{(j)}, \dots, c_{\min\{\phi_j, \tilde{m}\}}^{(j)}$ are chosen uniformly from $\mathcal{X}$ instead of $\mathcal{L}$. Note that the number $\min\{\phi_j, \tilde{m}\}$ is adaptively determined for each ciphertext $ct^{*(j)}$ according to the value of ϕ_j . On the other hand, since our construction contains only $\tilde{m}$ copies of TL-IP-HPS, we are only able to change at most $\tilde{m}$ instances at a time. So the number of instances to be changed is $\min\{\phi_j, \tilde{m}\}$.

- **Triggering Inner-level co-Hash via Functional Smoothness:** Since the first $\min\{\phi_j, \tilde{m}\}$ instances $c_1^{(j)}, \dots, c_{\min\{\phi_j, \tilde{m}\}}^{(j)}$ are chosen uniformly from

⁷ Multi-fold SMP is a multi-fold variant of SMP and requires $(c_i \leftarrow \mathcal{L})_i \stackrel{c}{\approx} (c_i \leftarrow \mathcal{X})_i$ for multiple instances. Its hardness can be (almost) tightly reduced to the hardness of SMP for our instantiations, by the random self-reducibility of the DDH/DCR/LWE assumptions.

$\mathcal{X}$, by the functional smoothness of TL-IP-HPS, the first $\min\{\phi_j, \tilde{m}\}$ terms

$\boxed{\mathbf{A}_{sk_1}(c_1^{(j)}), \dots, \mathbf{A}_{sk_{\min\{\phi_j, \tilde{m}\}}}(c_{\min\{\phi_j, \tilde{m}\}}^{(j)})}$ in $\boxed{\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})}$ can trigger inner-

level hash values $\Gamma_{s_1}(c_1^{(j)}), \dots, \Gamma_{s_{\min\{\phi_j, \tilde{m}\}}}(c_{\min\{\phi_j, \tilde{m}\}}^{(j)})$ in the subspace orthogonal to all $\mathbf{y} \in \mathcal{Q}$, respectively, conditioned on the projection keys and functional keys in (4). More precisely, by applying the (statistical) functional smoothness of TL-IP-HPS $\min\{\phi_j, \tilde{m}\}$ times, we have

$$\begin{aligned} \{\boxed{\mathbf{A}_{sk_1}(c_1^{(j)})}\}_{j \in [Q]} &\stackrel{s}{\approx} \{\boxed{\mathbf{A}_{sk_1}(c_1^{(j)})} + \Gamma_{s_1}(c_1^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_1)}\}_{j \in [Q]}, \dots, \\ \{\boxed{\mathbf{A}_{sk_{\min\{\phi_j, \tilde{m}\}}}(c_{\min\{\phi_j, \tilde{m}\}}^{(j)})}\}_{j \in [Q]} &\stackrel{s}{\approx} \{\boxed{\mathbf{A}_{sk_{\min\{\phi_j, \tilde{m}\}}}(c_{\min\{\phi_j, \tilde{m}\}}^{(j)})} + \Gamma_{s_{\min\{\phi_j, \tilde{m}\}}}(c_{\min\{\phi_j, \tilde{m}\}}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\min\{\phi_j, \tilde{m}\}})}\}_{j \in [Q]}, \end{aligned}$$

where $\tilde{\mathbf{x}}^{(\psi_1)}, \dots, \tilde{\mathbf{x}}^{(\psi_{\min\{\phi_j, \tilde{m}\}})}$ are orthogonal to all $\mathbf{y} \in \mathcal{Q}$ that $\mathcal{A}$ obtains functional secret keys since $\langle \mathbf{x}_0^{(\psi_i)}, \mathbf{y} \rangle = \langle \mathbf{x}_1^{(\psi_i)}, \mathbf{y} \rangle$ holds for all $\mathbf{y} \in \mathcal{Q}$ and all $\psi_i \in [Q]$. Consequently, for the j -th challenge ciphertext $ct^{*(j)}$, now we have

$$\mathbf{e}^{(j)} = \boxed{\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})} + \boxed{\sum_{i=1}^{\min\{\phi_j, \tilde{m}\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_i)}} + \mathbf{x}_\beta^{(j)}.$$

- **Randomness Amplifying via Multi-Key-Extracting:** Due to the fact that the first $\min\{\phi_j, \tilde{m}\}$ instances $c_1^{(j)}, \dots, c_{\min\{\phi_j, \tilde{m}\}}^{(j)}$ are chosen uniformly from $\mathcal{X}$, by the multi-key-extracting of TL-IP-HPS, the inner-level hash values $\{\Gamma_{s_i}(c_i^{(j)})\}_{i,j}$ in $\boxed{\sum_{i=1}^{\min\{\phi_j, \tilde{m}\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_i)}}$ are pseudorandom. Consequently, for all $j \in [Q]$, the $\mathbf{e}^{(j)}$ in the j -th challenge ciphertext $ct^{*(j)}$ can be further changed to

$$\mathbf{e}^{(j)} = \boxed{\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})} + \boxed{\sum_{i=1}^{\min\{\phi_j, \tilde{m}\}} \boxed{r_i^{(j)}} \cdot \tilde{\mathbf{x}}^{(\psi_i)}} + \mathbf{x}_\beta^{(j)}, \quad (6)$$

where $\boxed{r_1^{(j)}, \dots, r_{\min\{\phi_j, \tilde{m}\}}^{(j)}}$ are randomly chosen scalars.

- **Switching Language Back via SMP:** By the hardness of (multi-fold) SMP, for all $j \in [Q]$, the instances in the j -th challenge ciphertext $ct^{*(j)}$ are changed back to $\boxed{c_1^{(j)} \leftarrow \mathcal{L}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}}$, getting ready for the next hybrid.

At this point, for challenge ciphertexts $ct^{*(j)}$ whose $\phi_j \leq \tilde{m}$, the challenge bit β is completely hidden in them. To see this, observe that $\min\{\phi_j, \tilde{m}\} = \phi_j$ for such ciphertexts, the term $\boxed{\sum_{i=1}^{\min\{\phi_j, \tilde{m}\}} \boxed{r_i^{(j)}} \cdot \tilde{\mathbf{x}}^{(\psi_i)}}$ in (6) is a uniformly random element in V_j (by recalling that $\{\tilde{\mathbf{x}}^{(\psi_i)}\}_{i \in [\phi_j]}$ is a basis of V_j), and consequently, it perfectly hides $\beta \cdot \tilde{\mathbf{x}}^{(j)}$ contained in $\mathbf{x}_\beta^{(j)} = \beta \cdot \tilde{\mathbf{x}}^{(j)} + \mathbf{x}_0^{(j)}$ and hence the bit β .

In the second hybrid, we continue to extract entropy from $\boxed{\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})}$ for challenge ciphertexts $ct^{*(j)}$ whose $\phi_j > \tilde{m}$, through another language switching strategy. Namely, now the first $\min\{\phi_j - \tilde{m}, \tilde{m}\}$ instances $c_1^{(j)}, \dots, c_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}^{(j)}$

are chosen uniformly from $\mathcal{X}$ instead of $\mathcal{L}$, which is computationally indistinguishable by the hardness of (multi-fold) SMP. Accordingly, by the functional smoothness of TL-IP-HPS, the first $\min\{\phi_j - \tilde{m}, \tilde{m}\}$ terms in $\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)})$ can trigger inner-level hash values, and it holds that

$$\begin{aligned} \{\mathbf{A}_{sk_1}(c_1^{(j)})\}_{j \in [Q]} &\stackrel{s}{\approx} \{\mathbf{A}_{sk_1}(c_1^{(j)}) + \Gamma_{s_1}(c_1^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\tilde{m}+1})}\}_{j \in [Q]}, \dots, \\ \{\mathbf{A}_{sk_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}}(c_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}^{(j)})\}_{j \in [Q]} &\stackrel{s}{\approx} \{\mathbf{A}_{sk_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}}(c_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}^{(j)}) + \Gamma_{s_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}}(c_{\min\{\phi_j - \tilde{m}, \tilde{m}\}}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\tilde{m} + \min\{\phi_j - \tilde{m}, \tilde{m}\}})}\}_{j \in [Q]}, \end{aligned}$$

where now we adopt $\tilde{\mathbf{x}}^{(\psi_{\tilde{m}+1})}, \dots, \tilde{\mathbf{x}}^{(\psi_{\tilde{m} + \min\{\phi_j - \tilde{m}, \tilde{m}\}})}$ from the subspace orthogonal to all $\mathbf{y} \in \mathcal{Q}$. Then by the multi-key-extracting of TL-IP-HPS, the inner-level hash values $\{\Gamma_{s_i}(c_i^{(j)})\}_{i,j}$ can be further changed to truly random values, and now the $\mathbf{e}^{(j)}$ in the j -th challenge ciphertext $ct^{*(j)}$ is

$$\begin{aligned} \mathbf{e}^{(j)} &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \tilde{m}\}} \left[r_i^{(j)} \right] \cdot \tilde{\mathbf{x}}^{(\psi_i)} + \sum_{i=1}^{\min\{\phi_j - \tilde{m}, \tilde{m}\}} \left[r_{\tilde{m}+i}^{(j)} \right] \cdot \tilde{\mathbf{x}}^{(\psi_{\tilde{m}+i})} + \mathbf{x}_\beta^{(j)} \\ &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, 2 \cdot \tilde{m}\}} \left[r_i^{(j)} \right] \cdot \tilde{\mathbf{x}}^{(\psi_i)} + \mathbf{x}_\beta^{(j)}, \end{aligned}$$

where the last equality holds since $\min\{\phi_j, \tilde{m}\} + \min\{\phi_j - \tilde{m}, \tilde{m}\} = \min\{\phi_j, 2 \cdot \tilde{m}\}$ holds for ciphertexts $ct^{*(j)}$ whose $\phi_j > \tilde{m}$. We also change the instances in challenge ciphertexts back from $\mathcal{L}$. At this point, by a similar argument as that at the end of the first hybrid, for challenge ciphertexts $ct^{*(j)}$ whose $\phi_j \leq 2 \cdot \tilde{m}$, the challenge bit β is completely hidden in them.

Through a total of L hybrids, we can show that the challenge bit β is completely hidden in challenge ciphertexts $ct^{*(j)}$ whose $\phi_j \leq L \cdot \tilde{m}$. However, these include all challenge ciphertexts, since ϕ_j is the dimension of V_j and satisfies $\phi_j \leq m \leq L \cdot \lceil \frac{m}{L} \rceil = L \cdot \tilde{m}$ for all $j \in [Q]$. Consequently, the challenge bit β is perfectly hidden to the adversary $\mathcal{A}$ and the overall security loss is $O(L)$.

In summary, our economic tight proof strategy is to reuse the limited number of $\tilde{m} = \lceil \frac{m}{L} \rceil$ copies of TL-IP-HPS, by switching the languages back and forth and extracting entropy gradually during the hybrids. This greatly improves the efficiency of our IPFE construction, while still achieving tight security.

IMPLICATION: TIGHTER SECURITY FOR THE ALS SCHEME. Originally, the ALS scheme [8] was proved to be CPA-secure in the single-challenge setting. A standard hybrid argument implies that the ALS scheme is also mCPA-secure (in the multi-challenge setting), but it would incur a large loss factor Q with Q the number of ciphertexts. As such, the ALS scheme was only known to have loose mCPA security with security loss factor $O(Q)$ (e.g., as noted by Tomida [61]).

Now through our generic construction, we can interpret the DDH-based and the DCR-based ALS schemes as IPFE[TLHPS_{MDDH}, $L = m$] and IPFE[TLHPS_{DCR}, $L = m$], respectively. Interestingly, by using our economic proof strategy, it turns out that the DDH-based and the DCR-based ALS schemes have *tighter* mCPA security, with security loss factor only $O(m)$ instead of $O(Q)$.

3 Preliminaries

Notations. Let $\lambda \in \mathbb{N}$ denote the security parameter throughout the paper, and all algorithms, distributions, functions and adversaries take 1^λ as an implicit input. If x is defined by y or the value of y is assigned to x , we write $x := y$. For $n, m \in \mathbb{Z}$ with $n < m$, define $[n] := \{1, 2, \dots, n\}$ and $[n, m] := \{n, n+1, \dots, m\}$. For a real number $r \in \mathbb{R}$, $\lceil r \rceil \in \mathbb{Z}$ denotes the least integer greater than or equal to r , and $\lfloor r \rfloor \in \mathbb{Z}$ denotes the greatest integer less than or equal to r . For a set $\mathcal{X}$, denote by $x \leftarrow_s \mathcal{X}$ the procedure of sampling x from $\mathcal{X}$ uniformly at random. If $\mathcal{D}$ is distribution, $x \leftarrow_s \mathcal{D}$ means that x is sampled according to $\mathcal{D}$. All our algorithms are probabilistic unless stated otherwise. We use $y \leftarrow_s \mathcal{A}(x)$ to define the random variable y obtained by executing algorithm $\mathcal{A}$ on input x . If $\mathcal{A}$ is deterministic we write $y \leftarrow \mathcal{A}(x)$. “PPT” abbreviates probabilistic polynomial-time. Denote by poly some polynomial function and negl some negligible function.

Unless specifically noted, all vectors are column vectors in this paper. For a vector $\mathbf{x}$, $\mathbf{x}^\top$ denotes its transpose, $\|\mathbf{x}\|$ denotes its ℓ_2 norm, $\|\mathbf{x}\|_\infty$ denotes its infinity norm, namely the largest absolute value of all entries of $\mathbf{x}$, and $\text{gcd}(\mathbf{x})$ denotes the greatest common divisor of all entries of $\mathbf{x}$. For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^m$, let $\langle \mathbf{x}, \mathbf{y} \rangle$ denote their inner product $\mathbf{x}^\top \mathbf{y} \in \mathbb{Z}$.

Inner-Product Functional Encryption. Similar to prior works [8, 61], we consider inner-product functional encryption (IPFE) schemes for bounded norm inner product, i.e., $\langle \mathbf{x}, \mathbf{y} \rangle$ where $\|\mathbf{x}\|_\infty \leq X$ and $\|\mathbf{y}\|_\infty \leq Y$ for some polynomial bounds X and Y .

Definition 1 (IPFE). Let $X = \text{poly}(\lambda)$ and $Y = \text{poly}(\lambda)$ be polynomial bounds. An inner-product functional encryption (IPFE) scheme $\text{IPFE} = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ consists of four PPT algorithms:

- $(\text{mpk}, \text{msk}) \leftarrow_s \text{Setup}(1^m)$: Taking a vector dimension $m \in \mathbb{N}$ as input, the setup algorithm outputs a pair of master public key mpk and master secret key msk . Without loss of generality, we assume that mpk serves as an implicit input of KGen and Dec .
- $sk_{\mathbf{y}} \leftarrow_s \text{KGen}(\text{msk}, \mathbf{y})$: Taking as input msk and a vector $\mathbf{y} \in \mathbb{Z}^m$, the key generation algorithm outputs a secret key $sk_{\mathbf{y}}$.
- $ct \leftarrow_s \text{Enc}(\text{mpk}, \mathbf{x})$: Taking as input mpk and a vector $\mathbf{x} \in \mathbb{Z}^m$, the encryption algorithm outputs a ciphertext ct .
- $d/\perp \leftarrow \text{Dec}(sk_{\mathbf{y}}, ct)$: Taking as input a secret key $sk_{\mathbf{y}}$ and a ciphertext ct , the deterministic decryption algorithm outputs either a value $d \in \mathbb{Z}$ or a special symbol $\perp$ indicating the failure of decryption.

Correctness. For all $m \in \mathbb{N}$, $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^m$ satisfying $\|\mathbf{x}\|_\infty \leq X$ and $\|\mathbf{y}\|_\infty \leq Y$, it holds $\Pr \left[\begin{array}{l} (\text{mpk}, \text{msk}) \leftarrow_s \text{Setup}(1^m) \\ sk_{\mathbf{y}} \leftarrow_s \text{KGen}(\text{msk}, \mathbf{y}) \\ ct \leftarrow_s \text{Enc}(\text{mpk}, \mathbf{x}) \end{array} : \text{Dec}(sk_{\mathbf{y}}, ct) = \langle \mathbf{x}, \mathbf{y} \rangle \right] \geq 1 - \text{negl}(\lambda).$

Below we present the definition of indistinguishability under chosen-plaintext attacks in the (single-user and) multi-challenge setting (mCPA) for IPFE.

Definition 2 (mCPA Security for IPFE [61]). An IPFE scheme IPFE is mCPA secure, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}}(\lambda) := |\Pr[\text{Exp}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}} \Rightarrow 1] - \frac{1}{2}| \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}}$ is defined in Fig. 1, and we require that all $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0, \mathbf{x}_1)$ queries and all $\mathcal{O}_{\text{KGEN}}(\mathbf{y})$ queries made by $\mathcal{A}$ are restricted to satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$.

$\text{Exp}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}}:$ $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^m)$ $\beta \leftarrow \{0, 1\}$ //challenge bit $\beta' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{ENC}}, \mathcal{O}_{\text{KGEN}}}(\text{mpk})$ If $\beta' = \beta$: Return 1; Else: Return 0	$\mathcal{O}_{\text{ENC}}(\mathbf{x}_0 \in \mathbb{Z}^m, \mathbf{x}_1 \in \mathbb{Z}^m):$ $ct^* \leftarrow \text{Enc}(\text{mpk}, \mathbf{x}_\beta)$ Return ct^* $\mathcal{O}_{\text{KGEN}}(\mathbf{y} \in \mathbb{Z}^m):$ $sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}, \mathbf{y})$ Return $sk_{\mathbf{y}}$
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Fig. 1. The mCPA security experiment $\text{Exp}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}}$ for IPFE. To avoid trivial attacks, all $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0, \mathbf{x}_1)$ queries and all $\mathcal{O}_{\text{KGEN}}(\mathbf{y})$ queries made by $\mathcal{A}$ must satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$.

We will focus on mCPA security in the single-user and multi-challenge setting in the main body, and explain how to extend our results to the multi-user setting in Appendix D.

4 Two-Leveled Inner-Product Hash Proof System

In this section, we propose a new primitive called *Two-Leveled Inner-Product HPS* (TL-IP-HPS), which is made up of an outer-level Inner-Product HPS (IP-HPS) and an inner-level core hash function (co-Hash). We will first introduce the concepts of these two levels individually and then present their connections.

Firstly, the (outer-level) IP-HPS is a new variant of HPS [21], which enables computations of inner-product values of hash values with the help of extra functional keys. More precisely, the hash values $\mathbf{A}_{sk}(c)$ of IP-HPS are vectors over an (additive) group $\mathfrak{G}$, i.e., $\mathbf{A}_{sk}(c) \in \mathfrak{G}^m$ for some vector dimension m , and the inner-product of a hash value $\mathbf{A}_{sk}(c) = (\mathbf{g}_1, \dots, \mathbf{g}_m)^\top \in \mathfrak{G}^m$ with an integer vector $\mathbf{y} = (y_1, \dots, y_m)^\top \in \mathbb{Z}^m$ is defined by

$$\langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle := \sum_{i=1}^m y_i \cdot \mathbf{g}_i \in \mathfrak{G},$$

where “ $\cdot$ ” denotes scalar multiplication. A functional key $fk_{\mathbf{y}}$ w.r.t. $\mathbf{y} \in \mathbb{Z}^m$ allows the computations of $\langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$ for any instance $c \in \mathcal{X}$ via a functional private evaluation algorithm FPriv, namely, $\text{FPriv}(fk_{\mathbf{y}}, c) = \langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$. Moreover, the functional key $fk_{\mathbf{y}}$ itself can be derived by a functional key generation function $\mu(\cdot)$ from the hashing key sk and vector $\mathbf{y}$, namely, $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$. Formally, we present the syntax of IP-HPS.

Definition 3 (IP-HPS). An inner-product hash proof system (IP-HPS) scheme $\text{IPHPS} = (\text{Setup}_{\text{IPHPS}}, \alpha, \mu, \text{Pub}, \text{Priv}, \text{FPriv})$ consists of six PPT algorithms:

- $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$: Taking a vector dimension $m \in \mathbb{N}$ as input, the setup algorithm outputs a public parameter pp_{IPHPS} , which implicitly defines $(\mathfrak{G}, \mathcal{L}, \mathcal{X}, \mathcal{SK}, \mathcal{HV}, \Lambda_{(\cdot)})$, where $\mathfrak{G}$ is an (additive) group, $\mathcal{L} \subseteq \mathcal{X}$ is an NP-language with universe $\mathcal{X}$, $\mathcal{SK}$ is the hashing key space, $\mathcal{HV} := \mathfrak{G}^m$ is the hash value space, and $\Lambda_{(\cdot)} : \mathcal{X} \rightarrow \mathfrak{G}^m$ is a family of hash functions indexed by hashing keys $sk \in \mathcal{SK}$.

We require that $\Lambda_{(\cdot)}$ is efficiently computable and there are PPT algorithms for sampling $c \leftarrow \mathcal{L}$ together with a witness w , sampling $c \leftarrow \mathcal{X}$ and sampling $sk \leftarrow \mathcal{SK}$. The distributions of $c \leftarrow \mathcal{L}$, $c \leftarrow \mathcal{X}$ and $sk \leftarrow \mathcal{SK}$ are uniform by default, and can also be other efficiently sampleable distributions. We require pp_{IPHPS} to be an implicit input of other algorithms.

- $pk \leftarrow \alpha(sk)$: Taking a hashing key $sk \in \mathcal{SK}$ as input, the projection algorithm outputs a projection key pk .
- $fk_{\mathbf{y}} \leftarrow \mu(sk, \mathbf{y})$: Taking as input $sk \in \mathcal{SK}$ and a vector $\mathbf{y} \in \mathbb{Z}^m$, the functional key generation algorithm outputs a functional key $fk_{\mathbf{y}}$.
- $\mathbf{h} \leftarrow \text{Pub}(pk, c, w)$: Taking as input a projection key $pk = \alpha(sk)$, an instance $c \in \mathcal{L}$ and a witness w for $c \in \mathcal{L}$, the public evaluation algorithm outputs a hash value $\mathbf{h} = \Lambda_{sk}(c) \in \mathfrak{G}^m$.
- $\mathbf{h} \leftarrow \text{Priv}(sk, c)$: Taking as input a hashing key sk and an instance $c \in \mathcal{X}$, the private evaluation algorithm outputs a hash value $\mathbf{h} = \Lambda_{sk}(c) \in \mathfrak{G}^m$.
- $fv \leftarrow \text{FPriv}(fk_{\mathbf{y}}, c)$: Taking as input a functional key $fk_{\mathbf{y}}$ specified by vector $\mathbf{y} \in \mathbb{Z}^m$ and an instance $c \in \mathcal{X}$, the functional private evaluation algorithm outputs a function value $fv = \langle \Lambda_{sk}(c), \mathbf{y} \rangle \in \mathfrak{G}$.

Correctness. For all $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$, $sk \in \mathcal{SK}$, $c \in \mathcal{L}$ with witness w , and $pk := \alpha(sk)$, it holds that $\text{Pub}(pk, c, w) = \Lambda_{sk}(c) = \text{Priv}(sk, c)$.

Functional Correctness. For all $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$, $sk \in \mathcal{SK}$, $c \in \mathcal{X}$, $\mathbf{y} \in \mathbb{Z}^m$, and $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$, it holds that $\text{FPriv}(fk_{\mathbf{y}}, c) = \langle \Lambda_{sk}(c), \mathbf{y} \rangle$.

Similar to traditional HPS [21], IP-HPS is associated with a subset membership problem (SMP), which asks whether an element is uniformly chosen from $\mathcal{L}$ or $\mathcal{X}$. SMP can be extended to multi-fold SMP by considering multiple elements.

Definition 4 (SMP & Multi-fold SMP). The subset membership problem (SMP) related to IPHPS is hard, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{\text{IPHPS}, \mathcal{A}}^{\text{smp}}(\lambda) := |\Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, c) = 1] - \Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, c') = 1]| \leq \text{negl}(\lambda)$, where $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$, $c \leftarrow \mathcal{L}$ and $c' \leftarrow \mathcal{X}$.

The multi-fold SMP related to IPHPS is hard, if for any PPT adversary $\mathcal{A}$ and any polynomial $Q = \text{poly}(\lambda)$, it holds $\text{Adv}_{\text{IPHPS}, \mathcal{A}, Q}^{\text{msmp}}(\lambda) := |\Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, \{c_j\}_{j \in [Q]}) = 1] - \Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, \{c'_j\}_{j \in [Q]}) = 1]| \leq \text{negl}(\lambda)$, where $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$, $c_1, \dots, c_Q \leftarrow \mathcal{L}$ and $c'_1, \dots, c'_Q \leftarrow \mathcal{X}$.

Multi-fold SMP can generally be reduced to SMP with a security loss of Q with Q the number of folds. For some random-self reducible problems like matrix

DDH (MDDH), DCR or LWE, the hardness of multi-fold SMP can be (almost) tightly reduced to that of SMP. See Sect. 6 for more details.

Next, we present the syntax of the (inner-level) co-Hash.

Definition 5 (co-Hash). A core hash function (co-Hash) scheme coHash consists of a PPT setup algorithm $\text{Setup}_{\text{coHash}}$:

- $\text{pp}_{\text{coHash}} \leftarrow_s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$: Taking as input a public parameter pp_{IPHPS} of the (outer-level) IPHPS, the setup algorithm of co-Hash outputs a public parameter $\text{pp}_{\text{coHash}}$, which implicitly defines $(\mathfrak{G}_{\text{co}}, \mathfrak{g}_{\text{co}}, \text{ord}_{\text{co}}, \mathcal{SK}_{\text{co}}, \mathcal{HV}_{\text{co}}, \Gamma_{(\cdot)})$, where $\mathfrak{G}_{\text{co}}$ is a cyclic subgroup of the (additive) group $\mathfrak{G}$ defined by pp_{IPHPS} (as per Def. 3), $\mathfrak{g}_{\text{co}}$ is a generator of $\mathfrak{G}_{\text{co}}$, ord_{co} is the order of $\mathfrak{G}_{\text{co}}$, $\mathcal{SK}_{\text{co}}$ is the hashing key space, $\mathcal{HV}_{\text{co}} := \mathfrak{G}_{\text{co}}$ is the hash value space, and $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathfrak{G}_{\text{co}}$ is a family of hash functions indexed by hashing keys $s \in \mathcal{SK}_{\text{co}}$.

We do not require $\Gamma_{(\cdot)}$ to be efficiently computable. We require that there is a PPT algorithm for sampling $s \leftarrow_s \mathcal{SK}_{\text{co}}$, which is the uniform distribution by default and can also be other efficiently sampleable distribution. We require $\text{pp}_{\text{coHash}}$ to be an implicit input of other algorithms.

Remark 1 (On the formalization of co-Hash). We stress that the hash function $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathfrak{G}_{\text{co}}$ of coHash may not be efficiently computable. Namely, $\Gamma_s(c)$ does not need to be efficiently computable given any $s \in \mathcal{SK}_{\text{co}}$ and $c \in \mathcal{X}$.

Jumping ahead, we note that this relaxation is important and reasonable. On the one hand, this makes our two-leveled (IPHPS, coHash) weak enough to be instantiable from various assumptions ranging from MDDH, DCR to LWE, as will be shown in Sect. 6. On the other hand, this does not affect its usefulness in building more complex cryptographic primitives. For example, in Sect. 5, we will use a two-leveled (IPHPS, coHash) to construct an IPFE, where the construction of IPFE involves only the efficient (outer-level) IPHPS, while the inner-level coHash appears only in the security proof. During the security proof, we will carefully organize the game transitions, so that the reduction algorithm either does not need to compute the hash function $\Gamma_{(\cdot)}$, or that it is an unbounded algorithm (when reducing to statistical properties) and thus can compute $\Gamma_{(\cdot)}$ by brute force, so the inefficiency of coHash would not be a problem.

Now we are ready to present the definition of two-leveled IP-HPS.

Definition 6 (Two-Leveled IP-HPS). An IP-HPS scheme IPHPS (as per Def. 3) and a co-Hash scheme coHash (as per Def. 5) constitute a two-leveled IP-HPS (TL-IP-HPS) scheme, denoted by $\text{TLHPS} = (\text{IPHPS}, \text{coHash})$, if they satisfy the properties of functional smoothness and multi-key-extracting as defined in the following Def. 7 and Def. 8.

Definition 7 (Functional Smoothness of Two-Leveled IP-HPS). A two-leveled IP-HPS scheme $\text{TLHPS} = (\text{IPHPS}, \text{coHash})$ has functional smoothness, if for any (unbounded) adversary $\mathcal{A}$ and any polynomial $Q = \text{poly}(\lambda)$, it holds that $\epsilon_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) := |\Pr[\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, 0} \Rightarrow 1] - \Pr[\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, 1} \Rightarrow 1]| \leq \text{negl}(\lambda)$,

where the experiments $\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}$ ($b \in \{0, 1\}$) are defined in Fig. 2, and we require that $\mathcal{A}$ can make only a single query to $\mathcal{O}_{\text{CHAL}}^b$, and the (single) $\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}})$ query and all $\mathcal{O}_{\text{FKEY}}(\mathbf{y})$ queries made by $\mathcal{A}$ are restricted to satisfy $\langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0$.

$\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}:$ $\ b \in \{0, 1\}$ $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$ $\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$ $sk \leftarrow \mathcal{SK}, \quad pk := \alpha(sk)$ $s \leftarrow \mathcal{SK}_{\text{co}}$ $b' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{CHAL}}^b, \mathcal{O}_{\text{FKEY}}}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk)$ Return b'	$\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}} \in \mathbb{Z}^m):$ $\ \text{single query}$ For $j \in [Q]$: $c^{(j)} \leftarrow \mathcal{X}$ $\mathbf{h}^{(j)} := \begin{cases} \mathbf{A}_{sk}(c^{(j)}), & \text{if } b = 0; \\ \mathbf{A}_{sk}(c^{(j)}) + \Gamma_s(c^{(j)}) \cdot \tilde{\mathbf{x}}^8, & \text{if } b = 1 \end{cases}$ Return $(c^{(j)}, \mathbf{h}^{(j)})_{j \in [Q]}$ $\mathcal{O}_{\text{FKEY}}(\mathbf{y} \in \mathbb{Z}^m):$ Return $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$
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Fig. 2. The functional smoothness experiments $\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}$ ($b \in \{0, 1\}$) for two-leveled TLHPS = (IPHPS, coHash). The difference between $b = 0$ and $b = 1$ is highlighted in gray box. To avoid trivial attacks, we require that the (single) $\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}})$ query and all $\mathcal{O}_{\text{FKEY}}(\mathbf{y})$ queries made by $\mathcal{A}$ must satisfy $\langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0$.

Definition 8 (Multi-Key-Extracting of Two-Leveled IP-HPS). A two-leveled IP-HPS scheme TLHPS = (IPHPS, coHash) supports multi-key-extracting, if for any PPT $\mathcal{A}$, any polynomial T and any polynomial Q , it holds that

$$\begin{aligned} \text{Adv}_{\text{TLHPS}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) &:= |\Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, \{c_{i,j}, \Gamma_{s_i}(c_{i,j})\}_{i \in [T], j \in [Q]}) = 1] \\ &\quad - \Pr[\mathcal{A}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, \{c_{i,j}, r_{i,j} \cdot \mathbf{g}_{\text{co}}\}_{i \in [T], j \in [Q]}) = 1]| \leq \text{negl}(\lambda), \end{aligned}$$

where $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$, $\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$, $c_{1,1}, \dots, c_{T,Q} \leftarrow \mathcal{X}$, $s_1, \dots, s_T \leftarrow \mathcal{SK}_{\text{co}}$, $r_{1,1}, \dots, r_{T,Q} \leftarrow \mathbb{Z}_{\text{ord}_{\text{co}}}$. Recall that $\text{pp}_{\text{coHash}}$ defines the hash value space $\mathfrak{G}_{\text{co}}$ of $\Gamma_{(\cdot)}$, where $\mathfrak{G}_{\text{co}}$ is a cyclic group with generator $\mathbf{g}_{\text{co}}$ of order ord_{co} . Here $r_{i,j} \cdot \mathbf{g}_{\text{co}}$ with $r_{i,j} \leftarrow \mathbb{Z}_{\text{ord}_{\text{co}}}$ is a uniformly random element in $\mathfrak{G}_{\text{co}}$.

5 Tightly Secure IPFE from Two-Leveled IP-HPS

In this section, we present a generic construction of tightly mCPA secure IPFE from two-leveled IP-HPS. The building block and parameter are as follows.

- A two-leveled IP-HPS scheme TLHPS = (IPHPS, coHash):

⁸ Here “+” denotes vector addition and “ $\cdot$ ” denotes component-wise scalar multiplication. More precisely, let $\mathbf{A}_{sk}(c) = (\mathbf{g}_1, \dots, \mathbf{g}_m)^\top \in \mathfrak{G}^m$ and $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_m)^\top \in \mathbb{Z}^m$, then $\mathbf{A}_{sk}(c) + \Gamma_s(c) \cdot \tilde{\mathbf{x}}$ equals $(\mathbf{g}_1 + \Gamma_s(c) \cdot \tilde{x}_1, \dots, \mathbf{g}_m + \Gamma_s(c) \cdot \tilde{x}_m)^\top \in \mathfrak{G}^m$.

- the (outer-level) IP-HPS scheme is $\text{IPHPS} = (\text{Setup}_{\text{IPHPS}}, \alpha, \mu, \text{Pub}, \text{Priv}, \text{FPriv})$, with language $\mathcal{L}$, hashing key space $\mathcal{SK}$ and hash value space $\mathcal{HV} = \mathfrak{G}^m$, where $\mathfrak{G}$ is an (additive) group;
- the (inner-level) co-Hash scheme is $\text{coHash} = \text{Setup}_{\text{coHash}}$, with hash value space $\mathcal{HV}_{\text{co}} = \mathfrak{G}_{\text{co}}$, where $\mathfrak{G}_{\text{co}}$ is a subgroup of $\mathfrak{G}$ and has generator $\mathbf{g}_{\text{co}}$.
- A positive integer $L \in \mathbb{N}$ (e.g., $L = 5$), which will determine the security loss factor of the resulting IPFE scheme.

Let X and Y be polynomials in λ . Our generic construction of IPFE is built from TLHPS and parameterized by L , thus is also denoted by $\text{IPFE}[\text{TLHPS}, L]$. The formal description of $\text{IPFE} = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ is shown in Fig. 3.

$ \begin{aligned} &(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^m): \\ &\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m). \\ &\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}}). \\ &\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}. \\ &\text{For } i \in [\tilde{m}]: \\ &\quad sk_i \leftarrow \mathcal{SK}, pk_i := \alpha(sk_i). \\ &\text{mpk} := (\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk_1, \dots, pk_{\tilde{m}}). \\ &\text{msk} := (sk_1, \dots, sk_{\tilde{m}}). \\ &\text{Return } (\text{mpk}, \text{msk}). \\ \\ &sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}, \mathbf{y} \in \mathbb{Z}^m): \\ &\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}. \\ &\text{For } i \in [\tilde{m}]: \\ &\quad fk_{i,\mathbf{y}} := \mu(sk_i, \mathbf{y}). \\ &\text{Return } sk_{\mathbf{y}} := (fk_{1,\mathbf{y}}, \dots, fk_{\tilde{m},\mathbf{y}}). \end{aligned} $	$ \begin{aligned} &ct \leftarrow \text{Enc}(\text{mpk}, \mathbf{x} \in \mathbb{Z}^m): \\ &\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}. \\ &\text{For } i \in [\tilde{m}]: \\ &\quad c_i \leftarrow \mathcal{L} \text{ with witness } w_i. \\ &\quad \mathbf{e} := \sum_{i=1}^{\tilde{m}} \text{Pub}(pk_i, c_i, w_i) + \mathbf{x} \cdot \mathbf{g}_{\text{co}} \in \mathfrak{G}^m.^9 \\ &\text{Return } ct := (c_1, \dots, c_{\tilde{m}}, \mathbf{e}). \\ \\ &d/\perp \leftarrow \text{Dec}(sk_{\mathbf{y}}, ct): \\ &\text{Parse } sk_{\mathbf{y}} = (fk_{1,\mathbf{y}}, \dots, fk_{\tilde{m},\mathbf{y}}, \mathbf{y}), \\ &\quad ct = (c_1, \dots, c_{\tilde{m}}, \mathbf{e}). \\ &\mathfrak{d} := \langle \mathbf{e}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \text{FPriv}(fk_{i,\mathbf{y}}, c_i) \in \mathfrak{G}. \\ &\quad // \text{ solve discrete logarithm of } \mathfrak{d} \text{ wrt. } \mathbf{g}_{\text{co}} \\ &\quad // \text{ by exhaustive search} \\ &\text{For } d \in [-mXY, mXY]: \\ &\quad \text{If } \mathfrak{d} = d \cdot \mathbf{g}_{\text{co}}: \text{ Return } d. \\ &\text{Return } \perp. \end{aligned} $
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Fig. 3. Generic construction of $\text{IPFE}[\text{TLHPS}, L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ from two-leveled $\text{TLHPS} = (\text{IPHPS}, \text{coHash})$ and parameterized by $L \in \mathbb{N}$.

Note that the construction of our IPFE uses the (outer-level) IP-HPS scheme IPHPS extensively, like its projection algorithm α , functional key generation algorithm μ , public evaluation algorithm Pub and functional private evaluation algorithm FPriv . As for the inner-level coHash, our IPFE uses only the group generator $\mathbf{g}_{\text{co}}$ to encode vector $\mathbf{x} \in \mathbb{Z}^m$ to $\mathbf{x} \cdot \mathbf{g}_{\text{co}} \in \mathfrak{G}_{\text{co}}^m \subseteq \mathfrak{G}^m$ in the Enc algorithm and decode $\mathfrak{d} \in \mathfrak{G}$ to $d \in \mathbb{Z}$ in the Dec algorithm. As we will see, the inner-level coHash will be triggered during the security proof and play essential roles in achieving tight mCPA security for our IPFE.

Correctness of our IPFE follows directly from the correctness and functional correctness of IPHPS. More precisely, during $\text{Dec}(sk_{\mathbf{y}}, ct)$, it holds that

$$\begin{aligned}
\mathfrak{d} &= \langle \mathbf{e}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \text{FPriv}(fk_{i,\mathbf{y}}, c_i) = \langle \mathbf{e}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \langle \mathbf{A}_{sk_i}(c_i), \mathbf{y} \rangle \\
&= \langle \sum_{i=1}^{\tilde{m}} \text{Pub}(pk_i, c_i, w_i) + \mathbf{x} \cdot \mathbf{g}_{\text{co}}, \mathbf{y} \rangle - \sum_{i=1}^{\tilde{m}} \langle \mathbf{A}_{sk_i}(c_i), \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle \cdot \mathbf{g}_{\text{co}}.
\end{aligned}$$

⁹ Similar to Footnote 8, here “ $\sum$ ” and “ $+$ ” denote vector addition and “ $\cdot$ ” denotes component-wise scalar multiplication.

Therefore, if $\|\mathbf{x}\|_\infty \leq X$ and $\|\mathbf{y}\|_\infty \leq Y$, then $|\langle \mathbf{x}, \mathbf{y} \rangle| < mXY$ and $\text{Dec}(sk_{\mathbf{y}}, ct)$ can solve the discrete logarithm of $\mathfrak{d}$ wrt. $\mathfrak{g}_{\text{co}}$ efficiently in the range of $[-mXY, mXY]$ by exhaustive search and recover the inner product $d = \langle \mathbf{x}, \mathbf{y} \rangle$ correctly. Alternatively, if the discrete logarithm in $\mathfrak{G}_{\text{co}}$ is easy-to-solve (e.g., the DCR and LWE instantiations in Sect. 6), $\text{Dec}(sk_{\mathbf{y}}, ct)$ can compute the discrete logarithm of $\mathfrak{d}$ wrt. $\mathfrak{g}_{\text{co}}$ directly instead of using an exhaustive search.

Next we show the tight mCPA security of our IPFE.

Theorem 1 (Tight mCPA Security of IPFE[TLHPS, L]). *Assume that TLHPS is a two-leveled IP-HPS that has hard SMP, functional soundness and supports multi-key-extracting. Then the scheme IPFE[TLHPS, L] in Fig. 3 is mCPA secure.*

Concretely, for any adversary $\mathcal{A}$ who makes at most Q_e queries to $\mathcal{O}_{\text{ENC}}$ and Q_k queries to $\mathcal{O}_{\text{GEN}}$, there exist adversaries $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, such that $\mathbf{T}(\mathcal{B}_1) \approx \mathbf{T}(\mathcal{B}_2) \approx \mathbf{T}(\mathcal{A}) + (Q_e + Q_k) \cdot \text{poly}(\lambda)$, with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and

$$\begin{aligned} \text{Adv}_{\text{IPFE}[\text{TLHPS}, L], \mathcal{A}}^{\text{mCPA}}(\lambda) &\leq 2L \cdot \text{Adv}_{\text{IPHPS}, \mathcal{B}_1, Q_e, \tilde{m}}^{\text{msmp}}(\lambda) + L \cdot \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda) \\ &\quad + \text{statist. loss}, \end{aligned}$$

where $\text{statist. loss} = \tilde{m} \cdot L \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmth}}(\lambda) \approx m \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmth}}(\lambda)$ with $\tilde{m} := \lceil \frac{m}{L} \rceil$.

We refer to Subsect. 2.3 for an overview of the proof, and postpone the formal proof to Appendix C. According to Theorem 1, IPFE[TLHPS, L] has mCPA security with loss factor $3L$, as long as the multi-fold SMP has tight reduction, and TLHPS = (IPHPS, coHash) has functional soundness with an exponentially small statistical factor and has tight multi-key-extracting property.

In Appendix D, we also show how to extend the tight mCPA security of our IPFE to the multi-user setting.

6 Instantiations (based on LWE)

Now we present instantiations of two-leveled inner-product hash proof system (TL-IP-HPS) proposed in Sect. 4 with (almost) tight security properties based on the matrix DDH (MDDH) assumptions, the DCR assumptions and the LWE assumptions, respectively. Then by plugging them into our generic IPFE construction in Sect. 5, we immediately obtain MDDH-based, DCR-based and LWE-based IPFE schemes with (almost) tight mCPA security.

Due to space limitations, we only provide the LWE-based instantiation in the main body, while postpone the details of the MDDH-based and DCR-based instantiations to Appendix E and Appendix F, respectively. (We refer to Subsect. 2.2 for an overview of our MDDH-based and DCR-based instantiations.)

The organization of this section is as follows. We present the LWE-based instantiation of two-leveled IP-HPS $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ and the resulting IPFE scheme IPFE_{LWE} in Subsect. 6.1 and Subsect. 6.2, respectively.

Lattice Backgrounds and LWE Assumptions. Here we briefly recall some lattice notions and the LWE assumptions. We present more detailed lattice preliminaries in Appendix G.1, and give the definitions of more LWE variants (e.g.,

Multi-secret LWE, Multi-instance LWE) in Appendix G.2, where we also show several (almost) tight reductions among these LWE assumptions.

For $\sigma > 0$ and $\mathbf{c} \in \mathbb{R}^n$, we define the Gaussian function on $\mathbb{R}^n$ centered at $\mathbf{c}$ with parameter σ by $\rho_{\sigma,\mathbf{c}}(\mathbf{x}) := e^{-\pi\|\mathbf{x}-\mathbf{c}\|^2/\sigma^2}$. Let $\mathcal{Y} \subseteq \mathbb{R}^n$ be a lattice.¹⁰ The discrete Gaussian distribution $D_{\mathcal{Y},\sigma,\mathbf{c}}$ over $\mathcal{Y}$ is defined by $D_{\mathcal{Y},\sigma,\mathbf{c}}(\mathbf{x}) := \rho_{\sigma,\mathbf{c}}(\mathbf{x})/\rho_{\sigma,\mathbf{c}}(\mathcal{Y})$ for any lattice vector $\mathbf{x} \in \mathcal{Y}$, where $\rho_{\sigma,\mathbf{c}}(\mathcal{Y}) := \sum_{\mathbf{z} \in \mathcal{Y}} \rho_{\sigma,\mathbf{c}}(\mathbf{z})$. The subscript $\mathbf{c}$ is taken to be $\mathbf{0}$ when omitted.

Let $n, q, l \in \mathbb{N}$, and χ be a distribution over $\mathbb{Z}$. A distribution χ is B -bounded if its support is limited to $[-B, B]$. Let $\mathbb{Z}_q$ be the ring of integers modulo q , and its elements are represented by the integers in $(-q/2, q/2]$.

The $\text{LWE}_{n,q,\chi,l}$ -assumption postulates the hardness of distinguishing $(\mathbf{A}, \mathbf{Aw} + \mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$, where $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{w} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow_{\$} \chi^l$ and $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l$.

Lemma 1 (Tail Bound [51]). *For any $t > 0$, we have $\Pr_{x \leftarrow_{\$} D_{\mathbb{Z},\sigma}}[|x| \geq t \cdot \sigma] \leq 2e^{-\frac{t^2}{2}}$ and $\Pr_{\mathbf{x} \leftarrow_{\$} D_{\mathbb{Z}^l,\sigma}}[\|\mathbf{x}\|_\infty \geq t \cdot \sigma \sqrt{l}] \leq t^l \cdot e^{\frac{1}{2}(1-t^2)}$.*

Lemma 2 (Smudging Lemma [11, Lem. 2.1]). *Let $B, B' \in \mathbb{N}$, and $e \in [-B, B]$ a fixed integer. Then for $e' \leftarrow_{\$} [-B', B']$, it holds $\Delta(e + e', e') = B/B'$.*

6.1 Instantiation of Two-Leveled IP-HPS from LWE

Now we present the LWE-based $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$. Let $\text{pp}_{\text{LWE}} = (n, q, l, l', \sigma_{\mathcal{L}}, \sigma_K, \sigma_s, \sigma_{\text{LWE}}, \chi_{\mathcal{L}}, \chi_K, \chi_s, B_{\mathcal{L}}, B_K, B_s, B_{e'})$ be LWE-related parameters, where $n, q, l, l' \in \mathbb{N}$ are integers with q a prime number and $l > l' > n$, $\sigma_{\mathcal{L}}, \sigma_K, \sigma_s, \sigma_{\text{LWE}} > 0$ are parameters for Gaussians, $\chi_{\mathcal{L}} = D_{\mathbb{Z},\sigma_{\mathcal{L}}}$, $\chi_K = D_{\mathbb{Z},\sigma_K}$ and $\chi_s = D_{\mathbb{Z},\sigma_s}$ are discrete Gaussian distributions over $\mathbb{Z}$, and $B_{\mathcal{L}}, B_K, B_s, B_{e'} \in \mathbb{N}$ are error bounds such that $\chi_{\mathcal{L}}$ is $B_{\mathcal{L}}$ -bounded, χ_K is B_K -bounded and χ_s is B_s -bounded. According to the tail bound (Lemma 1), $\chi_{\mathcal{L}} = D_{\mathbb{Z},\sigma_{\mathcal{L}}}$, $\chi_K = D_{\mathbb{Z},\sigma_K}$ and $\chi_s = D_{\mathbb{Z},\sigma_s}$ are $\sqrt{\lambda} \cdot \sigma_{\mathcal{L}}$ -bounded, $\sqrt{\lambda} \cdot \sigma_K$ -bounded and $\sqrt{\lambda} \cdot \sigma_s$ -bounded, respectively, except with exponentially small probability $2^{-\lambda}$,¹¹ so we can set $B_{\mathcal{L}} := \sqrt{\lambda} \cdot \sigma_{\mathcal{L}}$, $B_K := \sqrt{\lambda} \cdot \sigma_K$ and $B_s := \sqrt{\lambda} \cdot \sigma_s$. Jumping ahead, for the properties of $\text{TLHPS}_{\text{LWE}}$ to hold, we associate the parameters pp_{LWE} with a value $\epsilon_{\text{pp}_{\text{LWE}}}$ as defined below and require $\epsilon_{\text{pp}_{\text{LWE}}}$ to be negligible in λ :

$$\epsilon_{\text{pp}_{\text{LWE}}} := \max \left\{ \begin{array}{l} l/2^\lambda, lB_K B_{\mathcal{L}}/B_{e'}, (l-l')B_s/\sigma_K, B_s/B_{e'}, n/q^{l'-n}, q^{-n}, 2^{-l'^2/6}, \\ (l-l')/q^{l-l'}, (l-l')^2 \cdot (l\sqrt{\lambda} \cdot \sigma_{\text{LWE}} + 1)/\sigma_s, (l-l')^2 \cdot 2^{-\lambda} \end{array} \right\} \leq \text{negl}(\lambda). \quad (7)$$

We refer to Table 2 in Subsect. 6.2 for a suggestion of parameters pp_{LWE} so that (7) holds. Moreover, the properties of $\text{TLHPS}_{\text{LWE}}$ will be (almost) tightly reduced to the $\text{LWE}_{\lambda,q,D_{\mathbb{Z},\sigma_{\text{LWE}}},l}$ -assumption. For simplicity, we assume that all instantiations in this subsection take pp_{LWE} as an *implicit input*.

¹⁰ We use the symbol $\mathcal{Y}$ to denote lattices instead of using the conventional symbol Λ , since the symbol $\mathbf{A}$ is preserved to denote the hash function of IP-HPS.

¹¹ We will not mention this exponentially small probability hereafter for simplicity, and take for granted that $\chi_{\mathcal{L}}$ is $B_{\mathcal{L}}$ -bounded, χ_K is B_K -bounded and χ_s is B_s -bounded.

(Outer-Level) IP-HPS Scheme. We first present the LWE-based (outer-level) IP-HPS scheme. To accommodate instantiations based on LWE, we follow a recent paradigm [39], and relax the syntax of (outer-level) IP-HPS as a *probabilistic* one, by allowing its public and private evaluation algorithms (i.e., Pub and Priv) to be *probabilistic* algorithms that may toss coins. Instead of requiring correctness, similar to [39], we require a statistical indistinguishability of the two probabilistic evaluation algorithms, i.e., $\text{Pub}(pk, c, w) \stackrel{s}{\approx} \text{Priv}(sk, c)$ for $c \in \mathcal{L}$ with witness w . Moreover, regarding functional smoothness, we require that the outputs of $\text{Priv}(sk, c)$ (instead of the real hash values $\mathbf{A}_{sk}(c)$) can trigger inner-level hash values when $c \leftarrow \mathcal{X}$. For completeness, we present the formal definitions of *probabilistic* IP-HPS and the (adapted) functional smoothness in Def. 17 and 19 in Appendix G.3.

We note that probabilistic IP-HPS suffices for the generic IPFE construction in Sect. 5 and does not affect its tightly mCPA security. Indeed, during the security proof of IPFE (i.e., the proof of Theorem 1 and we refer to Subsect. 2.3 for an overview of the proof), we need to switch the computations of IP-HPS hash values from Pub to Priv , and need to resort to the IP-HPS hash values to trigger inner-level hash values. These were accomplished by the correctness and the functional smoothness of (non-probabilistic) IP-HPS, respectively. Now with probabilistic IP-HPS, we can argue these steps with the statistical evaluation indistinguishability and the (adapted) functional smoothness, respectively.

$\text{ppIPHPS} \leftarrow \text{SetupIPHPS}(1^m):$ $\text{ppLWE} = (n, q, l, l', \sigma_{\mathcal{L}}, \sigma_K, \sigma_s, \sigma_{\text{LWE}}, \chi_{\mathcal{L}}, \chi_K, \chi_s, B_{\mathcal{L}}, B_K, B_s, B_{e'})$. $\mathbf{A} \leftarrow \mathbb{Z}_q^{l \times n}$. $\mathbf{B} := (\mathbf{I}_{l-l'}) \in \mathbb{Z}_q^{l \times (l-l')}$, where $\mathbf{I}_{l-l'}$ denotes the $(l-l') \times (l-l')$ identity matrix. Return $\text{ppIPHPS} := (\text{ppLWE}, \mathbf{A} \in \mathbb{Z}_q^{l \times n}, \mathbf{B} \in \mathbb{Z}_q^{l \times (l-l')})$, which implicitly defines $(\mathfrak{G}, \mathcal{L}, \mathcal{X}, \mathcal{SK}, \mathcal{HV}, \mathbf{A}_{(\cdot)})$: $\mathfrak{G} := \mathbb{Z}_q$. $\mathcal{L} := \{ \mathbf{c} \mid \exists \mathbf{w} \in \mathbb{Z}_q^n, \mathbf{e} \in [-B_{\mathcal{L}}, B_{\mathcal{L}}]^l \text{ s.t. } \mathbf{c} = \mathbf{Aw} + \mathbf{e} \}$. $\parallel \text{“} \mathbf{c} \leftarrow \mathcal{L} \text{”}$ denotes sampling $\mathbf{c} := \mathbf{Aw} + \mathbf{e} \in \mathcal{L}$ by picking $\mathbf{w} \leftarrow \mathbb{Z}_q^n$ and $\mathbf{e} \leftarrow \chi_{\mathcal{L}}^l$. $\mathcal{X} := \{ \mathbf{c} \mid \exists \mathbf{w} \in \mathbb{Z}_q^n, \mathbf{v} \in \mathbb{Z}_q^{l-l'}, \mathbf{e} \in [-B_{\mathcal{L}}, B_{\mathcal{L}}]^l \text{ s.t. } \mathbf{c} = \mathbf{Aw} + \mathbf{Bv} + \mathbf{e} \} \subseteq \mathbb{Z}_q^l$. $\parallel \text{“} \mathbf{c} \leftarrow \mathcal{X} \text{”}$ denotes sampling $\mathbf{c} := \mathbf{Aw} + \mathbf{Bv} + \mathbf{e} \in \mathcal{X}$ by picking $\mathbf{w} \leftarrow \mathbb{Z}_q^n$, $\mathbf{v} \leftarrow \mathbb{Z}_q^{l-l'}$, $\mathbf{e} \leftarrow \chi_{\mathcal{L}}^l$. $\mathcal{SK} := \mathbb{Z}^{m \times l}$. $\parallel \text{“} sk \leftarrow \mathcal{SK} \text{”}$ denotes sampling $sk = \mathbf{K}$ from $\mathcal{SK} = \mathbb{Z}^{m \times l}$ according to $\mathbf{K} \leftarrow \chi_K^{m \times l}$. $\mathcal{HV} := \mathbb{Z}_q^m$. $\mathbf{A}_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{Z}_q^m$ is defined as below: for any $sk = \mathbf{K} \in \mathcal{SK} = \mathbb{Z}^{m \times l}$ and any $\mathbf{c} \in \mathcal{X} \subseteq \mathbb{Z}_q^l$, $\mathbf{A}_{sk}(\mathbf{c}) := \mathbf{K} \cdot \mathbf{c}$.	
$pk \leftarrow \alpha(sk = \mathbf{K}):$ $\mathbf{P} := \mathbf{K} \cdot \mathbf{A} \in \mathbb{Z}_q^{m \times n}$. Return $pk := \mathbf{P}$. $fk_{\mathbf{y}} \leftarrow \mu(sk = \mathbf{K}, \mathbf{y} \in \mathbb{Z}^m):$ $\mathbf{f}^\top := \mathbf{y}^\top \cdot \mathbf{K} \in \mathbb{Z}^{1 \times l}$. Return $fk_{\mathbf{y}} := \mathbf{f}^\top$.	$\mathbf{h} \leftarrow \text{Pub}(pk = \mathbf{P}, \mathbf{c}, (\mathbf{w}, \mathbf{e})): \parallel \mathbf{c} \in \mathcal{L} \text{ with } \mathbf{c} = \mathbf{Aw} + \mathbf{e}$ $\mathbf{e}' \leftarrow [-B_{e'}, B_{e'}]^m$. $\parallel \text{probabilistic public evaluation}$ Return $\mathbf{h} := \mathbf{P} \cdot \mathbf{w} + \mathbf{e}' \in \mathbb{Z}_q^m$. $\mathbf{h} \leftarrow \text{Priv}(sk = \mathbf{K}, \mathbf{c}): \parallel \text{probabilistic private evaluation}$ $\mathbf{e}' \leftarrow [-B_{e'}, B_{e'}]^m$. Return $\mathbf{h} := \mathbf{K} \cdot \mathbf{c} + \mathbf{e}' \in \mathbb{Z}_q^m$. $fv \leftarrow \text{FPriv}(fk_{\mathbf{y}} = \mathbf{f}^\top, \mathbf{c}):$ Return $fv := \mathbf{f}^\top \cdot \mathbf{c} \in \mathbb{Z}_q$.

Fig. 4. The LWE-based (outer-level) IPHPS_{LWE}, with *probabilistic* Pub, Priv algorithms.

Formally, our LWE-based probabilistic IPHPS_{LWE} = (Setup_{IPHPS}, α , μ , Pub, Priv, FPriv) is presented in Fig. 4. It is easy to check the functional correctness of IPHPS_{LWE}, i.e., $\text{FPriv}(fk_{\mathbf{y}}, \mathbf{c}) = \mathbf{y}^\top \cdot \mathbf{K} \cdot \mathbf{c} = \langle \mathbf{A}_{sk}(\mathbf{c}), \mathbf{y} \rangle$ holds. Now we analyze the (statistical) evaluation indistinguishability of IPHPS_{LWE}. For any $\mathbf{c} = \mathbf{A}\mathbf{w} + \mathbf{e} \in \mathcal{L}$ with witness $(\mathbf{w}, \mathbf{e})$, the statistical distance $\Delta(\text{Pub}(pk, \mathbf{c}, (\mathbf{w}, \mathbf{e})), \text{Priv}(sk, \mathbf{c})) = \Delta((\mathbf{K}\mathbf{A}) \cdot \mathbf{w} + \mathbf{e}', \mathbf{K} \cdot (\mathbf{A}\mathbf{w} + \mathbf{e}) + \mathbf{e}') \leq \Delta(\mathbf{e}', \mathbf{K}\mathbf{e} + \mathbf{e}') \leq m \cdot lB_K B_{\mathcal{L}}/B_{e'}$, according to the smudging lemma (i.e., Lemma 2) and the fact that $\|\mathbf{K} \cdot \mathbf{e}\|_\infty \leq lB_K B_{\mathcal{L}}$ for $sk = \mathbf{K} \leftarrow \chi_K^{m \times l}$ with χ_K being B_K -bounded, $\mathbf{e} \in [-B_{\mathcal{L}}, B_{\mathcal{L}}]^l$ and $\mathbf{e}' \leftarrow [-B_{e'}, B_{e'}]^m$. This shows the $(mlB_K B_{\mathcal{L}}/B_{e'})$ -statistical evaluation indistinguishability of IPHPS_{LWE}, where $mlB_K B_{\mathcal{L}}/B_{e'} \leq m \cdot \varepsilon_{\text{ppLWE}}$ is negligible assuming $\varepsilon_{\text{ppLWE}}$ defined in (7) is negligible.

Next, we analyze the (multi-fold) SMP related to IPHPS_{LWE}, which asks to distinguish $\{\mathbf{c}_i = \mathbf{A}\mathbf{w}_i + \mathbf{e}_i \leftarrow \mathcal{L}\}_i$ from $\{\mathbf{c}_i = \mathbf{A}\mathbf{w}_i + \mathbf{B}\mathbf{v}_i + \mathbf{e}_i \leftarrow \mathcal{X}\}_i$, where each $\mathbf{w}_i \leftarrow \mathbb{Z}_q^n$, $\mathbf{v}_i \leftarrow \mathbb{Z}_q^{l-l'}$ and $\mathbf{e}_i \leftarrow \chi_{\mathcal{L}}^l$. It is not difficult to see that the multi-fold SMP related to IPHPS_{LWE} is hard based on the multi-secret LWE _{$n, q, \chi_{\mathcal{L}}, l$} -assumption, since $\{\mathbf{c}_i = \mathbf{A}\mathbf{w}_i + \mathbf{e}_i \leftarrow \mathcal{L}\}_i \stackrel{c}{\approx} \{\mathbf{c}_i = \mathbf{u}_i \leftarrow \mathbb{Z}_q^l\}_i \equiv \{\mathbf{c}_i = \mathbf{u}_i + \mathbf{B}\mathbf{v}_i \leftarrow \mathbb{Z}_q^l\}_i \stackrel{c}{\approx} \{\mathbf{c}_i = \mathbf{A}\mathbf{w}_i + \mathbf{e}_i + \mathbf{B}\mathbf{v}_i \leftarrow \mathcal{X}\}_i$. As for the multi-secret LWE _{$n, q, \chi_{\mathcal{L}}, l$} -assumption, a recent work [39] establishes an almost tight reduction from LWE to multi-secret LWE, building upon the work of [9]. In particular, it shows that the multi-secret LWE _{$n, q, \chi_{\mathcal{L}}, l$} -assumption can be almost tightly reduced to LWE _{$\lambda, q, D_{\mathbb{Z}, \sigma_{\text{LWE}}}, l$} w.r.t. a related error distribution $D_{\mathbb{Z}, \sigma_{\text{LWE}}}$. We recall this result in Theorem 11 in Appendix G.2. Consequently, the hardness of the multi-fold SMP of IPHPS_{LWE} can be almost tightly reduced to LWE _{$\lambda, q, D_{\mathbb{Z}, \sigma_{\text{LWE}}}, l$} , and we have the following theorem as a direct corollary of Theorem 11 in Appendix G.2.

Theorem 2 (Almost Tight Multi-fold SMP related to IPHPS_{LWE}). *Let $\chi_{\mathcal{L}} = D_{\mathbb{Z}, \sigma_{\mathcal{L}}}$ be a discrete Gaussian with parameter $\sigma_{\mathcal{L}}$ satisfying $\sigma_{\mathcal{L}} = \sqrt{\sigma_0^2 + r^2}$, $\sigma_0 > \sigma_{\text{LWE}} \cdot C \cdot \sqrt{l} \cdot \sigma_1$, $\frac{q}{\sigma_1} \geq \sqrt{\ln(4n)/\pi}$ and $r \geq \sqrt{\lambda}$ for some $\sigma_0, \sigma_1, r, \sigma_{\text{LWE}} > 0$, where C is the global constant from Lemma 19 in Appendix G.2. Then for any adversary $\mathcal{A}$ and any Q , there exist adversaries $\mathcal{B}'$ and $\mathcal{B}$ s.t. $\mathbf{T}(\mathcal{B}') \approx \mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + Q \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and*

$$\text{Adv}_{\text{IPHPS}_{\text{LWE}}, \mathcal{A}, Q}^{\text{msmp}}(\lambda) \leq 2 \cdot \text{Adv}_{[n, q, \chi_{\mathcal{L}}, l], \mathcal{B}'}^{Q\text{-seccr-LWE}}(\lambda) \leq 4cn \cdot \text{Adv}_{[\lambda, q, D_{\mathbb{Z}, \sigma_{\text{LWE}}}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + \frac{2Q(l+c+1)}{2^\lambda},$$

where c is an integer such that

$$n \geq (\lfloor \frac{l}{c} \rfloor \log q + \lambda \log q + 2\lambda + 1) / \log(\sigma_1). \quad (8)$$

(Inner-Level) co-Hash Scheme. Next we describe the LWE-based (inner-level) co-Hash scheme. Similarly, to accommodate instantiations based on LWE, we need to relax the syntax of (inner-level) co-Hash to be a *probabilistic* one, by allowing the core hash function $\Gamma_s(c)$ to be a *randomized* function that may toss coins. For completeness, we present the formal definition of *probabilistic* co-Hash in Def. 18 in Appendix G.3. We also note that probabilistic co-Hash suffices for the generic IPFE construction in Sect. 5, as it is only used for the mCPA security proof of IPFE, during which it does not matter whether the core hash function is deterministic or randomized.

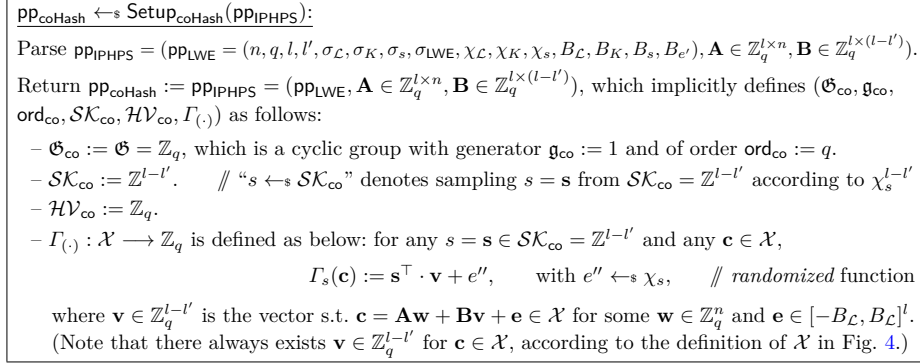


Fig. 5. The LWE-based (inner-level) $\text{coHash}_{\text{LWE}}$, where $\Gamma_s(\mathbf{c})$ is a *randomized* function.

Our LWE-based probabilistic $\text{coHash}_{\text{LWE}} = \text{Setup}_{\text{coHash}}$ is shown in Fig. 5.

Through the following two theorems, we prove the functional smoothness and the tight multi-key-extracting of our $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$.

Theorem 3 (Functional Smoothness of $\text{TLHPS}_{\text{LWE}}$). *Let $\chi_K = D_{\mathbb{Z}, \sigma_K}$ be a discrete Gaussian with parameter $\sigma_K \geq \sqrt{m} \cdot X \cdot l'$, and let $l' \geq 2n \log q$. Moreover, suppose that pp_{LWE} has a negligible $\epsilon_{\text{pp}_{\text{LWE}}}$ as shown in (7). Then the two-leveled scheme $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ proposed in Fig. 4 and 5 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, Q}^{\text{funSmt}}(\lambda) \leq 6QmX \cdot \epsilon_{\text{pp}_{\text{LWE}}}$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q , where $\mathcal{A}$ is required to query $\mathcal{O}_{\text{CHAL}}^b$ with vector $\tilde{\mathbf{x}}$ of bounded norm X , i.e., $\|\tilde{\mathbf{x}}\|_\infty \leq X$.*

Theorem 4 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{LWE}}$). *Let $\chi_s = D_{\mathbb{Z}, \sigma_s}$, $l \geq (l - l' + 1) \log q + 2\lambda$, and $l - l' \geq \lambda$. Moreover, suppose that pp_{LWE} has a negligible $\epsilon_{\text{pp}_{\text{LWE}}}$ as shown in (7). If the $\text{LWE}_{\lambda, q, D_{\mathbb{Z}, \sigma_{\text{LWE}}}, l}$ -assumption holds, then the two-leveled scheme $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ proposed in Fig. 4 and 5 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{[\lambda, q, D_{\mathbb{Z}, \sigma_{\text{LWE}}}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + 5TQ \cdot \epsilon_{\text{pp}_{\text{LWE}}}$.*

Due to space limitations, we postpone the proofs of Theorem 3 and Theorem 4 to Appendix G.4 and Appendix G.5, respectively. We note that the proof of Theorem 3 involves a fine-grained statistical analysis of discrete Gaussians, and Theorem 4 basically follows from the tight reductions from LWE to Multi-instance LWE (as presented in Appendix G.2).

6.2 Instantiation of IPFE from LWE

By plugging our $\text{TLHPS}_{\text{LWE}}$ proposed in Fig. 4 and 5 into the generic construction $\text{IPFE}[\text{TLHPS}, L]$ in Sect. 5, we immediately obtain an LWE-based IPFE scheme $\text{IPFE}_{\text{LWE}} = \text{IPFE}[\text{TLHPS}_{\text{LWE}}, L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$, as shown in Fig. 6.

$(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^m):$ $\text{pp}_{\text{LWE}} = (n, q, l, l', \sigma_{\mathcal{L}}, \sigma_K, \sigma_s, \sigma_{\text{LWE}}, \chi_{\mathcal{L}}, \chi_K, \chi_s, B_{\mathcal{L}}, B_K, B_s, B_{e'})$. $\mathbf{A} \leftarrow \mathbb{Z}_q^{l \times n}$. $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$. For $i \in [\tilde{m}]$: $\mathbf{K}_i \leftarrow \chi_K^{m \times l} = \mathcal{D}_{\mathbb{Z}, \sigma_K}^{m \times l}$. $\mathbf{P}_i := \mathbf{K}_i \cdot \mathbf{A} \in \mathbb{Z}_q^{m \times n}$. $\text{mpk} := (\text{pp}_{\text{LWE}}, \mathbf{A} \in \mathbb{Z}_q^{l \times n}, \{\mathbf{P}_i\}_{i \in [\tilde{m}]})$. $\text{msk} := \{\mathbf{K}_i\}_{i \in [\tilde{m}]}$. Return (mpk, msk) . $sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}, \mathbf{y} \in \mathbb{Z}_q^m):$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$. For $i \in [\tilde{m}]$: $\mathbf{f}_i^\top := \mathbf{y}^\top \cdot \mathbf{K}_i \in \mathbb{Z}_q^{1 \times l}$. Return $sk_{\mathbf{y}} := (\mathbf{f}_1, \dots, \mathbf{f}_{\tilde{m}}, \mathbf{y})$.	$ct \leftarrow \text{Enc}(\text{mpk}, \mathbf{x} \in \mathbb{Z}_q^m):$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$. For $i \in [\tilde{m}]$: $\mathbf{w}_i \leftarrow \mathbb{Z}_q^n$. $\mathbf{e}_i \leftarrow \chi_{\mathcal{L}}^l = \mathcal{D}_{\mathbb{Z}, \sigma_{\mathcal{L}}}^l$. $\mathbf{e}'_i \leftarrow [-B_{e'}, B_{e'}]_{\tilde{m}}$. $\mathbf{c}_i := \mathbf{A} \mathbf{w}_i + \mathbf{e}_i \in \mathbb{Z}_q^l$. $\hat{\mathbf{e}} := \sum_{i=1}^{\tilde{m}} (\mathbf{P}_i \cdot \mathbf{w}_i + \mathbf{e}'_i) + \lfloor \frac{q}{mXY} \rfloor \cdot \mathbf{x} \in \mathbb{Z}_q^m$. Return $ct := (\mathbf{c}_1, \dots, \mathbf{c}_{\tilde{m}}, \hat{\mathbf{e}})$. $d/\perp \leftarrow \text{Dec}(sk_{\mathbf{y}}, ct):$ Parse $sk_{\mathbf{y}} = (\mathbf{f}_1, \dots, \mathbf{f}_{\tilde{m}}, \mathbf{y})$, $ct = (\mathbf{c}_1, \dots, \mathbf{c}_{\tilde{m}}, \hat{\mathbf{e}})$. $\mathfrak{d} := \mathbf{y}^\top \cdot \hat{\mathbf{e}} - \sum_{i=1}^{\tilde{m}} \mathbf{f}_i^\top \cdot \mathbf{c}_i \in \mathbb{Z}_q$. Return $d \in [-mXY, mXY]$ that minimizes the value $ \mathfrak{d} - \lfloor \frac{q}{mXY} \rfloor \cdot d $.
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Fig. 6. The LWE-based IPFE scheme $\text{IPFE}_{\text{LWE}} = \text{IPFE}[\text{TLHPS}_{\text{LWE}}, L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ parameterized by $L \in \mathbb{N}$.

Since $\text{TLHPS}_{\text{LWE}}$ is a probabilistic one that has only approximate correctness, we analyze the correctness of IPFE_{LWE} specifically. Note that during decryption,

$$\begin{aligned}
\mathfrak{d} &:= \mathbf{y}^\top \cdot \hat{\mathbf{e}} - \sum_{i=1}^{\tilde{m}} \mathbf{f}_i^\top \cdot \mathbf{c}_i \\
&= \mathbf{y}^\top \cdot \left(\sum_{i=1}^{\tilde{m}} (\mathbf{K}_i \mathbf{A} \cdot \mathbf{w}_i + \mathbf{e}'_i) + \lfloor \frac{q}{mXY} \rfloor \cdot \mathbf{x} \right) - \sum_{i=1}^{\tilde{m}} \mathbf{y}^\top \cdot \mathbf{K}_i \cdot (\mathbf{A} \mathbf{w}_i + \mathbf{e}_i) \\
&= \lfloor \frac{q}{mXY} \rfloor \cdot \langle \mathbf{x}, \mathbf{y} \rangle + \mathbf{y}^\top \cdot \sum_{i=1}^{\tilde{m}} (\mathbf{e}'_i - \mathbf{K}_i \mathbf{e}_i).
\end{aligned}$$

The norm of the error term is $\|\mathbf{y}^\top \cdot \sum_{i=1}^{\tilde{m}} (\mathbf{e}'_i - \mathbf{K}_i \mathbf{e}_i)\|_\infty \leq mY \cdot \tilde{m}(B_{e'} + lB_K B_{\mathcal{L}})$. To ensure the correctness of IPFE_{LWE} , it suffices to set the parameters such that $mY \cdot \tilde{m}(B_{e'} + lB_K B_{\mathcal{L}}) \leq \lfloor \frac{q}{mXY} \rfloor / 4$ holds. We refer to Table 2 for a suggestion of parameters pp_{LWE} that satisfies the condition.

One may notice that the $\text{IPFE}_{\text{LWE}} = \text{IPFE}[\text{TLHPS}_{\text{LWE}}, L]$ in Fig. 6 is slightly different from the generic construction $\text{IPFE}[\text{TLHPS}, L]$ presented in Fig. 3 in Sect. 5: here we use $\lfloor \frac{q}{mXY} \rfloor \in \mathbb{Z}_q$, instead of the generator $\mathbf{g}_{\text{co}} = 1$ of $\mathfrak{G}_{\text{co}} = \mathbb{Z}_q$, to encode/decode $\mathbf{x}$ during encryption/decryption of IPFE_{LWE} . As we analyzed above, this change is intended to handle the LWE errors and ensure the correctness of IPFE_{LWE} , and it will not affect the security of IPFE_{LWE} . To see this, recall that the security proof of IPFE (cf. Subsect. 2.3 for an overview) first triggers inner-level hash values $\{\Gamma_{s_i}(\mathbf{c}_i^{(j)}) \cdot \mathbf{x}^{(j)}\}_{i,j}$ by the functional smoothness of TL-IP-HPS, and then changes them to uniformly random elements $\{(r_i^{(j)} \cdot \mathbf{g}_{\text{co}}) \cdot \mathbf{x}^{(j)}\}_{i,j} = \{(r_i^{(j)} \cdot 1) \cdot \mathbf{x}^{(j)}\}_{i,j}$ with $r_i^{(j)} \leftarrow \mathbb{Z}_q$ by the multi-key-extracting of TL-IP-HPS, which are sufficient to mask the $\{\lfloor \frac{q}{mXY} \rfloor \cdot \mathbf{x}^{(j)}\}_{i,j}$ in challenge ciphertexts.

By plugging Theorem 2, Theorem 3 and Theorem 4 regarding the (almost) tight properties of $\text{TLHPS}_{\text{LWE}}$ into Theorem 1, we have the following corollary showing the almost tight mCPA security of $\text{IPFE}_{\text{LWE}} = \text{IPFE}[\text{TLHPS}_{\text{LWE}}, L]$ based on the LWE assumption.

Corollary 1 (Almost Tight mCPA Security of IPFE_{LWE}). Assume that the $\text{LWE}_{\lambda, q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}, l}$ -assumption holds and the parameters pp_{LWE} have a negligible $\epsilon_{\text{pp}_{\text{LWE}}}$ as shown in (7) and satisfy all conditions in Theorems 2 to 4. Then the scheme IPFE_{LWE} in Fig. 6 is mCPA secure.

Concretely, for any adversary $\mathcal{A}$ who makes at most Q_e queries to $\mathcal{O}_{\text{ENC}}$ and Q_k queries to $\mathcal{O}_{\text{GEN}}$, there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + (Q_e + Q_k) \cdot \text{poly}(\lambda)$, with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and

$$\text{Adv}_{\text{IPFE}_{\text{LWE}}, \mathcal{A}}^{\text{mCPA}}(\lambda) \leq L \cdot (8cn + 1) \cdot \text{Adv}_{[\lambda, q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + \text{statist. loss},$$

where c is an integer satisfying (8) and $\text{statist. loss} = LQ_e \tilde{m}(9 + 7mX) \cdot \epsilon_{\text{pp}_{\text{LWE}}}$.

We note that the above statistical loss $LQ_e \tilde{m}(9 + 7mX) \cdot \epsilon_{\text{pp}_{\text{LWE}}}$ is sub-exponentially small assuming $\epsilon_{\text{pp}_{\text{LWE}}}$ is, and in particular less than $2^{-\sqrt{\lambda}}$ if $\epsilon_{\text{pp}_{\text{LWE}}} < 2^{-1.25\sqrt{\lambda}}$. This is because L is a constant and $Q_e, \tilde{m}, m, X$ are all polynomials in λ .

Setting the Parameters. We give a suggestion for parameters $\text{pp}_{\text{LWE}} = (n, q, l, l', \sigma_{\mathcal{L}}, \sigma_K, \sigma_s, \sigma_{\text{LWE}}, \chi_{\mathcal{L}}, \chi_K, \chi_s, B_{\mathcal{L}}, B_K, B_s, B_{e'})$ in Table 2, so that the $\epsilon_{\text{pp}_{\text{LWE}}}$ defined in (7) is sub-exponentially small ($< 2^{-1.25\sqrt{\lambda}}$), all conditions in Theorems 2 to 4 are met, the correctness of IPFE_{LWE} is guaranteed, and the statistical loss in Corollary 1 is sub-exponentially small ($< 2^{-\sqrt{\lambda}}$).

Table 2. Parameter setting, where λ denotes the security parameter.

Parameters pp_{LWE}	n	q	l	l'	$\sigma_{\mathcal{L}}$	σ_K	σ_s	σ_{LWE}	$B_{e'}$
Setting	11λ	$2^{7\sqrt{\lambda}}$	λ^2	$\lambda^2 - 2\lambda$	$\lambda^{2.5} \cdot 2^{\sqrt{\lambda}}$	$2^{3\sqrt{\lambda}}$	$2^{1.5\sqrt{\lambda}}$	λ	$2^{6\sqrt{\lambda}}$

Parameters pp_{LWE} (cont.)	$\chi_{\mathcal{L}}$	χ_K	χ_s	$B_{\mathcal{L}}$	B_K	B_s
Setting	$D_{\mathbb{Z}, \lambda^{2.5} \cdot 2^{\sqrt{\lambda}}}$	$D_{\mathbb{Z}, 2^{3\sqrt{\lambda}}}$	$D_{\mathbb{Z}, 2^{1.5\sqrt{\lambda}}}$	$\lambda^3 \cdot 2^{\sqrt{\lambda}}$	$\sqrt{\lambda} \cdot 2^{3\sqrt{\lambda}}$	$\sqrt{\lambda} \cdot 2^{1.5\sqrt{\lambda}}$

Parameters for Theorem 2	σ_0	σ_1	r	c
Setting	$\frac{\sqrt{2}}{2} \lambda^{2.5} 2^{\sqrt{\lambda}}$	$2^{\sqrt{\lambda}}$	$\frac{\sqrt{2}}{2} \lambda^{2.5} 2^{\sqrt{\lambda}}$	2λ

Under the parameters in Table 2, the mCPA security of our IPFE_{LWE} scheme can be almost tightly reduced to the $\text{LWE}_{\lambda, q, D_{\mathbb{Z}}, \lambda, \lambda^2}$ -assumption with security loss factor $O(L \cdot \lambda^2)$. By setting L small constants (say $L = 5$ or $L = 100$), our IPFE_{LWE} scheme constitutes the *first* almost tightly secure IPFE scheme based on the LWE assumption.

Note that the modulus q of our IPFE_{LWE} scheme is super-polynomial, and this is because we rely on the noise smudging technique (i.e., Lemma 2) for the security of the underlying TLHPS_{LWE}. It is always desirable to have a smaller modulus. However, we stress that our work takes the first theoretical step in investigating whether (almost) tightly secure IPFE schemes from lattices is possible and how to achieve it. We believe that our ideas may open the door to further improvements. Once one can construct (almost) tightly secure LWE-based

TLHPS with smaller modulus, then through our generic IPFE construction in Sect. 5, we can immediately obtain (almost) tightly secure LWE-based IPFE scheme with smaller modulus.

Besides, we note that the security loss of our IPFE_{LWE} scheme is $O(L \cdot \lambda^2)$ when reducing to the $\text{LWE}_{\lambda, q, D_{\mathbb{Z}}, \lambda, \lambda^2}$ -assumption. This quadratic security loss is incurred in Theorem 2, which in turn stems from the random self-reducibility of LWE (i.e., the reduction from LWE to Multi-secret LWE given in [39], as recalled in Theorem 11 in Appendix G.2). If one can come up with a tighter reduction from LWE to Multi-secret LWE, say $O(\lambda)$, then one can immediately get an LWE-based IPFE scheme with tighter security via our generic construction.

These reflect the theoretical merit of our generic IPFE construction, and we leave the above improvements as interesting future work.

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Appendix

A Comparison of Ciphertext Size: An Example

The size of ciphertexts might be the most important factor affecting the communication costs of IPFE. Below we compare the ciphertext size of our tightly secure IPFE schemes with the loosely secure ALS schemes [8], which are currently ones of the most efficient adaptively secure IPFE schemes.

Let us consider the scenario of using IPFE schemes for DNA analysis, as illustrated in [61], where the vector length m can be as large as 2^{20} for DNA sequences and the number of ciphertexts Q can be as large as 2^{27} . The comparison is conducted under a target security level of 128 bits.

The following comparison shows that

- Our DDH-based and DCR-based tightly-secure IPFE schemes have *shorter* ciphertexts than the DDH-based and DCR-based ALS schemes, which were ones of the most efficient adaptively secure IPFE schemes prior to our work.
- Our LWE-based almost-tightly secure IPFE scheme has *shorter* ciphertexts than the LWE-based ALS scheme in certain cases, which was one of the most efficient lattice-based, adaptively secure IPFE schemes prior to our work.

These suggest that our schemes significantly improve the efficiency of the existing tightly secure IPFE schemes [61, 48, 49], and even outperform the efficiency of the loosely secure ALS schemes [8] in some situations.

Comparison of DDH-based Schemes. We first compare our DDH-based IPFE scheme with the DDH-based ALS scheme [8], and show their ciphertext size and security loss factor.

- The DDH-based ALS scheme [8] has $m + 2$ group elements in ciphertext, and has security loss factor $O(Q)$. In our work, we give a tighter analysis on its security loss, reducing it from $O(Q)$ to $O(m) = 3m$.

To compensate the optimized security loss $3m \approx 2^{22}$, each group element should be of $2 \times (128 + 22) = 300$ bits. Therefore, the DDH-based ALS scheme has ciphertext of $\approx 300m$ bits.

- Our DDH-based scheme (under $L = 100$) has $1.02m$ group elements in ciphertext and has a tight security loss factor $O(1) = 300$ (cf. Table 1 in Sect. 1).

To compensate the security loss $300 \approx 2^9$, each group element should be of $2 \times (128 + 9) = 274$ bits. Therefore, our DDH-based scheme has *shorter* ciphertext of $1.02m \times 274 \approx 280m$ bits.

The comparison shows that our DDH-based scheme has *shorter* ciphertexts than the DDH-based ALS scheme after the compensation for security loss.

Comparison of DCR-based Schemes. Next we compare the ciphertext size of the DCR-based schemes.

- The DCR-based ALS scheme [8] has $m + 1$ elements in $\mathbb{Z}_{N^2}$ in ciphertext, and has security loss factor $O(Q)$. Our work optimizes their loss factor from $O(Q)$ to $O(m) = 3m$.

To compensate the optimized security loss $3m \approx 2^{22}$, the modulus N needs to be about 4120 bits (which corresponds to the security level of $128 + 22 = 150$ bits) according to NIST’s recommendation¹². Therefore, the DCR-based ALS scheme has ciphertext of $(m + 1) \times 2 \times 4120 \approx 8240m$ bits.

- Our DCR-based scheme (under $L = 100$) has $1.01m$ elements in $\mathbb{Z}_{N^2}$ in ciphertext, and has a tight security loss factor $O(1) = 300$ (cf. Table 1 in Sect. 1).

To compensate the security loss $300 \approx 2^9$, the modulus N needs to be about 3350 bits (which corresponds to the security level of $128 + 9 = 137$ bits) according to NIST’s recommendation (see Footnote 12). Therefore, our DCR-based scheme has *shorter* ciphertext of $1.01m \times 2 \times 3350 \approx 6767m$ bits.

The comparison shows that our DCR-based scheme has *shorter* ciphertexts than the DCR-based ALS scheme after the compensation for security loss.

Comparison of LWE-based Schemes. Finally we compare the ciphertext size of the LWE-based schemes.

- The LWE-based ALS scheme [8] has $m + l$ elements in $\mathbb{Z}_q$ in ciphertext, and has security loss factor $O(Q)$. Note that our economic proof strategy does not apply to the LWE-based ALS scheme to reduce its security loss, as noted in Footnote 1.

To compensate the security loss $O(Q) \approx 2^{27}$, according to the parameter conditions shown in [8, page 12], it requires that $q \geq \Omega(\sqrt{n}/\alpha')$ with $\alpha' := \Omega(\alpha/(n^6 K \log^2 q \log^{5/2} n))$ and $l \geq 4n \log q$. By simply taking the constants hidden in $\Omega(\cdot)$ as 1 and taking equality for “ $\geq$ ”, we give a conservative parameter setting for ALS:

$$q = \sqrt{n}/\alpha' = \sqrt{n}(n^6 K \log^2 q \log^{5/2} n)/\alpha \geq n^{6.5} \log^{5/2} n \quad \text{and} \quad l = 4n \log q. \quad (9)$$

Besides, for the security of the LWE-based ALS scheme to hold, as stated in [8, page 12], it further requires that the vector length m satisfies $m \leq (1-c) \cdot n$ for some $0 < c < 1$.¹³ This means that $n \geq m \approx 2^{20}$ for ALS. Then by plugging $n = 2^{20}$ into (9), we get a concrete conservative parameter setting for ALS:

$$q \approx (2^{20})^{6.5} \times (20)^{5/2} \approx 2^{140} \quad \text{and} \quad l \approx 4 \times 2^{20} \times 140 \approx 2^{29}.$$

¹² NIST’s recommendation for key length can be found in <https://csrc.nist.gov/csrc/media/projects/cryptographic-module-validation-program/documents/fips140-2/fips1402ig.pdf>. More precisely, here the values 4120 bits (and 3350 bits) are computed according to the equation (1) on page 126 of the NIST document.

¹³ Note that the vector length is denoted by l in [8, page 12], which is slightly different from our notations where the vector length is denoted by m .

Therefore, the LWE-based ALS scheme has ciphertext of $\approx (m + 2^{29}) \times 140 \approx 2^7 m + 2^{36}$ bits.

- Our LWE-based scheme (under $L = 100$) has $(0.01l + 1)m$ elements in $\mathbb{Z}_q$ in ciphertext and has an almost tight security loss factor $O(100 \cdot \lambda^2)$ (cf. Table 1 in Sect. 1 and Fig. 6 in Subsect. 6.2).

To compensate the security loss $O(100 \cdot \lambda^2) \approx 2^{20}$, according to our parameter setting shown in Table 2 in Subsect. 6.2, we can set

$$q \approx 2^{7\sqrt{\lambda}} \approx 2^{85} \quad \text{and} \quad l \approx \lambda^2 \approx 21904$$

(corresponding to the security level of $128 + 20 = 148$ bits). Thus, our LWE-based scheme has ciphertext of $\approx (0.01 \times 21904 + 1) \times m \times 85 \approx 2^{14} m$ bits.

The comparison shows that our LWE-based scheme has *shorter* ciphertexts than the LWE-based ALS scheme when the vector length $m \leq 2^{22}$, after the compensation for security loss.

B Comparison with Existing Frameworks from HPS & Discussions on Simulation-based Security of IPFE

COMPARISON WITH EXISTING IPFE FRAMEWORKS FROM HPS. There are existing frameworks for constructing IPFE from (variants of) HPS [66, 13, 17]. But all these frameworks establish the security of the resulting IPFE schemes in the *single-challenge* setting¹⁴, and it seems hard to extend their techniques to achieve tight security in the multi-challenge setting. Moreover, all these frameworks have instantiations only under the DDH, DCR or HSM (a class group variant of DCR) assumptions, but do not support instantiations from lattices.

As far as we understand, probably the most important reason why these frameworks can only work in the *single-challenge* setting is due to their proof strategies (and the properties they defined for their HPS variants). Roughly speaking, all these frameworks essentially prove the security of their IPFE in two main steps: they first change the HPS instance in the challenge ciphertext from $c \leftarrow_s \mathcal{L}$ to $c \leftarrow_s \mathcal{X}$ by the hardness of SMP; then they resort to various statistical properties of their HPS variants to argue that the remained entropy in the hashing key sk can be used to randomize the (single) challenge ciphertext. Their proof strategies resemble that in [21] and are similar to the proof strategy of the ALS scheme [8] (cf. Subsect. 2.1), which all consider a single challenge.

Such proof strategy can hardly be extended to the multi-challenge setting, since the entropy in the hashing key sk is bounded, and it can not be used to randomize multiple (Q) challenge ciphertexts simultaneously via statistical properties of HPS alone. Consequently, their proof strategies need a hybrid argument and incur a security loss of $O(Q)$.

¹⁴ We note that the “tighter proof” in the title of [17] refers to tighter proof in the single-challenge setting, rather than in the multi-challenge setting.

In contrast, our framework for constructing tightly secure IPFE from TL-IP-HPS is different from theirs. To achieve tight security, we equip our TL-IP-HPS with crucial properties of (statistical) functional smoothness¹⁵ and (computational) multi-key-extracting. Once the instances are changed to $\{c^{(j)} \leftarrow_s \mathcal{X}\}_j$, we can apply the statistical functional smoothness of TL-IP-HPS to trigger multiple inner-level core hash values $\{\Gamma_s(c^{(j)})\}_j$ simultaneously, and then apply the *computational* multi-key-extracting of TL-IP-HPS to amplify them to uniformly random values to randomize multiple challenge ciphertexts simultaneously.

Moreover, our framework also enables instantiations from LWE following a recent paradigm in [39]. In particular, we propose our LWE-based TL-IP-HPS by carefully design the (randomized) inner-level co-Hash, which on the one hand is “*weak*” enough to be triggered by the outer-level IP-HPS, but on the other hand “*strong*” enough to be amplified to uniformly random values based on LWE (cf. the explanations in Subsect. 2.2). Consequently, our framework from TL-IP-HPS yields the first almost tightly secure IPFE scheme based on the LWE assumption.

ON ACHIEVING TIGHT SIMULATION-BASED SECURITY IN MULTI-CHALLENGE SETTING. There are two types of security notions for IPFE: indistinguishability-based (IND) security [8] and simulation-based (SIM) security [7].

In our work, we focus on achieving tight adaptive IND-based security, which is in fact the strongest security notion for IPFE in multi-challenge setting. The reason is as follows.

- On one hand, as demonstrated in [16, 7], adaptive SIM-based security is impossible to achieve in the multi-challenge setting for IPFE, no matter tightly or non-tightly.
- On the other hand, as shown in [55, 3], non-adaptive SIM-based security is tightly implied by the adaptive IND-based security, even in the multi-user and multi-challenge setting.

Overall, adaptive IND-based security is the strongest notion that IPFE can achieve in the multi-challenge setting, as achieved by our tightly secure IPFE schemes.

Indeed, all existing tightly secure IPFE schemes [61, 48, 49] prove the adaptive IND-based security of their schemes. This is reasonable, as the adaptive IND-based security provides sufficient security guarantees for IPFE in realistic scenarios: according to the definition of adaptive IND-based security of IPFE

¹⁵ We note that in [17], Castagnos et al. defined for their HPS variant a statistical property called vector smoothness, which may look similar to the functional smoothness of our TL-IP-HPS. But we stress that these two notions of smoothness are fundamentally different: their vector smoothness considers the smoothness of only a *single* HPS hash value (which can be used to randomize a single challenge ciphertext), while our functional smoothness requires that *multiple* IP-HPS hash values $\{\mathbf{A}_{sk}(c^{(j)})\}_j$ can trigger multiple inner-level hash values $\{\Gamma_s(c^{(j)})\}_j$ simultaneously (which together with our multi-key-extracting property, can be used to randomize multiple challenge ciphertexts).

(cf. Def. 2 in Sect. 3), it naturally ensures that in the presence of secret keys $sk_{\mathbf{y}}$, the encryptions of vectors $\mathbf{x}$ leak nothing beyond the inner products $\langle \mathbf{x}, \mathbf{y} \rangle$, as one expected (since otherwise, if the encryptions leak additional information beyond $\langle \mathbf{x}, \mathbf{y} \rangle$, then with the additional leaked information, a PPT adversary could trivially distinguish the encryptions of $\mathbf{x}_0$ from encryptions of $\mathbf{x}_1$).

C Proof of Theorem 1 (Tight mCPA Security of IPFE[TLHPS, L])

Theorem 1. *Assume that TLHPS = (IPHPS, coHash) is a two-leveled IP-HPS that has hard SMP, functional soundness and supports multi-key-extracting. Then the scheme IPFE[TLHPS, L] proposed in Fig. 3 is mCPA secure.*

Concretely, for any adversary $\mathcal{A}$ who makes at most Q_e queries to $\mathcal{O}_{\text{ENC}}$ and Q_k queries to $\mathcal{O}_{\text{GEN}}$, there exist adversaries $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$, such that $\mathbf{T}(\mathcal{B}_1) \approx \mathbf{T}(\mathcal{B}_2) \approx \mathbf{T}(\mathcal{A}) + (Q_e + Q_k) \cdot \text{poly}(\lambda)$, with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and

$$\text{Adv}_{\text{IPFE}[\text{TLHPS}, L], \mathcal{A}}^{\text{mCPA}}(\lambda) \leq 2L \cdot \text{Adv}_{\text{IPHPS}, \mathcal{B}_1, Q_e, \tilde{m}}^{\text{msmp}}(\lambda) + L \cdot \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda) + \text{statist. loss},$$

where $\text{statist. loss} = \tilde{m} \cdot L \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmtH}}(\lambda) \approx m \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmtH}}(\lambda)$ with $\tilde{m} := \lceil \frac{m}{L} \rceil$.

Proof of Theorem 1. We prove the theorem by defining a sequence of games $G_{-1} - G_L$ and showing adjacent games indistinguishable. The differences between adjacent games are highlighted in gray boxes, and a brief description of the differences is summarized in Table 3. By $\Pr_i[\cdot]$ we denote the probability of a particular event occurring in game G_i .

Game G_{-1} : This is the $\text{Exp}_{\text{IPFE}, \mathcal{A}}^{\text{mCPA}}$ experiment (cf. Fig. 1). Let Win denote the event that $\beta' = \beta$. By definition, $\text{Adv}_{\text{IPFE}[\text{TLHPS}, L], \mathcal{A}}^{\text{mCPA}}(\lambda) = |\Pr_{-1}[\text{Win}] - \frac{1}{2}|$.

Let $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$. Let $\text{mpk} = (\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk_1, \dots, pk_{\tilde{m}})$ and $\text{msk} = (sk_1, \dots, sk_{\tilde{m}})$ with $pk_i = \alpha(sk_i)$ for each $i \in [\tilde{m}]$. In this game, when answering an $\mathcal{O}_{\text{ENC}}$ query $(\mathbf{x}_0, \mathbf{x}_1)$, the challenger samples $c_1, \dots, c_{\tilde{m}} \leftarrow_s \mathcal{L}$ with witnesses $w_1, \dots, w_{\tilde{m}}$ respectively, computes $\mathbf{e} := \sum_{i=1}^{\tilde{m}} \text{Pub}(pk_i, c_i, w_i) + \mathbf{x}_\beta \cdot \mathbf{g}_{\text{co}}$, and returns the challenge ciphertext $ct^* := (c_1, \dots, c_{\tilde{m}}, \mathbf{e})$ to $\mathcal{A}$. Upon an $\mathcal{O}_{\text{GEN}}$ query $(\mathbf{y})$, the challenger computes $fk_{i, \mathbf{y}} := \mu(sk_i, \mathbf{y})$ for all $i \in [\tilde{m}]$, and returns the secret key $sk_{\mathbf{y}} := (fk_{1, \mathbf{y}}, \dots, fk_{\tilde{m}, \mathbf{y}}, \mathbf{y})$ to $\mathcal{A}$. Recall that to avoid trivial attacks, all $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0, \mathbf{x}_1)$ queries and all $\mathcal{O}_{\text{GEN}}(\mathbf{y})$ queries must satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$.

Game G_0 : It is the same as G_{-1} , except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0, \mathbf{x}_1)$, the challenger computes $\mathbf{e}$ without using the witness $w_1, \dots, w_{\tilde{m}}$ for $c_1, \dots, c_{\tilde{m}} \in \mathcal{L}$:

$$\bullet \mathbf{e} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i) + \mathbf{x}_\beta \cdot \mathbf{g}_{\text{co}}.$$

By the correctness of IPHPS, it holds that $\text{Pub}(pk_i, c_i, w_i) = \mathbf{A}_{sk_i}(c_i)$ for all $i \in [\tilde{m}]$. Thus, the change is just conceptual, and we have $\Pr_{-1}[\text{Win}] = \Pr_0[\text{Win}]$.

Table 3. Brief Description of Games G_{-1} , G_0 , $\{G_\zeta\}_{\zeta \in [L]}$ for the tight mCPA security proof of IPFE[TLHPS, L]. Here column “ $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ ” suggests how the j -th ($j \in [Q_e]$) challenge ciphertext $ct^{*(j)} = (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ is generated: $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_s \mathcal{L}$ are uniformly chosen, and sub-column “ $\mathbf{e}^{(j)} =$ ” shows the computation of $\mathbf{e}^{(j)}$, where $\phi_j := \text{rank}(\tilde{\mathbf{x}}^{(1)} \parallel \dots \parallel \tilde{\mathbf{x}}^{(j)}) \in [0, m]$ denotes the rank of the linear subspace generated by $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$ with each $\tilde{\mathbf{x}}^{(k)} := \mathbf{x}_1^{(k)} - \mathbf{x}_0^{(k)}$, and for all $i \in [\min\{\phi_j, \zeta \cdot \tilde{m}\}]$, $r_i^{(j)} \leftarrow_s \mathbb{Z}_{\text{ord}_{\text{co}}}$ is uniformly sampled and $\tilde{\mathbf{x}}^{(\psi_i)}$ denotes the first $\tilde{\mathbf{x}}^{(\tau)}$ such that $\text{rank}(\tilde{\mathbf{x}}^{(1)} \parallel \dots \parallel \tilde{\mathbf{x}}^{(\tau)}) = i$. Column “ $\mathcal{O}_{\text{KGEN}}$ ” shows the output returned by $\mathcal{O}_{\text{KGEN}}$.

	$\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$	$\mathcal{O}_{\text{KGEN}}(\mathbf{y})$	Justification/Assumption
	$\mathbf{e}^{(j)} =$		
G_{-1}	$\sum_{i=1}^{\tilde{m}} \text{Pub}(pk_i, c_i^{(j)}, w_i^{(j)}) + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$	$sk_{\mathbf{y}} = \begin{pmatrix} fk_{1,\mathbf{y}} \\ \vdots \\ fk_{\tilde{m},\mathbf{y}} \\ \mathbf{y} \end{pmatrix}$ $= \begin{pmatrix} \mu(sk_1, \mathbf{y}) \\ \vdots \\ \mu(sk_{\tilde{m}}, \mathbf{y}) \\ \mathbf{y} \end{pmatrix}$	The mCPA security experiment
G_0	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By correctness of IPHPS
$\vdots$	$\vdots$		–
G_ζ	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \zeta \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		$G_\zeta = G_0$ when $\zeta = 0$
$G_{\zeta+1}$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By Lemma 3 (see Table 4)
$\vdots$	$\vdots$		–
G_L	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By the fact that $\phi_j \leq m \leq L \cdot \tilde{m}$ $\Pr[\text{Win}] = \frac{1}{2}$ in G_L

Game G_ζ , $0 \leq \zeta \leq L$: This is a sequence of games starting from G_0 . For the ease of exposition, we first define some notations. Let $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ denote the j -th $\mathcal{O}_{\text{ENC}}$ query ($j \in [Q_e]$) made by $\mathcal{A}$, and let $ct^{*(j)} = (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ denote the corresponding j -th challenge ciphertext. For $j \in [Q_e]$, we define $\tilde{\mathbf{x}}^{(j)} := \mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$ as the difference between $\mathbf{x}_0^{(j)}$ and $\mathbf{x}_1^{(j)}$ modulo ord_{co} , and define $\phi_j := \text{rank}(\tilde{\mathbf{x}}^{(1)} \parallel \dots \parallel \tilde{\mathbf{x}}^{(j)}) \in [0, m]$ as the rank of the linear subspace generated by $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$. For $i \in [\phi_{Q_e}]$, we define ψ_i as the first query index $\tau \in [Q_e]$ such that $\phi_\tau = \text{rank}(\tilde{\mathbf{x}}^{(1)} \parallel \dots \parallel \tilde{\mathbf{x}}^{(\tau)})$ equals i , and thus $\tilde{\mathbf{x}}^{(\psi_i)}$ denotes the first $\tilde{\mathbf{x}}^{(\tau)}$ such that $\text{rank}(\tilde{\mathbf{x}}^{(1)} \parallel \dots \parallel \tilde{\mathbf{x}}^{(\tau)}) = i$.

With the above notations, note that for any $j \in [Q_e]$, $\tilde{\mathbf{x}}^{(\psi_1)}, \dots, \tilde{\mathbf{x}}^{(\psi_{\phi_j})}$ constitute a basis of the linear subspace generated by $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$, whose rank is ϕ_j . Moreover, for any $j \in [Q_e]$, the values of $\phi_j, \psi_1, \dots, \psi_{\phi_j}$ and $\tilde{\mathbf{x}}^{(\psi_1)}, \dots, \tilde{\mathbf{x}}^{(\psi_{\phi_j})}$ can be efficiently computed from the first j queries of $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(\eta)}, \mathbf{x}_1^{(\eta)})_{\eta \in [j]}$ made by $\mathcal{A}$, and thus are known to the challenger at the point of answering the j -th $\mathcal{O}_{\text{ENC}}$ query $(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$.

Now we are ready to describe the games. This sequence of games is the same as G_0 , except that, in G_ζ , when answering the j -th $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ query ($j \in [Q_e]$), the challenger computes $\mathbf{e}^{(j)}$ as follows:

$$\bullet \mathbf{e}^{(j)} := \begin{cases} \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}, & \text{if } \phi_j \leq \zeta \cdot \tilde{m}; \\ \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}, & \text{if } \phi_j > \zeta \cdot \tilde{m}, \end{cases}$$

where $r_1^{(j)}, \dots, r_{\min\{\phi_j, \zeta \cdot \tilde{m}\}}^{(j)} \leftarrow_s \mathbb{Z}_{\text{ord}_{\text{co}}}$ are chosen uniformly and independently of other parts of the game. It can also be written succinctly as

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \zeta \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}.$$

When $\zeta = 0$, we have $\zeta \cdot \tilde{m} = 0$, thus the terms in the above gray boxes vanish, and consequently we recover the original description of $\mathbf{G}_0$.

We will prove that $\mathbf{G}_\zeta$ and $\mathbf{G}_{\zeta+1}$ are computationally indistinguishable for all $0 \leq \zeta < L$ via the following lemma.

Lemma 3 ($\mathbf{G}_\zeta \rightarrow \mathbf{G}_{\zeta+1}$). *For all $0 \leq \zeta < L$, we have $|\Pr_\zeta[\text{Win}] - \Pr_{\zeta+1}[\text{Win}]| \leq 2 \cdot \text{Adv}_{\text{IPHPS}, \mathcal{B}_1, Q_e, \tilde{m}}^{\text{msmp}}(\lambda) + \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda) + \tilde{m} \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmtH}}(\lambda)$.*

We postpone the proof to the end of Theorem 1 in Appendix C.1.

As the sequence of games $\{\mathbf{G}_\zeta\}_{0 \leq \zeta \leq L}$ proceeds, we finally arrive at $\mathbf{G}_L$.

Game $\mathbf{G}_L$: It is the same as $\mathbf{G}_0$, except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries, the challenger computes $\mathbf{e}^{(j)}$ as follows:

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, L \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}},$$

where $r_1^{(j)}, \dots, r_{\min\{\phi_j, L \cdot \tilde{m}\}}^{(j)} \leftarrow_s \mathbb{Z}_{\text{ord}_{\text{co}}}$ are uniformly chosen. Since we always have $\phi_j \leq m \leq L \cdot \lceil \frac{m}{L} \rceil = L \cdot \tilde{m}$, it can be simply written as

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}.$$

Notice that for any $j \in [Q_e]$, $\tilde{\mathbf{x}}^{(j)}$ always lies in the linear subspace generated by $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$, of which $\tilde{\mathbf{x}}^{(\psi_1)}, \dots, \tilde{\mathbf{x}}^{(\psi_{\phi_j})}$ constitute a basis. Then $\tilde{\mathbf{x}}^{(j)}$ can be represented as a linear combination of the basis, namely, $\tilde{\mathbf{x}}^{(j)} = \sum_{i=1}^{\phi_j} c_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)}$ for some $c_1^{(j)}, \dots, c_{\phi_j}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}$. Thus together with the fact that $\mathbf{x}_\beta^{(j)} = \mathbf{x}_0^{(j)} + \beta \cdot (\mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)}) = \mathbf{x}_0^{(j)} + \beta \cdot \tilde{\mathbf{x}}^{(j)}$, it follows that

$$\begin{aligned} \mathbf{e}^{(j)} &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + (\mathbf{x}_0^{(j)} + \beta \cdot \tilde{\mathbf{x}}^{(j)}) \cdot \mathbf{g}_{\text{co}} \\ &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} (r_i^{(j)} + \beta \cdot c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_0^{(j)} \cdot \mathbf{g}_{\text{co}}. \end{aligned}$$

Since for any $j \in [Q_e]$, $r_1^{(j)}, \dots, r_{\phi_j}^{(j)}$ are uniformly chosen from $\mathbb{Z}_{\text{ord}_{\text{co}}}$, they perfectly hide β in the above gray term. Thus the challenge bit β is completely hidden to $\mathcal{A}$, and we have that $\Pr_L[\text{Win}] = \frac{1}{2}$.

Taking all things together, Theorem 1 follows. $\square$

C.1 Proof of Lemma 3 ($G_\zeta \rightarrow G_{\zeta+1}$)

Lemma 3. For all $0 \leq \zeta < L$, $|\Pr_\zeta[\text{Win}] - \Pr_{\zeta+1}[\text{Win}]| \leq 2 \cdot \text{Adv}_{\text{IPHPS}, \mathcal{B}_1, Q_e \tilde{m}}^{\text{msmp}}(\lambda) + \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda) + \tilde{m} \cdot \epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmth}}(\lambda).$

Proof of Lemma 3. Let ζ be any integer satisfying $0 \leq \zeta < L$. Note that the difference between G_ζ and $G_{\zeta+1}$ lies in the computation of $\mathbf{e}^{(j)}$ for answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries: in G_ζ , $\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \zeta \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$, while in $G_{\zeta+1}$, $\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$. For $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries satisfying $\phi_j \leq \zeta \cdot \tilde{m}$, we have both $\min\{\phi_j, \zeta \cdot \tilde{m}\} = \phi_j$ and $\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\} = \phi_j$, thus the computations of $\mathbf{e}^{(j)}$ in G_ζ and $G_{\zeta+1}$ are the same for answering such queries. Consequently, the difference actually lies in the computation of $\mathbf{e}^{(j)}$ for answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries satisfying $\phi_j > \zeta \cdot \tilde{m}$.

To show that G_ζ and $G_{\zeta+1}$ are computationally indistinguishable, we define a sequence of intermediate games $G_{\zeta,0} - G_{\zeta,\tilde{m}+3}$ between G_ζ and $G_{\zeta+1}$, where a brief description of the differences is summarized in Table 4.

Game $G_{\zeta,0}$: It is the same as G_ζ , except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger samples $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ as follows:

- if $\zeta \cdot \tilde{m} < \phi_j < (\zeta+1) \cdot \tilde{m}$, $c_1^{(j)}, \dots, c_{\phi_j - \zeta \tilde{m}}^{(j)} \leftarrow \mathcal{X}$ and $c_{\phi_j - \zeta \tilde{m} + 1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$;
- if $\phi_j \geq (\zeta+1) \cdot \tilde{m}$, $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{X}$.

That is, in the case $\phi_j > \zeta \cdot \tilde{m}$, the first $\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}$ elements $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}$ are chosen from $\mathcal{X}$ instead of $\mathcal{L}$.

Note that the witnesses for $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ are not used at all (due to the game change in G_0). Thus by the multi-fold SMP related to IPHPS, the sampling of $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)} \leftarrow \mathcal{L}$ in G_ζ is computationally indistinguishable from the sampling of $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)} \leftarrow \mathcal{X}$ in $G_{\zeta,0}$. Consequently, we have the following claim.

Claim 1. $|\Pr_\zeta[\text{Win}] - \Pr_{\zeta,0}[\text{Win}]| \leq \text{Adv}_{\text{IPHPS}, \mathcal{B}_1, Q_e \tilde{m}}^{\text{msmp}}(\lambda).$

For completeness, we provide a proof for Claim 1 in Appendix C.2.

Game $G_{\zeta,\eta}$, $0 \leq \eta \leq \tilde{m}$: It is the same as $G_{\zeta,0}$, except that, at the beginning of these games, the challenger picks $\tilde{m}$ hashing keys $s_1, \dots, s_{\tilde{m}} \leftarrow \mathcal{SK}_{\text{co}}$ of the inner-level coHash uniformly at random. Moreover, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger computes $\mathbf{e}^{(j)}$ as follows:

- $\mathbf{e}^{(j)} := \begin{cases} \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\phi_j - \zeta \tilde{m}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}, & \text{if } \zeta \cdot \tilde{m} < \phi_j \leq \zeta \cdot \tilde{m} + \eta; \\ \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}, & \text{if } \phi_j > \zeta \cdot \tilde{m} + \eta. \end{cases}$

Table 4. Brief Description of Games G_ζ , $G_{\zeta,0}$, $\{G_{\zeta,\eta}\}_{\eta \in [\tilde{m}]}$, $G_{\zeta,\tilde{m}+1}$, $G_{\zeta,\tilde{m}+2}$, $G_{\zeta,\tilde{m}+3} = G_{\zeta+1}$ for the tight mCPA security proof of IPFE[TLHPS, L], where $0 \leq \zeta < L$. Here column “ $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ ” in the case that $\phi_j > \zeta \cdot \tilde{m}$ ” suggests how the j -th ($j \in [Q_e]$) challenge ciphertext $ct^{*(j)} = (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ is generated in the case that $\phi_j > \zeta \cdot \tilde{m}$: sub-column “ $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ ” shows the generation of $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$; sub-column “ $\mathbf{e}^{(j)}$ ” shows the computation of $\mathbf{e}^{(j)}$, where $\phi_j := \text{rank}(\tilde{\mathbf{x}}^{(1)} || \dots || \tilde{\mathbf{x}}^{(j)}) \in [0, m]$ denotes the rank of the linear subspace generated by $\tilde{\mathbf{x}}^{(1)}, \dots, \tilde{\mathbf{x}}^{(j)} \in \mathbb{Z}_{\text{ord}_{\text{co}}}^m$ with each $\tilde{\mathbf{x}}^{(k)} := \mathbf{x}_1^{(k)} - \mathbf{x}_0^{(k)}$, and for all $i \in [\min\{\phi_j, (\zeta + 1) \cdot \tilde{m}\}]$, $r_i^{(j)}, r_i'^{(j)} \leftarrow_{\$} \mathbb{Z}_{\text{ord}_{\text{co}}}$ are uniformly sampled and $\tilde{\mathbf{x}}^{(\psi_i)}$ denotes the first $\tilde{\mathbf{x}}^{(\tau)}$ such that $\text{rank}(\tilde{\mathbf{x}}^{(1)} || \dots || \tilde{\mathbf{x}}^{(\tau)}) = i$, and $s_1, \dots, s_{\tilde{m}} \leftarrow_{\$} \mathcal{SK}_{\text{co}}$ are uniformly random hashing keys of the inner-level coHash. Column “ $\mathcal{O}_{\text{KGEN}}$ ” shows the output returned by $\mathcal{O}_{\text{KGEN}}$.

	$\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ in the case that $\phi_j > \zeta \cdot \tilde{m}$		$\mathcal{O}_{\text{KGEN}}(\mathbf{y})$	Justification/Assumption
	$c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$	$\mathbf{e}^{(j)} =$		
G_ζ	$c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_{\$} \mathcal{L}$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \zeta \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$	$sk_{\mathbf{y}} = \begin{pmatrix} fk_{1,\mathbf{y}} \\ \dots \\ fk_{\tilde{m},\mathbf{y}} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mu(sk_1, \mathbf{y}) \\ \dots \\ \mu(sk_{\tilde{m}}, \mathbf{y}) \\ \mathbf{y} \end{pmatrix}$	–
$G_{\zeta,0}$	$c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)} \leftarrow_{\$} \mathcal{X}$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By <i>multi-fold SMP</i> related to IPHPS
$\vdots$	$\vdots$	$\vdots$		–
$G_{\zeta,\eta}$	$c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}+1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_{\$} \mathcal{L}$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \eta\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		$G_{\zeta,\eta} = G_{\zeta,0}$ when $\eta = 0$
$G_{\zeta,\eta+1}$	$\vdots$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \eta+1\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By <i>functional smoothness</i> of TLHPS
$\vdots$	$\vdots$	$\vdots$		–
$G_{\zeta,\tilde{m}}$	$\vdots$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \tilde{m}\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		–
$G_{\zeta,\tilde{m}+1}$	$\vdots$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \tilde{m}\}} \Gamma_{s_i}^{(j)}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By <i>multi-key-extracting</i> of TLHPS
$G_{\zeta,\tilde{m}+2}$	$\vdots$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By renaming $\{r_i'^{(j)}\}_{i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ to $\{r_{\zeta \tilde{m}+i}^{(j)}\}_{i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$
$G_{\zeta,\tilde{m}+3} = G_{\zeta+1}$	$c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_{\$} \mathcal{L}$	$\sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$		By <i>multi-fold SMP</i> related to IPHPS

It can also be written succinctly as

$$\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \eta\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}.$$

Note that the inner-level hash function $\Gamma_{(\cdot)}$ may not be efficiently computable, thus the challenger may not be PPT. This does not matter, since the following arguments (before the challenger is changed back to PPT) are statistical.

When $\eta = 0$, the terms in the above gray boxes vanish, and thus we recover the original description of $\mathbf{G}_{\zeta,0}$.

Below we argue that $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$ are computationally indistinguishable for all $0 \leq \eta < \tilde{m}$. Note that for $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries satisfying $\zeta \cdot \tilde{m} < \phi_j \leq \zeta \cdot \tilde{m} + \eta$, we have both $\min\{\phi_j - \zeta \cdot \tilde{m}, \eta\} = \phi_j - \zeta \cdot \tilde{m}$ and $\min\{\phi_j - \zeta \cdot \tilde{m}, \eta + 1\} = \phi_j - \zeta \cdot \tilde{m}$, thus the computations of $\mathbf{e}^{(j)}$ in $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$ are the same for answering such queries. Consequently, the difference between $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$ lies in the computations of $\mathbf{e}^{(j)}$ for answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries satisfying $\phi_j \geq \zeta \cdot \tilde{m} + \eta + 1$:

- in $\mathbf{G}_{\zeta,\eta}$, $\min\{\phi_j - \zeta \cdot \tilde{m}, \eta\} = \eta$, and

$$\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}},$$

- in $\mathbf{G}_{\zeta,\eta+1}$, $\min\{\phi_j - \zeta \cdot \tilde{m}, \eta + 1\} = \eta + 1$, and

$$\begin{aligned} \mathbf{e}^{(j)} := & \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} \\ & + \boxed{\Gamma_{s_{\eta+1}}(c_{\eta+1}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + \eta + 1})}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}. \end{aligned}$$

For such $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries satisfying $\phi_j \geq \zeta \cdot \tilde{m} + \eta + 1$, together with the fact that $0 \leq \eta < \tilde{m}$, it follows that $1 \leq \eta + 1 \leq \min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}$, thus $c_{\eta+1}^{(j)}$ is chosen from $\mathcal{X}$ due to the game change in $\mathbf{G}_{\zeta,0}$. Moreover, since $\tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + \eta + 1})} = \mathbf{x}_1^{(\psi_{\zeta \tilde{m} + \eta + 1})} - \mathbf{x}_0^{(\psi_{\zeta \tilde{m} + \eta + 1})}$, it holds that $\langle \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + \eta + 1})}, \mathbf{y} \rangle = \langle \mathbf{x}_1^{(\psi_{\zeta \tilde{m} + \eta + 1})}, \mathbf{y} \rangle - \langle \mathbf{x}_0^{(\psi_{\zeta \tilde{m} + \eta + 1})}, \mathbf{y} \rangle = 0$ for all $\mathbf{y}$ that $\mathcal{A}$ makes an $\mathcal{O}_{\text{KGEN}}(\mathbf{y})$ query. Then by the statistical functional smoothness of TLHPS (cf. Def. 7), in the computation of $\mathbf{e}^{(j)}$, $\mathbf{A}_{sk_{\eta+1}}(c_{\eta+1}^{(j)})$ will trigger the term $\boxed{\Gamma_{s_{\eta+1}}(c_{\eta+1}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + \eta + 1})}}$, even in the view of $\mathcal{A}$ who obtains $pk_{\eta+1} = \alpha(sk_{\eta+1})$ from mpk and obtains $fk_{\eta+1,\mathbf{y}} = \mu(sk_{\eta+1}, \mathbf{y})$ from $sk_{\mathbf{y}}$ via $\mathcal{O}_{\text{KGEN}}(\mathbf{y})$ queries. Thus, $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$ are statistically close, and more precisely, we have the following claim.

Claim 2. For all $0 \leq \eta < \tilde{m}$, we have $|\Pr_{\zeta,\eta}[\text{Win}] - \Pr_{\zeta,\eta+1}[\text{Win}]| \leq \varepsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmt}}(\lambda)$.

For completeness, we provide a proof for Claim 2 in Appendix C.3.

As the sequence of games $\{\mathbf{G}_{\zeta,\eta}\}_{0 \leq \eta \leq \tilde{m}}$ proceeds, we arrive at $\mathbf{G}_{\zeta,\tilde{m}}$. We restate $\mathbf{G}_{\zeta,\tilde{m}}$ as follows.

Game $G_{\zeta, \tilde{m}}$: It is the same as $G_{\zeta, 0}$, except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger computes $\mathbf{e}^{(j)}$ as follows:

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \tilde{m}\}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}.$$

Game $G_{\zeta, \tilde{m}+1}$: It is the same as $G_{\zeta, \tilde{m}}$, except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger computes $\mathbf{e}^{(j)}$ as follows:

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \cdot \tilde{m}, \tilde{m}\}} r_i'^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}},$$

where $r_1'^{(j)}, \dots, r_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}'^{(j)} \leftarrow_{\$} \mathbb{Z}_{\text{ord}_{\text{co}}}$ are uniformly chosen. We note that the challenger is now PPT again.

Note that in $G_{\zeta, \tilde{m}}$, the hashing keys $s_1, \dots, s_{\tilde{m}}$ are used only in the computations of $\{\Gamma_{s_i}(c_i^{(j)})\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ when computing $\{\mathbf{e}^{(j)}\}_{j \in [Q_e]}$ in $\mathcal{O}_{\text{ENC}}$. Since $\{c_i^{(j)}\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ are uniformly chosen from $\mathcal{X}$ (due to the game change in $G_{\zeta, 0}$), by the multi-key-extracting property of TLHPS (cf. Def. 8), the terms $\{\Gamma_{s_i}(c_i^{(j)})\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ involved in the computations of $\{\mathbf{e}^{(j)}\}_{j \in [Q_e]}$ in $G_{\zeta, \tilde{m}}$ are computationally indistinguishable from the uniformly chosen $\{r_i'^{(j)} \cdot \mathbf{g}_{\text{co}}\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ in $G_{\zeta, \tilde{m}+1}$. Consequently, we have the following claim.

Claim 3. $|\Pr_{\zeta, \tilde{m}}[\text{Win}] - \Pr_{\zeta, \tilde{m}+1}[\text{Win}]| \leq \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda).$

For completeness, we provide a proof for Claim 3 in Appendix C.4.

Game $G_{\zeta, \tilde{m}+2}$: It is the same as $G_{\zeta, \tilde{m}+1}$, except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger computes $\mathbf{e}^{(j)}$ as follows:

$$\bullet \mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, (\zeta+1) \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}},$$

where $r_1^{(j)}, \dots, r_{\min\{\phi_j, (\zeta+1) \tilde{m}\}}^{(j)} \leftarrow_{\$} \mathbb{Z}_{\text{ord}_{\text{co}}}$ are uniformly chosen.

Note that $G_{\zeta, \tilde{m}+2}$ is essentially the same as the previous $G_{\zeta, \tilde{m}+1}$, just renaming the $\{r_i'^{(j)}\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ in $G_{\zeta, \tilde{m}+1}$ to $\{r_{\zeta \tilde{m} + i}^{(j)}\}_{j \in [Q_e], i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$. Thus, the change is just conceptual, and we have that $\Pr_{\zeta, \tilde{m}+1}[\text{Win}] = \Pr_{\zeta, \tilde{m}+2}[\text{Win}]$.

Game $G_{\zeta, \tilde{m}+3}$: It is the same as $G_{\zeta, \tilde{m}+2}$, except that, when answering $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries that satisfy $\phi_j > \zeta \cdot \tilde{m}$, the challenger changes the sampling of $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ back to:

$$\bullet c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_{\$} \mathcal{L}.$$

The change from $G_{\zeta, \tilde{m}+2}$ to $G_{\zeta, \tilde{m}+3}$ is reverse to the change from G_ζ to $G_{\zeta, 0}$. Similar to Claim 1, we have the following claim.

Claim 4. $|\Pr_{\zeta, \tilde{m}+2}[\text{Win}] - \Pr_{\zeta, \tilde{m}+3}[\text{Win}]| \leq \text{Adv}_{\text{IPHPs}, \mathcal{B}_1, Q_e \tilde{m}}^{\text{msmp}}(\lambda)$.

It is not hard to see that $G_{\zeta, \tilde{m}+3}$ is in fact the same as $G_{\zeta+1}$. Thus we have $\Pr_{\zeta, \tilde{m}+3}[\text{Win}] = \Pr_{\zeta+1}[\text{Win}]$.

This completes the proof of computational indistinguishability between G_ζ and $G_{\zeta+1}$ for all $0 \leq \zeta < L$. Taking all things together, Lemma 3 follows. $\square$

C.2 Proof of Claim 1 ($G_\zeta \rightarrow G_{\zeta,0}$)

Claim 1. $|\Pr_\zeta[\text{Win}] - \Pr_{\zeta,0}[\text{Win}]| \leq \text{Adv}_{\text{IPHPs}, \mathcal{B}_1, Q_e \tilde{m}}^{\text{msmp}}(\lambda)$.

Proof. We construct an adversary $\mathcal{B}_1$ against the multi-fold SMP related to IPHPs (cf. Def. 4), where the number of folds is $Q_e \tilde{m}$, as follows.

$\mathcal{B}_1$ is given $(\text{pp}_{\text{IPHPs}}, \{c_{i,j}\}_{i \in [\tilde{m}], j \in [Q_e]})$, where $\text{pp}_{\text{IPHPs}} \leftarrow \text{Setup}_{\text{IPHPs}}(1^m)$, and $\mathcal{B}_1$ aims to decide whether $c_{1,1}, \dots, c_{\tilde{m}, Q_e} \leftarrow \mathcal{L}$ (say $b = 0$) or $c_{1,1}, \dots, c_{\tilde{m}, Q_e} \leftarrow \mathcal{X}$ (say $b = 1$). $\mathcal{B}_1$ will simulate G_ζ or $G_{\zeta,0}$ for $\mathcal{A}$, depending on the value of b .

- Firstly, $\mathcal{B}_1$ invokes $\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPs}})$, and for all $i \in [\tilde{m}]$, $\mathcal{B}_1$ samples $sk_i \leftarrow \mathcal{SK}$ itself and computes $pk_i := \alpha(sk_i)$. Then $\mathcal{B}_1$ sends $\text{mpk} := (\text{pp}_{\text{IPHPs}}, \text{pp}_{\text{coHash}}, pk_1, \dots, pk_{\tilde{m}})$ to $\mathcal{A}$, while keeps $\text{msk} := (sk_1, \dots, sk_{\tilde{m}})$ privately. $\mathcal{B}_1$ also picks a challenge bit $\beta \leftarrow \{0, 1\}$ for $\mathcal{A}$.
 - $\mathcal{B}_1$ has the master secret key msk , thus can honestly answer $\mathcal{O}_{\text{KGen}}$ queries made by $\mathcal{A}$, the same way as G_ζ and $G_{\zeta,0}$.
 - As for $\mathcal{O}_{\text{Enc}}$ queries, when answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{Enc}}$ query $\mathcal{O}_{\text{Enc}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, $\mathcal{B}_1$ computes $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ as follows:
 - if $\phi_j \leq \zeta \cdot \tilde{m}$, $\mathcal{B}_1$ samples $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$, the same as G_ζ and $G_{\zeta,0}$;
 - if $\phi_j > \zeta \cdot \tilde{m}$, $\mathcal{B}_1$ sets the first $\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}$ elements $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}$ as the $c_{1,j}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}, j}$ contained in its own input, and samples the remaining elements $c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}+1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$ uniformly.
- Then $\mathcal{B}_1$ samples $r_1^{(j)}, \dots, r_{\min\{\phi_j, \zeta \cdot \tilde{m}\}}^{(j)} \leftarrow \mathbb{Z}_{\text{ordco}}$, computes $\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\min\{\phi_j, \zeta \cdot \tilde{m}\}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_\beta^{(j)} \cdot \mathbf{g}_{\text{co}}$ in the same way as G_ζ and $G_{\zeta,0}$, and returns the challenge ciphertext $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.

In the case $b = 0$, $(c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}) := (c_{1,j}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}, j})$ are uniformly random over $\mathcal{L}$, thus $\mathcal{B}_1$ perfectly simulates G_ζ for $\mathcal{A}$; in the case $b = 1$, $(c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}) := (c_{1,j}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}, j})$ are uniformly random over $\mathcal{X}$, thus $\mathcal{B}_1$ perfectly simulates $G_{\zeta,0}$ for $\mathcal{A}$.

- Finally, $\mathcal{B}_1$ receives a bit β' from $\mathcal{A}$ and returns 1 to its own challenger if and only if $\beta' = \beta$.

Overall, $\mathcal{B}_1$ simulates G_ζ for $\mathcal{A}$ in the case $b = 0$ and simulates $G_{\zeta,0}$ for $\mathcal{A}$ in the case $b = 1$, thus $\mathcal{B}_1$ successfully distinguishes $b = 0$ from $b = 1$ as long as the probability that $\beta' = \beta$ in G_ζ differs non-negligibly from that in $G_{\zeta,0}$. Consequently, we have $\text{Adv}_{\text{IPHPs}, \mathcal{B}_1, Q_e \tilde{m}}^{\text{msmp}}(\lambda) \geq |\Pr_\zeta[\text{Win}] - \Pr_{\zeta,0}[\text{Win}]|$.

This completes the proof of Claim 1. $\blacksquare$

C.3 Proof of Claim 2 ($\mathbf{G}_{\zeta,\eta} \rightarrow \mathbf{G}_{\zeta,\eta+1}$)

Claim 2. For all $0 \leq \eta < \tilde{m}$, we have $|\Pr_{\zeta,\eta}[\text{Win}] - \Pr_{\zeta,\eta+1}[\text{Win}]| \leq \epsilon_{\text{TLHPS},\mathcal{B}_3,Q_e}^{\text{funSmth}}(\lambda)$.

Proof. We construct an unbounded adversary $\mathcal{B}_3$ against the functional smoothness of TLHPS (cf. Def. 7) as follows.

$\mathcal{B}_3$ is in the $\text{Exp}_{\text{TLHPS},\mathcal{A},Q}^{\text{funSmth},b}$ experiment defined in Fig. 2 and aims to guess the bit $b \in \{0, 1\}$ (i.e., identify whether it is in the $\text{Exp}_{\text{TLHPS},\mathcal{A},Q}^{\text{funSmth},0}$ experiment or in the $\text{Exp}_{\text{TLHPS},\mathcal{A},Q}^{\text{funSmth},1}$ experiment). Firstly, $\mathcal{B}_3$ is given $(\text{pp}_{\text{IPHPs}}, \text{pp}_{\text{coHash}}, pk = \alpha(sk))$, where $sk \leftarrow_s \mathcal{SK}$ is chosen by its own challenger. Then $\mathcal{B}_3$ has access to the oracles $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$ defined in Fig. 2, where $\mathcal{B}_3$ can only make a single query to $\mathcal{O}_{\text{CHAL}}^b$. $\mathcal{B}_3$ simulates $\mathbf{G}_{\zeta,\eta} / \mathbf{G}_{\zeta,\eta+1}$ for $\mathcal{A}$ as follows.

- Firstly, $\mathcal{B}_3$ samples $sk_i \leftarrow_s \mathcal{SK}$ and computes $pk_i := \alpha(sk_i)$ for all $i \in [\tilde{m}] \setminus \{\eta+1\}$. For $i = \eta+1$, $\mathcal{B}_3$ sets $sk_{\eta+1} := sk$ implicitly and defines $pk_{\eta+1} := pk$ explicitly, where sk is the hashing key chosen by $\mathcal{B}_3$'s own challenger and pk is part of $\mathcal{B}_3$'s own input.

Then $\mathcal{B}_3$ sends $\text{mpk} := (\text{pp}_{\text{IPHPs}}, \text{pp}_{\text{coHash}}, pk_1, \dots, pk_{\tilde{m}})$ to $\mathcal{A}$, and picks a challenge bit $\beta \leftarrow_s \{0, 1\}$ for $\mathcal{A}$. $\mathcal{B}_3$ also samples $s_i \leftarrow_s \mathcal{SK}_{\text{co}}$ for all $i \in [\tilde{m}] \setminus \{\eta+1\}$.

- When answering an $\mathcal{O}_{\text{KGEN}}$ query $(\mathbf{y})$ made by $\mathcal{A}$, $\mathcal{B}_3$ computes $fk_{i,\mathbf{y}} := \mu(sk_i, \mathbf{y})$ for all $i \in [\tilde{m}] \setminus \{\eta+1\}$, using the corresponding sk_i . Moreover, $\mathcal{B}_3$ submits $(\mathbf{y})$ to its own $\mathcal{O}_{\text{FKEY}}$ oracle, obtains $fk_{\mathbf{y}}$ from $\mathcal{O}_{\text{FKEY}}$, and sets $fk_{\eta+1,\mathbf{y}} := fk_{\mathbf{y}}$. Then $\mathcal{B}_3$ returns to $\mathcal{A}$.

Since the $fk_{\mathbf{y}}$ output by $\mathcal{O}_{\text{FKEY}}$ is $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$, and by the fact that $\mathcal{B}_3$ sets $sk_{\eta+1} := sk$ implicitly, it follows that $fk_{\eta+1,\mathbf{y}} = fk_{\mathbf{y}} = \mu(sk, \mathbf{y}) = \mu(sk_{\eta+1}, \mathbf{y})$. Therefore, $\mathcal{B}_3$'s simulation of $\mathcal{O}_{\text{KGEN}}$ oracle is the same as $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$.

- As for $\mathcal{O}_{\text{ENC}}$ queries, when answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{ENC}}$ query $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, in the case that $\phi_j \leq \zeta \cdot \tilde{m}$, $\mathcal{B}_3$ samples $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_s \mathcal{L}$ with witnesses $w_1^{(j)}, \dots, w_{\tilde{m}}^{(j)}$ respectively, samples $r_1^{(j)}, \dots, r_{\phi_j}^{(j)} \leftarrow_s \mathbb{Z}_{\text{ord}_{\text{co}}}$, and computes $\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}$, the same as $\mathbf{G}_{\zeta,\eta}$ and $\mathbf{G}_{\zeta,\eta+1}$. $\mathcal{B}_3$ returns $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.

Note that although $\mathcal{B}_3$ does not possess $sk_{\eta+1}$, $\mathcal{B}_3$ can compute the term $\mathbf{A}_{sk_{\eta+1}}(c_{\eta+1}^{(j)})$ via the public evaluation algorithm $\text{Pub}(pk_{\eta+1}, c_{\eta+1}^{(j)}, w_{\eta+1}^{(j)})$ by using the projection key $pk_{\eta+1}$ and a witness $w_{\eta+1}^{(j)}$ for $c_{\eta+1}^{(j)} \in \mathcal{L}$.

- When answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{ENC}}$ query $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, as the query index j grows, in the case that $\zeta \cdot \tilde{m} < \phi_j \leq \zeta \cdot \tilde{m} + \eta$, $\mathcal{B}_3$ samples $c_1^{(j)}, \dots, c_{\phi_j - \zeta \tilde{m}}^{(j)} \leftarrow_s \mathcal{X}$ and $c_{\phi_j - \zeta \tilde{m} + 1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow_s \mathcal{L}$ with witnesses $w_{\phi_j - \zeta \tilde{m} + 1}^{(j)}, \dots, w_{\tilde{m}}^{(j)}$ respectively, samples $r_1^{(j)}, \dots, r_{\zeta \cdot \tilde{m}}^{(j)} \leftarrow_s \mathbb{Z}_{\text{ord}_{\text{co}}}$, and computes

$$\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\phi_j - \zeta \tilde{m}} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}},$$

the same way as $G_{\zeta, \eta}$ and $G_{\zeta, \eta+1}$. $\mathcal{B}_3$ returns $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.

Note that $1 \leq \phi_j - \zeta \tilde{m} \leq \eta$ in this case, so $c_{\eta+1}^{(j)}$ is always uniformly chosen from $\mathcal{L}$ with $w_{\eta+1}^{(j)}$, and then $\mathcal{B}_3$ can compute the term $\mathbf{A}_{sk_{\eta+1}}(c_{\eta+1}^{(j)})$ via the public evaluation algorithm $\text{Pub}(pk_{\eta+1}, c_{\eta+1}^{(j)}, w_{\eta+1}^{(j)})$ by using the projection key $pk_{\eta+1}$. Moreover, $\mathcal{B}_3$ has all the $s_i \in \mathcal{SK}_{\text{co}}$ for $i = 1, \dots, \phi_j - \zeta \tilde{m} (\leq \eta)$, thus $\mathcal{B}_3$ can compute the terms $\Gamma_{s_i}(c_i^{(j)})$ from s_i and $c_i^{(j)}$ by brute force. Here we stress that although the inner-level hash function $\Gamma_{(\cdot)}$ may not be efficiently computable (as explained in Remark 1), the unbounded adversary $\mathcal{B}_3$ can compute it by brute force since we are reducing to the statistical property of functional smoothness.

- When answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{ENC}}$ query $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, as the query index j grows, in the case that $\phi_j \geq \zeta \cdot \tilde{m} + \eta + 1$, $\mathcal{B}_3$ computes the challenge ciphertext $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ as follows.

If this is the *first* Enc query such that $\phi_j = \zeta \cdot \tilde{m} + \eta + 1$, we know that $\psi_{\zeta \cdot \tilde{m} + \eta + 1} = j$ for this query index j according our notations, and $\mathcal{B}_3$ will do some preparations as follows. $\mathcal{B}_3$ computes $\tilde{\mathbf{x}}^{(j)} := \mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)} \in \mathbb{Z}_{\text{ordco}}^m$, submits $\tilde{\mathbf{x}}^{(j)}$ ($= \tilde{\mathbf{x}}^{(\psi_{\zeta \cdot \tilde{m} + \eta + 1})}$) to its own $\mathcal{O}_{\text{CHAL}}^b$ oracle, and obtains a challenge $(c^{(k)}, \mathbf{h}^{(k)})_{k \in [Q_e]}$, where $c^{(k)} \leftarrow \mathcal{X}$, and $\mathbf{h}^{(k)} := \mathbf{A}_{sk}(c^{(k)})$ (if $b = 0$)

or $\mathbf{h}^{(k)} := \mathbf{A}_{sk}(c^{(k)}) + \Gamma_s(c^{(k)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \cdot \tilde{m} + \eta + 1})}$ (if $b = 1$) for all $k \in [Q_e]$ with $s \leftarrow \mathcal{SK}_{\text{co}}$ chosen by $\mathcal{B}_3$'s own challenger. $\mathcal{B}_3$ sets $s_{\eta+1} := s$ implicitly, where $s \leftarrow \mathcal{SK}_{\text{co}}$ is the inner-level hashing key chosen by $\mathcal{B}_3$'s own challenger.

Recall that all $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$ queries and all $\mathcal{O}_{\text{KGEN}}(\mathbf{y})$ queries made by $\mathcal{A}$ must satisfy $\langle \mathbf{x}_0^{(j)}, \mathbf{y} \rangle = \langle \mathbf{x}_1^{(j)}, \mathbf{y} \rangle$, as required by the mCPA security of IPFE. Consequently, the single $\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}}^{(j)})$ query and all $\mathcal{O}_{\text{FKEY}}(\mathbf{y})$ queries made by $\mathcal{B}_3$ satisfies $\langle \tilde{\mathbf{x}}^{(j)}, \mathbf{y} \rangle = \langle \mathbf{x}_1^{(j)} - \mathbf{x}_0^{(j)}, \mathbf{y} \rangle = 0$, as required by the functional smoothness of TLHPS.

Next, we describe how $\mathcal{B}_3$ computes the challenge ciphertext $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$. Note that in this case, we have $\phi_j \geq \zeta \cdot \tilde{m} + \eta + 1$, together with the fact that $0 \leq \eta < \tilde{m}$, it follows that $1 \leq \eta + 1 \leq \min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}$. $\mathcal{B}_3$ sets $c_{\eta+1}^{(j)}$ as the $c^{(j)}$ contained in its own challenge, samples all $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}$ except $c_{\eta+1}^{(j)}$ uniformly from $\mathcal{X}$, and samples $c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\} + 1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$. Since $c_{\eta+1}^{(j)} := c^{(j)}$ contained in $\mathcal{B}_3$'s own input is uniformly chosen from $\mathcal{X}$, $\mathcal{B}_3$'s simulation of $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ is the same as $G_{\zeta, \eta}$ and $G_{\zeta, \eta+1}$.

Then $\mathcal{B}_3$ samples $r_1^{(j)}, \dots, r_{\zeta \cdot \tilde{m}}^{(j)} \leftarrow \mathbb{Z}_{\text{ordco}}$, and computes $\mathbf{e}^{(j)}$ using the $\mathbf{h}^{(j)}$ contained in its own challenge:

$$\begin{aligned} \mathbf{e}^{(j)} := & \sum_{i=1, i \neq \eta+1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m} + i})} \\ & + \mathbf{h}^{(j)} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}. \end{aligned}$$

$\mathcal{B}_3$ returns $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.

- In the case $b = 0$, $\mathcal{B}_3$'s challenge satisfies $\boxed{\mathbf{h}^{(j)} := \mathbf{A}_{sk}(c^{(j)})}$. Since $\mathcal{B}_3$ sets $sk_{\eta+1} := sk$ implicitly and sets $c_{\eta+1}^{(j)} := c^{(j)}$ explicitly, we have

$$\begin{aligned} \mathbf{e}^{(j)} &= \sum_{i=1, i \neq \eta+1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} \\ &\quad + \boxed{\mathbf{A}_{sk_{\eta+1}}(c_{\eta+1}^{(j)})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}} \\ &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}, \end{aligned}$$

the same as $\mathbf{G}_{\zeta, \eta}$.

- In the case $b = 1$, $\mathcal{B}_3$'s challenge satisfies $\boxed{\mathbf{h}^{(j)} := \mathbf{A}_{sk}(c^{(j)}) + \Gamma_s(c^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \cdot \tilde{m} + \eta + 1})}}$.

Since $\mathcal{B}_3$ sets $sk_{\eta+1} := sk$ and $s_{\eta+1} := s$ implicitly, and sets $c_{\eta+1}^{(j)} := c^{(j)}$ explicitly, we have

$$\begin{aligned} \mathbf{e}^{(j)} &= \sum_{i=1, i \neq \eta+1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} \\ &\quad + \boxed{\mathbf{A}_{sk_{\eta+1}}(c_{\eta+1}^{(j)}) + \Gamma_{s_{\eta+1}}(c_{\eta+1}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \cdot \tilde{m} + \eta + 1})}} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}} \\ &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\eta} \Gamma_{s_i}(c_i^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} \\ &\quad + \boxed{\Gamma_{s_{\eta+1}}(c_{\eta+1}^{(j)}) \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \cdot \tilde{m} + \eta + 1})}} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}, \end{aligned}$$

the same as $\mathbf{G}_{\zeta, \eta+1}$.

- Finally, $\mathcal{B}_3$ receives a bit β' from $\mathcal{A}$ and returns 1 to its own challenger if and only if $\beta' = \beta$.

Overall, $\mathcal{B}_3$ simulates $\mathbf{G}_{\zeta, \eta}$ for $\mathcal{A}$ in the case $b = 0$ and simulates $\mathbf{G}_{\zeta, \eta+1}$ for $\mathcal{A}$ in the case $b = 1$, thus $\mathcal{B}_3$ successfully distinguishes $b = 0$ from $b = 1$ as long as the probability that $\beta' = \beta$ in $\mathbf{G}_{\zeta, \eta}$ differs non-negligibly from that in $\mathbf{G}_{\zeta, \eta+1}$. Consequently, we have $\epsilon_{\text{TLHPS}, \mathcal{B}_3, Q_e}^{\text{funSmth}}(\lambda) \geq |\Pr_{\zeta, \eta}[\text{Win}] - \Pr_{\zeta, \eta+1}[\text{Win}]|$.

This completes the proof of Claim 2. $\blacksquare$

C.4 Proof of Claim 3 ($\mathbf{G}_{\zeta, \tilde{m}} \rightarrow \mathbf{G}_{\zeta, \tilde{m}+1}$)

Claim 3. $|\Pr_{\zeta, \tilde{m}}[\text{Win}] - \Pr_{\zeta, \tilde{m}+1}[\text{Win}]| \leq \text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda)$.

Proof. We construct an adversary $\mathcal{B}_2$ against the multi-key-extracting property of TLHPS (cf. Def. 8), where the number of keys is $\tilde{m}$ and the number of instances per key is Q_e , as follows.

$\mathcal{B}_2$ is given $(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, \{c_{i,j}, \mathbf{h}_{i,j}\}_{i \in [\tilde{m}], j \in [Q_e]})$, where $\text{pp}_{\text{IPHPS}} \leftarrow^s \text{Setup}_{\text{IPHPS}}(1^m)$, $\text{pp}_{\text{coHash}} \leftarrow^s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$, $c_{1,1}, \dots, c_{\tilde{m}, Q_e} \leftarrow^s \mathcal{X}$, and $\mathcal{B}_2$ aims to decide whether $\boxed{\mathbf{h}_{i,j} := \Gamma_{s_i}(c_{i,j})}$ for all $i \in [\tilde{m}], j \in [Q_e]$ with $s_1, \dots, s_{\tilde{m}} \leftarrow^s \mathcal{SK}_{\text{co}}$ (say $b = 0$) or $\boxed{\mathbf{h}_{i,j} := r'_{i,j} \cdot \mathbf{g}_{\text{co}}}$ for all $i \in [\tilde{m}], j \in [Q_e]$ with $r'_{1,1}, \dots, r'_{\tilde{m}, Q_e} \leftarrow^s \mathbb{Z}_{\text{ord}_{\text{co}}}$ (say $b = 1$). $\mathcal{B}_2$ will simulate $\mathbf{G}_{\zeta, \tilde{m}}$ or $\mathbf{G}_{\zeta, \tilde{m}+1}$ for $\mathcal{A}$, depending on the value of b .

- Firstly, $\mathcal{B}_2$ samples $sk_i \leftarrow \mathcal{SK}$ and computes $pk_i := \alpha(sk_i)$ for all $i \in [\tilde{m}]$. Then $\mathcal{B}_2$ sends $\text{mpk} := (\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk_1, \dots, pk_{\tilde{m}})$ to $\mathcal{A}$, while keeps $\text{msk} := (sk_1, \dots, sk_{\tilde{m}})$ privately. $\mathcal{B}_2$ also picks a challenge bit $\beta \leftarrow \{0, 1\}$ for $\mathcal{A}$.
- $\mathcal{B}_2$ has the master secret key msk , thus can honestly answer $\mathcal{O}_{\text{KGEN}}$ queries made by $\mathcal{A}$, the same way as $\mathcal{G}_{\zeta, \tilde{m}}$ and $\mathcal{G}_{\zeta, \tilde{m}+1}$.
- As for $\mathcal{O}_{\text{ENC}}$ queries, when answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{ENC}}$ query $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, in the case that $\phi_j \leq \zeta \cdot \tilde{m}$, $\mathcal{B}_2$ samples $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$, picks $r_1^{(j)}, \dots, r_{\phi_j}^{(j)} \leftarrow \mathbb{Z}_{\text{ord}_{\text{co}}}$, and computes $\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\phi_j} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}$ using the master secret key $\text{msk} = (sk_1, \dots, sk_{\tilde{m}})$, the same as $\mathcal{G}_{\zeta, \tilde{m}}$ and $\mathcal{G}_{\zeta, \tilde{m}+1}$. $\mathcal{B}_2$ returns $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.
- When answering the j -th ($j \in [Q_e]$) $\mathcal{O}_{\text{ENC}}$ query $\mathcal{O}_{\text{ENC}}(\mathbf{x}_0^{(j)}, \mathbf{x}_1^{(j)})$, as the query index j grows, in the case that $\phi_j > \zeta \cdot \tilde{m}$, $\mathcal{B}_2$ computes the challenge ciphertext $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ as follows.

$\mathcal{B}_2$ sets the first $\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}$ elements $c_1^{(j)}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}}^{(j)}$ as the $c_{1,j}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}, j}$ contained in its own input, and samples the remaining elements $c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}+1}^{(j)}, \dots, c_{\tilde{m}}^{(j)} \leftarrow \mathcal{L}$ uniformly. Since the $c_{1,j}, \dots, c_{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}, j}$ contained in $\mathcal{B}_2$'s own input are uniformly chosen from $\mathcal{X}$, $\mathcal{B}_2$'s simulation of $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ is the same as $\mathcal{G}_{\zeta, \eta}$ and $\mathcal{G}_{\zeta, \eta+1}$.

Then $\mathcal{B}_2$ computes $\mathbf{e}^{(j)}$ using the master secret key $\text{msk} = (sk_1, \dots, sk_{\tilde{m}})$ and the $\{\mathbf{h}_{i,j}\}_{i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]}$ contained in its own input:

$$\mathbf{e}^{(j)} := \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}} \boxed{\mathbf{h}_{i,j}} \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}.$$

$\mathcal{B}_2$ returns the challenge ciphertext $ct^{*(j)} := (c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}, \mathbf{e}^{(j)})$ to $\mathcal{A}$.

- In the case $b = 0$, $\boxed{\mathbf{h}_{i,j} := \Gamma_{s_i}(c_{i,j})}$ for all $i \in [\tilde{m}], j \in [Q_e]$ with $s_1, \dots, s_{\tilde{m}} \leftarrow \mathcal{SK}_{\text{co}}$. Since $\mathcal{B}_2$ sets $c_i^{(j)} := c_{i,j}$ for all $i \in [\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}]$ explicitly, we have

$$\mathbf{e}^{(j)} = \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}} \boxed{\Gamma_{s_i}(c_i^{(j)})} \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}},$$

the same as $\mathcal{G}_{\zeta, \tilde{m}}$.

- In the case $b = 1$, $\boxed{\mathbf{h}_{i,j} := r'_{i,j} \cdot \mathbf{g}_{\text{co}}}$ for all $i \in [\tilde{m}], j \in [Q_e]$ with $r'_{1,1}, \dots, r'_{\tilde{m}, Q_e} \leftarrow \mathbb{Z}_{\text{ord}_{\text{co}}}$. Thus we have

$$\begin{aligned} \mathbf{e}^{(j)} &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}} \boxed{r'_{i,j} \cdot \mathbf{g}_{\text{co}}} \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}} \\ &= \sum_{i=1}^{\tilde{m}} \mathbf{A}_{sk_i}(c_i^{(j)}) + \sum_{i=1}^{\zeta \cdot \tilde{m}} r_i^{(j)} \cdot \tilde{\mathbf{x}}^{(\psi_i)} \cdot \mathbf{g}_{\text{co}} + \sum_{i=1}^{\min\{\phi_j - \zeta \tilde{m}, \tilde{m}\}} \boxed{r'_{i,j}} \cdot \tilde{\mathbf{x}}^{(\psi_{\zeta \tilde{m}+i})} \cdot \boxed{\mathbf{g}_{\text{co}}} + \mathbf{x}_{\beta}^{(j)} \cdot \mathbf{g}_{\text{co}}, \end{aligned}$$

the same as $\mathcal{G}_{\zeta, \tilde{m}+1}$.

- Finally, $\mathcal{B}_2$ receives a bit β' from $\mathcal{A}$ and returns 1 to its own challenger if and only if $\beta' = \beta$.

Overall, $\mathcal{B}_2$ simulates $G_{\zeta, \tilde{m}}$ for $\mathcal{A}$ in the case $b = 0$ and simulates $G_{\zeta, \tilde{m}+1}$ for $\mathcal{A}$ in the case $b = 1$, thus $\mathcal{B}_2$ successfully distinguishes $b = 0$ from $b = 1$ as long as the probability that $\beta' = \beta$ in $G_{\zeta, \tilde{m}}$ differs non-negligibly from that in $G_{\zeta, \tilde{m}+1}$. Consequently, we have $\text{Adv}_{\text{TLHPS}, \mathcal{B}_2, \tilde{m}, Q_e}^{\text{mk-mext}}(\lambda) \geq |\Pr_{\zeta, \tilde{m}}[\text{Win}] - \Pr_{\zeta, \tilde{m}+1}[\text{Win}]|$.

This completes the proof of Claim 3. $\blacksquare$

D Extending Tight Security to The Multi-User Setting

Firstly, we give the formal definition of mCPA security in the multi-user setting.

Definition 9 (mCPA Security in The Multi-User Setting [61]). *An IPFE scheme IPFE is mCPA secure in the multi-user setting, if for any PPT adversary $\mathcal{A}$ and any polynomial n , it holds that $\text{Adv}_{\text{IPFE}, \mathcal{A}, n}^{\text{mCPA}}(\lambda) := |\Pr[\text{Exp}_{\text{IPFE}, \mathcal{A}, n}^{\text{mCPA}} \Rightarrow 1] - \frac{1}{2}| \leq \text{negl}(\lambda)$, where the experiment $\text{Exp}_{\text{IPFE}, \mathcal{A}, n}^{\text{mCPA}}$ is defined in Fig. 7, and we require that all $\mathcal{O}_{\text{ENC}}(i, \mathbf{x}_0, \mathbf{x}_1)$ queries and all $\mathcal{O}_{\text{KGEN}}(i, \mathbf{y})$ queries made by $\mathcal{A}$ w.r.t. a same user $i \in [n]$ are restricted to satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$.*

$\text{Exp}_{\text{IPFE}, \mathcal{A}, n}^{\text{mCPA}}:$ For $i \in [n]$: $(\text{mpk}^{(i)}, \text{msk}^{(i)}) \leftarrow \text{Setup}(1^m)$ $\beta \leftarrow \{0, 1\}$ // challenge bit $\beta' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{ENC}}, \mathcal{O}_{\text{KGEN}}}(\text{mpk})$ If $\beta' = \beta$: Return 1; Else: Return 0	$\mathcal{O}_{\text{ENC}}(i \in [n], \mathbf{x}_0 \in \mathbb{Z}^m, \mathbf{x}_1 \in \mathbb{Z}^m):$ $ct^* \leftarrow \text{Enc}(\text{mpk}^{(i)}, \mathbf{x}_\beta)$ Return ct^* $\mathcal{O}_{\text{KGEN}}(i \in [n], \mathbf{y} \in \mathbb{Z}^m):$ $sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}^{(i)}, \mathbf{y})$ Return $sk_{\mathbf{y}}$
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Fig. 7. The mCPA security experiment $\text{Exp}_{\text{IPFE}, \mathcal{A}, n}^{\text{mCPA}}$ in the multi-user setting for IPFE. To avoid trivial attacks, all $\mathcal{O}_{\text{ENC}}(i, \mathbf{x}_0, \mathbf{x}_1)$ queries and all $\mathcal{O}_{\text{KGEN}}(i, \mathbf{y})$ queries made by $\mathcal{A}$ w.r.t. a same user $i \in [n]$ must satisfy $\langle \mathbf{x}_0, \mathbf{y} \rangle = \langle \mathbf{x}_1, \mathbf{y} \rangle$.

Remark 2 (Extension to The Multi-User Setting). For better readability, we prove the tight mCPA security of our IPFE in the single-user setting in Theorem 1. Our result extends naturally to the multi-user setting. More precisely, the $(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}})$ contained in mpk can be set as the public parameter shared among all users, and each user i samples its own master secret key $\text{msk}^{(i)} = (sk_1^{(i)}, \dots, sk_m^{(i)})$ and computes its own master public key $\text{mpk}^{(i)} = (pk_1^{(i)}, \dots, pk_m^{(i)})$. In our single-user mCPA security proof (i.e., the proof of Theorem 1), all statistical arguments (like the use of the functional smoothness property of TLHPS in the transition $G_{\zeta, \eta} \rightarrow G_{\zeta, \eta+1}$) remain unchanged, since they introduce only statistically small differences and do not affect the tightness of reductions. The computational arguments include the use of the multi-fold

SMP (in the game transitions $G_\zeta \rightarrow G_{\zeta,0}$ and $G_{\zeta,\tilde{m}+2} \rightarrow G_{\zeta,\tilde{m}+3}$) and the multi-key-extracting property of TLHPS (in the game transition $G_{\zeta,\tilde{m}} \rightarrow G_{\zeta,\tilde{m}+1}$). We can adapt these computational steps easily to the multi-user setting: in the transitions $G_\zeta \rightarrow G_{\zeta,0}$ and $G_{\zeta,\tilde{m}+2} \rightarrow G_{\zeta,\tilde{m}+3}$, we can modify the sampling of $c_1^{(j)}, \dots, c_{\tilde{m}}^{(j)}$ of all users simultaneously, and in the transition $G_{\zeta,\tilde{m}} \rightarrow G_{\zeta,\tilde{m}+1}$, we can change the terms $\{F_{s_i}(c_i^{(j)})\}$ to $\{r_i'^{(j)} \cdot g_{\text{co}}\}$ for all users simultaneously. These game changes can still be reduced to the multi-fold SMP and the multi-key-extracting property of TLHPS, respectively, except that the number of folds/keys/instances would increase by a factor of the number of users. Nevertheless, this would not affect the tightness of the mCPA security ultimately, since in our instantiations in Sect. 6, the multi-fold SMP and the multi-key-extracting property of TLHPS can be further tightly reduced to the underlying computational assumption (MDDH, DCR or LWE), with loss factor independent of the number of folds/keys/instances.

E Postponed MDDH-based Instantiation

In this section, we present the MDDH-based instantiation of two-leveled IPHPS scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ and the resulting IPFE scheme $\text{IPFE}_{\text{MDDH}}$.

More precisely, we first recall the formal definition of MDDH assumptions, which cover the standard DDH and k -Linear (k -LIN) assumptions, as well as necessary notations in Appendix E.1. Then we present the MDDH-based instantiation $\text{TLHPS}_{\text{MDDH}}$ in Appendix E.2, and provide the proofs of its Functional Smoothness and Tight Multi-Key-Extracting in Appendix E.3 and Appendix E.4, respectively. Finally, by plugging our $\text{TLHPS}_{\text{MDDH}}$ into the generic construction $\text{IPFE}[\text{TLHPS}, L]$ proposed in Sect. 5, we show the resulting IPFE scheme $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}, L]$ in Appendix E.5, and give a detailed analysis of the DDH and k -LIN-based IPFE schemes $\text{IPFE}_{\text{MDDH}}$ in Appendix E.6.

E.1 MDDH Assumptions

Let GGen be a PPT algorithm outputting a description of cyclic group $\text{gpar} = (\mathbb{G}, p, P)$, where $\mathbb{G}$ is an additive cyclic group of prime order $p > 2^\lambda$ and with a generator P . We require gpar to be an implicit input of other algorithms.

We use implicit representation of group elements as in [26]. For $a \in \mathbb{Z}_p$, denote by $\llbracket a \rrbracket = aP \in \mathbb{G}$ as the implicit representation of a in $\mathbb{G}$. Similarly, for a matrix $\mathbf{A} = (a_{i,j}) \in \mathbb{Z}_p^{n \times m}$ we define $\llbracket \mathbf{A} \rrbracket$ as the implicit representation of $\mathbf{A}$ in $\mathbb{G}$. $\text{Span}(\mathbf{A}) := \{\mathbf{A}\mathbf{r} \mid \mathbf{r} \in \mathbb{Z}_p^m\} \subseteq \mathbb{Z}_p^n$ denotes the linear span of $\mathbf{A}$, and similarly $\text{Span}(\llbracket \mathbf{A} \rrbracket) := \{\llbracket \mathbf{A}\mathbf{r} \rrbracket \mid \mathbf{r} \in \mathbb{Z}_p^m\} \subseteq \mathbb{G}^n$. Note that given $\mathbf{A}$, $\llbracket \mathbf{B} \rrbracket$, $\llbracket \mathbf{C} \rrbracket$ and $\mathbf{D}$ with matching dimensions, one can efficiently compute $\llbracket \mathbf{AB} \rrbracket$, $\llbracket \mathbf{B} + \mathbf{C} \rrbracket$ and $\llbracket \mathbf{CD} \rrbracket$.

Let $l, k \in \mathbb{N}$ be integers with $l > k$. A probabilistic distribution $\mathcal{D}_{l,k}$ is called a *matrix distribution*, if it outputs matrices in $\mathbb{Z}_p^{l \times k}$ of full rank k in polynomial time. Without loss of generality, we assume that the first k rows

of $\mathbf{A} \leftarrow \mathcal{D}_{l,k}$ form an invertible matrix. Let $\mathcal{D}_k := \mathcal{D}_{k+1,k}$. Denote by $\mathcal{U}_{l,k}$ the uniform distribution over all matrices in $\mathbb{Z}_p^{l \times k}$. Let $\mathcal{U}_k := \mathcal{U}_{k+1,k}$.

In the following, we recall the Matrix DDH (MDDH), Q -fold MDDH assumptions, parameterized by a matrix distribution $\mathcal{D}_{l,k}$, as well as the random self-reducibility of the MDDH assumptions. The advantage functions for an adversary $\mathcal{A}$ against the $\mathcal{D}_{l,k}$ -MDDH and Q -fold $\mathcal{D}_{l,k}$ -MDDH assumptions over group $\mathbb{G}$ are denoted by $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{\text{mddh}}(\lambda)$ and $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{Q\text{-mddh}}(\lambda)$, respectively.

MDDH Assumptions. The $\mathcal{D}_{l,k}$ -Matrix DDH ($\mathcal{D}_{l,k}$ -MDDH) problem over group $\mathbb{G}$ is to distinguish the two distributions $([\mathbf{A}], [\mathbf{Aw}])$ and $([\mathbf{A}], [\mathbf{u}])$, where $\mathbf{A} \leftarrow \mathcal{D}_{l,k}$, $\mathbf{w} \leftarrow \mathbb{Z}_p^k$ and $\mathbf{u} \leftarrow \mathbb{Z}_p^l$.

Definition 10 ($\mathcal{D}_{l,k}$ -MDDH Assumption). The $\mathcal{D}_{l,k}$ -MDDH assumption holds over group $\mathbb{G}$, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{\text{mddh}}(\lambda) := |\Pr[\mathcal{A}([\mathbf{A}], [\mathbf{Aw}]) = 1] - \Pr[\mathcal{A}([\mathbf{A}], [\mathbf{u}]) = 1]| \leq \text{negl}(\lambda)$, where $\mathbf{A} \leftarrow \mathcal{D}_{l,k}$, $\mathbf{w} \leftarrow \mathbb{Z}_p^k$ and $\mathbf{u} \leftarrow \mathbb{Z}_p^l$.

MDDH assumption covers many well-studied assumptions, such as the DDH and the k -Linear (k -LIN) assumptions, by specifying the matrix distribution as

$$\mathcal{LIN}_1 \text{ and } \mathcal{LIN}_k \text{ respectively [26], where } \mathcal{LIN}_k : \mathbf{A} = \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_k \\ 1 & \cdots & 1 \end{pmatrix} \in \mathbb{Z}_p^{(k+1) \times k}.$$

Several relations among MDDH assumptions parameterized by different matrix distributions were established in [26, 27].

Lemma 4 ($\mathcal{D}_{l,k}$ -MDDH $\Rightarrow \mathcal{U}_k$ -MDDH [26] $\Rightarrow \mathcal{U}_{l',k}$ -MDDH [27]). For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$ such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A})$ and $\text{Adv}_{\mathcal{U}_k, \mathbb{G}, \mathcal{A}}^{\text{mddh}}(\lambda) \leq \text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda)$.

For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$ such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A})$ and $\text{Adv}_{\mathcal{U}_{l',k}, \mathbb{G}, \mathcal{A}}^{\text{mddh}}(\lambda) \leq \text{Adv}_{\mathcal{U}_k, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda)$.

Consequently, for any $l > k$, $\mathcal{U}_{l,k}$ -MDDH assumption is tightly implied by the k -LIN assumption (i.e., $\mathcal{LIN}_k$ -MDDH).

For $Q \geq 1$, consider the Q -fold $\mathcal{D}_{l,k}$ -MDDH problem over group $\mathbb{G}$, which is to distinguish two distributions $([\mathbf{A}], [\mathbf{AW}])$ and $([\mathbf{A}], [\mathbf{U}])$, where $\mathbf{A} \leftarrow \mathcal{D}_{l,k}$, $\mathbf{W} \leftarrow \mathbb{Z}_p^{k \times Q}$ and $\mathbf{U} \leftarrow \mathbb{Z}_p^{l \times Q}$. The distinguishing advantage of an adversary $\mathcal{A}$ is denoted by $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{Q\text{-mddh}}(\lambda) := |\Pr[\mathcal{A}([\mathbf{A}], [\mathbf{AW}]) = 1] - \Pr[\mathcal{A}([\mathbf{A}], [\mathbf{U}]) = 1]|$, where $\mathbf{A} \leftarrow \mathcal{D}_{l,k}$, $\mathbf{W} \leftarrow \mathbb{Z}_p^{k \times Q}$ and $\mathbf{U} \leftarrow \mathbb{Z}_p^{l \times Q}$. The Q -fold $\mathcal{D}_{l,k}$ -MDDH assumption over $\mathbb{G}$ assumes that $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{Q\text{-mddh}}(\lambda) \leq \text{negl}(\lambda)$ for any PPT $\mathcal{A}$.

$\mathcal{D}_{l,k}$ -MDDH is random self-reducible [26], namely Q -fold and (1-fold) $\mathcal{D}_{l,k}$ -MDDH can be tightly reduced to each other.

Lemma 5 (Random Self-Reducibility of MDDH [26]). Let $Q > l - k$. For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + Q \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{A}}^{Q\text{-mddh}}(\lambda) \leq (l - k) \cdot \text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda) + 1/(p - 1)$.

E.2 Instantiation of Two-Leveled IP-HPS from MDDH

Our MDDH-based instantiation of two-leveled IP-HPS scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ is parameterized by a matrix distribution $\mathcal{D}_{l,k}$ with $l > k$ and a positive integer $t \in \mathbb{N}$, thus is also denoted by $\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t]$.

(Outer-Level) IP-HPS Scheme. We first present the MDDH-based (outer-level) $\text{IPHPS}_{\text{MDDH}} = (\text{Setup}_{\text{IPHPS}}, \alpha, \mu, \text{Pub}, \text{Priv}, \text{FPriv})$ in Fig. 8. It is routine to check the correctness and the functional correctness of $\text{IPHPS}_{\text{MDDH}}$.

$\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m):$ $\text{gpar} = (\mathbb{G}, p, P) \leftarrow \text{GGen}. \quad \mathbf{A} \leftarrow \mathcal{D}_{l,k}, \text{ where } \mathbf{A} \in \mathbb{Z}_p^{l \times k}.$ Return $\text{pp}_{\text{IPHPS}} := (\text{gpar}, [\mathbf{A}] \in \mathbb{G}^{l \times k}, t)$, which implicitly defines $(\mathfrak{G}, \mathcal{L}, \mathcal{X}, \mathcal{SK}, \mathcal{HV}, \mathbf{A}_{(\cdot)})$ as follows: $\mathfrak{G} := \mathbb{G}.$ $\mathcal{L} := \{ [\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t]) \mid \exists (\mathbf{w}_1, \dots, \mathbf{w}_t) \in \mathbb{Z}_p^{k \times t}, \text{ s.t. } \forall i \in [t], [\mathbf{c}_i] = [\mathbf{A}\mathbf{w}_i] \} \subseteq \mathcal{X} := \mathbb{G}^{l \times t}.$ $\mathcal{SK} := (\mathbb{Z}_p^{m \times l})^{1 \times t}.$ $\mathcal{HV} := \mathbb{G}^m.$ $\mathbf{A}_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{G}^m$ is defined as below: for any $sk = (\mathbf{K}_1, \dots, \mathbf{K}_t) \in \mathcal{SK}$ with each $\mathbf{K}_i \in \mathbb{Z}_p^{m \times l}$ and any $[\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t]) \in \mathcal{X}$ with each $[\mathbf{c}_i] \in \mathbb{G}^l$, $\mathbf{A}_{sk}([\mathbf{C}]) := \sum_{i=1}^t \mathbf{K}_i \cdot [\mathbf{c}_i].$	
$pk \leftarrow \alpha(sk = (\mathbf{K}_1, \dots, \mathbf{K}_t)):$ For $i \in [t]:$ $[\mathbf{P}_i] := \mathbf{K}_i \cdot [\mathbf{A}] \in \mathbb{G}^{m \times k}.$ Return $pk := ([\mathbf{P}_1], \dots, [\mathbf{P}_t]).$	$[\mathbf{h}] \leftarrow \text{Pub}(pk = ([\mathbf{P}_1], \dots, [\mathbf{P}_t]), [\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t]), (\mathbf{w}_1, \dots, \mathbf{w}_t)):$ $\parallel [\mathbf{C}] \in \mathcal{L} \text{ with } [\mathbf{c}_i] = [\mathbf{A}\mathbf{w}_i] \text{ for } \forall i \in [t]$ Return $[\mathbf{h}] := \sum_{i=1}^t [\mathbf{P}_i] \cdot \mathbf{w}_i \in \mathbb{G}^m.$
$fk_{\mathbf{y}} \leftarrow \mu(sk = (\mathbf{K}_1, \dots, \mathbf{K}_t), \mathbf{y} \in \mathbb{Z}^m):$ For $i \in [t]:$ $\mathbf{f}_i^\top := \mathbf{y}^\top \cdot \mathbf{K}_i \in \mathbb{Z}_p^{1 \times l}.$ Return $fk_{\mathbf{y}} := (\mathbf{f}_1, \dots, \mathbf{f}_t).$	$[\mathbf{h}] \leftarrow \text{Priv}(sk = (\mathbf{K}_1, \dots, \mathbf{K}_t), [\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t])):$ Return $[\mathbf{h}] := \sum_{i=1}^t \mathbf{K}_i \cdot [\mathbf{c}_i] \in \mathbb{G}^m.$ $[fv] \leftarrow \text{FPriv}(fk_{\mathbf{y}} = (\mathbf{f}_1, \dots, \mathbf{f}_t), [\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t])):$ Return $[fv] := \sum_{i=1}^t \mathbf{f}_i^\top \cdot [\mathbf{c}_i] \in \mathbb{G}.$

Fig. 8. The MDDH-based (outer-level) IP-HPS scheme $\text{IPHPS}_{\text{MDDH}}$.

Below we analyze the SMP related to $\text{IPHPS}_{\text{MDDH}}$. Note that an instance $[\mathbf{C}]$ belongs to the language $\mathcal{L}$ if and only if all columns $[\mathbf{c}_1], \dots, [\mathbf{c}_t]$ of $[\mathbf{C}]$ lie in the linear span $\text{Span}([\mathbf{A}]) := \{[\mathbf{A}\mathbf{r}] \mid \mathbf{r} \in \mathbb{Z}_p^k\} \subseteq \mathbb{G}^l$. Consequently, the SMP related to $\text{IPHPS}_{\text{MDDH}}$ corresponds to the t -fold $\mathcal{D}_{l,k}$ -MDDH assumptions over $\mathbb{G}$, and generally, for any $Q \geq 1$, the Q -fold SMP related to $\text{IPHPS}_{\text{MDDH}}$ corresponds to the tQ -fold $\mathcal{D}_{l,k}$ -MDDH assumptions over $\mathbb{G}$. By the random self-reducibility of $\mathcal{D}_{l,k}$ -MDDH (cf. Lemma 5 in Appendix E.1), we have the following theorem.

Theorem 5 (Tight Multi-fold SMP related to $\text{IPHPS}_{\text{MDDH}}$). *Let $Q > (l - k)/t$. For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$ such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + tQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{IPHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{msmp}}(\lambda) \leq (l - k) \cdot \text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda) + 1/(p - 1).$*

(Inner-Level) co-Hash Scheme. Next we describe the MDDH-based inner-level $\text{coHash}_{\text{MDDH}} = \text{Setup}_{\text{coHash}}$ in Fig. 9.

$\text{pp}_{\text{coHash}} \leftarrow_s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}}):$
 Parse $\text{pp}_{\text{IPHPS}} = (\text{gpar} = (\mathbb{G}, p, P), [\mathbf{A}] \in \mathbb{G}^{l \times k}, t).$
 $\mathbf{B} \leftarrow_s \mathbb{Z}_p^{l \times (l-k)}.$
 Return $\text{pp}_{\text{coHash}} := (\text{gpar}, [\mathbf{A}] \in \mathbb{G}^{l \times k}, [\mathbf{B}] \in \mathbb{G}^{l \times (l-k)}, t)$, which implicitly defines $(\mathfrak{G}_{\text{co}}, \mathfrak{g}_{\text{co}}, \text{ord}_{\text{co}}, \mathcal{SK}_{\text{co}}, \mathcal{HV}_{\text{co}}, \Gamma_{(\cdot)})$ as follows:
 – $\mathfrak{G}_{\text{co}} := \mathfrak{G} = \mathbb{G}$, which is a cyclic group with generator $\mathfrak{g}_{\text{co}} := P$ and of order $\text{ord}_{\text{co}} := p$.
 – $\mathcal{SK}_{\text{co}} := \mathbb{Z}_p^{(l-k) \times t}.$
 – $\mathcal{HV}_{\text{co}} := \mathbb{G}.$
 – $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{G}$ is defined as below: for any $s = (s_1, \dots, s_t) \in \mathcal{SK}_{\text{co}}$ with each $s_i \in \mathbb{Z}_p^{l-k}$ and any $[\mathbf{C}] = ([\mathbf{c}_1], \dots, [\mathbf{c}_t]) \in \mathcal{X}$ with each $[\mathbf{c}_i] \in \mathbb{G}^l$,

$$\Gamma_s([\mathbf{C}]) := \sum_{i=1}^t s_i^\top \cdot [\mathbf{v}_i],$$
 where each $\mathbf{v}_i \in \mathbb{Z}_p^{l-k}$ is the (unique) vector s.t. $[\mathbf{c}_i] = [\mathbf{A}\mathbf{w}_i + \mathbf{B}\mathbf{v}_i]$ for some $\mathbf{w}_i \in \mathbb{Z}_p^k$.

Fig. 9. The MDDH-based (inner-level) co-Hash scheme $\text{coHash}_{\text{MDDH}}$.

We note that the inner-level hash function $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{G}$ in our $\text{coHash}_{\text{MDDH}}$ is well-defined, except with an exponentially small probability, as shown below. Since $\mathbf{A} \in \mathbb{Z}_p^{l \times k}$ is output by the matrix distribution $\mathcal{D}_{l,k}$, it is of full rank k . Besides, $\mathbf{B} \in \mathbb{Z}_p^{l \times (l-k)}$ is uniformly chosen during the generation of $\text{pp}_{\text{coHash}}$, so $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ forms an invertible matrix, except with probability at most $(l-k)/p$, which is exponentially small. Consequently, for each $[\mathbf{c}_i] \in \mathbb{G}^l$, there exist unique vectors $\mathbf{v}_i \in \mathbb{Z}_p^{l-k}$ and $\mathbf{w}_i \in \mathbb{Z}_p^k$ such that $[\mathbf{c}_i] = [\mathbf{A}\mathbf{w}_i + \mathbf{B}\mathbf{v}_i]$, and $\Gamma_{(\cdot)}$ is well-defined.

Through the following two theorems, we prove the functional smoothness and the tight multi-key-extracting of our $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$.

Theorem 6 (Functional Smoothness of $\text{TLHPS}_{\text{MDDH}}$). *The two-leveled scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ proposed in Fig. 8 and 9 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = 0$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q .*

Theorem 7 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{MDDH}}$). *If the $\mathcal{D}_{(l-k)t}$ -MDDH assumption holds over $\mathbb{G}$, then the two-leveled scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ proposed in Fig. 8 and 9 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\mathcal{D}_{(l-k)t}, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda) + 1/(p-1)$.*

We present their proofs in Appendix E.3 and Appendix E.4, respectively.

E.3 Proof of Theorem 6 (Functional Smoothness of $\text{TLHPS}_{\text{MDDH}}$)

Before presenting the proof of Theorem 6, we first show a useful lemma.

Lemma 6. For $\text{pp}_{\text{IPHPs}} = (\text{gpar}, [\mathbf{A}] \in \mathbb{G}^{l \times k}, t) \leftarrow_s \text{Setup}_{\text{IPHPs}}(1^m)$ and $\text{pp}_{\text{coHash}} = (\text{gpar}, [\mathbf{A}] \in \mathbb{G}^{l \times k}, [\mathbf{B}] \in \mathbb{G}^{l \times (l-k)}, t) \leftarrow_s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPs}})$, $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ forms an invertible matrix, except with probability at most $(l-k)/p$.

For the ease of exposition, we will not mention this exponentially small probability hereafter, and take for granted that $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ is invertible.

The lemma holds due to the following reasons. Note that $\mathbf{A} \in \mathbb{Z}_p^{l \times k}$ is output by the matrix distribution $\mathcal{D}_{l,k}$ during the generation of pp_{IPHPs} , thus it is of full rank k . Besides, $\mathbf{B} \in \mathbb{Z}_p^{l \times (l-k)}$ is uniformly chosen during the generation of $\text{pp}_{\text{coHash}}$. Consequently, $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ forms an invertible matrix except with probability at most $(l-k)/p$.

Next we present the proof of Theorem 6.

Theorem 6. The two-leveled scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ proposed in Fig. 8 and 9 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = 0$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q .

Proof of Theorem 6. To show that $\epsilon_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = 0$, we first elaborate the $\text{Exp}_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}, b}$ experiments ($b \in \{0, 1\}$) defined in Fig. 2.

Let $\text{pp}_{\text{IPHPs}} = (\text{gpar}, [\mathbf{A}], t)$ and $\text{pp}_{\text{coHash}} = (\text{gpar}, [\mathbf{A}], [\mathbf{B}], t)$. Let $sk = (\mathbf{K}_1, \dots, \mathbf{K}_t)$ with each $\mathbf{K}_i \leftarrow_s \mathbb{Z}_p^{m \times l}$ and $s = (s_1, \dots, s_t)$ with each $s_i \leftarrow_s \mathbb{Z}_p^{l-k}$. In the experiment, $\mathcal{A}$ first obtains $(\text{pp}_{\text{IPHPs}}, \text{pp}_{\text{coHash}}, pk)$ where

$$pk = ([\mathbf{P}_1], \dots, [\mathbf{P}_t]) \text{ with each } [\mathbf{P}_i] := \mathbf{K}_i \cdot [\mathbf{A}] \in \mathbb{G}^{m \times k}, \quad (10)$$

and has oracle access to $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$. Through $\mathcal{O}_{\text{FKEY}}$ queries ($\mathbf{y} \in \mathbb{Z}^m$), $\mathcal{A}$ learns functional keys

$$fk_{\mathbf{y}} = (\mathbf{f}_1, \dots, \mathbf{f}_t) \text{ with each } \mathbf{f}_i^\top := \mathbf{y}^\top \cdot \mathbf{K}_i \in \mathbb{Z}_p^{1 \times l}. \quad (11)$$

At some point, $\mathcal{A}$ makes a single $\mathcal{O}_{\text{CHAL}}^b$ query ($\tilde{\mathbf{x}} \in \mathbb{Z}^m$), and obtains $([\mathbf{C}^{(j)}]) = ([\mathbf{c}_1^{(j)}], \dots, [\mathbf{c}_t^{(j)}]), [\mathbf{h}^{(j)}]_{j \in [Q]}$, where $[\mathbf{C}^{(j)}] \leftarrow_s \mathcal{X} = \mathbb{G}^{l \times t}$ and

$$[\mathbf{h}^{(j)}] := \begin{cases} \mathbf{A}_{sk}([\mathbf{C}^{(j)}]) = \sum_{i=1}^t \mathbf{K}_i \cdot [\mathbf{c}_i^{(j)}], & \text{if } b = 0; \\ \mathbf{A}_{sk}([\mathbf{C}^{(j)}]) + \Gamma_s([\mathbf{C}^{(j)}]) \cdot \tilde{\mathbf{x}} = \sum_{i=1}^t \mathbf{K}_i \cdot [\mathbf{c}_i^{(j)}] + \sum_{i=1}^t s_i^\top \cdot [\mathbf{v}_i^{(j)}] \cdot \tilde{\mathbf{x}}, & \text{if } b = 1, \end{cases} \quad (12)$$

where each $\mathbf{v}_i^{(j)} \in \mathbb{Z}_p^{l-k}$ is the (unique) vector s.t. $[\mathbf{c}_i^{(j)}] = [\mathbf{A}\mathbf{w}_i^{(j)} + \mathbf{B}\mathbf{v}_i^{(j)}]$ for some $\mathbf{w}_i^{(j)} \in \mathbb{Z}_p^k$. Then $\mathcal{A}$ may continue to make $\mathcal{O}_{\text{FKEY}}$ queries ($\mathbf{y} \in \mathbb{Z}^m$) and learn functional keys $fk_{\mathbf{y}}$, similar to (11). Finally, $\mathcal{A}$ outputs b' , and the advantage of $\mathcal{A}$ is defined by $\epsilon_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = |\Pr[b' = 1 | b = 0] - \Pr[b' = 1 | b = 1]|$.

Below we analyze $\mathcal{A}$'s advantage. We will show that $\mathcal{A}$ has zero advantage in distinguishing $b = 0$ and $b = 1$. More precisely, we note that the only differences between $b = 0$ and $b = 1$ are the distributions of $[\mathbf{h}^{(j)}]$ in (12), and we will show that the $[\mathbf{h}^{(j)}] = \sum_{i=1}^t \mathbf{K}_i \cdot [\mathbf{c}_i^{(j)}]$ (if $b = 0$) is identically distributed as

the $\llbracket \mathbf{h}^{(j)} \rrbracket = \sum_{i=1}^t \mathbf{K}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \mathbf{s}_i^\top \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}}$ (if $b = 1$), from the point of view of $\mathcal{A}$. To see this, let $\mathbf{A}^\perp \in \mathbb{Z}_p^{(l-k) \times l}$ be a non-zero matrix in the kernel of $\mathbf{A}$ such that both $\mathbf{A}^\perp \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_p^{(l-k) \times k}$ and $\mathbf{A}^\perp \cdot \mathbf{B} = \mathbf{I} \in \mathbb{Z}_p^{(l-k) \times (l-k)}$ hold. For example, such $\mathbf{A}^\perp$ can be set as the last $l - k$ rows of $(\mathbf{A}, \mathbf{B})^{-1}$ (note that it is taken for granted that $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ forms an invertible matrix according to Lemma 6). For the convenience of our analysis, we sample $sk = (\mathbf{K}_1, \dots, \mathbf{K}_t)$ with each $\mathbf{K}_i \leftarrow \mathbb{Z}_p^{m \times l}$ equivalently via $\mathbf{K}_i := \tilde{\mathbf{K}}_i + \tilde{\mathbf{x}} \cdot \mathbf{r}_i^\top \cdot \mathbf{A}^\perp \in \mathbb{Z}_p^{m \times l}$, where $\tilde{\mathbf{K}}_i \leftarrow \mathbb{Z}_p^{m \times l}$, $\tilde{\mathbf{x}} \in \mathbb{Z}^m$ is the (single) $\mathcal{O}_{\text{CHAL}}^b$ query made by $\mathcal{A}$, and $\mathbf{r}_i \leftarrow \mathbb{Z}_p^{l-k}$ for all $i \in [t]$. Below we analyze the view of $\mathcal{A}$, and in particular, the information about $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ that $\mathcal{A}$ may obtain in the experiments. We highlight the parts related to $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ in [dashed boxes] for better readability.

- Firstly, by the fact that $\mathbf{A}^\perp \cdot \mathbf{A} = \mathbf{0}$, in (10) we have each

$$\llbracket \mathbf{P}_i \rrbracket = \mathbf{K}_i \cdot \llbracket \mathbf{A} \rrbracket = (\tilde{\mathbf{K}}_i + \tilde{\mathbf{x}} \cdot \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \mathbf{A}^\perp) \cdot \llbracket \mathbf{A} \rrbracket = \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{A} \rrbracket,$$

thus $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ is completely hidden in the projection key pk .

- Secondly, since all $\mathcal{O}_{\text{FKEY}}$ queries ($\mathbf{y} \in \mathbb{Z}^m$) made by $\mathcal{A}$ satisfy $\langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = \mathbf{y}^\top \cdot \tilde{\mathbf{x}} = 0$, thus in (11) we have each

$$\mathbf{f}_i^\top = \mathbf{y}^\top \cdot \mathbf{K}_i = \mathbf{y}^\top \cdot (\tilde{\mathbf{K}}_i + \tilde{\mathbf{x}} \cdot \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \mathbf{A}^\perp) = \mathbf{y}^\top \cdot \tilde{\mathbf{K}}_i,$$

and consequently, $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ is also completely hidden in all functional keys $fk_{\mathbf{y}}$ that $\mathcal{A}$ obtains.

- Finally, through making the single $\mathcal{O}_{\text{CHAL}}^b$ query ($\tilde{\mathbf{x}} \in \mathbb{Z}^m$), $\mathcal{A}$ obtains $(\llbracket \mathbf{C}^{(j)} \rrbracket = (\llbracket \mathbf{c}_1^{(j)} \rrbracket, \dots, \llbracket \mathbf{c}_t^{(j)} \rrbracket), \llbracket \mathbf{h}^{(j)} \rrbracket)_{j \in [Q]}$, where each $\llbracket \mathbf{c}_i^{(j)} \rrbracket \in \mathbb{Z}_p^l$ can be uniquely parsed as $\llbracket \mathbf{c}_i^{(j)} \rrbracket = \llbracket \mathbf{A} \mathbf{w}_i^{(j)} + \mathbf{B} \mathbf{v}_i^{(j)} \rrbracket$ for some $\mathbf{v}_i^{(j)} \in \mathbb{Z}_p^{l-k}$ and $\mathbf{w}_i^{(j)} \in \mathbb{Z}_p^k$. Moreover, in the case of $b = 0$, in (12), we have

$$\begin{aligned} \llbracket \mathbf{h}^{(j)} \rrbracket &= \sum_{i=1}^t \mathbf{K}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket = \sum_{i=1}^t (\tilde{\mathbf{K}}_i + \tilde{\mathbf{x}} \cdot \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \mathbf{A}^\perp) \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket \\ &= \sum_{i=1}^t \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \tilde{\mathbf{x}} \cdot \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \mathbf{A}^\perp \cdot \llbracket \mathbf{A} \mathbf{w}_i^{(j)} + \mathbf{B} \mathbf{v}_i^{(j)} \rrbracket \\ &= \sum_{i=1}^t \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \tilde{\mathbf{x}} \cdot \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \\ &= \sum_{i=1}^t \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}}, \end{aligned} \tag{13}$$

where the second last equality follows from the facts that $\mathbf{A}^\perp \cdot \mathbf{A} = \mathbf{0}$ and $\mathbf{A}^\perp \cdot \mathbf{B} = \mathbf{I}$, and the last equality holds since $\mathbf{r}_i^\top \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \in \mathbb{G}$ is a group element. Similarly, in the case of $b = 1$, in (12), we have

$$\begin{aligned} \llbracket \mathbf{h}^{(j)} \rrbracket &= \sum_{i=1}^t \mathbf{K}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \mathbf{s}_i^\top \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}} \\ &= \sum_{i=1}^t \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t \begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}} + \sum_{i=1}^t \mathbf{s}_i^\top \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}} \\ &= \sum_{i=1}^t \tilde{\mathbf{K}}_i \cdot \llbracket \mathbf{c}_i^{(j)} \rrbracket + \sum_{i=1}^t (\begin{bmatrix} \text{---} \\ \mathbf{r}_i^\top \\ \text{---} \end{bmatrix} + \mathbf{s}_i)^\top \cdot \llbracket \mathbf{v}_i^{(j)} \rrbracket \cdot \tilde{\mathbf{x}}. \end{aligned} \tag{14}$$

Since $\mathcal{A}$ obtains nothing about $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ in other parts of the experiments, $(\mathbf{r}_1, \dots, \mathbf{r}_t)$ completely hides $(\mathbf{s}_1, \dots, \mathbf{s}_t)$ in (14), namely $(\mathbf{r}_1 + \mathbf{s}_1, \dots, \mathbf{r}_t + \mathbf{s}_t)$ is identically distributed as $(\mathbf{r}_1, \dots, \mathbf{r}_t)$. Consequently, the distribution of $\llbracket \mathbf{h}^{(j)} \rrbracket$ in (14) (i.e, the case $b = 1$ in (12)) is identical to that in (13) (i.e, the case $b = 0$ in (12)), from the point of view of $\mathcal{A}$. Therefore, $\mathcal{A}$ has no advantage in distinguishing $b = 0$ and $b = 1$.

Overall, we get that $\epsilon_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = 0$, and Theorem 6 follows. $\square$

E.4 Proof of Theorem 7 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{MDDH}}$)

Theorem 7. *If the $\mathcal{D}_{(l-k)t}$ -MDDH assumption holds over $\mathbb{G}$, then the two-leveled scheme $\text{TLHPS}_{\text{MDDH}} = (\text{IPHPS}_{\text{MDDH}}, \text{coHash}_{\text{MDDH}})$ proposed in Fig. 8 and 9 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\mathcal{D}_{(l-k)t}, \mathbb{G}, \mathcal{B}}^{\text{mddh}}(\lambda)$.*

Proof of Theorem 7. Firstly, we construct an adversary $\mathcal{B}'$ against the (1-fold) $\mathcal{U}_{TQ, (l-k)t}$ -MDDH over $\mathbb{G}$, so that $\text{Adv}_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\mathcal{U}_{TQ, (l-k)t}, \mathbb{G}, \mathcal{B}'}^{\text{mddh}}(\lambda)$. Then by $\mathcal{D}_{(l-k)t}$ -MDDH $\Rightarrow \mathcal{U}_{TQ, (l-k)t}$ -MDDH (i.e., Lemma 4 in Appendix E.1), Theorem 7 follows.

Given a challenge $(\llbracket \mathbf{V} \rrbracket, \llbracket \mathbf{u} \rrbracket)$, $\mathcal{B}'$ wants to distinguish $\llbracket \mathbf{u} \rrbracket = \llbracket \mathbf{V}\mathbf{s} \rrbracket$ from $\llbracket \mathbf{u} \rrbracket \leftarrow_{\$} \mathbb{G}^{TQ}$, where $\mathbf{V} \leftarrow_{\$} \mathcal{U}_{TQ, (l-k)t}$ and $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_p^{(l-k)t}$.

Let us fix some *notations* used in this proof. We parse $\mathbf{V} \in \mathbb{Z}_p^{TQ \times (l-k)t}$ as $\mathbf{V} = \begin{pmatrix} \mathbf{V}^{(1)} \\ \vdots \\ \mathbf{V}^{(T)} \end{pmatrix}$ with each $\mathbf{V}^{(i)} \in \mathbb{Z}_p^{Q \times (l-k)t}$ further parsed as $\mathbf{V}^{(i)} = \begin{pmatrix} \mathbf{v}_1^{(i,1)\top} & \mathbf{v}_2^{(i,1)\top} & \dots & \mathbf{v}_t^{(i,1)\top} \\ \vdots & \vdots & & \vdots \\ \mathbf{v}_1^{(i,Q)\top} & \mathbf{v}_2^{(i,Q)\top} & \dots & \mathbf{v}_t^{(i,Q)\top} \end{pmatrix}$

where each $\mathbf{v}_n^{(i,j)} \in \mathbb{Z}_p^{l-k}$ for $i \in [T], j \in [Q], n \in [t]$, parse $\mathbf{s} \in \mathbb{Z}_p^{(l-k)t}$ as

$$\mathbf{s} = \begin{pmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \\ \vdots \\ \mathbf{s}_t \end{pmatrix} \text{ with each } \mathbf{s}_n \in \mathbb{Z}_p^{l-k} \text{ for } n \in [t], \text{ and parse } \mathbf{u} \in \mathbb{Z}_p^{TQ} \text{ as } \mathbf{u} = \begin{pmatrix} \mathbf{u}^{(1)} \\ \vdots \\ \mathbf{u}^{(T)} \end{pmatrix}$$

with each $\mathbf{u}^{(i)} \in \mathbb{Z}_p^Q$ further parsed as $\mathbf{u}^{(i)} = \begin{pmatrix} u^{(i,1)} \\ \vdots \\ u^{(i,Q)} \end{pmatrix}$ where each $u^{(i,j)} \in \mathbb{Z}_p$ for

$i \in [T], j \in [Q]$.

$\mathcal{B}'$ is constructed by invoking $\mathcal{A}$ as follows. $\mathcal{B}'$ invokes $\mathbf{A} \leftarrow_{\$} \mathcal{D}_{l,k}$ and $\mathbf{B} \leftarrow_{\$} \mathbb{Z}_p^{l \times (l-k)}$ itself, and sets $\text{pp}_{\text{IPHPS}} := (\text{gpar}, \llbracket \mathbf{A} \rrbracket, t)$ and $\text{pp}_{\text{coHash}} := (\text{gpar}, \llbracket \mathbf{A} \rrbracket, \llbracket \mathbf{B} \rrbracket, t)$. Then for all $i \in [T]$ and $j \in [Q]$, $\mathcal{B}'$ sets $\llbracket \mathbf{C}^{(i,j)} \rrbracket = (\llbracket \mathbf{c}_1^{(i,j)} \rrbracket, \dots, \llbracket \mathbf{c}_t^{(i,j)} \rrbracket)$ with each $\llbracket \mathbf{c}_n^{(i,j)} \rrbracket := \llbracket \mathbf{A}\mathbf{w}_n^{(i,j)} + \mathbf{B}\mathbf{v}_n^{(i,j)} \rrbracket$ for uniformly chosen $\mathbf{w}_n^{(i,j)} \leftarrow_{\$} \mathbb{Z}_p^k$ and the $\mathbf{v}_n^{(i,j)} \in \mathbb{Z}_p^{l-k}$ contained in $\mathbf{V}$ (for $n \in [t]$). $\mathcal{B}'$ can compute such $\llbracket \mathbf{c}_n^{(i,j)} \rrbracket$ by using $\mathbf{B}$ and the $\llbracket \mathbf{v}_n^{(i,j)} \rrbracket$ contained in its own challenge $\llbracket \mathbf{V} \rrbracket$. Moreover, for all $i \in [T]$, $\mathcal{B}'$

samples $\tilde{\mathbf{s}}_1^{(i)}, \dots, \tilde{\mathbf{s}}_t^{(i)} \leftarrow_{\mathcal{S}} \mathbb{Z}_p^{l-k}$ uniformly. Then for all $i \in [T]$ and $j \in [Q]$, $\mathcal{B}'$ sets

$$\llbracket h^{(i,j)} \rrbracket := \sum_{n \in [t]} \llbracket \mathbf{v}_n^{(i,j)\top} \rrbracket \cdot \tilde{\mathbf{s}}_n^{(i)} + \llbracket u^{(i,j)} \rrbracket \in \mathbb{G}$$

using the $\llbracket \mathbf{v}_n^{(i,j)} \rrbracket$ and the $\llbracket u^{(i,j)} \rrbracket$ contained in its own challenge $(\llbracket \mathbf{V} \rrbracket, \llbracket \mathbf{u} \rrbracket)$. Finally, $\mathcal{B}'$ invokes $\mathcal{A}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, \{\llbracket \mathbf{C}^{(i,j)} \rrbracket, \llbracket h^{(i,j)} \rrbracket\}_{i \in [T], j \in [Q]})$, and returns to its own challenger whatever $\mathcal{A}$ outputs.

Next, we analyze the advantage of $\mathcal{B}'$.

- Since $\mathbf{V} \leftarrow_{\mathcal{S}} \mathcal{U}_{TQ, (l-k)t}$, all $\mathbf{v}_n^{(i,j)}$ are independently and uniformly random over $\mathbb{Z}_p^{l-k}$. Then all $\llbracket \mathbf{c}_n^{(i,j)} \rrbracket := \llbracket \mathbf{A} \mathbf{w}_n^{(i,j)} + \mathbf{B} \mathbf{v}_n^{(i,j)} \rrbracket$ with uniformly chosen $\mathbf{w}_n^{(i,j)} \leftarrow_{\mathcal{S}} \mathbb{Z}_p^k$ are independently and uniformly distributed over $\mathbb{G}^l$ (note that it is taken for granted that $(\mathbf{A}, \mathbf{B}) \in \mathbb{Z}_p^{l \times l}$ forms an invertible matrix according to Lemma 6 in Appendix E.3). Thus, all $\llbracket \mathbf{C}^{(i,j)} \rrbracket = (\llbracket \mathbf{c}_1^{(i,j)} \rrbracket, \dots, \llbracket \mathbf{c}_t^{(i,j)} \rrbracket)$ are independently and uniformly distributed over $\mathcal{X} = \mathbb{G}^{l \times t}$.
- In the case that $\llbracket \mathbf{u} \rrbracket = \llbracket \mathbf{V} \mathbf{s} \rrbracket$, for each $i \in [T]$ and $j \in [Q]$, we have that $\llbracket \mathbf{u}^{(i)} \rrbracket = \llbracket \mathbf{V}^{(i)} \mathbf{s} \rrbracket$ and more concretely, $\llbracket u^{(i,j)} \rrbracket = \llbracket \sum_{n \in [t]} \mathbf{v}_n^{(i,j)\top} \cdot \mathbf{s}_n \rrbracket$. Consequently, it follows that

$$\llbracket h^{(i,j)} \rrbracket := \sum_{n \in [t]} \llbracket \mathbf{v}_n^{(i,j)\top} \rrbracket \cdot \tilde{\mathbf{s}}_n^{(i)} + \llbracket u^{(i,j)} \rrbracket = \llbracket \sum_{n \in [t]} \mathbf{v}_n^{(i,j)\top} \cdot (\mathbf{s}_n + \tilde{\mathbf{s}}_n^{(i)}) \rrbracket.$$

By setting $s^{(i)} := (\mathbf{s}_1 + \tilde{\mathbf{s}}_1^{(i)}, \dots, \mathbf{s}_t + \tilde{\mathbf{s}}_t^{(i)})$, it follows that $\llbracket h^{(i,j)} \rrbracket = \Gamma_{s^{(i)}}(\llbracket \mathbf{C}^{(i,j)} \rrbracket)$, where all $s^{(i)}$ ($i \in [T]$) are independently and uniformly distributed over $\mathcal{SK}_{\text{co}} = \mathbb{Z}_p^{(l-k) \times t}$ due to the uniformity of $\tilde{\mathbf{s}}_1^{(i)}, \dots, \tilde{\mathbf{s}}_t^{(i)} \leftarrow_{\mathcal{S}} \mathbb{Z}_p^{l-k}$.

- In the case that $\llbracket \mathbf{u} \rrbracket \leftarrow_{\mathcal{S}} \mathbb{G}^{TQ}$, all $\llbracket u^{(i,j)} \rrbracket$ are independently and uniformly distributed over $\mathbb{G}$, and so all

$$\llbracket h^{(i,j)} \rrbracket := \sum_{n \in [t]} \llbracket \mathbf{v}_n^{(i,j)\top} \rrbracket \cdot \tilde{\mathbf{s}}_n^{(i)} + \llbracket u^{(i,j)} \rrbracket$$

are also independently and uniformly distributed over $\mathfrak{G}_{\text{co}} = \mathbb{G}$.

Overall, $\mathcal{B}'$ can successfully distinguish $\llbracket \mathbf{u} \rrbracket = \llbracket \mathbf{V} \mathbf{s} \rrbracket$ from $\llbracket \mathbf{u} \rrbracket \leftarrow_{\mathcal{S}} \mathbb{G}^{TQ}$ as long as $\mathcal{A}$ distinguishes $\{\llbracket \mathbf{C}^{(i,j)} \rrbracket \leftarrow_{\mathcal{S}} \mathcal{X}, \llbracket h^{(i,j)} \rrbracket = \Gamma_{s^{(i)}}(\llbracket \mathbf{C}^{(i,j)} \rrbracket)\}_{i \in [T], j \in [Q]}$ from

$\{\llbracket \mathbf{C}^{(i,j)} \rrbracket \leftarrow_{\mathcal{S}} \mathcal{X}, \llbracket h^{(i,j)} \rrbracket \leftarrow_{\mathcal{S}} \mathfrak{G}_{\text{co}}\}_{i \in [T], j \in [Q]}$, and thus we get that $\text{Adv}_{\text{TLHPS}_{\text{MDDH}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\mathcal{U}_{TQ, (l-k)t}, \mathbb{G}, \mathcal{B}'}^{\text{mddh}}(\lambda)$, as desired. This completes the proof of Theorem 7. $\square$

E.5 Instantiation of IPFE from MDDH

By plugging our $\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t]$ (parameterized by matrix distribution $\mathcal{D}_{l,k}$ and integer $t \in \mathbb{N}$) proposed in Fig. 8 and 9 in Appendix E.2 into the generic construction $\text{IPFE}[\text{TLHPS}, L]$ presented in Fig. 3 in Sect. 5, we immediately obtain an MDDH-based IPFE scheme $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t], L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$, as shown in Fig. 10.

$(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^m):$ $\text{gpar} = (\mathbb{G}, p, P) \leftarrow \text{GGen}.$ $\mathbf{A} \leftarrow \mathcal{D}_{l,k}.$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}], j \in [t]:$ $\mathbf{K}_{i,j} \leftarrow \mathbb{Z}_p^{m \times l}.$ $\llbracket \mathbf{P}_{i,j} \rrbracket := \mathbf{K}_{i,j} \cdot \mathbf{A} \in \mathbb{G}^{m \times k}.$ $\text{mpk} := (\text{gpar}, \llbracket \mathbf{A} \rrbracket \in \mathbb{G}^{l \times k}, \{\llbracket \mathbf{P}_{i,j} \rrbracket\}_{i \in [\tilde{m}], j \in [t]}).$ $\text{msk} := \{\mathbf{K}_{i,j}\}_{i \in [\tilde{m}], j \in [t]}.$ Return $(\text{mpk}, \text{msk}).$ $sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}, \mathbf{y} \in \mathbb{Z}^m):$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}], j \in [t]:$ $\mathbf{f}_{i,j}^\top := \mathbf{y}^\top \cdot \mathbf{K}_{i,j} \in \mathbb{Z}_p^{1 \times l}.$ Return $sk_{\mathbf{y}} := (\{\mathbf{f}_{i,j}\}_{i \in [\tilde{m}], j \in [t]}, \mathbf{y}).$	$ct \leftarrow \text{Enc}(\text{mpk}, \mathbf{x} \in \mathbb{Z}^m):$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}], j \in [t]:$ $\mathbf{w}_{i,j} \leftarrow \mathbb{Z}_p^k.$ $\llbracket \mathbf{c}_{i,j} \rrbracket := \llbracket \mathbf{A} \mathbf{w}_{i,j} \rrbracket \in \mathbb{G}^l.$ $\llbracket \mathbf{e} \rrbracket := \sum_{i=1}^{\tilde{m}} \sum_{j=1}^t \llbracket \mathbf{P}_{i,j} \rrbracket \cdot \mathbf{w}_{i,j} + \llbracket \mathbf{x} \rrbracket \in \mathbb{G}^m.$ Return $ct := (\{\llbracket \mathbf{c}_{i,j} \rrbracket\}_{i \in [\tilde{m}], j \in [t]}, \llbracket \mathbf{e} \rrbracket).$ $d/\perp \leftarrow \text{Dec}(sk_{\mathbf{y}}, ct):$ Parse $sk_{\mathbf{y}} = (\{\mathbf{f}_{i,j}\}_{i \in [\tilde{m}], j \in [t]}, \mathbf{y}),$ $ct = (\{\llbracket \mathbf{c}_{i,j} \rrbracket\}_{i \in [\tilde{m}], j \in [t]}, \llbracket \mathbf{e} \rrbracket).$ $\mathfrak{d} := \mathbf{y}^\top \cdot \llbracket \mathbf{e} \rrbracket - \sum_{i=1}^{\tilde{m}} \sum_{j=1}^t \mathbf{f}_{i,j}^\top \cdot \llbracket \mathbf{c}_{i,j} \rrbracket \in \mathbb{G}.$ For $d \in [-mXY, mXY]:$ If $\mathfrak{d} = \llbracket d \rrbracket$: Return $d.$ Return $\perp.$
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Fig. 10. The MDDH-based IPFE scheme $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t], L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ parameterized by matrix distribution $\mathcal{D}_{l,k}$ and integers $t, L \in \mathbb{N}$.

Let $x \cdot \mathbb{G}$ (resp., $x \cdot \mathbb{Z}_p$) denote x elements in $\mathbb{G}$ (resp., $\mathbb{Z}_p$). Under parameters $(\mathcal{D}_{l,k}, t, L)$, the MDDH-based $\text{IPFE}_{\text{MDDH}}$ has master public key $\text{mpk} : (lk + \tilde{m}tmk) \cdot \mathbb{G}$, master secret key $\text{msk} : \tilde{m}tml \cdot \mathbb{Z}_p$, functional secret key $sk_{\mathbf{y}} : \tilde{m}tl \cdot \mathbb{Z}_p^{16}$, and ciphertext $ct : (\tilde{m}tl + m) \cdot \mathbb{G}$, where $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$.

By plugging Theorem 5, Theorem 6 and Theorem 7 regarding the tight properties of the MDDH-based $\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t]$ in Appendix E.2 into Theorem 1 in Sect. 5, we have the following corollary showing the tight mCPA security of $\text{IPFE}_{\text{MDDH}}$ based on the MDDH assumptions.

Corollary 2 (Tight mCPA Security of $\text{IPFE}_{\text{MDDH}}$). *Assume that the $\mathcal{D}_{l,k}$ -MDDH assumption and the $\mathcal{D}_{(l-k)t}$ -MDDH assumption both hold over $\mathbb{G}$. Then the scheme $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t], L]$ in Fig. 10 is mCPA secure.*

Concretely, for any adversary $\mathcal{A}$ who makes at most Q_e queries to $\mathcal{O}_{\text{ENC}}$ and Q_k queries to $\mathcal{O}_{\text{KGEN}}$, there exist adversaries $\mathcal{B}_1, \mathcal{B}_2$, such that $\mathbf{T}(\mathcal{B}_1) \approx \mathbf{T}(\mathcal{B}_2) \approx \mathbf{T}(\mathcal{A}) + (Q_e + Q_k) \cdot \text{poly}(\lambda)$, with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and

$$\text{Adv}_{\text{IPFE}_{\text{MDDH}}, \mathcal{A}}^{\text{mCPA}}(\lambda) \leq 2L \cdot (l - k) \cdot \text{Adv}_{\mathcal{D}_{l,k}, \mathbb{G}, \mathcal{B}_1}^{\text{mddh}}(\lambda) + L \cdot \text{Adv}_{\mathcal{D}_{(l-k)t}, \mathbb{G}, \mathcal{B}_2}^{\text{mddh}}(\lambda) + 3L/(p - 1).$$

Concrete DDH and k -LIN-Based IPFE Schemes. Below we consider three concrete parameter settings of our $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t], L]$, and compare them with one of the most efficient adaptively secure IPFE schemes proposed by Agrawal, Libert and Stehlé (named ALS) [8] and currently the most efficient tightly secure IPFE scheme proposed by Tomida (named Tomida) [61] based on the DDH or k -LIN assumptions in the standard model.

- **Scheme 1 (The ALS Scheme and Tighter Analysis on Its Security Loss):** $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{L}\mathcal{I}\mathcal{N}_1, 1], m]$. It is in fact the DDH-based ALS

¹⁶ Here we do not count $\mathbf{y}$ in $sk_{\mathbf{y}}$, as it can be made public.

scheme [8]. Previously, the ALS scheme was only known to have loose mCPA security with security factor $O(Q_e)$ (e.g., as noted by Tomida [61]). Now with our Corollary 2, $\text{ALS} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], m]$ is turned out to have *tighter* mCPA security with loss factor only $O(m) = 3m$.

- **Scheme 2 (More Efficient Scheme with Tight Security compared with the Tomida Scheme):** $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_k, k], L]$. It is tightly mCPA secure based on the k -LIN (i.e., $\mathcal{LIN}_k$ -MDDH) assumption, with loss factor $3L$. When $L = 1$, it is in fact the k -LIN-based Tomida scheme [61].

However, under a general L (say $L = 5$ or $L = 100$), it is *more efficient* than the Tomida scheme in all aspects: mpk , msk and sk_y are about $L \times$ *shorter*, ct is about $2 \times \sim 2.9 \times$ *shorter*, and all algorithms (Setup, KGen, Enc, Dec) are about $\approx L \times$ *faster*. The only price is the concrete larger (yet still constant) loss factor $3L$, which is $L \times$ larger than that of Tomida. Nevertheless, this will decrease the security level concretely by at most $\log L$ bits (e.g., $\log 5 \approx 3$ bits when $L = 5$ and $\log 100 \approx 7$ bits when $L = 100$), and even after the compensation of the extra security loss L , it is still about $L \times$ *more efficient*.

We refer to Table 1 in Sect. 1 for comparison of this scheme under $L = 5$ and $L = 100$ with the Tomida scheme.

- **Scheme 3 (Even More Efficient Scheme with Tight Security based on k -LIN for $k \geq 2$):** $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{U}_{2k,k}, 1], L]$. It is also tightly mCPA secure based on k -LIN with loss factor $3L$, the same as **Scheme 2**.

However, it is *even more efficient* than **Scheme 2** for $k \geq 2$. For example, when $L = 5$ and $k = 2$, compared with **Scheme 2**: its mpk and ct are about $1.2 \times$ *shorter*, and its msk and sk_y are about $1.5 \times$ *shorter*. We refer to Table 5 for comparison of this scheme with **Scheme 2** and the Tomida scheme.

We present the detailed analysis and explanations in Appendix E.6.

Table 5. Comparison of our tightly secure k -LIN-based IPFE schemes with currently the most efficient tightly secure scheme Tomida [61] in the standard model.

IPFE Scheme (k -LIN-based)	$ \text{mpk} $	$ \text{msk} $	$ \text{sk}_y $	$ ct $	Ciphertext Expansion	Security Loss
Tomida [61]	$m^2k^2 + k + 1$	$m^2k^2 + m^2k$	$mk^2 + mk$	$mk^2 + mk + m$	$k^2 + k + 1$	$O(1) = 3$
Scheme 2	$\frac{m^2k^2}{L} + k + 1$	$\frac{m^2k^2 + m^2k}{L}$	$\frac{mk^2 + mk}{L}$	$\frac{mk^2 + mk}{L} + m$	$\frac{k^2 + k}{L} + 1$	$O(1) = 3L$
Scheme 3	$\frac{m^2k}{L} + 2k^2$	$\frac{2m^2k}{L}$	$\frac{2mk}{L}$	$\frac{2mk}{L} + m$	$\frac{2k}{L} + 1$	$O(1) = 3L$

E.6 Concrete DDH and k -LIN-Based IPFE Schemes

In Appendix E.5, we present three concrete parameter settings of our $\text{IPFE}_{\text{MDDH}} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{D}_{l,k}, t], L]$, and compare them with the ALS scheme [8] and the Tomida scheme [61] based on the DDH or k -LIN assumptions in the standard model. Here we give a detailed analysis of the efficiency and security loss of the resulting schemes.

- **Scheme 1 (The ALS Scheme and Tighter Analysis on Its Security Loss):** Under the parameter setting $(\mathcal{D}_{l,k} = \mathcal{LIN}_1, t = 1, L = m)$, we obtain the first scheme $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], m]$, which is in fact the DDH-based ALS scheme [8].

Originally, the ALS scheme was proved to be CPA-secure in [8] but only under the single-challenge setting. A standard hybrid argument implies that the ALS scheme is also mCPA-secure (in the multi-challenge setting), but it would incur a large loss factor Q_e with Q_e the number of ciphertexts. As such, the ALS scheme was only known to have loose mCPA security (e.g., as noted by Tomida [61]).

Now with our Corollary 2, $\text{ALS} = \text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], m]$ is turned out to be mCPA-secure with loss factor only $O(m) = 3m$ instead of $O(Q_e)$. Therefore, our new analysis shows that the ALS scheme achieves *tighter* mCPA security.

- **Scheme 2 (More Efficient Scheme with Tight Security compared with The Tomida Scheme):** Next we present tightly secure IPFE schemes. For any constant integers $k, L \in \mathbb{N}$, under the parameter setting $(\mathcal{D}_{l,k} = \mathcal{LIN}_k, t = k, L)$, we obtain the second scheme $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_k, k], L]$. According to Corollary 2, its mCPA security has loss factor $3L$ based on the k -LIN (i.e., $\mathcal{LIN}_k$ -MDDH) assumption, while its ciphertext ct consists of $(\frac{mk^2+mk}{L} + m) \cdot \mathbb{G}$. Consequently, the parameter L controls the trade-off between the efficiency and the loss factor of the resulting scheme.

In particular, when $L = 1$, the $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_k, k], 1]$ is in fact the k -LIN-based Tomida scheme [61] with ciphertext $ct : (mk^2 + mk + m) \cdot \mathbb{G}$, and furthermore, when $k = 1$, the $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], 1]$ is in fact the DDH-based Tomida scheme [61] with ciphertext $ct : 3m \cdot \mathbb{G}$. The security loss factor of the Tomida scheme is about $3 = O(1)$, and the ciphertext expansion (defined as the ratio of ciphertext length to input vector length) of the DDH-based Tomida scheme is $\frac{3m}{m} = 3$. Recall that the ALS scheme has ciphertext $ct : (m + 2) \cdot \mathbb{G}$ and ciphertext expansion $\frac{m+2}{m} = 1 + \frac{2}{m}$ which is very close to 1. So there is a gap ≈ 3 between the ciphertext size of the tightly secure Tomida scheme and the (loosely secure) ALS scheme. It was left as an open problem in [61] to construct more compact tightly secure IPFE schemes.

Now through our generic construction, we can get more compact tightly secure IPFE schemes by setting larger L . For example, when setting $L = 5$, we get k -LIN-based $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_k, k], 5]$ with ciphertext $ct : (\frac{mk^2+mk}{5} + m) \cdot \mathbb{G}$ and DDH-based $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], 5]$ with ciphertext $ct : (\frac{2}{5} + 1)m \cdot \mathbb{G}$. The security loss factor is about $3L = 15 = O(1)$, and the ciphertext expansion of the DDH-based scheme is $\frac{(\frac{2}{5}+1)m}{m} = 1.4$. Thus the ciphertext is at least $2\times$ shorter than that of Tomida, and the gap with ALS is reduced from ≈ 3 to ≈ 1.4 .

If one is willing to set larger L to get even more efficient scheme (at the cost of larger yet still constant security loss), say $L = 100$, we get k -LIN-based $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_k, k], 100]$ with ciphertext $ct : (\frac{mk^2+mk}{100} + m) \cdot \mathbb{G}$ and DDH-based $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{LIN}_1, 1], 100]$ with ciphertext $ct : (\frac{1}{50} + 1)m \cdot \mathbb{G}$.

The security loss factor is about $3L = 300 = O(1)$, and the ciphertext expansion of the DDH-based scheme is $\frac{1}{50} + 1 = 1.02$. Thus the ciphertext is at least $2.9\times$ *shorter* than that of Tomida, and the gap with ALS is further reduced to ≈ 1.02 , which is quite close to 1.

In fact, our tightly secure schemes are more efficient than the Tomida scheme in all aspects, not just ciphertext size: the master public key mpk is about $L\times$ *shorter*, both the master secret key msk and the functional secret key sk_y are $L\times$ *shorter*, and all algorithms (Setup , KGen , Enc , Dec) are about $\approx L\times$ *faster*. The only price is the concrete larger (yet still constant) loss factor $3L$, which is about $L\times$ larger than that of Tomida. Nevertheless, this will decrease the security level concretely by at most $\log L$ bits (e.g., $\log 5 \approx 3$ bits when $L = 5$ and $\log 100 \approx 7$ bits when $L = 100$), and even after the compensation of the extra security loss L , our schemes are still about $L\times$ *more efficient* than the Tomida scheme.

- **Scheme 3 (Even More Efficient Scheme with Tight Security based on k -LIN for $k \geq 2$):** Our construction is very generic and admits more tightly secure IPFE schemes. To see this, for any constant integers $k, L \in \mathbb{N}$, we provide another parameter setting $(\mathcal{D}_{l,k} = \mathcal{U}_{2k,k}, t = 1, L)$, under which we obtain the third scheme $\text{IPFE}[\text{TLHPS}_{\text{MDDH}}[\mathcal{U}_{2k,k}, 1], L]$. According to Corollary 2, its mCPA security has loss factor $3L$ based on the $\mathcal{U}_{2k,k}$ -MDDH and $\mathcal{U}_k$ -MDDH assumptions, both of which are in turn tightly implied by the k -LIN (i.e., $\mathcal{LIN}_k$ -MDDH) assumption by Lemma 4 in Appendix E.1. Thus this scheme also has loss factor $3L$ under the k -LIN assumption, the same as the second scheme.

As for efficiency, its ciphertext ct consists of $(\frac{2mk}{L} + m) \cdot \mathbb{G}$, which is *even shorter* than that of the second scheme whose ciphertext is $ct : (\frac{mk^2+mk}{L} + m) \cdot \mathbb{G}$ for $k \geq 2$ (see Table 5). For example, when $L = 5$ and $k = 2$, the third scheme has $ct : (\frac{4m}{5} + m) \cdot \mathbb{G}$ while the second scheme has $ct : (\frac{6m}{5} + m) \cdot \mathbb{G}$, so the ciphertext of the third scheme is about $1.2\times$ *shorter*. As for other parameter sizes, the master public key mpk of the third scheme is comparable to that of the second scheme (e.g., about $1.2\times$ shorter when $L = 5$, $k = 2$ and $m = 5$, and even shorter for larger m), and the master secret key msk and the functional secret key sk_y of the third scheme are also *even shorter* than those of the second scheme (e.g., about $1.5\times$ shorter when $k = 2$).

F Postponed DCR-based Instantiation

In this section, we present the DCR-based instantiation of two-leveled IP-HPS $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$ and the resulting IPFE scheme IPFE_{DCR} .

More precisely, we first recall the formal definition of DCR assumption and some useful lattice backgrounds in Appendix F.1. Then we present the DCR-based instantiation $\text{TLHPS}_{\text{DCR}}$ in Appendix F.2, and provide the proofs of its Functional Smoothness and Tight Multi-Key-Extracting in Appendix F.3 and Appendix F.4, respectively. Finally, by plugging our $\text{TLHPS}_{\text{DCR}}$ into the generic

construction $\text{IPFE}[\text{TLHPS}, L]$ proposed in Sect. 5, we show the resulting IPFE scheme $\text{IPFE}_{\text{DCR}} = \text{IPFE}[\text{TLHPS}_{\text{DCR}}, L]$ in Appendix F.5.

F.1 DCR Assumption and Lattice Backgrounds

In this subsection, we recall the Decisional Composite Residuosity (DCR) assumption according to [56, 21] and provide some necessary lattice backgrounds.

Let NGen be a PPT algorithm outputting (N, p, q) , where $N = pq$ and p, q are distinct safe primes of same bit-length, i.e., $p = 2p' + 1$ and $q = 2q' + 1$ with primes $p', q' > 2^{2\lambda+1}$. Let $N' := \phi(N)/4 = p'q'$.

Then the multiplicative group $\mathbb{Z}_{N^2}^*$ can be decomposed as an internal direct product

$$\mathbb{Z}_{N^2}^* = \mathbb{G}_N \cdot \mathbb{G}_{N'} \cdot \mathbb{G}_2 \cdot \{-1, 1\},$$

where $\mathbb{G}_N$ (resp., $\mathbb{G}_{N'}, \mathbb{G}_2$) is a cyclic group of order N (resp., $N', 2$).

Note that $g_N := 1 + N$ is a generator of $\mathbb{G}_N$. Moreover, for $X \in \mathbb{G}_N$, the discrete logarithm $\text{dlog}_{g_N} X \in \mathbb{Z}_N$ can be efficiently computed given only N and X [56].

On the other hand, for a uniformly sampled $g \leftarrow_s \mathbb{Z}_{N^2}^*$, $g_{N'} := g^{2N} \bmod N^2$ is a generator of $\mathbb{G}_{N'}$ except with an exponentially small probability $1 - \frac{\phi(N')}{N'} = \frac{p'+q'-1}{p'q'} < \frac{2}{\min\{p', q'\}} < 2^{-2\lambda}$.

In the following, we recall the DCR assumption. The original DCR problem [56] asks to distinguish $u \leftarrow_s \mathbb{Z}_{N^2}^*$ and $v := u^N \bmod N^2$. Below we adopt an equivalent version of DCR, which asks to distinguish $u \leftarrow_s \mathbb{G}_N \cdot \mathbb{G}_{N'}$ and $v \leftarrow_s \mathbb{G}_{N'}$.¹⁷ We also define multi-fold DCR by considering multiple elements.

Definition 11 (DCR Assumption). *The DCR assumption holds w.r.t. NGen , if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{\text{NGen}, \mathcal{A}}^{\text{dcr}}(\lambda) := |\Pr[\mathcal{A}(N, u) = 1] - \Pr[\mathcal{A}(N, v) = 1]| \leq \text{negl}(\lambda)$, where $(N, p, q) \leftarrow_s \text{NGen}$, $u \leftarrow_s \mathbb{G}_N \cdot \mathbb{G}_{N'}$ and $v \leftarrow_s \mathbb{G}_{N'}$.*

Definition 12 (Q -fold DCR Assumption). *For $Q \geq 1$, the Q -fold DCR assumption holds w.r.t. NGen , if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{\text{NGen}, \mathcal{A}}^{Q\text{-dcr}}(\lambda) := |\Pr[\mathcal{A}(N, \{u_i\}_{i \in [Q]}) = 1] - \Pr[\mathcal{A}(N, \{v_i\}_{i \in [Q]}) = 1]| \leq \text{negl}(\lambda)$, where $(N, p, q) \leftarrow_s \text{NGen}$, $u_1, \dots, u_Q \leftarrow_s \mathbb{G}_N \cdot \mathbb{G}_{N'}$ and $v_1, \dots, v_Q \leftarrow_s \mathbb{G}_{N'}$.*

¹⁷ It is folklore that this is an equivalent version of DCR (e.g., see [21, 50]). More precisely, for any adversary $\mathcal{A}$ against this version of DCR, we can construct an adversary $\mathcal{B}$ against the original DCR: on input x , $\mathcal{B}$ simply feeds $x^2 \bmod N^2$ to $\mathcal{A}$ and outputs whatever $\mathcal{A}$ outputs. If $x \leftarrow_s \mathbb{Z}_{N^2}^*$, then $x^2 \bmod N^2$ is uniformly distributed over $\mathbb{G}_N \cdot \mathbb{G}_{N'}$; if $x = u^N \bmod N^2$ with $u \leftarrow_s \mathbb{Z}_{N^2}^*$, then $x^2 \bmod N^2$ is uniformly distributed over $\mathbb{G}_{N'}$. Consequently, $\mathcal{B}$ can distinguish $x \leftarrow_s \mathbb{Z}_{N^2}^*$ and $x = u^N \bmod N^2$ with $u \leftarrow_s \mathbb{Z}_{N^2}^*$ (i.e., break the original DCR) as long as $\mathcal{A}$ can distinguish the uniform distributions over $\mathbb{G}_N \cdot \mathbb{G}_{N'}$ and $\mathbb{G}_{N'}$ (i.e., break this version of DCR). This shows that this version of DCR is as hard as the original DCR. The other direction is also easy to prove.

DCR assumption is random self-reducible [56], namely Q -fold DCR and (1-fold) DCR can be tightly reduced to each other. Formally, we have the following lemma, and we provide a formal proof of it for completeness.

Lemma 7 (Random Self-Reducibility of DCR [56]). *For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + Q \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{NGen}, \mathcal{A}}^{Q\text{-dcr}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) + (Q + 3) \cdot 2^{-2\lambda}$.*

Proof of Lemma 7. We construct an adversary $\mathcal{B}$ against the DCR assumption by invoking the adversary $\mathcal{A}$ against the Q -fold DCR assumption.

Given a challenge (N, u) , $\mathcal{B}$ wants to distinguish $u \leftarrow_{\$} \mathbb{G}_N \cdot \mathbb{G}_{N'}$ from $u \leftarrow_{\$} \mathbb{G}_{N'}$. $\mathcal{B}$ is constructed by invoking $\mathcal{A}$ as follows. For all $i \in [Q]$, $\mathcal{B}$ samples $r_i \leftarrow_{\$} \mathbb{Z}_{\lfloor \frac{N^2}{4} \rfloor}$ uniformly at random and sets $u_i := (u)^{r_i} \bmod N^2$. Then, $\mathcal{B}$ invokes $\mathcal{A}(N, \{u_i\}_{i \in [Q]})$, and returns to its own challenger whatever $\mathcal{A}$ outputs.

Next, we analyze the advantage of $\mathcal{B}$. Let $g_N := 1 + N$ which is a generator of $\mathbb{G}_N$, and let $g_{N'}$ be any generator of $\mathbb{G}_{N'}$. Since each r_i is uniformly chosen from $\mathbb{Z}_{\lfloor \frac{N^2}{4} \rfloor}$, it is statistically close to the uniform distribution over $\mathbb{Z}_{NN'}$ with statistical distance $1 - \frac{NN'}{\lfloor \frac{N^2}{4} \rfloor} \leq 1 - \frac{4N'}{N} = 1 - \frac{4p'q'}{(2p'+1)(2q'+1)} = \frac{2p'+2q'+1}{(2p'+1)(2q'+1)} < \frac{2}{2\min\{p', q'\}+1} < 2^{-2\lambda}$. For the ease of analysis, in the following analysis, we assume that all r_i are uniformly chosen from $\mathbb{Z}_{NN'}$ instead of $\mathbb{Z}_{\lfloor \frac{N^2}{4} \rfloor}$, and this will affect the advantage of $\mathcal{B}$ by at most $Q \cdot 2^{-2\lambda}$.

- In the case that $u \leftarrow_{\$} \mathbb{G}_N \cdot \mathbb{G}_{N'}$, u can be uniquely parsed as $u = (g_N)^s \cdot (g_{N'})^w \bmod N^2$ for uniformly distributed $s \leftarrow_{\$} \mathbb{Z}_N$ and $w \leftarrow_{\$} \mathbb{Z}_{N'}$. Then we have that each $u_i = (u)^{r_i} = (g_N)^{s \cdot r_i \bmod N} \cdot (g_{N'})^{w \cdot r_i \bmod N'} \bmod N^2$. On the one hand, for $s \leftarrow_{\$} \mathbb{Z}_N$ and $w \leftarrow_{\$} \mathbb{Z}_{N'}$, we have $s \in \mathbb{Z}_N^*$ and $w \in \mathbb{Z}_{N'}^*$, except with probability at most $1 - \phi(N)/N + 1 - \phi(N')/N' = (p+q-1)/(pq) + (p'+q'-1)/(p'q') < 2 \cdot 2^{-2\lambda}$. On the other hand, since each r_i is uniformly chosen from $\mathbb{Z}_{NN'}$, by the Chinese Remainder Theorem (CRT), $r_i \bmod N$ and $r_i \bmod N'$ are independently and uniformly distributed over $\mathbb{Z}_N$ and $\mathbb{Z}_{N'}$, respectively. Therefore, as long as $s \in \mathbb{Z}_N^*$ and $w \in \mathbb{Z}_{N'}^*$ hold, it follows that all $s \cdot r_i \bmod N$ and $w \cdot r_i \bmod N'$ are independently and uniformly distributed over $\mathbb{Z}_N$ and $\mathbb{Z}_{N'}$, respectively. Consequently, in this case, all u_i are independently and uniformly distributed over $\mathbb{G}_N \cdot \mathbb{G}_{N'}$, except with probability at most $2 \cdot 2^{-2\lambda}$.
- In the case that $u \leftarrow_{\$} \mathbb{G}_{N'}$, u can be uniquely parsed as $u = (g_{N'})^w \bmod N^2$ for uniformly distributed $w \leftarrow_{\$} \mathbb{Z}_{N'}$. Then we have that each $u_i = (u)^{r_i} = (g_{N'})^{w \cdot r_i \bmod N'} \bmod N^2$. Similar to the above case, we have that $w \in \mathbb{Z}_{N'}^*$ holds except with probability at most $1 - \phi(N')/N' < 2^{-2\lambda}$, and as long as $w \in \mathbb{Z}_{N'}^*$ holds, all $w \cdot r_i \bmod N'$ are independently and uniformly distributed over $\mathbb{Z}_{N'}$. Consequently, in this case, all u_i are independently and uniformly distributed over $\mathbb{G}_{N'}$, except with probability at most $2^{-2\lambda}$.

Overall, $\mathcal{B}$ successfully distinguishes $u \leftarrow_{\$} \mathbb{G}_N \cdot \mathbb{G}_{N'}$ from $u \leftarrow_{\$} \mathbb{G}_{N'}$ as long as $\mathcal{A}$ is able to distinguish $\{u_i \leftarrow_{\$} \mathbb{G}_N \cdot \mathbb{G}_{N'}\}_{i \in [Q]}$ from $\{u_i \leftarrow_{\$} \mathbb{G}_{N'}\}_{i \in [Q]}$, except with

probability at most $Q \cdot 2^{-2\lambda} + 2 \cdot 2^{-2\lambda} + 2^{-2\lambda} = (Q + 3) \cdot 2^{-2\lambda}$ by collecting all the statistical terms. Consequently, we have that $\text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) \geq \text{Adv}_{\text{NGen}, \mathcal{A}}^{Q\text{-dcr}}(\lambda) - (Q + 3) \cdot 2^{-2\lambda}$, and Lemma 7 follows. $\square$

Next we provide some necessary lattice backgrounds and notations.

Discrete Gaussians. For $\sigma > 0$ and $\mathbf{c} \in \mathbb{R}^n$, we define the Gaussian function on $\mathbb{R}^n$ centered at $\mathbf{c}$ with parameter σ by $\rho_{\sigma, \mathbf{c}}(\mathbf{x}) := e^{-\pi \|\mathbf{x} - \mathbf{c}\|^2 / \sigma^2}$. Let $\mathcal{Y} \subseteq \mathbb{R}^n$ be a lattice.¹⁸ The discrete Gaussian distribution $D_{\mathcal{Y}, \sigma, \mathbf{c}}$ over $\mathcal{Y}$ is defined by $D_{\mathcal{Y}, \sigma, \mathbf{c}}(\mathbf{x}) := \rho_{\sigma, \mathbf{c}}(\mathbf{x}) / \rho_{\sigma, \mathbf{c}}(\mathcal{Y})$ for any lattice vector $\mathbf{x} \in \mathcal{Y}$, where $\rho_{\sigma, \mathbf{c}}(\mathcal{Y}) := \sum_{\mathbf{z} \in \mathcal{Y}} \rho_{\sigma, \mathbf{c}}(\mathbf{z})$. The subscript $\mathbf{c}$ is taken to be $\mathbf{0}$ when omitted.

Smoothing Parameter. For any lattice $\mathcal{Y} \subseteq \mathbb{R}^n$ and any positive real $\epsilon > 0$, the *smoothing parameter* $\eta_\epsilon(\mathcal{Y})$ is defined as the smallest real $s > 0$ such that $\rho_{1/s}(\mathcal{Y}^* \setminus \{\mathbf{0}\}) \leq \epsilon$, where $\mathcal{Y}^* := \{\mathbf{y} \in \mathbb{R}^n \mid \forall \mathbf{x} \in \mathcal{Y}, \langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z}\}$ refers to the dual lattice of $\mathcal{Y}$.

Below we recall the following lemmas from [33].

Lemma 8 ([33, Cor. 2.8]). *Let $\mathcal{Y}' \subseteq \mathcal{Y} \subseteq \mathbb{R}^n$ be two lattices. Then for any $\epsilon \in (0, \frac{1}{2})$, any $\sigma \geq \eta_\epsilon(\mathcal{Y}')$ and any $\mathbf{c} \in \mathbb{R}^n$, the distribution $D_{\mathcal{Y}, \sigma, \mathbf{c}} \bmod \mathcal{Y}'$ is within statistical distance at most 2ϵ from the uniform distribution over $\mathcal{Y} \bmod \mathcal{Y}'$.*

Lemma 9 ([33, Lem. 3.1]). *For any n -dimensional lattice $\mathcal{Y}$ and any real $\epsilon > 0$, we have $\eta_\epsilon(\mathcal{Y}) \leq \tilde{bl}(\mathcal{Y}) \cdot \sqrt{\log(2n(1 + 1/\epsilon)) / \pi}$, where $\tilde{bl}(\mathcal{Y}) := \min_{\mathbf{B}} \|\tilde{\mathbf{B}}\| = \min_{\mathbf{B}} \max_{i \in [n]} \|\tilde{\mathbf{b}}_i\|$ denotes the Gram-Schmidt minimum of $\mathcal{Y}$ with the minimum taken over all bases $\mathbf{B}$ of $\mathcal{Y}$ and $\tilde{\mathbf{B}} = (\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n)$ being the Gram-Schmidt orthogonalization of $\mathbf{B}$.*

F.2 Instantiation of Two-Leveled IP-HPS from DCR

Now we present the DCR-based instantiation $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$.

(Outer-Level) IP-HPS Scheme. We first present the DCR-based (outer-level) $\text{IPHPS}_{\text{DCR}} = (\text{Setup}_{\text{IPHPS}}, \alpha, \mu, \text{Pub}, \text{Priv}, \text{FPriv})$ in Fig. 11. It is routine to check the correctness and the functional correctness of $\text{IPHPS}_{\text{DCR}}$.

Clearly, the (multi-fold) SMP related to $\text{IPHPS}_{\text{DCR}}$ corresponds to the (multi-fold) DCR assumptions w.r.t NGen. By the random self-reducibility of DCR (cf. Lemma 7 in Appendix F.1), we have the following theorem.

Theorem 8 (Tight Multi-fold SMP related to $\text{IPHPS}_{\text{DCR}}$). *Let $Q \geq 1$. For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$ such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + Q \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{IPHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{msmp}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) + (Q + 3) \cdot 2^{-2\lambda}$.*

¹⁸ As noted in Footnote 10, we use the symbol $\mathcal{Y}$ to denote lattices instead of using the conventional symbol Λ , since the symbol Λ is preserved to denote the hash function of IP-HPS.

$\text{ppIPHPS} \leftarrow \text{SetupIPHPS}(1^m):$ $(N, p, q) \leftarrow \text{NGen. } g \leftarrow \mathbb{Z}_{N^2}^*, g_N := 1 + N, g_{N'} := g^{2N} \bmod N^2.$ Return $\text{ppIPHPS} := (N, g_N, g_{N'})$, which implicitly defines $(\mathfrak{G}, \mathcal{L}, \mathcal{X}, \mathcal{SK}, \mathcal{HV}, \mathbf{A}_{(\cdot)})$ as follows: $\mathfrak{G} := \mathbb{G}_N \cdot \mathbb{G}_{N'}.$ $\mathcal{L} := \mathbb{G}_{N'} = \{c \mid \exists w \in \mathbb{Z}, \text{ s.t. } c = (g_{N'})^w \bmod N^2\} \subseteq \mathcal{X} := \mathbb{G}_N \cdot \mathbb{G}_{N'}.$ $\mathcal{SK} := \mathbb{Z}^m.$ $\mathcal{HV} := (\mathbb{G}_N \cdot \mathbb{G}_{N'})^m.$ $\mathbf{A}_{(\cdot)} : \mathcal{X} \rightarrow (\mathbb{G}_N \cdot \mathbb{G}_{N'})^m$ is defined as below: for any $sk = (k_1, \dots, k_m)^\top \in \mathcal{SK} = \mathbb{Z}^m$ and any $c \in \mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$, $\mathbf{A}_{sk}(c) := (c^{k_1}, \dots, c^{k_m})^\top \bmod N^2.$	
$pk \leftarrow \alpha(sk = (k_1, \dots, k_m)^\top):$ $(p_1, \dots, p_m) := ((g_{N'})^{k_1}, \dots, (g_{N'})^{k_m}) \bmod N^2.$ Return $pk := (p_1, \dots, p_m).$ $fk_{\mathbf{y}} \leftarrow \mu(sk = (k_1, \dots, k_m)^\top, \mathbf{y} \in \mathbb{Z}^m):$ Parse $\mathbf{y} = (y_1, \dots, y_m)^\top \in \mathbb{Z}^m.$ Return $fk_{\mathbf{y}} := \sum_{i=1}^m y_i \cdot k_i \in \mathbb{Z}.$	$\mathbf{h} \leftarrow \text{Pub}(pk = (p_1, \dots, p_m), c, w):$ $\parallel c \in \mathcal{L} \text{ with } c = (g_{N'})^w \bmod N^2$ Return $\mathbf{h} := (p_1^w, \dots, p_m^w)^\top \bmod N^2.$ $\mathbf{h} \leftarrow \text{Priv}(sk = (k_1, \dots, k_m)^\top, c):$ Return $\mathbf{h} := (c^{k_1}, \dots, c^{k_m})^\top \bmod N^2.$ $fv \leftarrow \text{FPriv}(fk_{\mathbf{y}} \in \mathbb{Z}, c):$ Return $fv := c^{fk_{\mathbf{y}}} \bmod N^2.$

Fig. 11. The DCR-based (outer-level) IP-HPS scheme $\text{IPHPS}_{\text{DCR}}$.

(Inner-Level) co-Hash Scheme. Next we describe the DCR-based inner-level $\text{coHash}_{\text{DCR}} = \text{Setup}_{\text{coHash}}$ in Fig. 12.

$\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{ppIPHPS}):$ Parse $\text{ppIPHPS} = (N, g_N, g_{N'}).$ Return $\text{pp}_{\text{coHash}} := \text{ppIPHPS} = (N, g_N, g_{N'})$, which implicitly defines $(\mathfrak{G}_{\text{co}}, \mathfrak{g}_{\text{co}}, \text{ord}_{\text{co}}, \mathcal{SK}_{\text{co}}, \mathcal{HV}_{\text{co}}, \Gamma_{(\cdot)})$ as follows: $\mathfrak{G}_{\text{co}} := \mathbb{G}_N (\subseteq \mathfrak{G} = \mathbb{G}_N \cdot \mathbb{G}_{N'})$, which is a cyclic group of order $\text{ord}_{\text{co}} := N$ and with generator $\mathfrak{g}_{\text{co}} := g_N = 1 + N$, and the discrete logarithm in $\mathfrak{G}_{\text{co}} = \mathbb{G}_N$ is easy to solve [56]. $\mathcal{SK}_{\text{co}} := \mathbb{Z}_N^*.$ $\mathcal{HV}_{\text{co}} := \mathbb{G}_N.$ $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{G}_N$ is defined as below: for any $s \in \mathcal{SK}_{\text{co}} = \mathbb{Z}_N^*$ and any $c \in \mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$, $\Gamma_s(c) := (g_N)^{s \cdot v} \bmod N^2,$ where $v \in \mathbb{Z}_N$ is the (unique) element s.t. $c = (g_N)^v \cdot (g_{N'})^w \bmod N^2$ for some $w \in \mathbb{Z}.$

Fig. 12. The DCR-based (inner-level) co-Hash scheme $\text{coHash}_{\text{DCR}}$.

We note that the inner-level hash function $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathbb{G}_N$ in our $\text{coHash}_{\text{DCR}}$ is well-defined, except with an exponentially small probability, as shown below. Note that $g_N := 1 + N$ is a generator of $\mathbb{G}_N$, and $g_{N'}$ defined by $g_{N'} := g^{2N} \bmod N^2$ for $g \leftarrow \mathbb{Z}_{N^2}^*$ is a generator of $\mathbb{G}_{N'}$ except with probability at most $2^{-2\lambda}$. Consequently, for each $c \in \mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$, there exist unique elements $v \in \mathbb{Z}_N$ and $w \in \mathbb{Z}_{N'}$ such that $c = (g_N)^v \cdot (g_{N'})^w \bmod N^2$, and $\Gamma_{(\cdot)}$ is well-defined.

Finally, through the following two theorems, we prove the functional smoothness and the tight multi-key-extracting of our $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$. In particular, to prove the functional soundness, we use the notation “ $sk \leftarrow_s \mathcal{SK}$ ” to denote sampling sk from $\mathcal{SK} = \mathbb{Z}^m$ according to the discrete Gaussian distribution $D_{\mathbb{Z}^m, \sigma}$, similar to the secret key distribution of the DCR-based IPFE scheme in [8] (i.e., the ALS scheme). We refer to Appendix F.1 for the definition of discrete Gaussian and useful properties for it.

Theorem 9 (Functional Smoothness of $\text{TLHPS}_{\text{DCR}}$). *Let “ $sk \leftarrow_s \mathcal{SK}$ ” denote sampling sk from $\mathcal{SK} = \mathbb{Z}^m$ according to the discrete Gaussian distribution $D_{\mathbb{Z}^m, \sigma}$ with $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$. Then the two-leveled scheme $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$ proposed in Fig. 11 and 12 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmtH}}(\lambda) \leq 2^{-2\lambda}$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q , where $\mathcal{A}$ is required to query $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$ with vectors $\tilde{\mathbf{x}}$ and $\mathbf{y}$ of bounded norm, i.e., $\|\tilde{\mathbf{x}}\|_{\infty} \leq X$ and $\|\mathbf{y}\|_{\infty} \leq Y$ for bounds X and Y , and the bounds satisfy $X, Y < \sqrt{N/m}$ (so that $mXY < N$)¹⁹.*

Theorem 10 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{DCR}}$). *If the DCR assumption holds w.r.t. NGen , then the two-leveled scheme $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$ proposed in Fig. 11 and 12 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) + (TQ + 3) \cdot 2^{-2\lambda}$.*

We present their proofs in Appendix F.3 and Appendix F.4, respectively.

F.3 Proof of Theorem 9 (Functional Smoothness of $\text{TLHPS}_{\text{DCR}}$)

Before presenting the proof of Theorem 9, we first show a useful lemma.

Lemma 10. *For $\text{pp}_{\text{IPHPS}} = (N, g_N, g_{N'}) \leftarrow_s \text{Setup}_{\text{IPHPS}}(1^m)$ and $\text{pp}_{\text{coHash}} = (N, g_N, g_{N'}) \leftarrow_s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$, $g_{N'}$ is a generator of $\mathbb{G}_{N'}$, except with probability at most $2^{-2\lambda}$.*

For the ease of exposition, we will not mention this exponentially small probability hereafter, and take for granted that $g_{N'}$ is a generator of $\mathbb{G}_{N'}$.

The lemma holds due to the following reasons. Note that $g_{N'}$ is defined by $g_{N'} := g^{2^N} \bmod N^2$ with $g \leftarrow_s \mathbb{Z}_{N^2}^*$ during the generation of pp_{IPHPS} , thus it is a generator of $\mathbb{G}_{N'}$ except with probability at most $1 - \frac{\phi(N')}{N'} = \frac{p' + q' - 1}{p'q'} < \frac{2}{\min\{p', q'\}} < 2^{-2\lambda}$.

Next we present the proof of Theorem 9.

¹⁹ Jumping ahead, this requirement naturally translates to the mCPA security of our DCR-based IPFE scheme as will be described later in Appendix F.5, where the adversary against the mCPA security is required to query $\mathcal{O}_{\text{ENC}}$ and $\mathcal{O}_{\text{KGEN}}$ with bounded norm vectors as well. We note that the DCR-based IPFE scheme in [8] (i.e., the ALS scheme) also makes such requirement.

Theorem 9. Let “ $sk \leftarrow_s SK$ ” denote sampling sk from $SK = \mathbb{Z}^m$ according to the discrete Gaussian distribution $D_{\mathbb{Z}^m, \sigma}$ with $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$. Then the two-leveled scheme $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$ proposed in Fig. 11 and 12 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) \leq 2^{-2\lambda}$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q , where $\mathcal{A}$ is required to query $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$ with vectors $\tilde{\mathbf{x}}$ and $\mathbf{y}$ of bounded norm, i.e., $\|\tilde{\mathbf{x}}\|_{\infty} \leq X$ and $\|\mathbf{y}\|_{\infty} \leq Y$ for bounds X and Y , and the bounds satisfy $X, Y < \sqrt{N/m}$ (so that $mXY < N$).

Proof of Theorem 9. To show that $\epsilon_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) \leq 2^{-2\lambda}$, we first elaborate the $\text{Exp}_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmth}, b}$ experiments ($b \in \{0, 1\}$) defined in Fig. 2.

Let $\text{pp}_{\text{IPHPS}} = \text{pp}_{\text{coHash}} = (N, g_N, g_{N'})$, where $g_N := 1 + N$ and $g_{N'} := g^{2N} \bmod N^2$ with uniformly chosen $g \leftarrow_s \mathbb{Z}_{N^2}^*$. Let $sk = (k_1, \dots, k_m)^\top \leftarrow_s D_{\mathbb{Z}^m, \sigma}$ and $s \leftarrow_s \mathbb{Z}_N^*$. In the experiment, $\mathcal{A}$ first obtains $(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk)$ where

$$pk = (p_1, \dots, p_m) = ((g_{N'})^{k_1}, \dots, (g_{N'})^{k_m}) \bmod N^2, \quad (15)$$

and has oracle access to $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$. Through $\mathcal{O}_{\text{FKEY}}$ queries $(\mathbf{y} = (y_1, \dots, y_m)^\top \in \mathbb{Z}^m)$, $\mathcal{A}$ learns functional keys

$$fk_{\mathbf{y}} = \sum_{i=1}^m y_i \cdot k_i \in \mathbb{Z}. \quad (16)$$

At some point, $\mathcal{A}$ makes a single $\mathcal{O}_{\text{CHAL}}^b$ query $(\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_m)^\top \in \mathbb{Z}^m)$, and obtains $(c^{(j)}, \mathbf{h}^{(j)})_{j \in [Q]}$, where $c^{(j)} \leftarrow_s \mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$ and

$$\mathbf{h}^{(j)} := \begin{cases} \mathbf{A}_{sk}(c^{(j)}) = ((c^{(j)})^{k_1}, \dots, (c^{(j)})^{k_m})^\top \bmod N^2, & \text{if } b = 0; \\ \mathbf{A}_{sk}(c^{(j)}) \cdot \Gamma_s(c^{(j)})^{\tilde{\mathbf{x}}} = ((c^{(j)})^{k_1} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_1}, \dots, (c^{(j)})^{k_m} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_m})^\top, & \text{if } b = 1, \end{cases} \quad (17)$$

where each $v^{(j)} \in \mathbb{Z}_N$ is the (unique) element s.t. $c^{(j)} = (g_N)^{v^{(j)}} \cdot (g_{N'})^{w^{(j)}} \bmod N^2$ for some $w^{(j)} \in \mathbb{Z}$. Then $\mathcal{A}$ may continue to make $\mathcal{O}_{\text{FKEY}}$ queries $(\mathbf{y} \in \mathbb{Z}^m)$ and learn functional keys $fk_{\mathbf{y}}$, similar to (16). Finally, $\mathcal{A}$ outputs b' , and the advantage of $\mathcal{A}$ is $\epsilon_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) = |\Pr[b' = 1 | b = 0] - \Pr[b' = 1 | b = 1]|$.

Below we analyze $\mathcal{A}$'s advantage in distinguishing $b = 0$ and $b = 1$. Note that the only differences between $b = 0$ and $b = 1$ are the distributions of $\mathbf{h}^{(j)}$ in (17). We will show that the $\mathbf{h}^{(j)} = ((c^{(j)})^{k_1}, \dots, (c^{(j)})^{k_m})^\top \bmod N^2$ (if $b = 0$) is statistically indistinguishable from the $\mathbf{h}^{(j)} = ((c^{(j)})^{k_1} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_1}, \dots, (c^{(j)})^{k_m} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_m})^\top \bmod N^2$ (if $b = 1$), from the point of view of $\mathcal{A}$. To see this, since each $c^{(j)}$ is uniformly chosen from $\mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$, $c^{(j)}$ can be uniquely parsed as $c^{(j)} = (g_N)^{v^{(j)}} \cdot (g_{N'})^{w^{(j)}} \bmod N^2$ for some $v^{(j)} \in \mathbb{Z}_N$ and $w^{(j)} \in \mathbb{Z}_{N'}$ (note that it is taken for granted that $g_{N'}$ is a generator of $\mathbb{G}_{N'}$ by Lemma 10). Consequently, in (17), in the case of $b = 0$, we have

$$\begin{aligned} \mathbf{h}^{(j)} &= ((c^{(j)})^{k_1}, \dots, (c^{(j)})^{k_m})^\top \bmod N^2 \\ &= ((g_N)^{v^{(j)} \cdot k_1} \cdot (g_{N'})^{w^{(j)} \cdot k_1}, \dots, (g_N)^{v^{(j)} \cdot k_m} \cdot (g_{N'})^{w^{(j)} \cdot k_m})^\top \bmod N^2, \end{aligned} \quad (18)$$

and in the case of $b = 1$, we have

$$\begin{aligned} \mathbf{h}^{(j)} &= ((c^{(j)})^{k_1} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_1}, \dots, (c^{(j)})^{k_m} \cdot (g_N)^{sv^{(j)} \cdot \tilde{x}_m})^\top \bmod N^2 \\ &= ((g_N)^{v^{(j)} \cdot (k_1 + \tilde{s}\tilde{x}_1)} \cdot (g_{N'})^{w^{(j)} \cdot k_1}, \dots, (g_N)^{v^{(j)} \cdot (k_m + \tilde{s}\tilde{x}_m)} \cdot (g_{N'})^{w^{(j)} \cdot k_m})^\top \bmod N^2. \end{aligned} \quad (19)$$

Since g_N is of order N , to show that the $\{\mathbf{h}^{(j)}\}_{j \in [Q]}$ in (18) are statistically close to the $\{\mathbf{h}^{(j)}\}_{j \in [Q]}$ in (19), it suffices to prove that $(k_1, \dots, k_m) \bmod N$ is statistically close to $(k_1 + \tilde{s}\tilde{x}_1, \dots, k_m + \tilde{s}\tilde{x}_m) \bmod N$, in the view of $\mathcal{A}$.

Now we analyze the view of $\mathcal{A}$, and in particular, the distribution of $sk = (k_1, \dots, k_m)^\top$ conditioned on the pk in (15) and the $fk_{\mathbf{y}}$ in (16) obtained from $\mathcal{O}_{\text{FKEY}}$ queries. For the convenience of our analysis, we assume that $\mathcal{A}$ makes $\mathcal{O}_{\text{FKEY}}$ queries for all $\mathbf{y} \in \mathbb{Z}^m$ orthogonal to $\tilde{\mathbf{x}} \in \mathbb{Z}^m$, i.e., for all $\mathbf{y} \in \mathcal{Q} := \{\mathbf{y} \in \mathbb{Z}^m \mid \langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0\}$ ²⁰, through which $\mathcal{A}$ can get as much information as possible about $sk = (k_1, \dots, k_m)^\top$. Formally, we have the following two claims.

Claim 5. Conditioned on $pk = ((g_{N'})^{k_1}, \dots, (g_{N'})^{k_m})^\top \bmod N^2$ and $\{fk_{\mathbf{y}} = \sum_{i=1}^m y_i \cdot k_i\}_{\mathbf{y} \in \mathcal{Q}}$, the conditional distribution of $sk = (k_1, \dots, k_m)^\top \leftarrow D_{\mathbb{Z}^m, \sigma}$ is $\mathbf{k}_0 + D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$, where $\mathbf{k}_0 = (k_{01}, \dots, k_{0m})^\top \in \mathbb{Z}^m$ is any particular vector satisfying the conditions and $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z} \subseteq \mathbb{Z}^m$ is a lattice with $\gcd(\tilde{\mathbf{x}})$ the greatest common divisor of all entries of $\tilde{\mathbf{x}}$.

Proof. Let $\Delta = (\delta_1, \dots, \delta_m)^\top := sk - \mathbf{k}_0 = (k_1 - k_{01}, \dots, k_m - k_{0m})^\top \in \mathbb{Z}^m$.

We will first prove that Δ is in the lattice $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z}$. Since both sk and $\mathbf{k}_0$ satisfy the conditions, i.e., $pk = ((g_{N'})^{k_1}, \dots, (g_{N'})^{k_m})^\top \bmod N^2 = ((g_{N'})^{k_{01}}, \dots, (g_{N'})^{k_{0m}})^\top \bmod N^2$ and $\{fk_{\mathbf{y}} = \sum_{i=1}^m y_i \cdot k_i = \sum_{i=1}^m y_i \cdot k_{0i}\}_{\mathbf{y} \in \mathcal{Q}}$, it follows that Δ satisfies

$$(\delta_1, \dots, \delta_m) \bmod N' = (0, \dots, 0), \quad (20)$$

$$\text{and } \{\sum_{i=1}^m y_i \cdot \delta_i = 0\}_{\mathbf{y} \in \mathcal{Q}}. \quad (21)$$

Note that (20) implies that $\delta_1, \dots, \delta_m$ are multiples of N' , and thus we get that $N' \mid \gcd(\Delta)$. On the other hand, (21) implies that Δ is orthogonal to all $\mathbf{y} \in \mathcal{Q}$. Since $\mathcal{Q}$ contains all $\mathbf{y}$ orthogonal to $\tilde{\mathbf{x}}$, it follows that Δ is in the span of $\tilde{\mathbf{x}}$, i.e., $\Delta = w \cdot \tilde{\mathbf{x}}$ for some $w \in \mathbb{Q}$, which further implies that $\gcd(\Delta) = w \cdot \gcd(\tilde{\mathbf{x}})$. Putting $N' \mid \gcd(\Delta)$ and $\gcd(\Delta) = w \cdot \gcd(\tilde{\mathbf{x}})$ together, we know that $\gcd(\Delta) = w \cdot \gcd(\tilde{\mathbf{x}}) = N' \cdot z$ for some $z \in \mathbb{Z}$. Consequently, we get that $w = \frac{1}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot z$, and it follows that $\Delta = w \cdot \tilde{\mathbf{x}} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot z$ for some $z \in \mathbb{Z}$, i.e., $\Delta \in \mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z}$.

Then we have that $sk = \mathbf{k}_0 + \Delta \in \mathbf{k}_0 + \mathcal{Y} \subseteq \mathbb{Z}^m$. Consequently, the conditional distribution of $sk = (k_1, \dots, k_m)^\top \leftarrow D_{\mathbb{Z}^m, \sigma}$ is $D_{\mathbf{k}_0 + \mathcal{Y}, \sigma}$, which equals $\mathbf{k}_0 + D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$ by the definition of discrete Gaussian, and Claim 5 follows. $\blacksquare$

²⁰ Note that $\mathcal{Q}$ contains an infinite number of vectors, so it is not possible for $\mathcal{A}$ to query all vectors $\mathbf{y} \in \mathcal{Q}$ to $\mathcal{O}_{\text{FKEY}}$. Nevertheless, since $fk_{\mathbf{y}}$ is linear w.r.t. $\mathbf{y}$ in our scheme (cf. Fig. 11), the unbounded adversary $\mathcal{A}$ can make $\mathcal{O}_{\text{FKEY}}$ queries for a basis of $\mathcal{Q}$, and with the obtained $fk_{\mathbf{y}}$ for the basis, $\mathcal{A}$ can effectively get $fk_{\mathbf{y}}$ for all $\mathbf{y} \in \mathcal{Q}$.

Claim 6. Let $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z}$ and $\mathcal{Y}' := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot N \cdot \mathbb{Z} \subseteq \mathcal{Y}$. Then for any $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$ and any $\mathbf{k}_0 \in \mathbb{Z}^m$, the distribution of $D_{\mathcal{Y}, \sigma, -\mathbf{k}_0} \bmod \mathcal{Y}'$ is within statistical distance at most $2^{-(2\lambda+1)}$ from the uniform distribution over $\mathcal{Y} \bmod \mathcal{Y}'$, which denotes the quotient lattice $\{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot u \mid u \in \mathbb{Z}_N\}$.

Proof. According to Lemma 8 in Appendix F.1, it suffices to prove that any $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$ must satisfy $\sigma \geq \eta_\epsilon(\mathcal{Y}')$ for $\epsilon := 2^{-(2\lambda+2)}$. To this end, we analyze $\eta_\epsilon(\mathcal{Y}')$ for the lattice $\mathcal{Y}' = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot N \cdot \mathbb{Z} \subseteq \mathcal{Y}$. Note that $\mathcal{Y}'$ is in fact a 1-dimensional lattice in $\mathbb{Z}^m$ with basis $\mathbf{b} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot N$. Then by Lemma 9 in Appendix F.1, we get that $\tilde{bl}(\mathcal{Y}') \leq \|\mathbf{b}\| = \frac{\|\tilde{\mathbf{x}}\|}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot N \leq \sqrt{m} \cdot \|\tilde{\mathbf{x}}\|_\infty \cdot N^2 \leq \sqrt{m} \cdot X \cdot N^2 < N^{\frac{5}{2}}$, where the last two inequalities follow from the constrain $\|\tilde{\mathbf{x}}\|_\infty \leq X < \sqrt{N/m}$, and it further implies that $\eta_\epsilon(\mathcal{Y}') \leq \tilde{bl}(\mathcal{Y}') \cdot \sqrt{\log(2(1+1/\epsilon))/\pi} \leq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$ for $\epsilon = 2^{-(2\lambda+2)}$. Consequently, we obtain that $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda} \geq \eta_\epsilon(\mathcal{Y}')$, and by Lemma 8, $D_{\mathcal{Y}, \sigma, -\mathbf{k}_0} \bmod \mathcal{Y}'$ is within statistical distance at most $2\epsilon = 2^{-(2\lambda+1)}$ from the uniform distribution over $\mathcal{Y} \bmod \mathcal{Y}'$, and Claim 6 follows. $\blacksquare$

Now we analyze the distributions of $(k_1, \dots, k_m) \bmod N$ and $(k_1 + s\tilde{x}_1, \dots, k_m + s\tilde{x}_m) \bmod N$ from the view of point of $\mathcal{A}$. We first analyze the distribution of $(k_1, \dots, k_m) \bmod N$. According to Claim 5, the distribution of $sk = (k_1, \dots, k_m)^\top$ conditioned on the pk in (15) and the fk_y in (16) obtained from $\mathcal{O}_{\text{FKY}}$ queries follows $\mathbf{k}_0 + D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$ where $\mathbf{k}_0 \in \mathbb{Z}^m$ is any particular vector satisfying the conditions and $\mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z} \subseteq \mathbb{Z}^m$, thus we have $(k_1, \dots, k_m)^\top \bmod N \equiv (\mathbf{k}_0 + \Delta) \bmod N$ with $\Delta \leftarrow_s D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$. For $\Delta \leftarrow_s D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$, $\Delta \in \mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot \mathbb{Z}$, we can further write $\Delta = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot z$ for some $z \in \mathbb{Z}$. Then we have that

$$\begin{aligned} & (k_1, \dots, k_m)^\top \bmod N \\ & \equiv (\mathbf{k}_0 + \Delta) \bmod N = (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot z) \bmod N \\ & = (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot (z \bmod N)) \bmod N \\ & \stackrel{s}{\approx} (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot u) \bmod N \quad \text{with } u \leftarrow_s \mathbb{Z}_N. \end{aligned} \tag{22}$$

Here we explain the last step “ $\stackrel{s}{\approx}$ ” of (22). Note that for $\Delta = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot z$ following the distribution $D_{\mathcal{Y}, \sigma, -\mathbf{k}_0}$, the term $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot (z \bmod N)$ exactly follows the distribution $D_{\mathcal{Y}, \sigma, -\mathbf{k}_0} \bmod \mathcal{Y}'$ with $\mathcal{Y}' = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot N \cdot \mathbb{Z} \subseteq \mathcal{Y}$. Then by Claim 6, the term $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot (z \bmod N)$ is statistically indistinguishable from the uniform distribution over $\mathcal{Y} \bmod \mathcal{Y}' = \{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot u \mid u \in \mathbb{Z}_N\}$, with statistical distance at most $2^{-(2\lambda+1)}$. This shows the last step “ $\stackrel{s}{\approx}$ ” of (22). Similarly, for

the distribution of $(k_1 + \tilde{s}\tilde{x}_1, \dots, k_m + \tilde{s}\tilde{x}_m) \bmod N$, we have that

$$\begin{aligned}
& (k_1 + \tilde{s}\tilde{x}_1, \dots, k_m + \tilde{s}\tilde{x}_m)^\top \bmod N = ((k_1, \dots, k_m)^\top + \tilde{s}\tilde{\mathbf{x}}) \bmod N \\
& \stackrel{s}{\approx} (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot u + \tilde{s}\tilde{\mathbf{x}}) \bmod N \quad \text{with } u \leftarrow_s \mathbb{Z}_N \\
& = (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot (u + \gcd(\tilde{\mathbf{x}}) \cdot (N')^{-1} \cdot s)) \bmod N \quad \text{with } u \leftarrow_s \mathbb{Z}_N \\
& \equiv (\mathbf{k}_0 + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot N' \cdot u') \bmod N \quad \text{with } u' \leftarrow_s \mathbb{Z}_N,
\end{aligned} \tag{23}$$

where the second last step “=” follows from the fact that $\gcd(N, N') = 1$ and $(N')^{-1}$ denotes the inverse of N' modulo N , and the last step “ $\equiv$ ” holds since $u \leftarrow_s \mathbb{Z}_N$ perfectly hides $\gcd(\tilde{\mathbf{x}}) \cdot (N')^{-1} \cdot s$ when modulo N .

By combining (22) and (23) together, we get the statistical indistinguishability of $(k_1, \dots, k_m) \bmod N$ and $(k_1 + \tilde{s}\tilde{x}_1, \dots, k_m + \tilde{s}\tilde{x}_m) \bmod N$ from the point of view of $\mathcal{A}$, with statistical distance at most $2 \cdot 2^{-(2\lambda+1)} = 2^{-2\lambda}$. This in turn shows the statistical indistinguishability of $\{\mathbf{h}^{(j)}\}_{j \in [Q]}$ in (18) (i.e, the case $b = 0$ in (17)) and the $\{\mathbf{h}^{(j)}\}_{j \in [Q]}$ in (19) (i.e, the case $b = 1$ in (17)), therefore $\mathcal{A}$ has advantage at most $2^{-2\lambda}$ in distinguishing $b = 0$ and $b = 1$.

Overall, we get that $\epsilon_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) \leq 2^{-2\lambda}$, and Theorem 9 follows. $\square$

F.4 Proof of Theorem 10 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{DCR}}$)

Theorem 10. *If the DCR assumption holds w.r.t. NGen , then the two-leveled scheme $\text{TLHPS}_{\text{DCR}} = (\text{IPHPS}_{\text{DCR}}, \text{coHash}_{\text{DCR}})$ proposed in Fig. 11 and 12 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) + (TQ + 3) \cdot 2^{-2\lambda}$.*

Proof of Theorem 10. Firstly, we construct an adversary $\mathcal{B}'$ against the TQ -fold DCR w.r.t. NGen , so that $\text{Adv}_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}'}^{TQ\text{-dcr}}(\lambda)$. Then by the random self-reducibility of DCR (i.e., Lemma 7 in Appendix F.1), Theorem 10 follows.

Given a challenge $(N, \{u_{i,j}\}_{i \in [T], j \in [Q]})$, $\mathcal{B}'$ wants to distinguish $\{u_{i,j} \leftarrow_s \mathbb{G}_N \cdot \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$ from $\{u_{i,j} \leftarrow_s \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$.

$\mathcal{B}'$ is constructed by invoking $\mathcal{A}$ as follows. $\mathcal{B}'$ samples $g \leftarrow_s \mathbb{Z}_{N^2}^*$, computes $g_N := 1 + N$, $g_{N'} := g^{2N} \bmod N^2$, and sets $\text{pp}_{\text{IPHPS}} := \text{pp}_{\text{coHash}} := (N, g_N, g_{N'})$. Then for all $i \in [T]$ and $j \in [Q]$, $\mathcal{B}'$ samples $s_i \leftarrow_s \mathbb{Z}_N^*$ and $r_{i,j} \leftarrow_s \mathbb{Z}_N$ uniformly at random, and sets $c_{i,j} := (g_N)^{r_{i,j}} \cdot u_{i,j} \bmod N^2$ and $h_{i,j} := (g_N)^{s_i \cdot r_{i,j}} \bmod N^2$.

Finally, $\mathcal{B}'$ invokes $\mathcal{A}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, \{c_{i,j}, \boxed{h_{i,j}}\}_{i \in [T], j \in [Q]})$, and returns to its own challenger whatever $\mathcal{A}$ outputs.

Next, we analyze $\mathcal{B}'$'s advantage. Note that $g_N = 1 + N$ is a generator of $\mathbb{G}_N$ and it is taken for granted that $g_{N'}$ is a generator of $\mathbb{G}_{N'}$ according to Lemma 10 in Appendix F.3.

- In the case that $\{u_{i,j} \leftarrow \mathbb{G}_N \cdot \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$, each $u_{i,j}$ can be uniquely parsed as $u_{i,j} = (g_N)^{v_{i,j}} \cdot (g_{N'})^{w_{i,j}} \bmod N^2$ for uniformly distributed $v_{i,j} \leftarrow \mathbb{Z}_N$ and $w_{i,j} \leftarrow \mathbb{Z}_{N'}$. Then we have that each $c_{i,j} = (g_N)^{r_{i,j}} \cdot u_{i,j} = (g_N)^{r_{i,j} + v_{i,j}} \cdot (g_{N'})^{w_{i,j}} \bmod N^2$ and $h_{i,j} = (g_N)^{s_i \cdot r_{i,j}} \bmod N^2$. On the one hand, all $c_{i,j}$ are independently and uniformly distributed over $\mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$ since $r_{i,j} + v_{i,j}$ is uniformly distributed over $\mathbb{Z}_N$ and $w_{i,j}$ is uniformly distributed over $\mathbb{Z}_{N'}$. On the other hand, the uniformity of $v_{i,j} \leftarrow \mathbb{Z}_N$ hides $r_{i,j}$ in $c_{i,j}$, then by the uniformity of $r_{i,j} \leftarrow \mathbb{Z}_N$ and by the fact that $s_i \in \mathbb{Z}_N^*$, it follows that all $h_{i,j} = (g_N)^{s_i \cdot r_{i,j}} \bmod N^2$ are uniformly distributed over $\mathfrak{G}_{\text{co}} = \mathbb{G}_N$ and are independent of $c_{i,j}$.
- In the case that $\{u_{i,j} \leftarrow \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$, each $u_{i,j}$ can be uniquely parsed as $u_{i,j} = (g_{N'})^{w_{i,j}} \bmod N^2$ for uniformly distributed $w_{i,j} \leftarrow \mathbb{Z}_{N'}$. Then we have that each $c_{i,j} = (g_N)^{r_{i,j}} \cdot u_{i,j} = (g_N)^{r_{i,j}} \cdot (g_{N'})^{w_{i,j}} \bmod N^2$ and $h_{i,j} = (g_N)^{s_i \cdot r_{i,j}} \bmod N^2$. On the one hand, all $c_{i,j}$ are independently and uniformly distributed over $\mathcal{X} = \mathbb{G}_N \cdot \mathbb{G}_{N'}$ since $r_{i,j}$ is uniformly distributed over $\mathbb{Z}_N$ and $w_{i,j}$ is uniformly distributed over $\mathbb{Z}_{N'}$. On the other hand, by the definition of $\Gamma(\cdot)$, it is clear to see that $h_{i,j} = (g_N)^{s_i \cdot r_{i,j}} \bmod N^2 = \Gamma_{s_i}(c_{i,j})$ in this case, with s_i uniformly chosen from $\mathcal{SK}_{\text{co}} = \mathbb{Z}_N^*$.

Overall, $\mathcal{B}'$ can distinguish $\{u_{i,j} \leftarrow \mathbb{G}_N \cdot \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$ from $\{u_{i,j} \leftarrow \mathbb{G}_{N'}\}_{i \in [T], j \in [Q]}$ as long as $\mathcal{A}$ is able to distinguish $\{c_{i,j} \leftarrow \mathcal{X}, \boxed{h_{i,j} \leftarrow \mathfrak{G}_{\text{co}}}\}_{i \in [T], j \in [Q]}$ from $\{c_{i,j} \leftarrow \mathcal{X}, \boxed{h_{i,j} = \Gamma_{s_i}(c_{i,j})}\}_{i \in [T], j \in [Q]}$, and thus we get that $\text{Adv}_{\text{TLHPS}_{\text{DCR}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{\text{NGen}, \mathcal{B}'}^{TQ\text{-dcr}}(\lambda)$, as desired. This completes the proof of Theorem 10. $\square$

F.5 Instantiation of IPFE from DCR

By plugging our $\text{TLHPS}_{\text{DCR}}$ proposed in Fig. 11 and 12 in Appendix F.2 into the generic construction $\text{IPFE}[\text{TLHPS}, L]$ presented in Fig. 3 in Sect. 5, we immediately obtain a DCR-based IPFE scheme $\text{IPFE}_{\text{DCR}} = \text{IPFE}[\text{TLHPS}_{\text{DCR}}, L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$, as shown in Fig. 13.

Let $x \cdot \mathbb{Z}$ (resp., $x \cdot \mathbb{Z}_{N^2}$) denote x elements in $\mathbb{Z}$ (resp., $\mathbb{Z}_{N^2}$). Under parameter L , the DCR-based IPFE_{DCR} has master public key $\text{mpk} : (\tilde{m}m + 1) \cdot \mathbb{Z}_{N^2}$, master secret key $\text{msk} : \tilde{m}m \cdot \mathbb{Z}$, functional secret key $sk_{\mathbf{y}} : \tilde{m} \cdot \mathbb{Z}$, and ciphertext $ct : (\tilde{m} + m) \cdot \mathbb{Z}_{N^2}$, where $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}$.

By plugging Theorem 8, Theorem 9 and Theorem 10 regarding the tight properties of $\text{TLHPS}_{\text{DCR}}$ in Appendix F.2 into Theorem 1, we have the following corollary showing the tight mCPA security of IPFE_{DCR} based on the DCR assumption.

Corollary 3 (Tight mCPA Security of IPFE_{DCR}). *Assume that the DCR assumption holds w.r.t. NGen and the discrete Gaussian distribution $D_{\mathbb{Z}^m, \sigma}$ satisfies $\sigma \geq N^{\frac{5}{2}} \cdot \sqrt{\lambda}$. Then the scheme IPFE_{DCR} in Fig. 13 is mCPA secure.*

Concretely, for any adversary $\mathcal{A}$ who makes at most Q_e queries to $\mathcal{O}_{\text{ENC}}$ and Q_k queries to $\mathcal{O}_{\text{KGEN}}$, there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) +$

$(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^m):$ $(N, p, q) \leftarrow \text{NGen}.$ $g \leftarrow \mathbb{Z}_{N^2}^*, g_N := 1 + N, g_{N'} := g^{2N} \bmod N^2.$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}]$: $(k_{i,1}, \dots, k_{i,m})^\top \leftarrow D_{\mathbb{Z}^m, \sigma}.$ $(p_{i,1}, \dots, p_{i,m}) := ((g_{N'})^{k_{i,1}}, \dots, (g_{N'})^{k_{i,m}}) \bmod N^2.$ $\text{mpk} := (N, g_N, g_{N'}, \{p_{i,j}\}_{i \in [\tilde{m}], j \in [m]}).$ $\text{msk} := \{k_{i,j}\}_{i \in [\tilde{m}], j \in [m]}.$ Return $(\text{mpk}, \text{msk}).$ $sk_{\mathbf{y}} \leftarrow \text{KGen}(\text{msk}, \mathbf{y} \in \mathbb{Z}^m):$ Parse $\mathbf{y} = (y_1, \dots, y_m)^\top \in \mathbb{Z}^m.$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}]$: $fk_{i,\mathbf{y}} := \sum_{j=1}^m y_j \cdot k_{i,j} \in \mathbb{Z}.$ Return $sk_{\mathbf{y}} := (fk_{1,\mathbf{y}}, \dots, fk_{\tilde{m},\mathbf{y}}).$	$ct \leftarrow \text{Enc}(\text{mpk}, \mathbf{x} \in \mathbb{Z}^m):$ Parse $\mathbf{x} = (x_1, \dots, x_m)^\top \in \mathbb{Z}^m.$ $\tilde{m} := \lceil \frac{m}{L} \rceil \in \mathbb{N}.$ For $i \in [\tilde{m}]$: $w_i \leftarrow \mathbb{Z}_{\lfloor \frac{N}{4} \rfloor}.$ $c_i := (g_{N'})^{w_i} \bmod N^2.$ For $j \in [m]$: $e_j := \prod_{i=1}^{\tilde{m}} (p_{i,j})^{w_i} \cdot (g_N)^{x_j} \bmod N^2.$ Return $ct := (c_1, \dots, c_{\tilde{m}}, e_1, \dots, e_m).$ $d/\perp \leftarrow \text{Dec}(sk_{\mathbf{y}}, ct):$ Parse $sk_{\mathbf{y}} = (fk_{1,\mathbf{y}}, \dots, fk_{\tilde{m},\mathbf{y}}),$ $ct = (c_1, \dots, c_{\tilde{m}}, e_1, \dots, e_m).$ $\mathfrak{d} := \prod_{j=1}^m (e_j)^{y_j} / \prod_{i=1}^{\tilde{m}} (c_i)^{fk_{i,\mathbf{y}}} \bmod N^2.$ $d := \text{dlog}_{g_N} \mathfrak{d} \bmod N.$ Return $d.$
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Fig. 13. The DCR-based IPFE scheme $\text{IPFE}_{\text{DCR}} = \text{IPFE}[\text{TLHPS}_{\text{DCR}}, L] = (\text{Setup}, \text{KGen}, \text{Enc}, \text{Dec})$ parameterized by $L \in \mathbb{N}$.

$(Q_e + Q_k) \cdot \text{poly}(\lambda)$, with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and

$$\text{Adv}_{\text{IPFE}_{\text{DCR}}, \mathcal{A}}^{\text{mCPA}}(\lambda) \leq 3L \cdot \text{Adv}_{\text{NGen}, \mathcal{B}}^{\text{dcr}}(\lambda) + L \cdot (3Q_e \tilde{m} + \tilde{m} + 9) \cdot 2^{-2\lambda}.$$

We note that the above statistical loss $L \cdot (3Q_e \tilde{m} + \tilde{m} + 9) \cdot 2^{-2\lambda}$ is exponentially small, and in particular less than $2^{-\lambda}$. This is because L is a constant and $Q_e, \tilde{m}$ are polynomials in the security parameter λ .

Concrete DCR-Based IPFE Schemes. Below we consider two concrete parameter settings of our $\text{IPFE}_{\text{DCR}} = \text{IPFE}[\text{TLHPS}_{\text{DCR}}, L]$.

- **Scheme 1 (The ALS Scheme and Tighter Analysis on Its Security Loss):** $\text{IPFE}[\text{TLHPS}_{\text{DCR}}, m]$. It is in fact the DCR-based ALS scheme [8]. Similar to the DDH-based ALS scheme (cf. Appendix E.5), the DCR-based ALS scheme was only known to have loose mCPA security prior to our work. Now with our Corollary 3, $\text{ALS} = \text{IPFE}[\text{TLHPS}_{\text{DCR}}, m]$ is turned out to have *tighter* mCPA security with loss factor only $O(m) = 3m$ instead of $O(Q_e)$.
- **Scheme 2 (The First Tightly Secure IPFE Scheme based on DCR):** $\text{IPFE}[\text{TLHPS}_{\text{DCR}}, L]$. It constitutes the *first* tightly secure IPFE scheme based on the DCR assumption, with loss factor $3L$, by setting L small constants (say $L = 5$ or $L = 100$). Moreover, it enjoys very compact ciphertexts and efficient algorithms, similar to our MDDH-based IPFE schemes in Appendix E.5. We refer to Table 1 in Sect. 1 for the efficiency of this scheme under $L = 5$ and $L = 100$.

G Missing Details in Sect. 6 (LWE-based Instantiation)

In this section, we provide missing details in Sect. 6 regarding the LWE-based instantiation of our two-leveled IP-HPS $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ and the resulting IPFE scheme IPFE_{LWE} .

More precisely, we first present some necessary lattice preliminaries and useful lemmas in Appendix G.1, give the formal definition of LWE assumptions as well as variants including the Multi-secret LWE and the Multi-instance LWE, and show several (almost) tight reductions among these LWE assumptions in Appendix G.2. Then we present the formal definition of *probabilistic* two-leveled IP-HPS including probabilistic outer-level IP-HPS and probabilistic inner-level co-Hash in Appendix G.3. Finally, we provide the proof of Theorem 3 (Functional Smoothness of $\text{TLHPS}_{\text{LWE}}$) and the proof of Theorem 4 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{LWE}}$) in Appendix G.4 and Appendix G.5, respectively.

G.1 Lattice Preliminaries

Discrete Gaussians. For $\sigma > 0$ and $\mathbf{c} \in \mathbb{R}^n$, we define the Gaussian function on $\mathbb{R}^n$ centered at $\mathbf{c}$ with parameter σ by $\rho_{\sigma, \mathbf{c}}(\mathbf{x}) := e^{-\pi \|\mathbf{x} - \mathbf{c}\|^2 / \sigma^2}$. Let $\Upsilon \subseteq \mathbb{R}^n$ be a lattice. The *discrete* Gaussian distribution $D_{\Upsilon, \sigma, \mathbf{c}}$ over Υ is defined by $D_{\Upsilon, \sigma, \mathbf{c}}(\mathbf{x}) := \rho_{\sigma, \mathbf{c}}(\mathbf{x}) / \rho_{\sigma, \mathbf{c}}(\Upsilon)$ for any lattice vector $\mathbf{x} \in \Upsilon$, where $\rho_{\sigma, \mathbf{c}}(\Upsilon) := \sum_{\mathbf{z} \in \Upsilon} \rho_{\sigma, \mathbf{c}}(\mathbf{z})$. The subscript $\mathbf{c}$ is taken to be $\mathbf{0}$ when omitted.

Lemma 11 ([24, 19, Lem. 3.2]). *For any $\sigma > 0, \mathbf{y} \in \mathbb{R}^m$, it holds that $\Delta(D_{\mathbb{Z}^m, \sigma}, D_{\mathbb{Z}^m, \sigma, \mathbf{y}}) \leq m \cdot \|\mathbf{y}\|_{\infty} / \sigma$.*

Based on Lemma 11, we present another smudging lemma, which shows that discrete Gaussians with large enough parameter σ can swallow any small vectors.

Lemma 12 (Smudging Lemma for Discrete Gaussians). *For any $\sigma > 0, \mathbf{y} \in \mathbb{Z}^m, \mathbf{c} \in \mathbb{R}^m$, it holds that*

$$\Delta(D_{\mathbb{Z}^m, \sigma, \mathbf{c}}, \mathbf{y} + D_{\mathbb{Z}^m, \sigma, \mathbf{c}}) \leq m \cdot (\|\mathbf{y}\|_{\infty} + 1) / \sigma.$$

Proof. Note that for any lattice $\Upsilon \subseteq \mathbb{R}^m$, $\mathbf{t}, \mathbf{c} \in \mathbb{R}^m$ and $\sigma > 0$, we have $D_{\Upsilon, \sigma, \mathbf{c}} = \mathbf{t} + D_{\Upsilon - \mathbf{t}, \sigma, \mathbf{c} - \mathbf{t}}$ according to the definition of discrete Gaussian. For $\mathbf{c} \in \mathbb{R}^m$, let $\lceil \mathbf{c} \rceil \in \mathbb{Z}^m$ denote a nearest integer vector to $\mathbf{c}$ so that $\|\mathbf{c} - \lceil \mathbf{c} \rceil\|_{\infty} \leq \frac{1}{2}$. Then

$$\begin{aligned} \Delta(D_{\mathbb{Z}^m, \sigma, \mathbf{c}}, \mathbf{y} + D_{\mathbb{Z}^m, \sigma, \mathbf{c}}) &= \Delta(\lceil \mathbf{c} \rceil + D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}, \mathbf{y} + \lceil \mathbf{c} \rceil + D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}) \\ &= \Delta(\lceil \mathbf{c} \rceil + D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}, \mathbf{y} + \lceil \mathbf{c} \rceil + D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}) \\ &= \Delta(D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}, \mathbf{y} + D_{\mathbb{Z}^m - \lceil \mathbf{c} \rceil, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}) \\ &= \Delta(D_{\mathbb{Z}^m, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}, D_{\mathbb{Z}^m, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil + \mathbf{y}}) \\ &\leq \Delta(D_{\mathbb{Z}^m, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil}, D_{\mathbb{Z}^m, \sigma}) + \Delta(D_{\mathbb{Z}^m, \sigma, \mathbf{c} - \lceil \mathbf{c} \rceil + \mathbf{y}}, D_{\mathbb{Z}^m, \sigma}) \\ &\leq m \cdot \frac{1}{2} / \sigma + m \cdot (\|\mathbf{y}\|_{\infty} + \frac{1}{2}) / \sigma = m \cdot (\|\mathbf{y}\|_{\infty} + 1) / \sigma, \end{aligned}$$

where the last inequality follows from Lemma 11. $\square$

Smoothing Parameter. For any lattice $\mathcal{Y} \subseteq \mathbb{R}^n$ and any positive real $\epsilon > 0$, the *smoothing parameter* $\eta_\epsilon(\mathcal{Y})$ is defined as the smallest real $s > 0$ such that $\rho_{1/s}(\mathcal{Y}^* \setminus \{\mathbf{0}\}) \leq \epsilon$, where $\mathcal{Y}^* := \{\mathbf{y} \in \mathbb{R}^n \mid \forall \mathbf{x} \in \mathcal{Y}, \langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z}\}$ refers to the dual lattice of $\mathcal{Y}$.

Below we recall some useful facts on smoothing parameter.

Lemma 13 ([54, 64, Lem. 2.4]). *For any lattice $\mathcal{Y}$, $\epsilon \in (0, 1)$, $\sigma \geq \eta_\epsilon(\mathcal{Y})$ and coset $\mathbf{c} + \mathcal{Y}$, it holds that $\rho_\sigma(\mathbf{c} + \mathcal{Y}) \in [\frac{1-\epsilon}{1+\epsilon}, 1] \cdot \rho_\sigma(\mathcal{Y})$.*

Lemma 14 (Smoothing Parameter for $\mathcal{Y}_1 \otimes \mathcal{Y}_2$ [53, Cor. 2.7]). *For any lattices $\mathcal{Y}_1, \mathcal{Y}_2$ and $1 + \epsilon' = (1 + \epsilon)^n$ with n the rank of $\mathcal{Y}_1$, we have*

$$\eta_{\epsilon'}(\mathcal{Y}_1 \otimes \mathcal{Y}_2) \leq \tilde{bl}(\mathcal{Y}_1) \cdot \eta_\epsilon(\mathcal{Y}_2),$$

where $\tilde{bl}(\mathcal{Y}_1) := \min_{\mathbf{B}} \|\tilde{\mathbf{B}}\| = \min_{\mathbf{B}} \max_{i \in [n]} \|\tilde{\mathbf{b}}_i\|$ denotes the Gram-Schmidt minimum of $\mathcal{Y}_1$ with the minimum taken over all bases $\mathbf{B}$ of $\mathcal{Y}_1$ and $\tilde{\mathbf{B}} = (\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n)$ being the Gram-Schmidt orthogonalization of $\mathbf{B}$.

Epecially, when $\mathcal{Y}_1 = \mathbf{x} \cdot \mathbb{Z}$, it holds that $\eta_\epsilon(\mathbf{x} \cdot \mathbb{Z} \otimes \mathcal{Y}_2) \leq \|\mathbf{x}\| \cdot \eta_\epsilon(\mathcal{Y}_2)$.

Next we present a bound on the smoothing parameter for a specific lattice $\Lambda^\perp(\mathbf{A}) := \{\boldsymbol{\delta}^\top \in \mathbb{Z}^{1 \times l} \mid \boldsymbol{\delta}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}\}$ with $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$ uniformly sampled. Its proof essentially follows from [33, 64], and here we provide a proof for completeness.

Lemma 15 (Smoothing Parameter for $\Lambda^\perp(\mathbf{A})$ [33, 64, Lem. 2.5]). *Let $n, q, l \in \mathbb{N}$ where q is prime and $l \geq 2n \log q$. Then, there exists a negligible function $\epsilon(\lambda) = 2^{-l^2/6}$, such that*

$$\Pr[\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n} : \eta_\epsilon(\Lambda^\perp(\mathbf{A})) \leq l] \geq 1 - q^{-n}.$$

Proof. The proof follows from the same argument as [33, Lem. 5.3]. Let $\mathcal{Q} := \{\mathbf{x} \in \mathbb{Z}^l : \|\mathbf{x}\|_\infty \leq q/4\}$ be the set of integer vectors with norm at most $q/4$. For any $\mathbf{s} \in \mathbb{Z}_q^n \setminus \{\mathbf{0}\}$, we have that

$$\Pr[\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n} : \mathbf{A} \cdot \mathbf{s} \bmod q \in \mathcal{Q}] \leq \frac{(q/2)^l}{q^l} = 2^{-l} \leq q^{-2n}.$$

By a union bound, it follows that

$$\Pr[\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n} : \exists \mathbf{s} \in \mathbb{Z}_q^n \setminus \{\mathbf{0}\} \text{ s.t. } \mathbf{A} \cdot \mathbf{s} \bmod q \in \mathcal{Q}] \leq q^{-n}.$$

Let $\Lambda(\mathbf{A}) := \{\mathbf{y} \in \mathbb{Z}^l \mid \exists \mathbf{s} \in \mathbb{Z}_q^n \text{ s.t. } \mathbf{y} = \mathbf{A} \cdot \mathbf{s} \bmod q\}$ and $(\Lambda^\perp(\mathbf{A}))^* := \{\mathbf{y} \in \mathbb{R}^l \mid \forall \boldsymbol{\delta}^\top \in \Lambda^\perp(\mathbf{A}), \boldsymbol{\delta}^\top \cdot \mathbf{y} \in \mathbb{Z}\}$. Then with the probability $1 - q^{-n}$ over the choice of $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, the length of the shortest nonzero vector $\lambda_1^\infty(\Lambda(\mathbf{A}))$ in $\Lambda(\mathbf{A})$ satisfies $\lambda_1^\infty(\Lambda(\mathbf{A})) \geq q/4$. Next, we note that $(\Lambda^\perp(\mathbf{A}))^* = 1/q \cdot \Lambda(\mathbf{A})$, and thus $\lambda_1^\infty((\Lambda^\perp(\mathbf{A}))^*) = 1/q \cdot \lambda_1^\infty(\Lambda(\mathbf{A})) \geq 1/4$.

Then according to [33, Lem. 2.6], for any $\epsilon > 0$, it holds that

$$\eta_\epsilon(\Lambda^\perp(\mathbf{A})) \leq \frac{\sqrt{\log(2l \cdot (1 + 1/\epsilon(\lambda)))}/\pi}{\lambda_1^\infty((\Lambda^\perp(\mathbf{A}))^*)} \leq 4 \cdot \sqrt{\log(2l \cdot (1 + 1/\epsilon(\lambda)))}/\pi.$$

By setting $\epsilon(\lambda) = 2^{-l^2/6}$, it is not hard to check that the following holds

$$\eta_\epsilon(\Lambda^\perp(\mathbf{A})) \leq 4 \cdot \sqrt{\log(2l \cdot (1 + 1/\epsilon(\lambda)))}/\pi \leq 4 \cdot (l/4) = l,$$

and then the lemma follows. $\square$

Lemma 16 (Products of Correlated Discrete Gaussians [59, Fact 2.1]).

Let $\sigma_0, \sigma_1, \sigma_2, \sigma_3 > 0$ s.t. $\sigma_0^2 = \sigma_1^2 + \sigma_2^2$, $1/\sigma_3^2 = 1/\sigma_1^2 + 1/\sigma_2^2$, and let $\mathbf{x}, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3 \in \mathbb{R}^n$ be such that $\mathbf{c}_3/\sigma_3^2 = \mathbf{c}_1/\sigma_1^2 + \mathbf{c}_2/\sigma_2^2$. Then it holds that

$$\rho_{\sigma_1}(\mathbf{x} - \mathbf{c}_1) \cdot \rho_{\sigma_2}(\mathbf{x} - \mathbf{c}_2) = \rho_{\sigma_0}(\mathbf{c}_1 - \mathbf{c}_2) \cdot \rho_{\sigma_3}(\mathbf{x} - \mathbf{c}_3).$$

Iteratively, for any $\sigma > 0$ and $\mathbf{x}, \mathbf{c}_1, \dots, \mathbf{c}_m \in \mathbb{R}^n$, there exist $\mathbf{t} \in \mathbb{R}^n$ and $c > 0$ (that depend on $\sigma, \mathbf{c}_1, \dots, \mathbf{c}_m$ but are independent of $\mathbf{x}$), s.t.

$$\prod_{i=1}^m \rho_\sigma(\mathbf{x} - \mathbf{c}_i) = c \cdot \rho_{\sigma/\sqrt{m}}(\mathbf{x} - \mathbf{t}).$$

Based on Lemma 16, we analyze the distribution of $\mathbf{z}$ for $\mathbf{y} \cdot \mathbf{z}^\top$ chosen according to a specific discrete Gaussian $D_{\mathbf{y} \cdot \mathbb{Z}^{1 \times l}, \sigma, \mathbf{K}}$.

Lemma 17. Let $\sigma > 0, \mathbf{y} \in \mathbb{R}^m \setminus \{\mathbf{0}\}, \mathbf{K} \in \mathbb{R}^{m \times l}$. Then for $\mathbf{y} \cdot \mathbf{z}^\top = \mathbf{T} \leftarrow_{\$} D_{\mathbf{y} \cdot \mathbb{Z}^{1 \times l}, \sigma, \mathbf{K}}$, the distribution of $\mathbf{z}$ is $D_{\mathbb{Z}^l, \sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}$, where $n_{\mathbf{y}}$ ($1 \leq n_{\mathbf{y}} \leq m$) denotes the number of non-zero entries in $\mathbf{y}$ and $\mathbf{t} \in \mathbb{R}^l$ is a vector that depends on $\sigma, \mathbf{y}, \mathbf{K}$.

Proof. Parse $\mathbf{y} = (y_1, \dots, y_m)^\top$ and $\mathbf{K} = \begin{pmatrix} \mathbf{k}_1^\top \\ \vdots \\ \mathbf{k}_m^\top \end{pmatrix} \in \mathbb{R}^{m \times l}$ with each $\mathbf{k}_i^\top \in \mathbb{R}^{1 \times l}$.

Since the rank of $\mathbf{T} = \mathbf{y} \cdot \mathbf{z}^\top$ is at most 1 and $\mathbf{y}$ is non-zero, the distribution of $\mathbf{T}$ can be determined by the distribution of $\mathbf{z}^\top$. Specifically, for any $\mathbf{z}_0 \in \mathbb{Z}^l$, $\Pr[\mathbf{z} = \mathbf{z}_0]$ equals $\Pr[\mathbf{T} = \mathbf{y} \cdot \mathbf{z}^\top = \mathbf{y} \cdot \mathbf{z}_0^\top]$, and we have

$$\begin{aligned} \Pr[\mathbf{z} = \mathbf{z}_0] &= \Pr[\mathbf{T} = \mathbf{y} \cdot \mathbf{z}_0^\top] = \frac{\rho_{\sigma, \mathbf{K}}(\mathbf{y} \cdot \mathbf{z}_0^\top)}{\sum_{\mathbf{z} \in \mathbb{Z}^l} \rho_{\sigma, \mathbf{K}}(\mathbf{y} \cdot \mathbf{z}^\top)} \\ &= \frac{\prod_{i=1}^m \rho_{\sigma, \mathbf{k}_i}(y_i \cdot \mathbf{z}_0)}{\sum_{\mathbf{z} \in \mathbb{Z}^l} \prod_{i=1}^m \rho_{\sigma, \mathbf{k}_i}(y_i \cdot \mathbf{z})} \end{aligned} \quad (24)$$

$$= \frac{\prod_{i: y_i \neq 0} \rho_{\sigma, \mathbf{k}_i/y_i}(\mathbf{z}_0)}{\sum_{\mathbf{z} \in \mathbb{Z}^l} \prod_{i: y_i \neq 0} \rho_{\sigma, \mathbf{k}_i/y_i}(\mathbf{z})} \quad (25)$$

$$= \frac{\rho_{\sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}(\mathbf{z}_0)}{\sum_{\mathbf{z} \in \mathbb{Z}^l} \rho_{\sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}(\mathbf{z})} = D_{\mathbb{Z}^l, \sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}(\mathbf{z}_0), \quad (26)$$

where the “=” in (24), (25) and (26) are explained as follows.

- By the definition of $\rho_{\sigma, \mathbf{K}}(\mathbf{y} \cdot \mathbf{z}_0^\top)$, it can be decomposed row by row, yielding the product of $\{\rho_{\sigma, \mathbf{k}_i}(y_i \cdot \mathbf{z}_0)\}_{i \in [m]}$. This shows the “=” in (24).
- For indices i s.t. $y_i = 0$, we have $\rho_{\sigma, \mathbf{k}_i}(y_i \cdot \mathbf{z}_0) = 1$, and for indices i s.t. $y_i \neq 0$, we have $\rho_{\sigma, \mathbf{k}_i}(y_i \cdot \mathbf{z}_0) = \rho_{\sigma, \mathbf{k}_i/y_i}(\mathbf{z}_0)$. This shows the “=” in (25).
- According to Lemma 16, there exist $\mathbf{t} \in \mathbb{R}^m$ and $c > 0$ (that depend on $\sigma, \mathbf{y}, \mathbf{K}$ and are independent of $\mathbf{z}_0$ and $\mathbf{z}$), s.t.

$$\prod_{i: y_i \neq 0} \rho_{\sigma, \mathbf{k}_i/y_i}(\mathbf{z}_0) = \prod_{i: y_i \neq 0} \rho_{\sigma}(\mathbf{z}_0 - \mathbf{k}_i/y_i) = c \cdot \rho_{\sigma/\sqrt{n_{\mathbf{y}}}}(\mathbf{z}_0 - \mathbf{t}) = c \cdot \rho_{\sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}(\mathbf{z}_0)$$

and similarly $\prod_{i: y_i \neq 0} \rho_{\sigma, \mathbf{k}_i/y_i}(\mathbf{z}) = c \cdot \rho_{\sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}(\mathbf{z})$. This shows the “=” in (26).

Consequently, $\mathbf{z}$ follows the distribution $D_{\mathbb{Z}^l, \sigma/\sqrt{n_{\mathbf{y}}}, \mathbf{t}}$ and the lemma follows. $\square$

We will use the following variant of the leftover hash lemma.

Lemma 18 (Leftover Hash Lemma [25]). *Let $n, l, q \in \mathbb{N}$ be integers with q a prime number and $l \geq n \log q + 2\lambda$. Then for uniformly chosen $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{x} \leftarrow_{\$} \{0, 1\}^l$ and $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^n$, we have that: $\Delta((\mathbf{A}, \mathbf{x}^\top \mathbf{A}), (\mathbf{A}, \mathbf{u}^\top)) \leq 2^{-\lambda}$.*

G.2 LWE, Multi-secret LWE, Multi-instance LWE Assumptions, and (Almost) Tight Reductions among Them

In this subsection, we present the definition of Learning-With-Errors (LWE) assumption (Def. 13), as well as some useful variants including the Multi-secret LWE (Def. 14), the Multi-instance LWE (Def. 15) and the Multi-instance LWE with noisy secrets (Def. 16). We also show several (almost) tight reduction among these LWE assumptions, including $\text{LWE} \Rightarrow \text{Multi-secret LWE}$ (Theorem 11), $\text{Multi-instance LWE} \Rightarrow \text{Multi-instance LWE with noisy secrets}$ (Theorem 12), and $\text{LWE} \Rightarrow \text{Multi-instance LWE with unbounded samples}$ (Theorem 13).

Let $n, q, l \in \mathbb{N}$, and χ be a distribution over $\mathbb{Z}$. Below we recall the LWE assumption and the multi-secret LWE assumption.

Definition 13 (LWE Assumption). *The $\text{LWE}_{n, q, \chi, l}$ -assumption holds, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{[n, q, \chi, l], \mathcal{A}}^{\text{LWE}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{A}, \mathbf{A}\mathbf{w} + \mathbf{e}) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{u}) = 1]| \leq \text{negl}(\lambda)$, where $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{w} \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow_{\$} \chi^l$, $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l$.*

Definition 14 (Multi-secret LWE Assumption). *Let $Q \geq 1$. The Q -secret $\text{LWE}_{n, q, \chi, l}$ -assumption holds, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{[n, q, \chi, l], \mathcal{A}}^{Q\text{-secre-LWE}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{A}, \{\mathbf{A}\mathbf{w}_i + \mathbf{e}_i\}_{i \in [Q]}) = 1] - \Pr[\mathcal{A}(\mathbf{A}, \{\mathbf{u}_i\}_{i \in [Q]}) = 1]| \leq \text{negl}(\lambda)$, where $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, and each $\mathbf{w}_i \leftarrow_{\$} \mathbb{Z}_q^n$, $\mathbf{e}_i \leftarrow_{\$} \chi^l$, $\mathbf{u}_i \leftarrow_{\$} \mathbb{Z}_q^l$ for $i \in [Q]$.*

A simple hybrid argument can show that $\text{Adv}_{[n, q, \chi, l]}^{Q\text{-LWE}}(\lambda) \leq Q \cdot \text{Adv}_{[n, q, \chi, l]}^{\text{LWE}}(\lambda)$. In a recent work [39], it shows an almost tight security reduction from LWE to multi-secret LWE, building upon the work of [9]. Below we recall the almost tight reduction, and to this end, we will first recall a useful lemma presenting the spectral norm upper bound of discrete Gaussian matrices.

Lemma 19 ([52, Lem. 2.8, Lem. 2.9]). Let $\mathbf{F} \leftarrow_s D_{\mathbb{Z}, \gamma}^{n \times l}$ and $l \geq n$. Then with all but 2^{-l} probability it holds that the spectral norm $\sigma_{\mathbf{F}}$ of $\mathbf{F}$ satisfies $\sigma_{\mathbf{F}} \leq \gamma \cdot C \cdot \sqrt{l}$ where C is a global constant.

Theorem 11 (LWE $\Rightarrow$ Multi-secret LWE with Prime Modulus [39]). Let $n, q, l \in \mathbb{N}$ with q a prime. Let $\sigma, \sigma_0, \sigma_1, r, \gamma > 0$ such that $\sigma = \sqrt{\sigma_0^2 + r^2}$, $\sigma_0 > \gamma \cdot C \cdot \sqrt{l} \cdot \sigma_1$, $\frac{q}{\sigma_1} \geq \sqrt{\ln(4n)/\pi}$ and $r \geq \sqrt{\lambda}$, where C is the global constant from Lemma 19. For any adversary $\mathcal{A}$ and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + Q \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{[n, q, D_{\mathbb{Z}, \sigma}, l], \mathcal{A}}^{Q\text{-secre-LWE}}(\lambda) \leq 2cn \cdot \text{Adv}_{[\lambda, q, D_{\mathbb{Z}, \gamma}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + \frac{Q(l+c+1)}{2^\lambda}$, where c is an integer such that $n \geq (\lfloor \frac{l}{c} \rfloor \log q + \lambda \log q + 2\lambda + 1) / \log(\sigma_1)$.

In the following, we also define a *multi-instance* version of LWE assumption, which stipulates the computational indistinguishability of multiple independent LWE instances $\{\mathbf{A}_i, \mathbf{A}_i \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}$ and $\{\mathbf{A}_i, \mathbf{u}_i\}_{i \in [T]}$. We note that this variant is different from the multi-secret LWE formalized in Def. 14, which considers the computational indistinguishability of $(\mathbf{A}, \{\mathbf{A} \mathbf{w}_i + \mathbf{e}_i\}_{i \in [Q]})$ and $(\mathbf{A}, \{\mathbf{u}_i\}_{i \in [Q]})$ with $\mathbf{A}$ shared among the multiple instances.

Definition 15 (Multi-instance LWE Assumption). Let $T \geq 1$. The T -instance $\text{LWE}_{n, q, \chi, l}$ -assumption holds, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{[n, q, \chi, l], \mathcal{A}}^{T\text{-inst-LWE}}(\lambda) := |\Pr[\mathcal{A}(\{\mathbf{A}_i, \mathbf{A}_i \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}) = 1] - \Pr[\mathcal{A}(\{\mathbf{A}_i, \mathbf{u}_i\}_{i \in [T]}) = 1]| \leq \text{negl}(\lambda)$, where each $\mathbf{A}_i \leftarrow_s \mathbb{Z}_q^{l \times n}$, $\mathbf{w}_i \leftarrow_s \mathbb{Z}_q^n$, $\mathbf{e}_i \leftarrow_s \chi^l$, $\mathbf{u}_i \leftarrow_s \mathbb{Z}_q^l$ for $i \in [T]$.

We also define a variant of multi-instance LWE, where both the secret vectors and the error vectors are sampled from the same distribution χ . This variant was formalized by Applebaum et al. [10] for (single-instance) LWE and was proved at least as hard as the usual definition of LWE where the secret vector is sampled uniformly at random, up to a small difference in the number of samples. It is not hard to see that similar result holds for multi-instance LWE, by applying the techniques in [10] to each LWE instance independently, and we state the result in the following Theorem 12, where we also provide a proof for completeness.

Definition 16 (Multi-instance LWE Assumption with Noisy Secrets). Let $T \geq 1$. The T -instance $\text{LWE}_{n, q, \chi, l}$ -assumption with noisy secrets holds, if for any PPT adversary $\mathcal{A}$, it holds that $\text{Adv}_{[n, q, \chi, l], \mathcal{A}}^{T\text{-inst-LWE-noi-secr}}(\lambda) := |\Pr[\mathcal{A}(\{\mathbf{A}_i, \mathbf{A}_i \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}) = 1] - \Pr[\mathcal{A}(\{\mathbf{A}_i, \mathbf{u}_i\}_{i \in [T]}) = 1]| \leq \text{negl}(\lambda)$, where each $\mathbf{A}_i \leftarrow_s \mathbb{Z}_q^{l \times n}$, $\mathbf{w}_i \leftarrow_s \chi^n$, $\mathbf{e}_i \leftarrow_s \chi^l$, $\mathbf{u}_i \leftarrow_s \mathbb{Z}_q^l$ for $i \in [T]$.

Theorem 12 (Multi-instance LWE $\Rightarrow$ Multi-instance LWE with Noisy Secrets [10]). Let $T \geq 1$. For any adversary $\mathcal{A}$, there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + T \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{[n, q, \chi, l], \mathcal{A}}^{T\text{-inst-LWE-noi-secr}}(\lambda) \leq \text{Adv}_{[n, q, \chi, l+n^2], \mathcal{B}}^{T\text{-inst-LWE}}(\lambda) + Tn/q^{n^2-n}$.

Proof. Let $\mathcal{A}$ be an adversary against the T -instance $\text{LWE}_{n, q, \chi, l}$ -assumption with noisy secrets. To prove the theorem, we will construct an adversary $\mathcal{B}$ against

the T -instance $\text{LWE}_{n,q,\chi,l+n^2}$ -assumption (with *uniform secrets*) by invoking $\mathcal{A}$, as described below.

Given a challenge $\{\mathbf{A}_i, \mathbf{u}_i\}_{i \in [T]}$, $\mathcal{B}$ wants to distinguish $\{\mathbf{u}_i = \mathbf{A}_i \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}$ from $\{\mathbf{u}_i \leftarrow \mathbb{Z}_q^{l+n^2}\}_{i \in [T]}$, where each $\mathbf{A}_i \leftarrow \mathbb{Z}_q^{(l+n^2) \times n}$, $\mathbf{w}_i \leftarrow \mathbb{Z}_q^n$, $\mathbf{e}_i \leftarrow \chi^{l+n^2}$ for $i \in [T]$.

Let us fix some *notations* used in this proof. For each $i \in [T]$, we parse $\mathbf{A}_i = \begin{pmatrix} \overline{\mathbf{A}}_i \\ \underline{\mathbf{A}}_i \end{pmatrix} \in \mathbb{Z}_q^{(l+n^2) \times n}$ with $\overline{\mathbf{A}}_i \in \mathbb{Z}_q^{n^2 \times n}$ and $\underline{\mathbf{A}}_i \in \mathbb{Z}_q^{l \times n}$, parse $\mathbf{e}_i = \begin{pmatrix} \overline{\mathbf{e}}_i \\ \underline{\mathbf{e}}_i \end{pmatrix} \in \mathbb{Z}_q^{l+n^2}$ with $\overline{\mathbf{e}}_i \in \mathbb{Z}_q^{n^2}$, $\underline{\mathbf{e}}_i \in \mathbb{Z}_q^l$, and parse $\mathbf{u}_i = \begin{pmatrix} \overline{\mathbf{u}}_i \\ \underline{\mathbf{u}}_i \end{pmatrix} \in \mathbb{Z}_q^{l+n^2}$ with $\overline{\mathbf{u}}_i \in \mathbb{Z}_q^{n^2}$, $\underline{\mathbf{u}}_i \in \mathbb{Z}_q^l$.

$\mathcal{B}$ is constructed by invoking $\mathcal{A}$ as follows.

- Firstly, $\mathcal{B}$ checks if the first n^2 rows of $\mathbf{A}_i$, i.e., $\overline{\mathbf{A}}_i$, are of all full rank for all $i \in [T]$ (e.g., by using Gaussian elimination), and aborts if the check fails.
- Then, for each $i \in [T]$, $\mathcal{B}$ chooses an invertible sub-matrix $\overline{\overline{\mathbf{A}}}_i \in \mathbb{Z}_q^{n^2 \times n}$ of $\overline{\mathbf{A}}_i \in \mathbb{Z}_q^{n^2 \times n}$ (given that $\overline{\mathbf{A}}_i$ is of full rank), and chooses the corresponding sub-vector $\overline{\overline{\mathbf{u}}}_i \in \mathbb{Z}_q^{n^2}$ of $\overline{\mathbf{u}}_i \in \mathbb{Z}_q^{n^2}$.
- Finally, for each $i \in [T]$, $\mathcal{B}$ sets $\widetilde{\mathbf{A}}_i := -\underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \in \mathbb{Z}_q^{l \times n}$ and $\widetilde{\mathbf{u}}_i := \underline{\mathbf{u}}_i - \underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \cdot \overline{\overline{\mathbf{u}}}_i \in \mathbb{Z}_q^l$, and invokes $\mathcal{A}(\{\widetilde{\mathbf{A}}_i, \widetilde{\mathbf{u}}_i\}_{i \in [T]})$.

$\mathcal{B}$ returns to its own challenger whatever $\mathcal{A}$ outputs.

Now we analyze the advantage of $\mathcal{B}$.

- Since each $\mathbf{A}_i \leftarrow \mathbb{Z}_q^{(l+n^2) \times n}$, all $\overline{\mathbf{A}}_i$ are uniformly distributed over $\mathbb{Z}_q^{n^2 \times n}$ and all $\underline{\mathbf{A}}_i$ are uniformly distributed over $\mathbb{Z}_q^{l \times n}$. Thus by a union bound, all $\overline{\mathbf{A}}_i$ are of full rank except with negligible probability at most Tn/q^{n^2-n} , so the check in the first step will succeed overwhelmingly. Moreover, in the case that the check succeeds, all $\widetilde{\mathbf{A}}_i := -\underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \in \mathbb{Z}_q^{l \times n}$ are independently and uniformly distributed over $\mathbb{Z}_q^{l \times n}$, thanks to the randomness of $\underline{\mathbf{A}}_i$ and the invertibility of $\overline{\overline{\mathbf{A}}}_i$.
- In the case that $\{\mathbf{u}_i = \mathbf{A}_i \cdot \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}$, we have that $\overline{\mathbf{u}}_i = \overline{\mathbf{A}}_i \cdot \mathbf{w}_i + \overline{\mathbf{e}}_i$ and $\underline{\mathbf{u}}_i = \underline{\mathbf{A}}_i \cdot \mathbf{w}_i + \underline{\mathbf{e}}_i$ for all $i \in [T]$, where $\overline{\mathbf{e}}_i \in \mathbb{Z}_q^{n^2}$ denotes the sub-vector of $\overline{\mathbf{e}}_i \in \mathbb{Z}_q^{n^2}$ corresponding to $\overline{\overline{\mathbf{A}}}_i$. It follows that

$$\begin{aligned} \widetilde{\mathbf{u}}_i &:= \underline{\mathbf{u}}_i - \underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \cdot \overline{\overline{\mathbf{u}}}_i = \underline{\mathbf{A}}_i \cdot \mathbf{w}_i + \underline{\mathbf{e}}_i - \underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \cdot (\overline{\overline{\mathbf{A}}}_i \cdot \mathbf{w}_i + \overline{\overline{\mathbf{e}}}_i) \\ &= \underline{\mathbf{e}}_i - \underline{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{A}}}_i^{-1} \cdot \overline{\overline{\mathbf{e}}}_i = \underline{\mathbf{e}}_i + \widetilde{\mathbf{A}}_i \cdot \overline{\overline{\mathbf{e}}}_i. \end{aligned}$$

By setting $\widetilde{\mathbf{w}}_i := \overline{\overline{\mathbf{e}}}_i$ and $\widetilde{\mathbf{e}}_i := \underline{\mathbf{e}}_i$, we have that $\widetilde{\mathbf{u}}_i = \widetilde{\mathbf{A}}_i \cdot \widetilde{\mathbf{w}}_i + \widetilde{\mathbf{e}}_i$, where all $\widetilde{\mathbf{w}}_i := \overline{\overline{\mathbf{e}}}_i$ ($i \in [T]$) are independently distributed according to χ^{n^2} and all $\widetilde{\mathbf{e}}_i := \underline{\mathbf{e}}_i$ ($i \in [T]$) are independently distributed according to χ^l , due to the randomness of $\mathbf{e}_i \leftarrow \chi^{l+n^2}$.

- In the case that $\{\mathbf{u}_i \leftarrow \mathbb{Z}_q^l\}_{i \in [T]}$, all $\tilde{\mathbf{u}}_i := \mathbf{u}_i - \mathbf{A}_i \cdot \overline{\mathbf{A}_i}^{-1} \cdot \overline{\mathbf{u}_i}$ are independently and uniformly distributed over $\mathbb{Z}_q^l$, due to the randomness of $\mathbf{u}_i$.

Overall, $\mathcal{B}$ can distinguish $\{\mathbf{A}_i, \mathbf{u}_i = \mathbf{A}_i \cdot \mathbf{w}_i + \mathbf{e}_i\}_{i \in [T]}$ from $\{\mathbf{A}_i, \mathbf{u}_i \leftarrow \mathbb{Z}_q^{l+n^2}\}_{i \in [T]}$ as long as $\mathcal{A}$ distinguishes $\{\tilde{\mathbf{A}}_i, \tilde{\mathbf{u}}_i = \tilde{\mathbf{A}}_i \cdot \tilde{\mathbf{w}}_i + \tilde{\mathbf{e}}_i\}$ from $\{\tilde{\mathbf{A}}_i, \tilde{\mathbf{u}}_i \leftarrow \mathbb{Z}_q^l\}$, except with probability at most Tn/q^{n^2-n} , and thus we get that $\text{Adv}_{[n,q,\chi,l],\mathcal{A}}^{T\text{-inst-LWE-noi-secr}}(\lambda) \leq \text{Adv}_{[n,q,\chi,l+n^2],\mathcal{B}}^{T\text{-inst-LWE}}(\lambda) + Tn/q^{n^2-n}$. This completes the proof of Theorem 12. $\square$

As shown in [33, 10, 31] (based on a core technique from [60]), when the error distribution χ is discrete Gaussian, there is a method for generating unboundedly many “fresh” LWE samples for a fixed secret and a somewhat wider Gaussian error distribution, given a sufficiently large number of initial samples.

Below we show that one can in fact generate multiple instances of LWE and produce unbounded samples for each LWE instance, given sufficiently many initial samples of a single LWE instance, as described below.

Theorem 13 (LWE $\Rightarrow$ Multi-instance LWE with Unbounded Samples).

Let χ' be a B' -bounded distribution over $\mathbb{Z}$, let $\chi = D_{\mathbb{Z},t}$, and let $l \geq (n+1)\log q + 2\lambda$. For any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{[n,q,\chi,Q],\mathcal{A}}^{T\text{-inst-LWE}}(\lambda) \leq \text{Adv}_{[n,q,\chi',l],\mathcal{B}}^{\text{LWE}}(\lambda) + TQ \cdot (lB' + 1)/t + 2TQ \cdot 2^{-\lambda}$.

Proof. Let $\mathcal{A}$ be an adversary against the T -instance $\text{LWE}_{n,q,\chi,Q}$ -assumption with error distribution χ and Q samples per instance.

The proof proceeds in two steps. Firstly, we will construct an adversary $\mathcal{B}'$ against the (single-instance) $\text{LWE}_{n,q,\chi,TQ}$ -assumption with the same error distribution χ and TQ samples, by invoking $\mathcal{A}$, such that

$$\text{Adv}_{[n,q,\chi,Q],\mathcal{A}}^{T\text{-inst-LWE}}(\lambda) \leq \text{Adv}_{[n,q,\chi,TQ],\mathcal{B}'}^{\text{LWE}}(\lambda). \quad (27)$$

Then, we will construct an adversary $\mathcal{B}$ against the (single-instance) $\text{LWE}_{n,q,\chi',l}$ -assumption with error distribution χ' and l samples, by invoking $\mathcal{B}'$, such that

$$\text{Adv}_{[n,q,\chi,TQ],\mathcal{B}'}^{\text{LWE}}(\lambda) \leq \text{Adv}_{[n,q,\chi',l],\mathcal{B}}^{\text{LWE}}(\lambda) + TQ \cdot (lB' + 1)/t + 2TQ \cdot 2^{-\lambda}. \quad (28)$$

Taking (27) and (28) together, Theorem 13 follows.

THE DESCRIPTION OF $\mathcal{B}'$. We first give the description of $\mathcal{B}'$ so that (27) holds, by invoking $\mathcal{A}$. Given a challenge $(\tilde{\mathbf{A}}, \tilde{\mathbf{u}})$ of $\text{LWE}_{n,q,\chi,TQ}$, $\mathcal{B}'$ wants to distinguish $\tilde{\mathbf{u}} = \tilde{\mathbf{A}}\tilde{\mathbf{w}} + \tilde{\mathbf{e}}$ from $\tilde{\mathbf{u}} \leftarrow \mathbb{Z}_q^{TQ}$, where $\tilde{\mathbf{A}} \leftarrow \mathbb{Z}_q^{TQ \times n}$, $\tilde{\mathbf{w}} \leftarrow \mathbb{Z}_q^n$, $\tilde{\mathbf{e}} \leftarrow \chi^{TQ}$.

For the ease of exposition, we parse $\tilde{\mathbf{A}} = \begin{pmatrix} \mathbf{A}^{(1)} \\ \vdots \\ \mathbf{A}^{(T)} \end{pmatrix} \in \mathbb{Z}_q^{TQ \times n}$ with each $\mathbf{A}^{(i)} \in \mathbb{Z}_q^{Q \times n}$, parse $\tilde{\mathbf{e}} = \begin{pmatrix} \mathbf{e}^{(1)} \\ \vdots \\ \mathbf{e}^{(T)} \end{pmatrix} \in \mathbb{Z}_q^{TQ}$ with each $\mathbf{e}^{(i)} \in \mathbb{Z}_q^Q$, and parse $\tilde{\mathbf{u}} = \begin{pmatrix} \mathbf{u}^{(1)} \\ \vdots \\ \mathbf{u}^{(T)} \end{pmatrix} \in \mathbb{Z}_q^{TQ}$ with each $\mathbf{u}^{(i)} \in \mathbb{Z}_q^Q$.

$\mathcal{B}'$ is constructed by invoking $\mathcal{A}$ as follows. For all $i \in [T]$, $\mathcal{B}'$ picks $\mathbf{s}^{(i)} \leftarrow_{\$} \mathbb{Z}_q^n$ uniformly at random, and sets $\mathbf{v}^{(i)} := \mathbf{A}^{(i)} \cdot \mathbf{s}^{(i)} + \mathbf{u}^{(i)}$ by using the $(\mathbf{A}^{(i)}, \mathbf{u}^{(i)})$ contained in its own challenge $(\tilde{\mathbf{A}}, \tilde{\mathbf{u}})$. Finally, $\mathcal{B}'$ invokes $\mathcal{A}(\{\mathbf{A}^{(i)}, \mathbf{v}^{(i)}\}_{i \in [T]})$, and returns to its own challenger whatever $\mathcal{A}$ outputs.

Next, we analyze the advantage of $\mathcal{B}'$.

- Since $\tilde{\mathbf{A}} \leftarrow_{\$} \mathbb{Z}_q^{TQ \times n}$, all $\mathbf{A}^{(i)}$ are independently and uniformly random over $\mathbb{Z}_q^{Q \times n}$.
- In the case that $\tilde{\mathbf{u}} = \tilde{\mathbf{A}}\tilde{\mathbf{w}} + \tilde{\mathbf{e}}$, for all $i \in [T]$, we have that $\mathbf{u}^{(i)} = \mathbf{A}^{(i)} \cdot \tilde{\mathbf{w}} + \mathbf{e}^{(i)}$, and it follows that $\mathbf{v}^{(i)} := \mathbf{A}^{(i)} \cdot \mathbf{s}^{(i)} + \mathbf{u}^{(i)} = \mathbf{A}^{(i)} \cdot (\mathbf{s}^{(i)} + \tilde{\mathbf{w}}) + \mathbf{e}^{(i)}$. By setting $\mathbf{w}^{(i)} := \mathbf{s}^{(i)} + \tilde{\mathbf{w}}$, it follows that $\mathbf{v}^{(i)} = \mathbf{A}^{(i)} \cdot \mathbf{w}^{(i)} + \mathbf{e}^{(i)}$, where all $\mathbf{w}^{(i)} = \mathbf{s}^{(i)} + \tilde{\mathbf{w}}$ ($i \in [T]$) are independently and uniformly random over $\mathbb{Z}_q^n$ due to the randomness of $\mathbf{s}^{(i)} \leftarrow_{\$} \mathbb{Z}_q^n$, and all $\mathbf{e}^{(i)}$ ($i \in [T]$) are independently distributed according to χ^Q due to the randomness of $\tilde{\mathbf{e}} \leftarrow_{\$} \chi^{TQ}$.
- In the case that $\tilde{\mathbf{u}} \leftarrow_{\$} \mathbb{Z}_q^{TQ}$, all $\mathbf{u}^{(i)}$ are independently and uniformly random over $\mathbb{Z}_q^Q$, so are the $\mathbf{v}^{(i)} := \mathbf{A}^{(i)} \cdot \mathbf{s}^{(i)} + \mathbf{u}^{(i)}$.

Overall, $\mathcal{B}'$ can distinguish $(\tilde{\mathbf{A}} \leftarrow_{\$} \mathbb{Z}_q^{TQ \times n}, \tilde{\mathbf{u}} = \tilde{\mathbf{A}}\tilde{\mathbf{w}} + \tilde{\mathbf{e}})$ from $(\tilde{\mathbf{A}} \leftarrow_{\$} \mathbb{Z}_q^{TQ \times n}, \tilde{\mathbf{u}} \leftarrow_{\$} \mathbb{Z}_q^{TQ})$ as long as $\mathcal{A}$ distinguishes $\{\mathbf{A}^{(i)} \leftarrow_{\$} \mathbb{Z}_q^{Q \times n}, \mathbf{v}^{(i)} = \mathbf{A}^{(i)} \cdot \mathbf{w}^{(i)} + \mathbf{e}^{(i)}\}_{i \in [T]}$ from $\{\mathbf{A}^{(i)} \leftarrow_{\$} \mathbb{Z}_q^{Q \times n}, \mathbf{v}^{(i)} \leftarrow_{\$} \mathbb{Z}_q^Q\}_{i \in [T]}$, and thus we get that $\text{Adv}_{[n,q,\chi,Q],\mathcal{A}}^{T\text{-inst-LWE}}(\lambda) \leq \text{Adv}_{[n,q,\chi,TQ],\mathcal{B}'}^{\text{LWE}}(\lambda)$. This shows that (27) holds.

THE DESCRIPTION OF $\mathcal{B}$. Now we give the description of $\mathcal{B}$ so that (28) holds, by invoking $\mathcal{B}'$. Given a challenge $(\mathbf{A}, \mathbf{u})$ of $\text{LWE}_{n,q,\chi',l}$, $\mathcal{B}$ wants to distinguish $\mathbf{u} = \mathbf{A}\mathbf{w} + \mathbf{e}$ from $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l$, where $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{w} \leftarrow_{\$} \mathbb{Z}_q^n$ and $\mathbf{e} \leftarrow_{\$} \chi^l$.

$\mathcal{B}$ is constructed by invoking $\mathcal{B}'$ as follows. For all $i \in [TQ]$, $\mathcal{B}$ samples $\mathbf{x}_i \leftarrow_{\$} \{0,1\}^l$, $\tilde{e}_i \leftarrow_{\$} D_{\mathbb{Z},t}$, and sets $\mathbf{a}_i^\top := \mathbf{x}_i^\top \mathbf{A} \in \mathbb{Z}_q^{1 \times n}$, $b_i := \mathbf{x}_i^\top \mathbf{u} + \tilde{e}_i \in \mathbb{Z}_q$ by using its own challenge $(\mathbf{A}, \mathbf{u})$. Then $\mathcal{B}$ defines

$$\tilde{\mathbf{A}} := \begin{pmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_{TQ}^\top \end{pmatrix} \in \mathbb{Z}_q^{TQ \times n} \quad \text{and} \quad \tilde{\mathbf{b}} := \begin{pmatrix} b_1 \\ \vdots \\ b_{TQ} \end{pmatrix} \in \mathbb{Z}_q^{TQ}. \quad (29)$$

Finally, $\mathcal{B}$ invokes $\mathcal{B}'(\tilde{\mathbf{A}}, \tilde{\mathbf{b}})$, and returns to its own challenger whatever $\mathcal{B}'$ outputs.

Next, we analyze the advantage of $\mathcal{B}$.

- In the case that $\mathbf{u} = \mathbf{A}\mathbf{w} + \mathbf{e}$, for each $i \in [TQ]$, we have the following

$$\begin{aligned} (\mathbf{a}_i^\top, b_i) &:= (\mathbf{x}_i^\top \mathbf{A}, \mathbf{x}_i^\top \mathbf{u} + \tilde{e}_i) \\ &= (\mathbf{x}_i^\top \mathbf{A}, \mathbf{x}_i^\top (\mathbf{A}\mathbf{w} + \mathbf{e}) + \tilde{e}_i) \\ &= (\mathbf{x}_i^\top \mathbf{A}, (\mathbf{x}_i^\top \mathbf{A})\mathbf{w} + \mathbf{x}_i^\top \mathbf{e} + \tilde{e}_i) \\ &\stackrel{s}{\approx} (\mathbf{x}_i^\top \mathbf{A}, (\mathbf{x}_i^\top \mathbf{A})\mathbf{w} + \cancel{\mathbf{x}_i^\top \mathbf{e}} + \tilde{e}_i) \end{aligned} \quad (30)$$

$$\stackrel{s}{\approx} (\mathbf{a}_i^\top \leftarrow_{\$} \mathbb{Z}_q^{1 \times n}, b_i = \mathbf{a}_i^\top \mathbf{w} + \tilde{e}_i), \quad (31)$$

where the “ $\overset{s}{\approx}$ ” in (30) and (31) are explained as follows.

- Since $\mathbf{x}_i \in \{0,1\}^l$ and $\mathbf{e} \leftarrow_{\$} \chi'^l$ with χ' being B' -bounded, we have that $\|\mathbf{x}_i^\top \mathbf{e}\|_\infty \leq lB'$. Then by noting that $\tilde{e}_i$ follows $D_{\mathbb{Z},t}$, according to the smudging lemma for discrete Gaussians (i.e., Lemma 12 in Appendix G.1), we get that $\mathbf{x}_i^\top \mathbf{e} + \tilde{e}_i \overset{s}{\approx} \tilde{e}_i$ with statistical distance at most $(lB' + 1)/t$. This shows the “ $\overset{s}{\approx}$ ” in (30).
- Since $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{x}_i \leftarrow_{\$} \{0,1\}^l$ with $l \geq n \log q + 2\lambda$ and q a prime number, by the leftover hash lemma (i.e., Lemma 18 in Appendix G.1), we have that $\mathbf{x}_i^\top \mathbf{A} \overset{s}{\approx} (\mathbf{a}_i^\top \leftarrow_{\$} \mathbb{Z}_q^{1 \times n})$ with statistical distance at most $2^{-\lambda}$. This shows the “ $\overset{s}{\approx}$ ” in (31).

Then by a hybrid argument over $i \in [TQ]$, we know that $\tilde{\mathbf{A}}$ defined in (29) is statistically close to the uniform distribution over $\mathbb{Z}_q^{TQ \times n}$, and $\tilde{\mathbf{b}}$ defined in (29) is statistically close to $\tilde{\mathbf{b}} = \tilde{\mathbf{A}}\mathbf{w} + \tilde{\mathbf{e}}$ with $\tilde{\mathbf{e}} := (\tilde{e}_1, \dots, \tilde{e}_{TQ})^\top \in \mathbb{Z}_q^{TQ}$ following $D_{\mathbb{Z},t}^{TQ} = \chi^{TQ}$, and the statistical distance is at most $TQ \cdot (lB' + 1)/t + TQ \cdot 2^{-\lambda}$.

- In the case that $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l$, for each $i \in [TQ]$, we have the following

$$\begin{aligned}
(\mathbf{a}_i^\top, b_i) &:= (\mathbf{x}_i^\top \mathbf{A}, \mathbf{x}_i^\top \mathbf{u} + \tilde{e}_i) \\
&= \mathbf{x}_i^\top (\mathbf{A}, \mathbf{u}) + (\mathbf{0}^\top, \tilde{e}_i) \\
&\overset{s}{\approx} (\mathbf{a}_i^\top \leftarrow_{\$} \mathbb{Z}_q^{1 \times n}, b_i \leftarrow_{\$} \mathbb{Z}_q) + (\mathbf{0}^\top, \tilde{e}_i) \\
&\equiv (\mathbf{a}_i^\top \leftarrow_{\$} \mathbb{Z}_q^{1 \times n}, b_i \leftarrow_{\$} \mathbb{Z}_q),
\end{aligned} \tag{32}$$

where the “ $\overset{s}{\approx}$ ” in (32) is explained as follows.

- Since $\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}$, $\mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l$, $\mathbf{x}_i \leftarrow_{\$} \{0,1\}^l$ with $l \geq (n+1) \log q + 2\lambda$ and q a prime number, by applying the leftover hash lemma (i.e., Lemma 18 in Appendix G.1) but viewing $(\mathbf{A}, \mathbf{u})$ as the matrix $\mathbf{A}$ in the leftover hash lemma, we have that $\mathbf{x}_i^\top (\mathbf{A}, \mathbf{u}) \overset{s}{\approx} (\mathbf{a}_i^\top \leftarrow_{\$} \mathbb{Z}_q^{1 \times n}, b_i \leftarrow_{\$} \mathbb{Z}_q)$, with statistical distance at most $2^{-\lambda}$. This shows the “ $\overset{s}{\approx}$ ” in (32).

Then by a hybrid argument over $i \in [TQ]$, we know that both $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{b}}$ defined in (29) are statistically close to the uniform distributions, and the statistical distance is at most $TQ \cdot 2^{-\lambda}$.

Overall, $\mathcal{B}$ can successfully distinguish $(\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}, \mathbf{u} = \mathbf{A}\mathbf{w} + \mathbf{e})$ from $(\mathbf{A} \leftarrow_{\$} \mathbb{Z}_q^{l \times n}, \mathbf{u} \leftarrow_{\$} \mathbb{Z}_q^l)$ as long as $\mathcal{B}'$ distinguishes $(\tilde{\mathbf{A}} \leftarrow_{\$} \mathbb{Z}_q^{TQ \times n}, \tilde{\mathbf{b}} = \tilde{\mathbf{A}}\mathbf{w} + \tilde{\mathbf{e}})$ from $(\tilde{\mathbf{A}} \leftarrow_{\$} \mathbb{Z}_q^{TQ \times n}, \tilde{\mathbf{b}} \leftarrow_{\$} \mathbb{Z}_q^{TQ})$, except with probability at most $TQ \cdot (lB' + 1)/t + 2TQ \cdot 2^{-\lambda}$, and thus we get $\text{Adv}_{[n,q,\chi,TQ],\mathcal{B}'}^{\text{LWE}}(\lambda) \leq \text{Adv}_{[n,q,\chi',l],\mathcal{B}}^{\text{LWE}}(\lambda) + TQ \cdot (lB' + 1)/t + 2TQ \cdot 2^{-\lambda}$. This shows that (28) holds.

Finally, Theorem 13 holds by combining (27) and (28) together. $\square$

G.3 Probabilistic Two-Leveled IP-HPS: Definitions

In this subsection, we relax TL-IP-HPS to *probabilistic* TL-IP-HPS, following a recent paradigm [39]. More precisely, we present the formal definition of *probabilistic* (outer-level) IP-HPS in Def. 17 and the formal definition of *probabilistic* (inner-level) co-Hash in Def. 18, which together constitute a *probabilistic* Two-Leveled IP-HPS.

We also give the formal definition of *functional smoothness* for probabilistic TL-IP-HPS in Def. 19, which is almost identical to that for (standard) TL-IP-HPS as defined in Def. 7 in Sect. 4, except that now we require the outputs of $\{\text{Priv}(sk, c^{(j)})\}_j$ (instead of the real hash values $\{\mathbf{A}_{sk}(c^{(j)})\}_j$) to trigger inner-level hash values $\{\Gamma_s(c^{(j)})\}_j$. As for the property of *multi-key-extracting*, it is the same as that for (standard) TL-IP-HPS, and we refer to Def. 8 in Sect. 4 for its formal definition.

Definition 17 (Probabilistic IP-HPS). A probabilistic *inner-product hash proof system* (IP-HPS) scheme $\text{IPHPS} = (\text{Setup}_{\text{IPHPS}}, \alpha, \mu, \text{Pub}, \text{Priv}, \text{FPriv})$ is an IP-HPS scheme as per Def. 3, except that Pub and Priv are probabilistic algorithms:

- $\mathbf{h} \leftarrow_s \text{Pub}(pk, c, w)$: Taking as input a projection key $pk = \alpha(sk)$, an instance $c \in \mathcal{L}$ and a witness w for $c \in \mathcal{L}$, the probabilistic public evaluation algorithm outputs a hash value $\mathbf{h} \in \mathfrak{G}^m$.
- $\mathbf{h} \leftarrow_s \text{Priv}(sk, c)$: Taking as input a hashing key sk and an instance $c \in \mathcal{X}$, the probabilistic private evaluation algorithm outputs a hash value $\mathbf{h} \in \mathfrak{G}^m$.

Moreover, there exists a negligible function $\epsilon_{\text{evalInd}}$ (in λ) such that the following property holds:

- $\epsilon_{\text{evalInd}}$ -**Statistical Evaluation Indistinguishability:** For all $\text{pp}_{\text{IPHPS}} \leftarrow_s \text{Setup}_{\text{IPHPS}}(1^m)$, $sk \in SK$, $c \in \mathcal{L}$ with witness w , and $pk := \alpha(sk)$, it holds that

$$\Delta(\text{Pub}(pk, c, w), \text{Priv}(sk, c)) \leq \epsilon_{\text{evalInd}},$$

where the probability is only over the inner coin tosses of Pub and Priv .

As for the functional private evaluation algorithm FPriv , we still require it to be a deterministic algorithm that satisfies **Functional Correctness**, i.e., $\text{FPriv}(fk_{\mathbf{y}}, c) = \langle \mathbf{A}_{sk}(c), \mathbf{y} \rangle$ holds for all $c \in \mathcal{X}$ and $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$.

Definition 18 (Probabilistic co-Hash). A probabilistic *core hash function* (co-Hash) scheme coHash consists of a PPT setup algorithm $\text{Setup}_{\text{coHash}}$:

- $\text{pp}_{\text{coHash}} \leftarrow_s \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$: Taking as input a public parameter pp_{IPHPS} of the (outer-level) IPHPS, the setup algorithm of co-Hash outputs a public parameter $\text{pp}_{\text{coHash}}$, which implicitly defines $(\mathfrak{G}_{\text{co}}, \mathbf{g}_{\text{co}}, \text{ord}_{\text{co}}, SK_{\text{co}}, \mathcal{HV}_{\text{co}}, \Gamma_{(\cdot)})$ as per Def. 5, except that $\Gamma_{(\cdot)} : \mathcal{X} \rightarrow \mathfrak{G}_{\text{co}}$ is a family of randomized hash functions indexed by hashing keys $s \in SK_{\text{co}}$.

Definition 19 (Functional Smoothness of Probabilistic Two-Leveled IP-HPS). A probabilistic two-leveled IP-HPS scheme $\text{TLHPS} = (\text{IPHPS}, \text{coHash})$ has functional smoothness, if for any (unbounded) adversary $\mathcal{A}$ and any polynomial $Q = \text{poly}(\lambda)$, it holds that $\epsilon_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) := |\Pr[\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, 0} \Rightarrow 1] - \Pr[\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, 1} \Rightarrow 1]| \leq \text{negl}(\lambda)$, where the experiments $\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}$ ($b \in \{0, 1\}$) are defined in Fig. 14, and we require that $\mathcal{A}$ can make only a single query to $\mathcal{O}_{\text{CHAL}}^b$, and the (single) $\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}})$ query and all $\mathcal{O}_{\text{FKEY}}(\mathbf{y})$ queries made by $\mathcal{A}$ are restricted to satisfy $\langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0$.

$\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b} \quad // b \in \{0, 1\}$ $\text{pp}_{\text{IPHPS}} \leftarrow \text{Setup}_{\text{IPHPS}}(1^m)$ $\text{pp}_{\text{coHash}} \leftarrow \text{Setup}_{\text{coHash}}(\text{pp}_{\text{IPHPS}})$ $sk \leftarrow \mathcal{SK}, \quad pk := \alpha(sk)$ $s \leftarrow \mathcal{SK}_{\text{co}}$ $b' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{CHAL}}^b, \mathcal{O}_{\text{FKEY}}}(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk)$ Return b'	$\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}} \in \mathbb{Z}^m):$ // single query For $j \in [Q]:$ $c^{(j)} \leftarrow \mathcal{X}$ $\mathbf{h}^{(j)} := \begin{cases} \text{Priv}(sk, c^{(j)}), & \text{if } b = 0; \\ \text{Priv}(sk, c^{(j)}) + \Gamma_s(c^{(j)}) \cdot \tilde{\mathbf{x}}, & \text{if } b = 1 \end{cases}$ Return $(c^{(j)}, \mathbf{h}^{(j)})_{j \in [Q]}$ $\mathcal{O}_{\text{FKEY}}(\mathbf{y} \in \mathbb{Z}^m):$ Return $fk_{\mathbf{y}} := \mu(sk, \mathbf{y})$
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Fig. 14. The functional smoothness experiments $\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}$ ($b \in \{0, 1\}$) for probabilistic two-leveled $\text{TLHPS} = (\text{IPHPS}, \text{coHash})$. The difference between $b = 0$ and $b = 1$ is highlighted in gray box. To avoid trivial attacks, we require that the (single) $\mathcal{O}_{\text{CHAL}}^b(\tilde{\mathbf{x}})$ query and all $\mathcal{O}_{\text{FKEY}}(\mathbf{y})$ queries made by $\mathcal{A}$ must satisfy $\langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0$.

G.4 Proof of Theorem 3 (Functional Smoothness of $\text{TLHPS}_{\text{LWE}}$)

Theorem 3. Let $\chi_K = D_{\mathbb{Z}, \sigma_K}$ be a discrete Gaussian with parameter $\sigma_K \geq \sqrt{m} \cdot X \cdot l'$, and let $l' \geq 2n \log q$. Moreover, suppose that pp_{LWE} has a negligible $\epsilon_{\text{pp}_{\text{LWE}}}$ as shown in (7). Then the two-leveled scheme $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ proposed in Fig. 4 and 5 has functional smoothness with $\epsilon_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) \leq 6QmX \cdot \epsilon_{\text{pp}_{\text{LWE}}}$ for any (unbounded) adversary $\mathcal{A}$ and any polynomial Q , where $\mathcal{A}$ is required to query $\mathcal{O}_{\text{CHAL}}^b$ with vector $\tilde{\mathbf{x}}$ of bounded norm X , i.e., $\|\tilde{\mathbf{x}}\|_{\infty} \leq X$.

Proof of Theorem 3. To show that $\epsilon_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, Q}^{\text{funSmth}}(\lambda) \leq 6QmX \cdot \epsilon_{\text{pp}_{\text{LWE}}}$, we first elaborate the $\text{Exp}_{\text{TLHPS}, \mathcal{A}, Q}^{\text{funSmth}, b}$ experiments ($b \in \{0, 1\}$) defined in Fig. 2.

Let $\text{pp}_{\text{IPHPS}} = \text{pp}_{\text{coHash}} = (\text{pp}_{\text{LWE}}, \mathbf{A}, \mathbf{B})$. Let $sk = \mathbf{K} \leftarrow D_{\mathbb{Z}, \sigma_K}^{m \times l}$ and $s = s \leftarrow D_{\mathbb{Z}, \sigma_s}^{l-l'}$. In the experiment, $\mathcal{A}$ first obtains $(\text{pp}_{\text{IPHPS}}, \text{pp}_{\text{coHash}}, pk)$ where

$$pk = \mathbf{P} = \mathbf{K} \cdot \mathbf{A} \in \mathbb{Z}_q^{m \times n}, \quad (33)$$

and has oracle access to $\mathcal{O}_{\text{CHAL}}^b$ and $\mathcal{O}_{\text{FKEY}}$. Through $\mathcal{O}_{\text{FKEY}}$ queries ($\mathbf{y} \in \mathbb{Z}^m$), $\mathcal{A}$ learns functional keys

$$fk_{\mathbf{y}} = \mathbf{f}^{\top} = \mathbf{y}^{\top} \cdot \mathbf{K} \in \mathbb{Z}^{1 \times l}. \quad (34)$$

At some point, $\mathcal{A}$ makes a single $\mathcal{O}_{\text{CHAL}}^b$ query ($\tilde{\mathbf{x}} \in \mathbb{Z}^m$), and obtains $(\mathbf{c}^{(j)}, \mathbf{h}^{(j)})_{j \in [Q]}$, where $\mathbf{c}^{(j)} \leftarrow_{\mathcal{S}} \mathcal{X}$ and

$$\mathbf{h}^{(j)} := \begin{cases} \text{Priv}(sk, \mathbf{c}^{(j)}) = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}^{(j)}, & \text{if } b = 0; \\ \text{Priv}(sk, \mathbf{c}^{(j)}) + \Gamma_s(\mathbf{c}^{(j)}) \cdot \tilde{\mathbf{x}} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}}, & \text{if } b = 1, \end{cases} \quad (35)$$

where each $\mathbf{v}^{(j)} \in \mathbb{Z}_q^{l-l'}$ is the vector s.t. $\mathbf{c}^{(j)} = \mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)} + \mathbf{e}^{(j)} \in \mathcal{X}$ for some $\mathbf{w}^{(j)} \in \mathbb{Z}_q^n$, $\mathbf{e}^{(j)} \in [B_{\mathcal{L}}, B_{\mathcal{L}}]^l$, each $\mathbf{e}^{(j)} \leftarrow_{\mathcal{S}} [-B_{e'}, B_{e'}]^m$ and each $e''^{(j)} \leftarrow_{\mathcal{S}} \chi_s$. Then $\mathcal{A}$ may continue to make $\mathcal{O}_{\text{FKEY}}$ queries ($\mathbf{y} \in \mathbb{Z}^m$) and learn functional keys $fk_{\mathbf{y}}$, similar to (34). Finally, $\mathcal{A}$ outputs b' , and the advantage of $\mathcal{A}$ is defined by $\epsilon_{\text{TLHPS}_{\text{LWE}, \mathcal{A}, Q}}^{\text{funSmth}}(\lambda) = |\Pr[b' = 1 | b = 0] - \Pr[b' = 1 | b = 1]|$.

Below we analyze $\mathcal{A}$'s advantage. Note that the only differences between $b = 0$ and $b = 1$ are the distributions of $\mathbf{h}^{(j)}$ in (35). We will show that the $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}^{(j)}\}_{j \in [Q]}$ ($b = 0$) are statistically indistinguishable from the $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}}\}_{j \in [Q]}$ ($b = 1$), from the point of view of $\mathcal{A}$. In the following, we may assume that $\tilde{\mathbf{x}}$ is non-zero w.l.o.g., since otherwise if $\tilde{\mathbf{x}} = \mathbf{0}$, then the terms in gray box in (35) vanish, and it is clear that $\mathcal{A}$ has no advantage in distinguishing $b = 0$ and $b = 1$ and the theorem trivially holds.

To this end, we first analyze the view of $\mathcal{A}$, and in particular, the distribution of $sk = \mathbf{K}$ conditioned on the pk in (33) and the $fk_{\mathbf{y}}$ in (34) obtained from $\mathcal{O}_{\text{FKEY}}$ queries. For the convenience of our analysis, we assume that $\mathcal{A}$ makes $\mathcal{O}_{\text{FKEY}}$ queries for all $\mathbf{y} \in \mathbb{Z}^m$ orthogonal to $\tilde{\mathbf{x}} \in \mathbb{Z}^m$, i.e., for all $\mathbf{y} \in \mathcal{Q} := \{\mathbf{y} \in \mathbb{Z}^m \mid \langle \tilde{\mathbf{x}}, \mathbf{y} \rangle = 0\}^{21}$, through which $\mathcal{A}$ can get as much information as possible about $sk = \mathbf{K}$. Formally, we have the following two claims.

Claim 7 (Conditional Distribution of $sk = \mathbf{K}$). Conditioned on $pk = \mathbf{K} \cdot \mathbf{A}$ and $\{fk_{\mathbf{y}} = \mathbf{y}^\top \cdot \mathbf{K}\}_{\mathbf{y} \in \mathcal{Q}}$, the conditional distribution of $sk = \mathbf{K} \leftarrow_{\mathcal{S}} D_{\mathbb{Z}, \sigma_K}^{m \times l}$ is $\mathbf{K}_0 + D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$, where $\mathbf{K}_0 \in \mathbb{Z}^{m \times l}$ is any particular matrix satisfying the conditions and $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}) \subseteq \mathbb{Z}^{m \times l}$ is a lattice with $\gcd(\tilde{\mathbf{x}})$ the greatest common divisor of all entries of $\tilde{\mathbf{x}} \in \mathbb{Z}^m$ and $\Lambda^\perp(\mathbf{A}) := \{\mathbf{z}^\top \in \mathbb{Z}^{1 \times l} \mid \mathbf{z}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}\}$.

Proof. Let $\mathbf{K}_0 \in \mathbb{Z}^{m \times l}$ be any particular matrix satisfying the conditions and let $\Delta := \mathbf{K} - \mathbf{K}_0 \in \mathbb{Z}^{m \times l}$.

We will first prove that Δ is in the lattice $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A})$. Since both $sk = \mathbf{K}$ and $\mathbf{K}_0$ satisfy the conditions, i.e., $pk = \mathbf{K} \cdot \mathbf{A} = \mathbf{K}_0 \cdot \mathbf{A} \in \mathbb{Z}_q^{m \times n}$ and $\{fk_{\mathbf{y}} = \mathbf{y}^\top \cdot \mathbf{K} = \mathbf{y}^\top \cdot \mathbf{K}_0 \in \mathbb{Z}^{1 \times l}\}_{\mathbf{y} \in \mathcal{Q}}$, it follows that Δ satisfies

$$\Delta \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{m \times n} \quad \text{and} \quad \{\mathbf{y}^\top \cdot \Delta = \mathbf{0} \in \mathbb{Z}^{1 \times l}\}_{\mathbf{y} \in \mathcal{Q}}.$$

²¹ Note that $\mathcal{Q}$ contains an infinite number of vectors, so it is not possible for $\mathcal{A}$ to query all vectors $\mathbf{y} \in \mathcal{Q}$ to $\mathcal{O}_{\text{FKEY}}$. Nevertheless, since $fk_{\mathbf{y}}$ is linear w.r.t. $\mathbf{y}$ in our scheme (cf. Fig. 4), the unbounded adversary $\mathcal{A}$ can make $\mathcal{O}_{\text{FKEY}}$ queries for a basis of $\mathcal{Q}$, and with the obtained $fk_{\mathbf{y}}$ for the basis, $\mathcal{A}$ can effectively get $fk_{\mathbf{y}}$ for all $\mathbf{y} \in \mathcal{Q}$. We note that the security proofs of many existing LWE-based IPFE schemes (e.g., [8, 63]) also make such arguments.

Note that $\{\mathbf{y}^\top \cdot \boldsymbol{\Delta} = \mathbf{0}\}_{\mathbf{y} \in \mathcal{Q}}$ implies that each column of $\boldsymbol{\Delta}$ is orthogonal to all $\mathbf{y} \in \mathcal{Q}$. Since $\mathcal{Q}$ contains all $\mathbf{y}$ orthogonal to $\tilde{\mathbf{x}}$, it follows that each column of $\boldsymbol{\Delta}$ is in the span of $\tilde{\mathbf{x}}$, i.e., $\boldsymbol{\Delta} = (w_1 \cdot \tilde{\mathbf{x}}, \dots, w_l \cdot \tilde{\mathbf{x}})$ for some $w_1, \dots, w_l \in \mathbb{Q}$, or $\boldsymbol{\Delta} = \tilde{\mathbf{x}} \cdot \mathbf{w}^\top$ for some $\mathbf{w}^\top = (w_1, \dots, w_l) \in \mathbb{Q}^{1 \times l}$ in short. Since $\boldsymbol{\Delta} = \tilde{\mathbf{x}} \cdot \mathbf{w}^\top \in \mathbb{Z}^{m \times l}$ is an integer matrix, $\mathbf{w}^\top \in \mathbb{Q}^{1 \times l}$ must be in the form of $\mathbf{w}^\top = \frac{1}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top$ for some $\mathbf{z}^\top \in \mathbb{Z}^{1 \times l}$. So we have $\boldsymbol{\Delta} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top$ for some $\mathbf{z}^\top \in \mathbb{Z}^{1 \times l}$. Then together with $\boldsymbol{\Delta} \cdot \mathbf{A} = \mathbf{0}$, we have $\boldsymbol{\Delta} \cdot \mathbf{A} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{m \times n}$, which further implies that $\mathbf{z}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}$ since $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}$ is assumed to be non-zero w.l.o.g. and q is a prime number. Consequently, we have $\boldsymbol{\Delta} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top$ for some $\mathbf{z}^\top \in \Lambda^\perp(\mathbf{A}) := \{\mathbf{z}^\top \in \mathbb{Z}^{1 \times l} \mid \mathbf{z}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}\}$, i.e., $\boldsymbol{\Delta} \in \mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A})$.

Then we have that $sk = \mathbf{K} = \mathbf{K}_0 + \boldsymbol{\Delta} \in \mathbf{K}_0 + \mathcal{Y} \subseteq \mathbb{Z}^{m \times l}$. Consequently, the conditional distribution of $sk = \mathbf{K} \leftarrow_{\$} D_{\mathbb{Z}, \sigma_K}^{m \times l}$ is $D_{\mathbf{K}_0 + \mathcal{Y}, \sigma_K}$, which equals $\mathbf{K}_0 + D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$ by the definition of discrete Gaussian, and Claim 7 follows. $\blacksquare$

Claim 8 (Distribution of $D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0} \cdot \mathbf{B}$). Let $\mathcal{Y} := \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}) \subseteq \mathbb{Z}^{m \times l}$ with $\Lambda^\perp(\mathbf{A}) := \{\mathbf{z}^\top \in \mathbb{Z}^{1 \times l} \mid \mathbf{z}^\top \cdot \mathbf{A} = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}\}$, let $\mathbf{B} := \begin{pmatrix} \mathbf{I}_{l-l'} \\ \mathbf{0} \end{pmatrix} \in \mathbb{Z}_q^{l \times (l-l')}$, and let $l' \geq 2n \log q$. Then for any $\sigma_K \geq \sqrt{m} \cdot X \cdot l'$ and any $\mathbf{K}_0 \in \mathbb{Z}^{m \times l}$, the distribution of $D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0} \cdot \mathbf{B}$ is within statistical distance at most $n/q^{l'-n} + q^{-n} + 2^{-l'^2/6+2}$ from the distribution $D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$, where $\mathbf{K}_{01} \in \mathbb{Z}_q^{m \times (l-l')}$ is made up of the first $l-l'$ columns of $\mathbf{K}_0$.

Proof. Note that for the specific $\mathbf{B} = \begin{pmatrix} \mathbf{I}_{l-l'} \\ \mathbf{0} \end{pmatrix} \in \mathbb{Z}_q^{l \times (l-l')}$, $D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0} \cdot \mathbf{B}$ in fact consists of the first $l-l'$ columns of $D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$, and the claim essentially analyzes this marginal distribution of discrete Gaussian. The proof is essentially implicit in the preimage sampling algorithm of Gentry et al. [33] and is explicitly provided by Wee and Wu [64, Lem. 2.7] using slightly different notations. Below we give a full proof of this claim for completeness.

Let $\boldsymbol{\Delta} \leftarrow_{\$} D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$. We analyze the distribution of $\boldsymbol{\Delta} \cdot \mathbf{B}$ as follows.

Firstly, for $\boldsymbol{\Delta} \leftarrow_{\$} D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$, we can write $\boldsymbol{\Delta} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top$ for some $\mathbf{z}^\top \in \Lambda^\perp(\mathbf{A})$. For ease of analysis, we parse $\mathbf{z}^\top = (\mathbf{z}_1^\top \in \mathbb{Z}^{1 \times (l-l')}, \mathbf{z}_2^\top \in \mathbb{Z}^{1 \times l'}) \in \mathbb{Z}^{1 \times l}$, parse $\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \in \mathbb{Z}_q^{l \times n}$ with $\mathbf{A}_1 \in \mathbb{Z}_q^{(l-l') \times n}$ and $\mathbf{A}_2 \in \mathbb{Z}_q^{l' \times n}$, and parse $\mathbf{K}_0 = (\mathbf{K}_{01} \in \mathbb{Z}_q^{m \times (l-l')}, \mathbf{K}_{02} \in \mathbb{Z}_q^{m \times l'}) \in \mathbb{Z}_q^{m \times l}$. Then we have

$$\boldsymbol{\Delta} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top = \left(\underbrace{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top}_{:= \boldsymbol{\Delta}_1 \in \mathbb{Z}^{m \times (l-l')}}, \underbrace{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_2^\top}_{:= \boldsymbol{\Delta}_2 \in \mathbb{Z}^{m \times l'}} \right)$$

and

$$\boldsymbol{\Delta} \cdot \mathbf{B} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \cdot \begin{pmatrix} \mathbf{I}_{l-l'} \\ \mathbf{0} \end{pmatrix} = \underbrace{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top}_{= \boldsymbol{\Delta}_1}.$$

Moreover, $\mathbf{z}^\top \in \Lambda^\perp(\mathbf{A})$ means that $\mathbf{z}_1^\top \cdot \mathbf{A}_1 + \mathbf{z}_2^\top \cdot \mathbf{A}_2 = \mathbf{0}$.

Next we consider the distribution $D_{\mathcal{R}, \sigma_K, -\mathbf{K}_0}$ of Δ :

$$\begin{aligned}
D(\Delta) &= \frac{\rho_{\sigma_K, -\mathbf{K}_0}(\Delta)}{\sum_{\Delta' \in \mathcal{R}} \rho_{\sigma_K, -\mathbf{K}_0}(\Delta')} = \frac{\rho_{\sigma_K, -\mathbf{K}_0}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top)}{\sum_{\mathbf{z}'^\top \in \Lambda^\perp(\mathbf{A})} \rho_{\sigma_K, -\mathbf{K}_0}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}'^\top)} \\
&= \frac{\rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_2^\top)}{\sum_{\mathbf{z}_1'^\top \in \mathbb{Z}^{1 \times (l-l')}} \sum_{\mathbf{z}_2'^\top : \mathbf{z}_2'^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1'^\top \cdot \mathbf{A}_1} \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1'^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_2'^\top)}. \tag{36}
\end{aligned}$$

Since $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{l \times n}$ is uniformly chosen during the generation of pp_{IPHP5} , its submatrix $\mathbf{A}_2 \in \mathbb{Z}_q^{l' \times n}$ is of full rank except with negligible probability at most $n/q^{l'-n}$. Consequently, for every $\mathbf{z}_1'^\top \in \mathbb{Z}^{1 \times (l-l')}$, there exists some $\mathbf{u}_{\mathbf{z}_1'^\top}^\top \in \mathbb{Z}^{1 \times l'}$ such that $\mathbf{u}_{\mathbf{z}_1'^\top}^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1'^\top \cdot \mathbf{A}_1$, and all $\mathbf{z}_2'^\top$ satisfying $\mathbf{z}_2'^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1'^\top \cdot \mathbf{A}_1$ must lie in $\mathbf{u}_{\mathbf{z}_1'^\top}^\top + \Lambda^\perp(\mathbf{A}_2)$ with $\Lambda^\perp(\mathbf{A}_2) := \{\delta_2^\top \in \mathbb{Z}^{1 \times l'} \mid \delta_2^\top \cdot \mathbf{A}_2 = \mathbf{0} \in \mathbb{Z}_q^{1 \times n}\}$. Let Denom be the denominator in (36). Then

$$\begin{aligned}
\text{Denom} &= \sum_{\mathbf{z}_1'^\top \in \mathbb{Z}^{1 \times (l-l')}} \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1'^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot (\mathbf{u}_{\mathbf{z}_1'^\top}^\top + \Lambda^\perp(\mathbf{A}_2))) \\
&= \sum_{\mathbf{z}_1'^\top \in \mathbb{Z}^{1 \times (l-l')}} \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1'^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{u}_{\mathbf{z}_1'^\top}^\top + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)).
\end{aligned}$$

We claim that $\sigma_K \geq \eta_\epsilon(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))$ for a negligible $\epsilon := 2^{-l'^2/6}$, which will be shown later. Then by applying Lemma 13, we know that for all $\mathbf{u}_{\mathbf{z}_1'^\top}^\top \in \mathbb{Z}^{1 \times l'}$, it holds that

$$\begin{aligned}
&\rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{u}_{\mathbf{z}_1'^\top}^\top + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)) \\
&= \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{u}_{\mathbf{z}_1'^\top}^\top + \mathbf{K}_{02} + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)) \in [\frac{1-\epsilon}{1+\epsilon}, 1] \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)).
\end{aligned}$$

Thus, we can write

$$\text{Denom} = c_Q \cdot \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}) \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)) \tag{37}$$

for some $c_Q \in [\frac{1-\epsilon}{1+\epsilon}, 1]$.

Now we consider the marginal distribution D_1 of $\Delta \cdot \mathbf{B} = \Delta_1 = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top$:

$$\begin{aligned}
D_1(\Delta_1) &= D_1(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top) = \sum_{\mathbf{z}_2^\top : \mathbf{z}_2^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1^\top \cdot \mathbf{A}_1} D(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top, \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_2^\top) \\
&= \frac{\sum_{\mathbf{z}_2^\top : \mathbf{z}_2^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1^\top \cdot \mathbf{A}_1} \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_2^\top)}{c_Q \cdot \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}) \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))} \\
&= \frac{\rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top) \cdot \rho_{\sigma_K, -\mathbf{K}_{02}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot (\mathbf{u}_{\mathbf{z}_1^\top}^\top + \Lambda^\perp(\mathbf{A}_2)))}{c_Q \cdot \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}) \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))} \tag{38}
\end{aligned}$$

$$\begin{aligned}
&= \frac{c \cdot \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top) \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))}{c_Q \cdot \rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}) \cdot \rho_{\sigma_K}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))} \tag{39} \\
&= \frac{c}{c_Q} \cdot \frac{\rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top)}{\rho_{\sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')})} = \frac{c}{c_Q} \cdot D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}_1^\top),
\end{aligned}$$

where (38) holds for an arbitrary vector $\mathbf{u}_{\mathbf{z}_1^\top}^\top \in \mathbb{Z}^{1 \times l'}$ satisfying $\mathbf{u}_{\mathbf{z}_1^\top}^\top \cdot \mathbf{A}_2 = -\mathbf{z}_1^\top \cdot \mathbf{A}_1$ and (39) follows from a similar analysis as (37) for $c \in [\frac{1-\epsilon}{1+\epsilon}, 1]$ guaranteed by Lemma 13. Since $\frac{c}{c_Q} \in [\frac{1-\epsilon}{1+\epsilon}, \frac{1+\epsilon}{1-\epsilon}] \subseteq [1-2\epsilon, 1+4\epsilon]$, it follows that

$$D_1(\mathbf{\Delta}_1) \in [1-2\epsilon, 1+4\epsilon] \cdot D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}, \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}(\mathbf{\Delta}_1),$$

so the statistical distance between D_1 and $D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}, \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$ is bounded by 4ϵ . This shows that the distribution of $\mathbf{\Delta} \cdot \mathbf{B} = \mathbf{\Delta}_1$ is statistically close to $D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}, \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$ with statistical distance $4\epsilon = 2^{-l'^2/6+2}$.

It remains to show the claim that $\sigma_K \geq \eta_\epsilon(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2))$ for $\epsilon = 2^{-l'^2/6}$. Firstly, according to Lemma 14, we know that $\eta_\epsilon(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)) = \eta_\epsilon(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z} \otimes \Lambda^\perp(\mathbf{A}_2)) \leq \|\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}\| \cdot \eta_\epsilon(\Lambda^\perp(\mathbf{A}_2)) \leq \sqrt{m} \cdot \|\tilde{\mathbf{x}}\|_\infty \cdot \eta_\epsilon(\Lambda^\perp(\mathbf{A}_2))$. Then by Lemma 15, since $\mathbf{A}_2 \in \mathbb{Z}_q^{l' \times n}$ as a sub-matrix of $\mathbf{A} \leftarrow_s \mathbb{Z}_q^{l \times n}$ is uniformly distributed over $\mathbb{Z}_q^{l' \times n}$ and $l' \geq 2n \log q$, it holds that $\eta_\epsilon(\Lambda^\perp(\mathbf{A}_2)) \leq l'$ for $\epsilon = 2^{-l'^2/6}$, except with negligible probability q^{-n} . By combining the above facts together, we have that $\eta_\epsilon(\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A}_2)) \leq \sqrt{m} \cdot X \cdot l' \leq \sigma_K$.

Finally, by collecting the all statistical terms, Claim 8 follows. $\blacksquare$

Now based on the above two claims, we analyze the distributions of $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)}\}_{j \in [Q]}$ (i.e., the case $b = 0$ in (35)) and $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}}\}_{j \in [Q]}$ (i.e., the case $b = 1$ in (35)) from the view of $\mathcal{A}$. We first analyze the distribution of $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)}\}_{j \in [Q]}$. According to Claim 7, the distribution of $sk = \mathbf{K}$ conditioned on the pk in (33) and the fk_y in (34) obtained from $\mathcal{O}_{\text{FKY}}$ queries follows $\mathbf{K}_0 + \mathbf{\Delta}$, where $\mathbf{K}_0 \in \mathbb{Z}^{m \times l}$ is any particular matrix satisfying the conditions and $\mathbf{\Delta} \leftarrow_s D_{\mathcal{Y}, \sigma_K, -\mathbf{K}_0}$ follows the discrete Gaussian over the lattice $\mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A})$. Then we have that

$$\begin{aligned} \text{Case } b = 0 \text{ in (35): } \mathbf{h}^{(j)} &= \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)} \\ &= \mathbf{K} \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)} + \mathbf{e}^{(j)}) + \mathbf{e}'^{(j)} \end{aligned} \quad (40)$$

$$\stackrel{s}{\approx} \mathbf{K} \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)} + \mathbf{e}^{(j)}) + \mathbf{e}'^{(j)} \quad (41)$$

$$\begin{aligned} &\equiv (\mathbf{K}_0 + \mathbf{\Delta}) \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \mathbf{e}'^{(j)} \\ &= \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \mathbf{\Delta} \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \mathbf{e}'^{(j)} \end{aligned} \quad (42)$$

$$\stackrel{s}{\approx} \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \cdot \mathbf{v}^{(j)} + \mathbf{e}'^{(j)}, \quad (43)$$

where (40) holds since $\mathbf{c}^{(j)} = \mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)} + \mathbf{e}^{(j)} \in \mathcal{X}$, (41) holds since $\mathbf{e}'^{(j)} \leftarrow_s [-B_{e'}, B_{e'}]^m$ smudges $\mathbf{K} \cdot \mathbf{e}^{(j)}$ with statistical distance at most $mlB_K B_{\mathcal{L}}/B_{e'}$ according to the smudging lemma (i.e., Lemma 2) and the fact that $\|\mathbf{K} \cdot \mathbf{e}^{(j)}\|_\infty \leq lB_K B_{\mathcal{L}}$, (42) holds since $\mathbf{\Delta} \in \mathcal{Y} = \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \Lambda^\perp(\mathbf{A})$ satisfies $\mathbf{\Delta} \cdot \mathbf{A} = \mathbf{0}$, and (43) holds for $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \leftarrow_s D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}, \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$ since $\mathbf{\Delta} \cdot \mathbf{B}$ is within statistical distance at most $n/q^{l'-n} + q^{-n} + 2^{-l'^2/6+2}$ from $D_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}, \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$ ac-

cording to Claim 8. Similarly, for the distribution of $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}}\}_{j \in [Q]}$, we have that

$$\begin{aligned} \text{Case } b = 1 \text{ in (35): } \mathbf{h}^{(j)} &= \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}} \\ &\stackrel{s}{\approx} \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \cdot \mathbf{v}^{(j)} + \mathbf{e}'^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}} \\ &= \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot (\mathbf{z}^\top + \gcd(\tilde{\mathbf{x}}) \cdot \mathbf{s}^\top) \cdot \mathbf{v}^{(j)} + \mathbf{e}'^{(j)} + e''^{(j)} \cdot \tilde{\mathbf{x}} \\ &\stackrel{s}{\approx} \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot (\mathbf{z}^\top + \cancel{\gcd(\tilde{\mathbf{x}}) \cdot \mathbf{s}^\top}) \cdot \mathbf{v}^{(j)} + \mathbf{e}'^{(j)} + e''^{(j)} \cdot \tilde{\mathbf{x}} \quad (44) \end{aligned}$$

$$\stackrel{s}{\approx} \mathbf{K}_0 \cdot (\mathbf{A}\mathbf{w}^{(j)} + \mathbf{B}\mathbf{v}^{(j)}) + \frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \cdot \mathbf{v}^{(j)} + \mathbf{e}'^{(j)} + \cancel{e''^{(j)} \cdot \tilde{\mathbf{x}}}, \quad (45)$$

where the “ $\stackrel{s}{\approx}$ ” in (44) and (45) are explained in detail as follows.

- Since $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbf{z}^\top \leftarrow_{\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})} \cdot \mathbb{Z}^{1 \times (l-l')}, \sigma_K, -\mathbf{K}_{01}}$ and $\frac{\tilde{\mathbf{x}}}{\gcd(\tilde{\mathbf{x}})}$ is assumed to be non-zero, by Lemma 17 in Appendix G.1 (where we give a fine-grained analysis of the distribution of $\mathbf{z}^\top$), we know that $\mathbf{z}^\top$ follows $D_{\mathbb{Z}^{1 \times (l-l')}, \sigma_K / \sqrt{n_{\tilde{\mathbf{x}}}}, \mathbf{t}^\top}$ with $n_{\tilde{\mathbf{x}}} \in [m]$ the number of non-zero entries in $\tilde{\mathbf{x}}$ and $\mathbf{t}^\top \in \mathbb{R}^{1 \times (l-l')}$ a vector that depends on $\sigma_K, \tilde{\mathbf{x}}, \mathbf{K}_{01}$. Moreover, since $\|\tilde{\mathbf{x}}\|_\infty \leq X$ and $\mathbf{s} \leftarrow_{\chi_s^{l-l'}}$, it holds that $\|\gcd(\tilde{\mathbf{x}}) \cdot \mathbf{s}^\top\|_\infty \leq X \cdot B_s$. Then according to the smudging lemma for discrete Gaussians (i.e., Lemma 12 in Appendix G.1), we get that $\mathbf{z}^\top + \gcd(\tilde{\mathbf{x}}) \cdot \mathbf{s}^\top \stackrel{s}{\approx} \mathbf{z}^\top$ with statistical distance at most $(l-l') \cdot (XB_s + 1) / (\sigma_K / \sqrt{n_{\tilde{\mathbf{x}}}}) \leq (l-l') \cdot (XB_s + 1) \sqrt{m} / \sigma_K$. This shows the “ $\stackrel{s}{\approx}$ ” in (44).
- Since $\mathbf{e}'^{(j)} \leftarrow_{[-B_{e'}, B_{e'}]^m}$, and $\|e''^{(j)} \cdot \tilde{\mathbf{x}}\|_\infty \leq X \cdot B_s$ for $e''^{(j)} \leftarrow_{\chi_s}$, by the smudging lemma (i.e., Lemma 2), we have $\mathbf{e}'^{(j)} + e''^{(j)} \cdot \tilde{\mathbf{x}} \stackrel{s}{\approx} \mathbf{e}'^{(j)}$ with statistical distance at most $mXB_s/B_{e'}$. This shows the “ $\stackrel{s}{\approx}$ ” in (45).

By combining (43) and (45) together and collecting all the statistical terms, we get the statistical indistinguishability of $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)}\}_{j \in [Q]}$ (i.e., the case $b = 0$ in (35)) and $\{\mathbf{h}^{(j)} = \mathbf{K} \cdot \mathbf{c}^{(j)} + \mathbf{e}'^{(j)} + (\mathbf{s}^\top \cdot \mathbf{v}^{(j)} + e''^{(j)}) \cdot \tilde{\mathbf{x}}\}_{j \in [Q]}$ (i.e., the case $b = 1$ in (35)) from the point of view of $\mathcal{A}$, with statistical distance at most

$$\begin{aligned} &2 \cdot (Q \cdot mlB_K B_{\mathcal{L}} / B_{e'} + n/q^{l'-n} + q^{-n} + 2^{-l'^2/6+2}) \\ &+ (l-l') \cdot (XB_s + 1) \sqrt{m} / \sigma_K + Q \cdot mXB_s / B_{e'}. \end{aligned}$$

Finally, according to the definition of $\varepsilon_{\text{ppLWE}}$ in (7), it is not hard to see that the above statistical distance is no more than $6QmX \cdot \varepsilon_{\text{ppLWE}}$. Therefore, $\mathcal{A}$ has advantage $\varepsilon_{\text{TLHPS}_{\text{LWE}, \mathcal{A}, Q}}^{\text{funSmth}}(\lambda) \leq 6QmX \cdot \varepsilon_{\text{ppLWE}}$ in distinguishing $b = 0$ and $b = 1$, and Theorem 3 follows. $\square$

G.5 Proof of Theorem 4 (Tight Multi-Key-Extracting of $\text{TLHPS}_{\text{LWE}}$)

Theorem 4. Let $\chi_s = D_{\mathbb{Z}, \sigma_s}$, $l \geq (l-l'+1) \log q + 2\lambda$, and $l-l' \geq \lambda$. Moreover, suppose that ppLWE has a negligible $\varepsilon_{\text{ppLWE}}$ as shown in (7). If the

$\text{LWE}_{\lambda,q,D_{\mathbb{Z}},\sigma_{\text{LWE}}}$, l -assumption holds, then the two-leveled scheme $\text{TLHPS}_{\text{LWE}} = (\text{IPHPS}_{\text{LWE}}, \text{coHash}_{\text{LWE}})$ proposed in Fig. 4 and 5 supports multi-key-extracting. Concretely, for any adversary $\mathcal{A}$, any T and any Q , there exists an adversary $\mathcal{B}$, such that $\mathbf{T}(\mathcal{B}) \approx \mathbf{T}(\mathcal{A}) + TQ \cdot \text{poly}(\lambda)$ with $\text{poly}(\lambda)$ independent of $\mathbf{T}(\mathcal{A})$, and $\text{Adv}_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{[\lambda, q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + 5TQ \cdot \varepsilon_{\text{ppLWE}}$.

Proof of Theorem 4. Firstly, we will construct an adversary $\mathcal{B}'$ against the T -instance $\text{LWE}_{l-l', q, \chi_s, Q}$ -assumption with *noisy secrets* (cf. Def. 16), so that

$$\text{Adv}_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{[l-l', q, \chi_s, Q], \mathcal{B}'}^{T\text{-inst-LWE-noi-secr}}(\lambda). \quad (46)$$

Then by Theorem 12 in Appendix G.2 (Multi-instance LWE $\Rightarrow$ Multi-instance LWE with Noisy Secrets), the following holds for an adversary $\mathcal{B}''$:

$$\begin{aligned} \text{Adv}_{[l-l', q, \chi_s, Q], \mathcal{B}'}^{T\text{-inst-LWE-noi-secr}}(\lambda) &\leq \text{Adv}_{[l-l', q, \chi_s, Q+(l-l')^2], \mathcal{B}''}^{T\text{-inst-LWE}}(\lambda) + T(l-l')/q^{l-l'} \\ &\leq \text{Adv}_{[l-l', q, \chi_s, Q+(l-l')^2], \mathcal{B}''}^{T\text{-inst-LWE}}(\lambda) + T \cdot \varepsilon_{\text{ppLWE}}, \end{aligned} \quad (47)$$

where the last inequality holds by the definition of $\varepsilon_{\text{ppLWE}}$ in (7). Moreover, according to the tail bound (Lemma 1), $D_{\mathbb{Z}, \sigma_{\text{LWE}}}$ is $\sqrt{\lambda} \cdot \sigma_{\text{LWE}}$ -bounded except with probability $2^{-\lambda}$, then by the condition that $l \geq (l-l'+1) \log q + 2\lambda$ and according to Theorem 13 in Appendix G.2 (LWE $\Rightarrow$ Multi-instance LWE with Unbounded Samples), the following holds for an adversary $\mathcal{B}'''$:

$$\begin{aligned} &\text{Adv}_{[l-l', q, \chi_s, Q+(l-l')^2], \mathcal{B}''}^{T\text{-inst-LWE}}(\lambda) \\ &\leq \text{Adv}_{[l-l', q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}, l], \mathcal{B}'''}^{\text{LWE}}(\lambda) + T(Q + (l-l')^2) \cdot (l\sqrt{\lambda} \cdot \sigma_{\text{LWE}} + 1)/\sigma_s \\ &\quad + 2T(Q + (l-l')^2) \cdot 2^{-\lambda} \\ &\leq \text{Adv}_{[\lambda, q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}, l], \mathcal{B}}^{\text{LWE}}(\lambda) + 4TQ \cdot \varepsilon_{\text{ppLWE}}, \end{aligned} \quad (48)$$

where the last inequality holds since the $\text{LWE}_{\lambda, q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}}$, l -assumption tightly implies the $\text{LWE}_{l-l', q, D_{\mathbb{Z}}, \sigma_{\text{LWE}}}$, l -assumption for $l-l' \geq \lambda$, and by the definition of $\varepsilon_{\text{ppLWE}}$ in (7). Finally, Theorem 4 holds by putting (46), (47) and (48) together.

It remains to present the description of $\mathcal{B}'$ such that (46) holds. Given a challenge $(\{\mathbf{V}_i, \mathbf{u}_i\}_{i \in [T]})$, $\mathcal{B}'$ wants to distinguish $\{\mathbf{u}_i = \mathbf{V}_i \mathbf{s}_i + \mathbf{e}_i\}_{i \in [T]}$ from $\{\mathbf{u}_i \leftarrow \mathbb{Z}_q^Q\}_{i \in [T]}$, where each $\mathbf{V}_i \leftarrow \mathbb{Z}_q^{Q \times (l-l')}$, $\mathbf{s}_i \leftarrow \chi_s^{l-l'}$, $\mathbf{e}_i \leftarrow \chi_s^Q$ for $i \in [T]$.

Let us fix some *notations* used in this proof. For each $i \in [T]$, we parse

$$\mathbf{V}_i = \begin{pmatrix} \mathbf{v}_i^{(1)\top} \\ \vdots \\ \mathbf{v}_i^{(Q)\top} \end{pmatrix} \in \mathbb{Z}_q^{Q \times (l-l')} \text{ with each } \mathbf{v}_i^{(j)} \in \mathbb{Z}_q^{l-l'}, \text{ parse } \mathbf{e}_i = \begin{pmatrix} e_i^{(1)} \\ \vdots \\ e_i^{(Q)} \end{pmatrix} \in \mathbb{Z}_q^Q$$

$$\text{with each } e_i^{(j)} \in \mathbb{Z}_q, \text{ and parse } \mathbf{u}_i = \begin{pmatrix} u_i^{(1)} \\ \vdots \\ u_i^{(Q)} \end{pmatrix} \in \mathbb{Z}_q^Q \text{ with each } u_i^{(j)} \in \mathbb{Z}_q.$$

$\mathcal{B}'$ is constructed by invoking $\mathcal{A}$ as follows. $\mathcal{B}'$ invokes $\mathbf{A} \leftarrow \mathbb{Z}_q^{l \times n}$ itself, defines $\mathbf{B} := \begin{pmatrix} \mathbf{I}_{l-l'} \\ \mathbf{0} \end{pmatrix} \in \mathbb{Z}_q^{l \times (l-l')}$, and sets $\text{pp}_{\text{IPHPS}} := \text{pp}_{\text{coHash}} := (\text{pp}_{\text{LWE}}, \mathbf{A}, \mathbf{B})$.

Then for all $i \in [T]$ and $j \in [Q]$, $\mathcal{B}'$ sets $\mathbf{c}_i^{(j)} := \mathbf{A}\mathbf{w}_i^{(j)} + \mathbf{B}\mathbf{v}_i^{(j)} + \mathbf{e}_i^{(j)}$ by choosing $\mathbf{w}_i^{(j)} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e}_i^{(j)} \leftarrow \chi_{\mathcal{L}}^l$ randomly and using the $\mathbf{v}_i^{(j)} \in \mathbb{Z}_q^{l-l'}$ contained in its own challenge $\mathbf{V}_i$. Moreover, for all $i \in [T]$ and $j \in [Q]$, $\mathcal{B}'$ sets $h_i^{(j)} := u_i^{(j)} \in \mathbb{Z}_q$ which is contained in its own challenge $\mathbf{u}_i$. Finally, $\mathcal{B}'$ invokes $\mathcal{A}(\text{pp}_{\text{IPHS}}, \text{pp}_{\text{coHash}}, \{\mathbf{c}_i^{(j)}, \boxed{h_i^{(j)}}\}_{i \in [T], j \in [Q]})$, and returns to its own challenger whatever $\mathcal{A}$ outputs.

Next, we analyze the advantage of $\mathcal{B}'$.

- Since each $\mathbf{V}_i \leftarrow \mathbb{Z}_q^{Q \times (l-l')}$, all $\mathbf{v}_i^{(j)}$ are independently and uniformly random over $\mathbb{Z}_q^{l-l'}$. Then all $\mathbf{c}_i^{(j)} := \mathbf{A}\mathbf{w}_i^{(j)} + \mathbf{B}\mathbf{v}_i^{(j)} + \mathbf{e}_i^{(j)}$ with randomly chosen $\mathbf{w}_i^{(j)} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e}_i^{(j)} \leftarrow \chi_{\mathcal{L}}^l$ are independently distributed over $\mathcal{X}$, and follow the distribution $\mathbf{c}_i^{(j)} \leftarrow \mathcal{X}$.
- In the case that $\{\mathbf{u}_i = \mathbf{V}_i \mathbf{s}_i + \mathbf{e}_i\}_{i \in [T]}$, we have that $h_i^{(j)} := u_i^{(j)} = \mathbf{v}_i^{(j)\top} \cdot \mathbf{s}_i + e_i^{(j)}$ for all $i \in [T]$ and $j \in [Q]$. By setting $s^{(i)} := \mathbf{s}^{(i)}$, it follows that $h_i^{(j)} = \mathbf{v}_i^{(j)\top} \cdot \mathbf{s}_i + e_i^{(j)} = \Gamma_{s^{(i)}}(\mathbf{c}_i^{(j)})$, where all $s^{(i)} = \mathbf{s}^{(i)}$ ($i \in [T]$) are independently distributed over $\mathcal{SK}_{\text{co}} = \mathbb{Z}^{l-l'}$ and follow the distribution $\chi_s^{l-l'}$, and all $e_i^{(j)}$ ($i \in [T], j \in [Q]$) are independently distributed according to χ_s due to the randomness of $\mathbf{e}_i \leftarrow \chi_s^Q$.
- In the case that $\{\mathbf{u}_i \leftarrow \mathbb{Z}_q^Q\}_{i \in [T]}$, all $h_i^{(j)} := u_i^{(j)}$ are independently and uniformly distributed over $\mathfrak{G}_{\text{co}} = \mathbb{Z}_q$.

Overall, $\mathcal{B}'$ can successfully distinguish $\{\mathbf{u}_i = \mathbf{V}_i \mathbf{s}_i + \mathbf{e}_i\}_{i \in [T]}$ from $\{\mathbf{u}_i \leftarrow \mathbb{Z}_q^Q\}_{i \in [T]}$

as long as $\mathcal{A}$ distinguishes $\{\mathbf{c}_i^{(j)} \leftarrow \mathcal{X}, \boxed{h_i^{(j)} = \Gamma_{s^{(i)}}(\mathbf{c}_i^{(j)})}\}_{i \in [T], j \in [Q]}$ from $\{\mathbf{c}_i^{(j)} \leftarrow \mathcal{X},$

$\boxed{h_i^{(j)} \leftarrow \mathfrak{G}_{\text{co}}}\}_{i \in [T], j \in [Q]}$, and thus we get $\text{Adv}_{\text{TLHPS}_{\text{LWE}}, \mathcal{A}, T, Q}^{\text{mk-mext}}(\lambda) \leq \text{Adv}_{[l-l', q, \chi_s, Q], \mathcal{B}'}^{T\text{-inst-LWE-noi-secr}}(\lambda)$.

This shows that (46) holds as desired, and completes the proof of Theorem 4. $\square$

Table of Contents

1	Introduction	2
1.1	Our Contributions	4
2	Technical Overview	7
2.1	Recap: The ALS Scheme and The Tomida Scheme	7
2.2	Technique 1: Two-Leveled Inner-Product Hash Proof System	8
2.3	Technique 2: Compact IPFE Design & Economic Proof Strategy	12
3	Preliminaries	17
4	Two-Leveled Inner-Product Hash Proof System	18
5	Tightly Secure IPFE from Two-Leveled IP-HPS	21
6	Instantiations (based on LWE)	23
6.1	Instantiation of Two-Leveled IP-HPS from LWE	24
6.2	Instantiation of IPFE from LWE	27
A	Comparison of Ciphertext Size: An Example	35
B	Comparison with Existing Frameworks from HPS & Discussions on Simulation-based Security of IPFE	37
C	Proof of Theorem 1 (Tight $\mathbf{mCPA}$ Security of $\mathbf{IPFE[TLHPS, L]}$)	39
C.1	Proof of Lemma 3 ($\mathbf{G}_\zeta \rightarrow \mathbf{G}_{\zeta+1}$)	42
C.2	Proof of Claim 1 ($\mathbf{G}_\zeta \rightarrow \mathbf{G}_{\zeta,0}$)	46
C.3	Proof of Claim 2 ($\mathbf{G}_{\zeta,\eta} \rightarrow \mathbf{G}_{\zeta,\eta+1}$)	47
C.4	Proof of Claim 3 ($\mathbf{G}_{\zeta,\tilde{m}} \rightarrow \mathbf{G}_{\zeta,\tilde{m}+1}$)	49
D	Extending Tight Security to The Multi-User Setting	51
E	Postponed MDDH-based Instantiation	52
E.1	MDDH Assumptions	52
E.2	Instantiation of Two-Leveled IP-HPS from MDDH	54
E.3	Proof of Theorem 6 (Functional Smoothness of $\mathbf{TLHPS}_{\mathbf{MDDH}}$)	55
E.4	Proof of Theorem 7 (Tight Multi-Key-Extracting of $\mathbf{TLHPS}_{\mathbf{MDDH}}$)	58
E.5	Instantiation of IPFE from MDDH	59
E.6	Concrete DDH and k -LIN-Based IPFE Schemes	61
F	Postponed DCR-based Instantiation	63
F.1	DCR Assumption and Lattice Backgrounds	64
F.2	Instantiation of Two-Leveled IP-HPS from DCR	66
F.3	Proof of Theorem 9 (Functional Smoothness of $\mathbf{TLHPS}_{\mathbf{DCR}}$)	68
F.4	Proof of Theorem 10 (Tight Multi-Key-Extracting of $\mathbf{TLHPS}_{\mathbf{DCR}}$)	72
F.5	Instantiation of IPFE from DCR	73
G	Missing Details in Sect. 6 (LWE-based Instantiation)	75
G.1	Lattice Preliminaries	75
G.2	LWE, Multi-secret LWE, Multi-instance LWE Assumptions, and (Almost) Tight Reductions among Them	78
G.3	Probabilistic Two-Leveled IP-HPS: Definitions	84
G.4	Proof of Theorem 3 (Functional Smoothness of $\mathbf{TLHPS}_{\mathbf{LWE}}$)	85
G.5	Proof of Theorem 4 (Tight Multi-Key-Extracting of $\mathbf{TLHPS}_{\mathbf{LWE}}$)	90